

# Machine

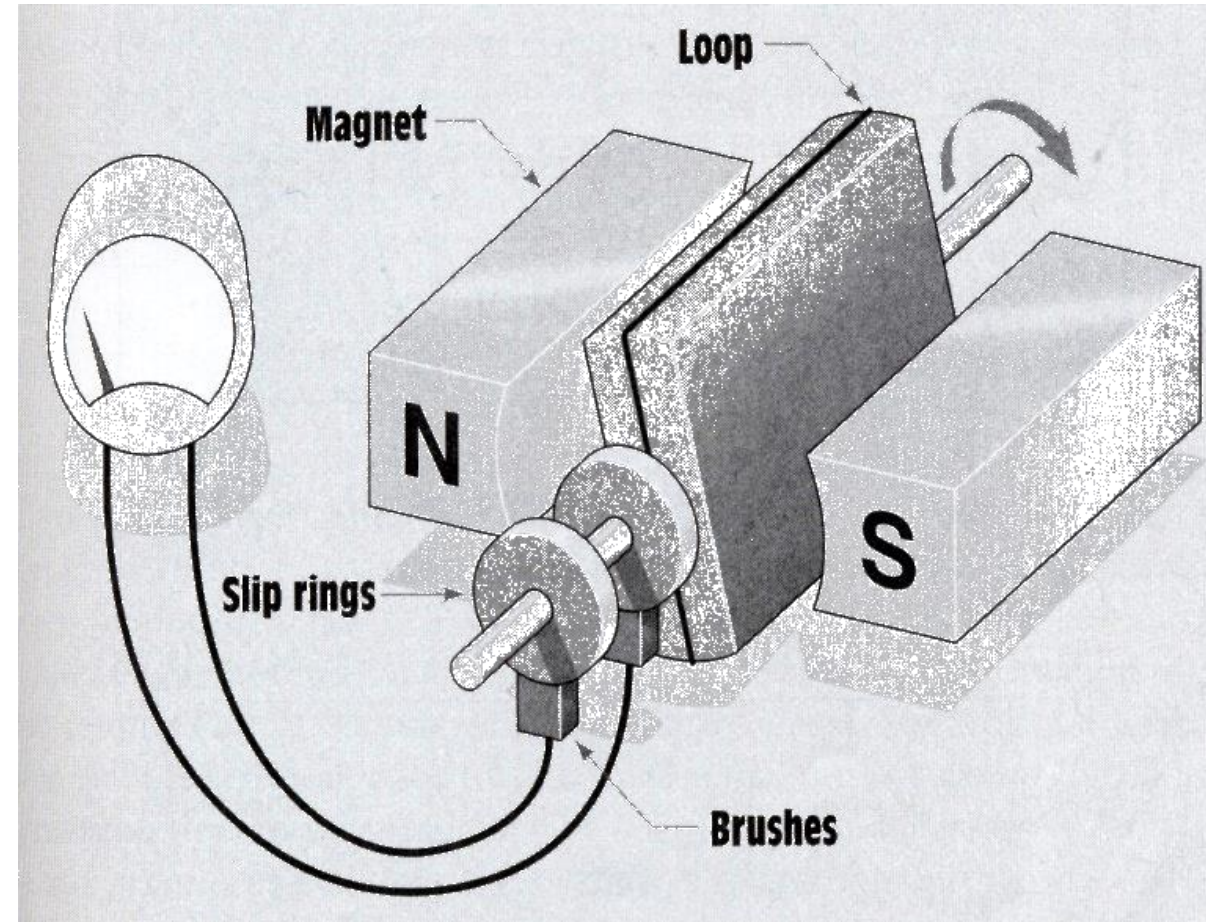
DC Machine

# Overview of Direct Current Machines

- Direct-current (DC) machines are divided into dc generators and dc motors.
- Most DC machines are similar to AC machines: i.e. they have AC voltages and current within them.
- DC machines have DC outputs just because they have a mechanism converting AC voltages to DC voltages at their terminals.
- This mechanism is called a **commutator**; therefore, DC machines are also called **commutating machines**.
- DC generators **are not as common as they used to be**, because direct current, when required, is mainly produced by **electronic rectifiers**.
- While dc motors are widely used, such automobile, aircraft, and portable electronics, in speed control applications...

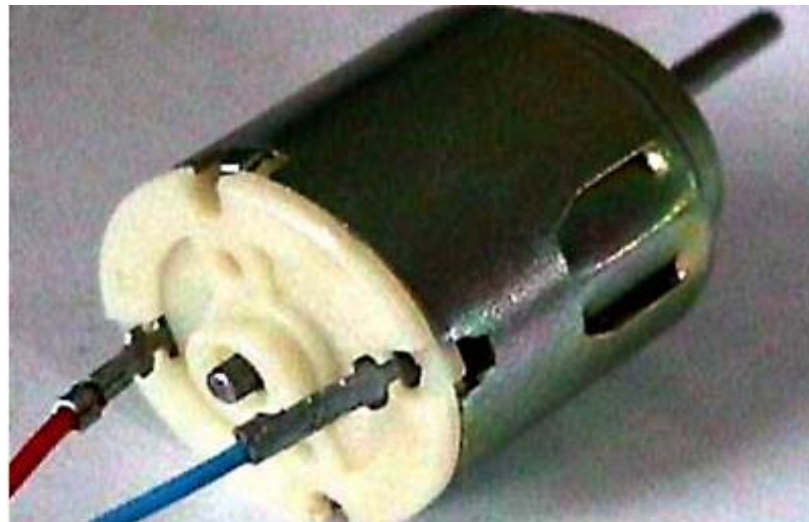
# DC Generator

- A dc generator is a machine that converts **mechanical energy into electrical energy (dc voltage and current)** by using the principle of magnetic induction.
- In this example, the ends of the **wire loop have been connected to two slip rings** mounted on the shaft, while **brushes** are used to carry the current from the loop to the outside of the circuit.

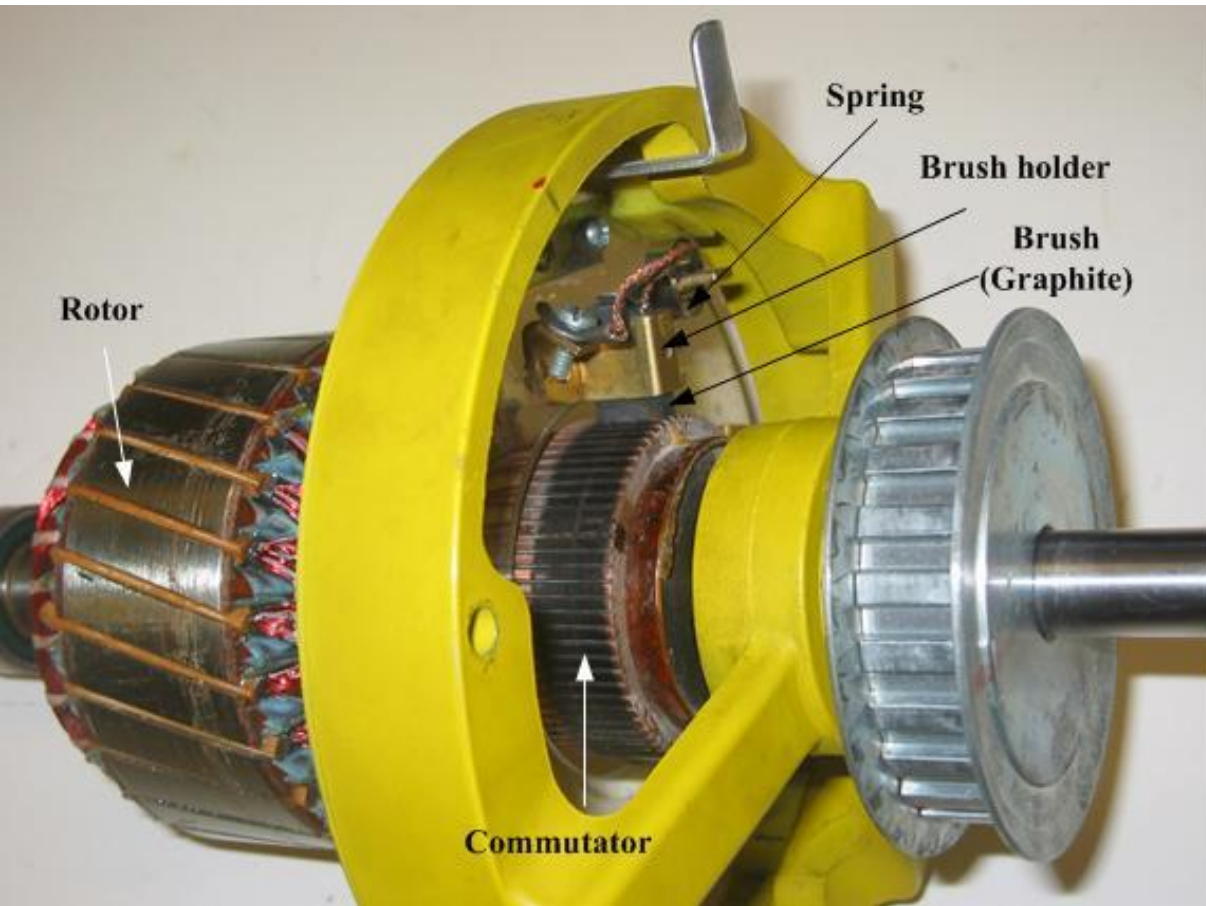


# DC Motor

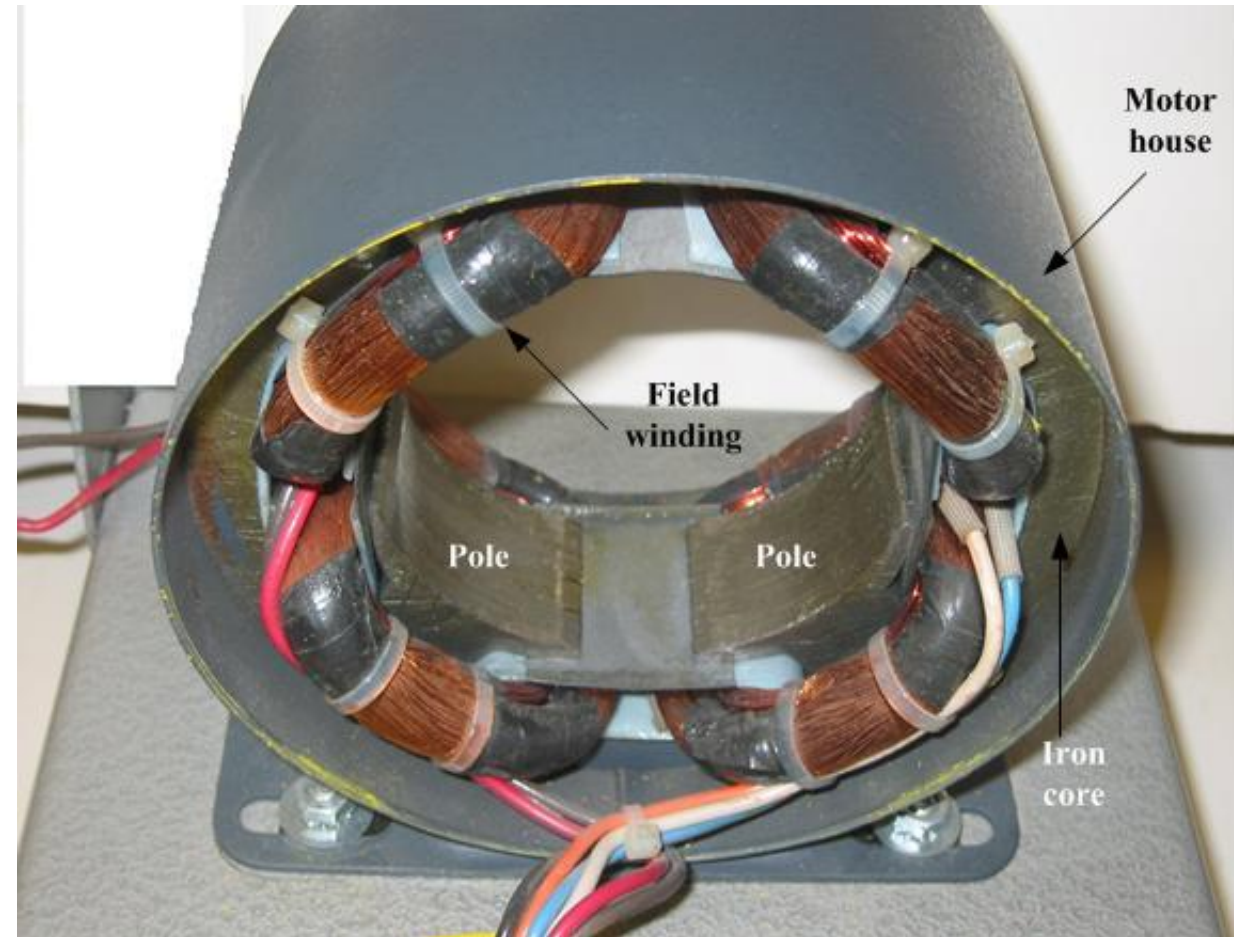
- DC motors are everywhere! In a house, almost every mechanical movement that you see around you is caused by an DC (direct current) motor.
- An dc motor is a machine that converts electrical energy into mechanical energy by supplying a dc power (voltage and current).
- An advantage of DC motors is that it is easy to control their speed in a wide Range.



# Construction of DC Machine

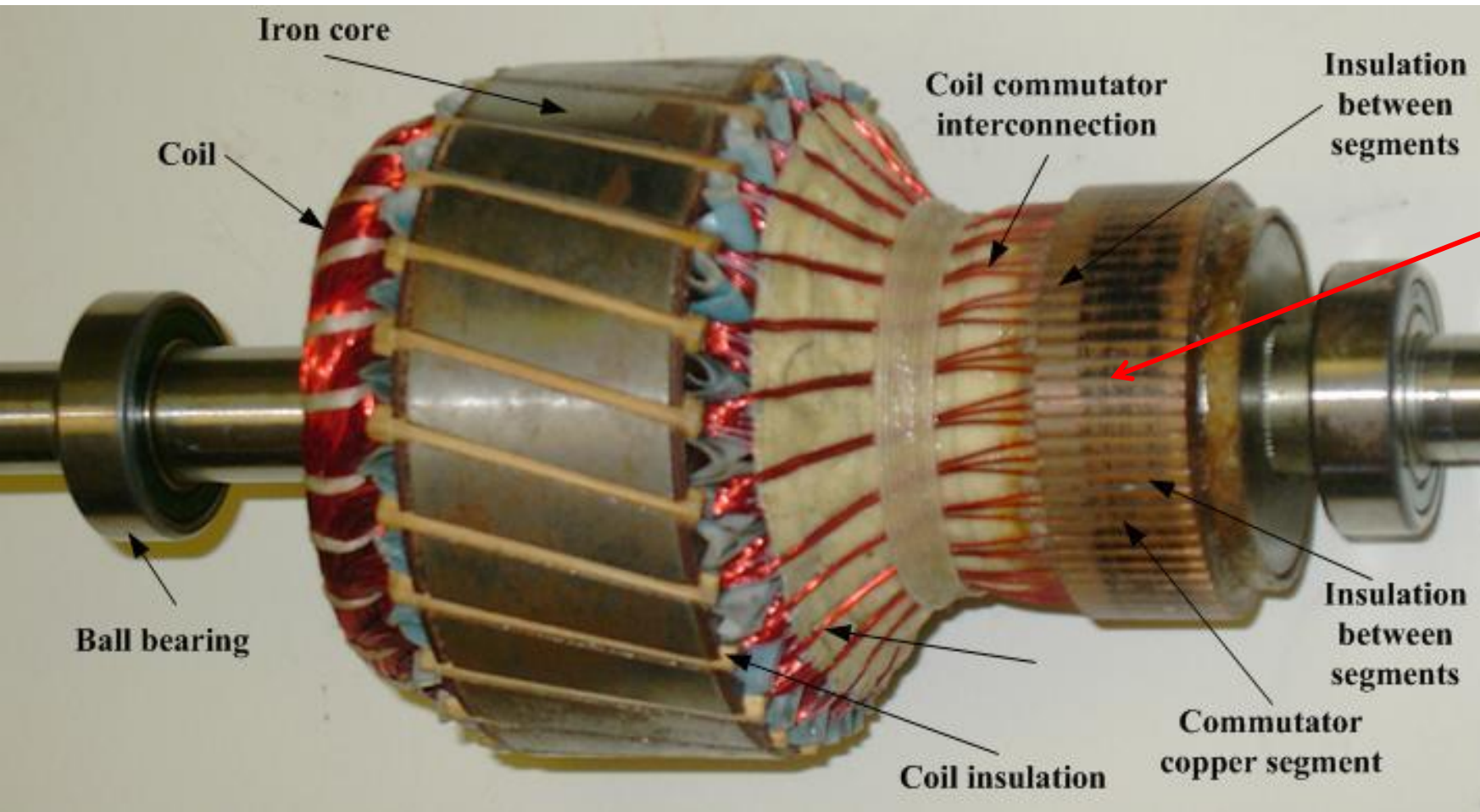


Cutaway view of a dc motor



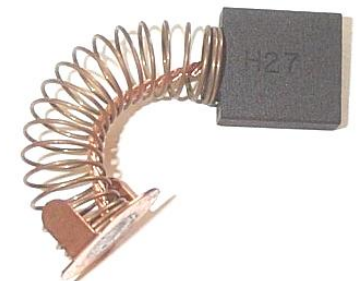
Stator with poles visible

# Construction of DC Machine

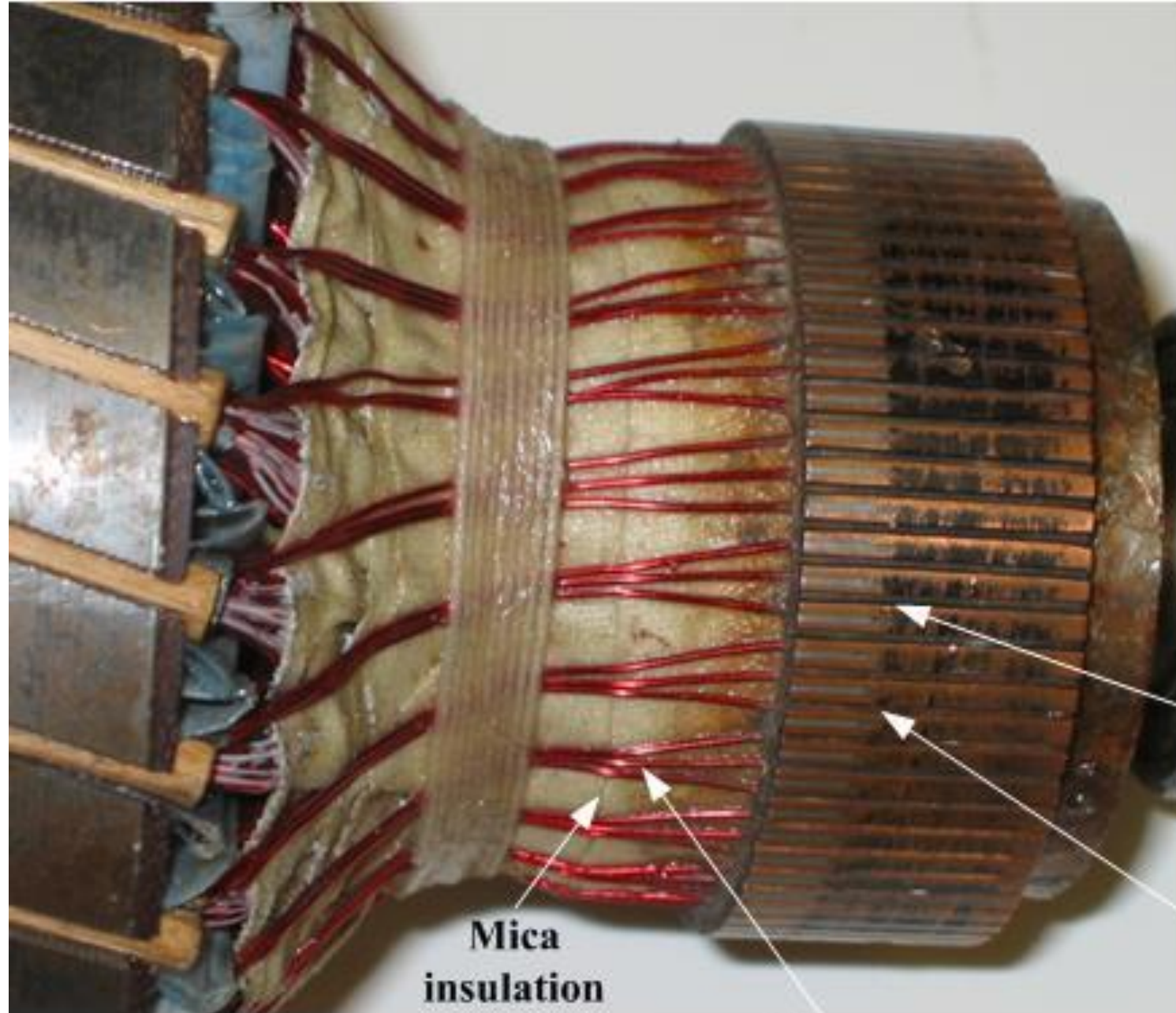


**Rotor of a dc motor.**

segments



brushes



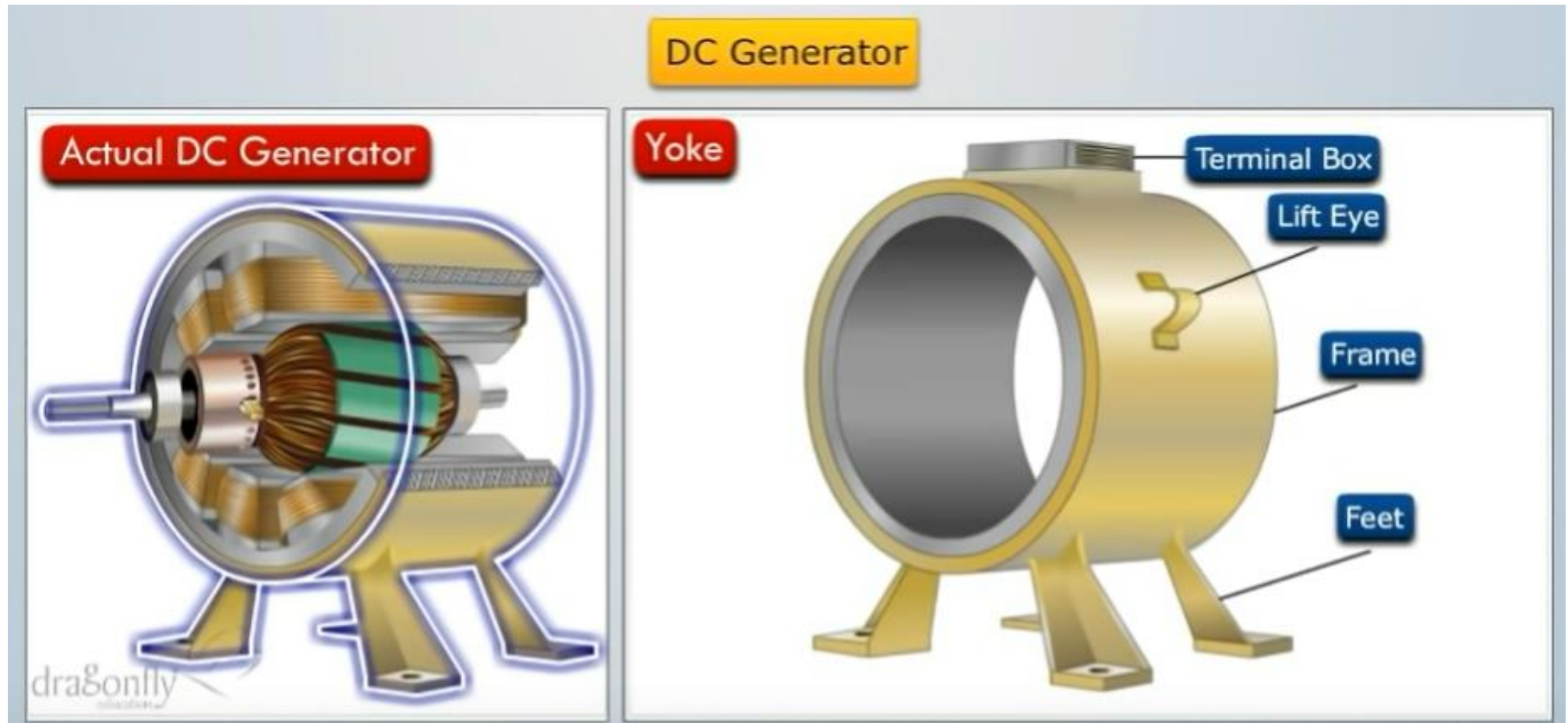
**Mica  
insulation**

**Copper  
conductors**

**Mica Insulation  
between segments**

**Copper  
segment**

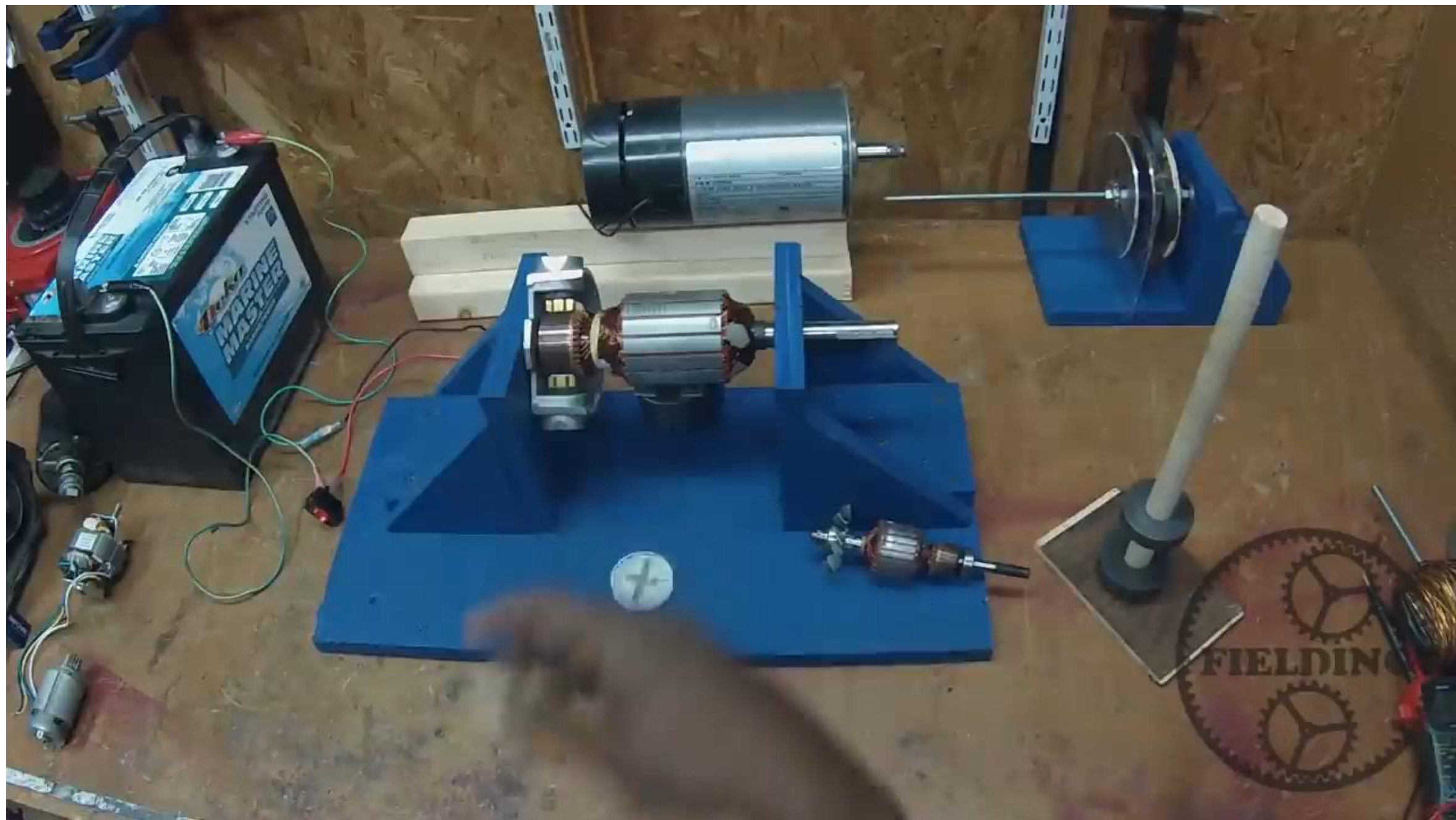
# DC Generator



Yoke

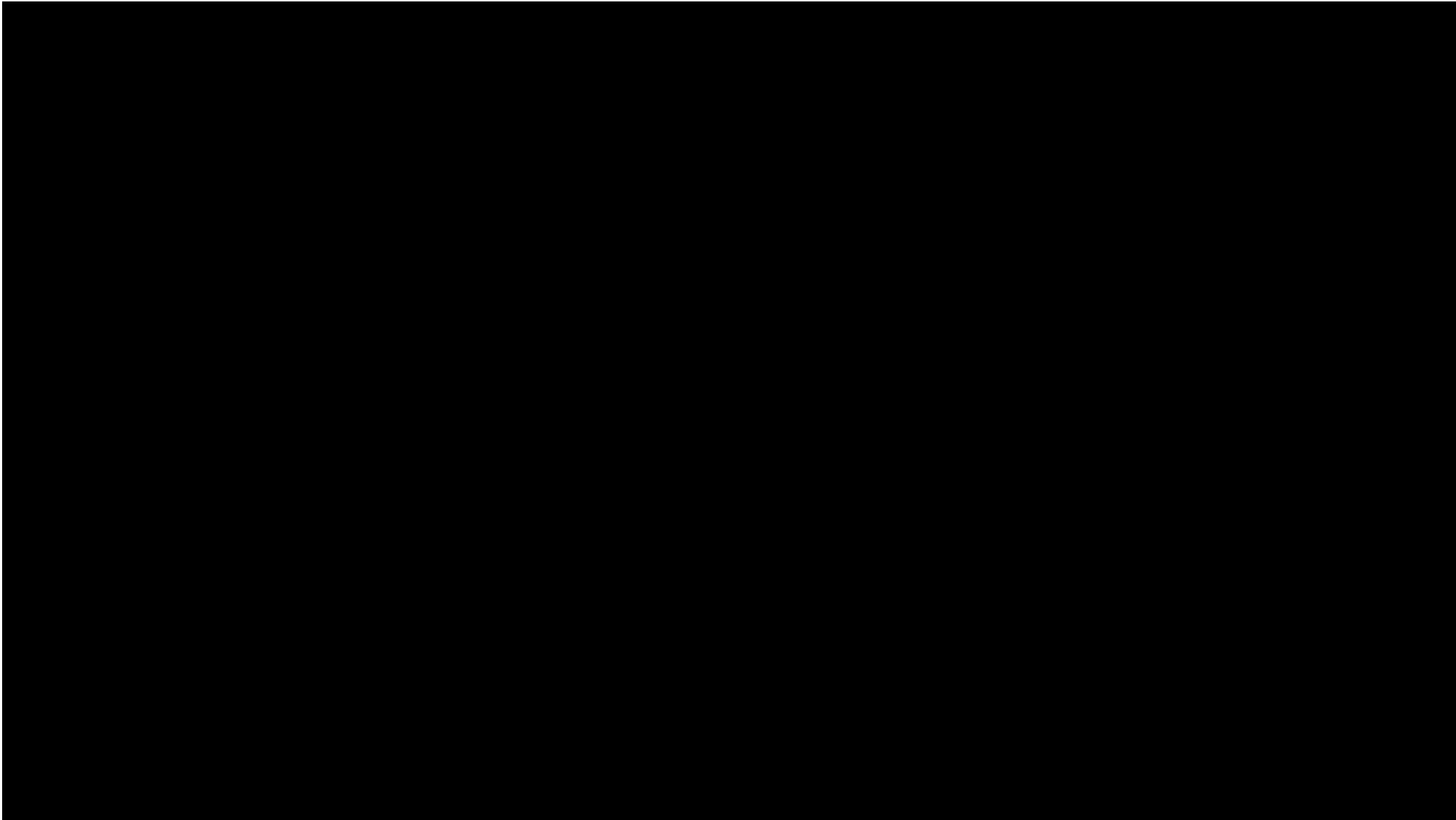


- <https://www.youtube.com/watch?v=uQq8yWfqIDM>
- <https://www.youtube.com/watch?v=T8wgX0RrQyk>
- <https://www.youtube.com/watch?v=hFPZDd-5SJA>





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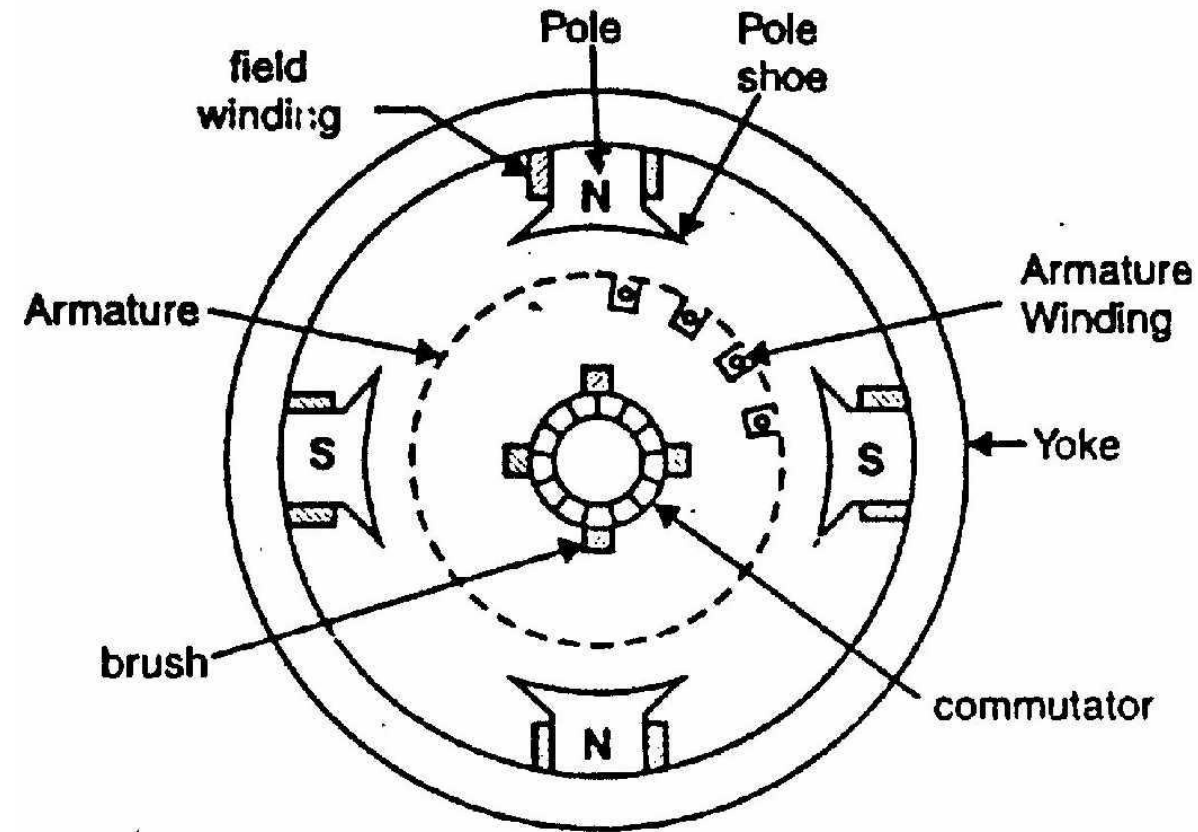
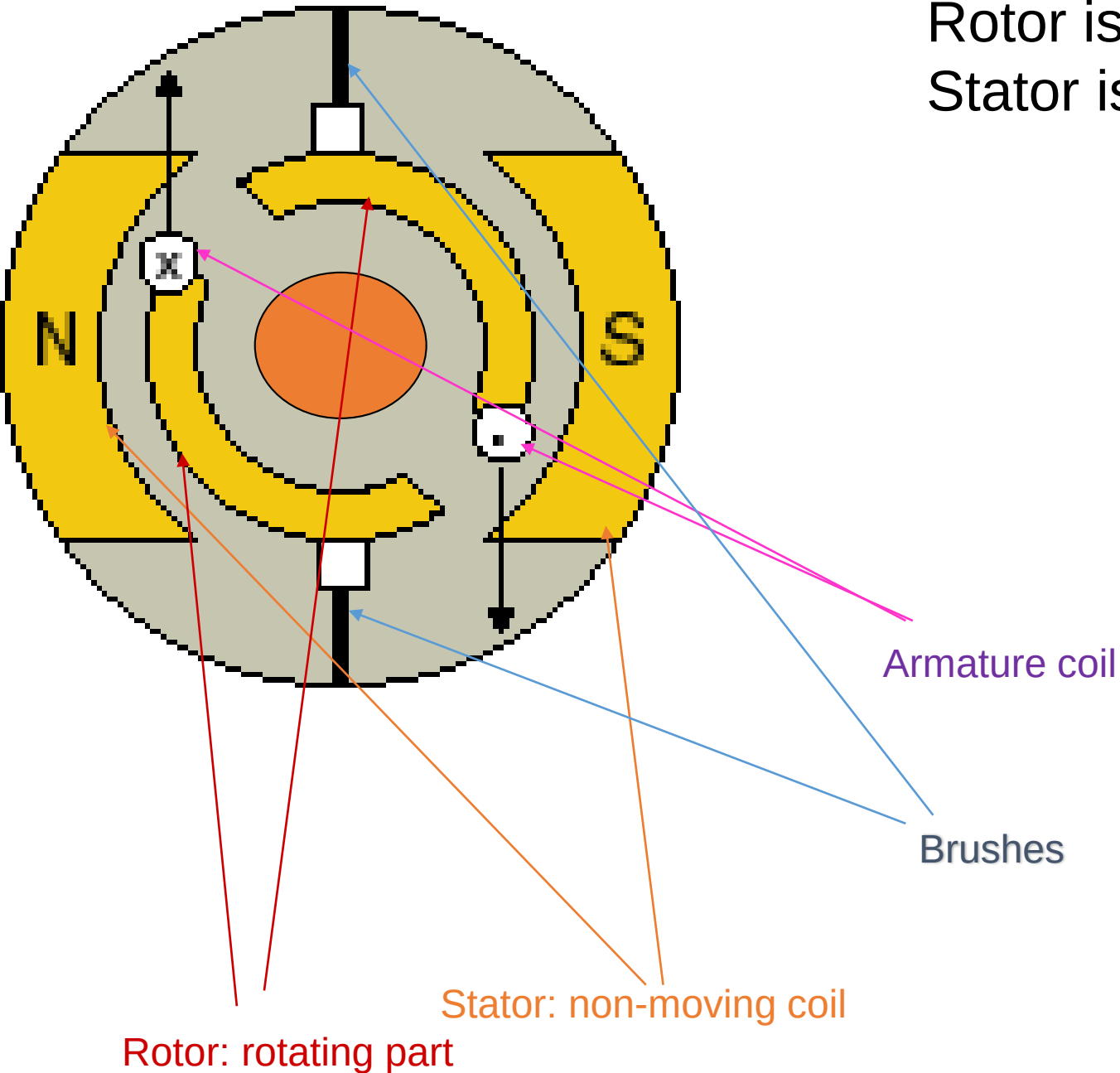


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# Construction of DC Machine (Generator)

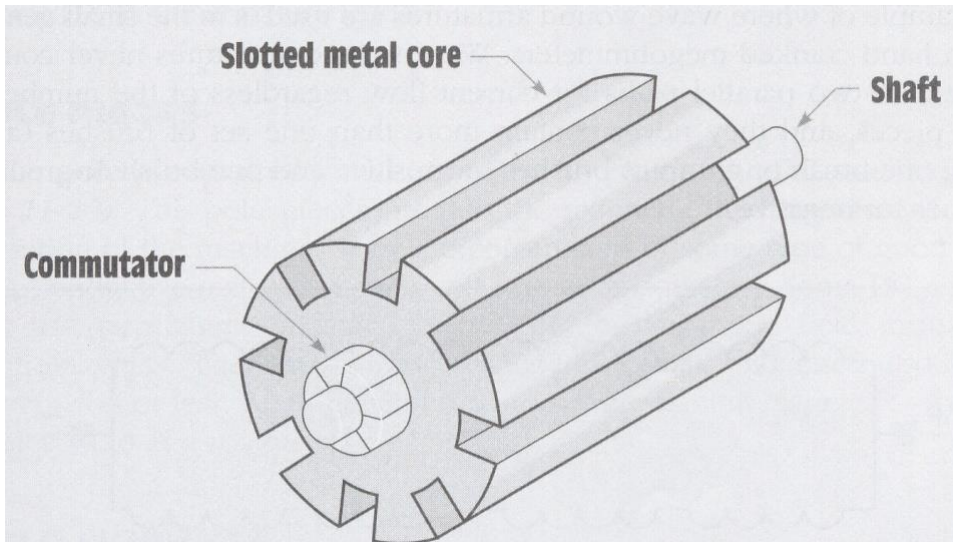
Rotor is the rotating part - armature

Stator is the stationary part - field

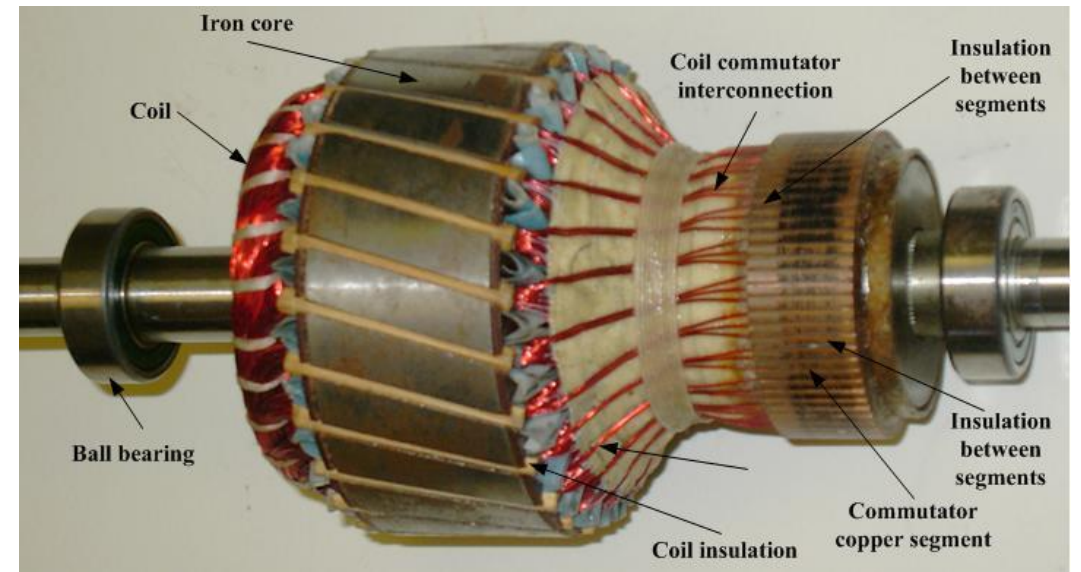


# ARMATURE

- More loops of wire = higher rectified voltage
- In practical, loops are generally placed in slots of an iron core
- The iron acts as a magnetic conductor by providing a low-reluctance path for magnetic lines of flux to increase the inductance of the loops and provide a higher induced voltage.
- The commutator is connected to the slotted iron core.
- The entire assembly of iron core, commutator, and windings is called the armature.
- The windings of armatures are connected in different ways depending on the requirements of the machine.



Loops of wire are wound around slot in a metal core

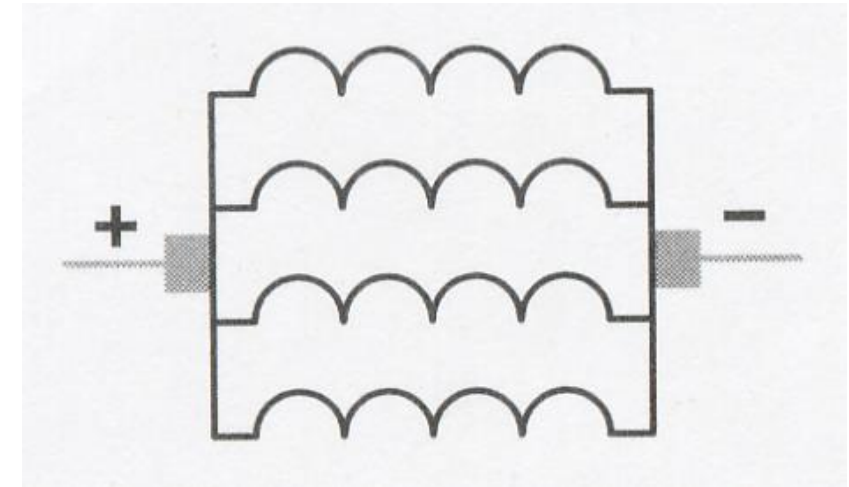
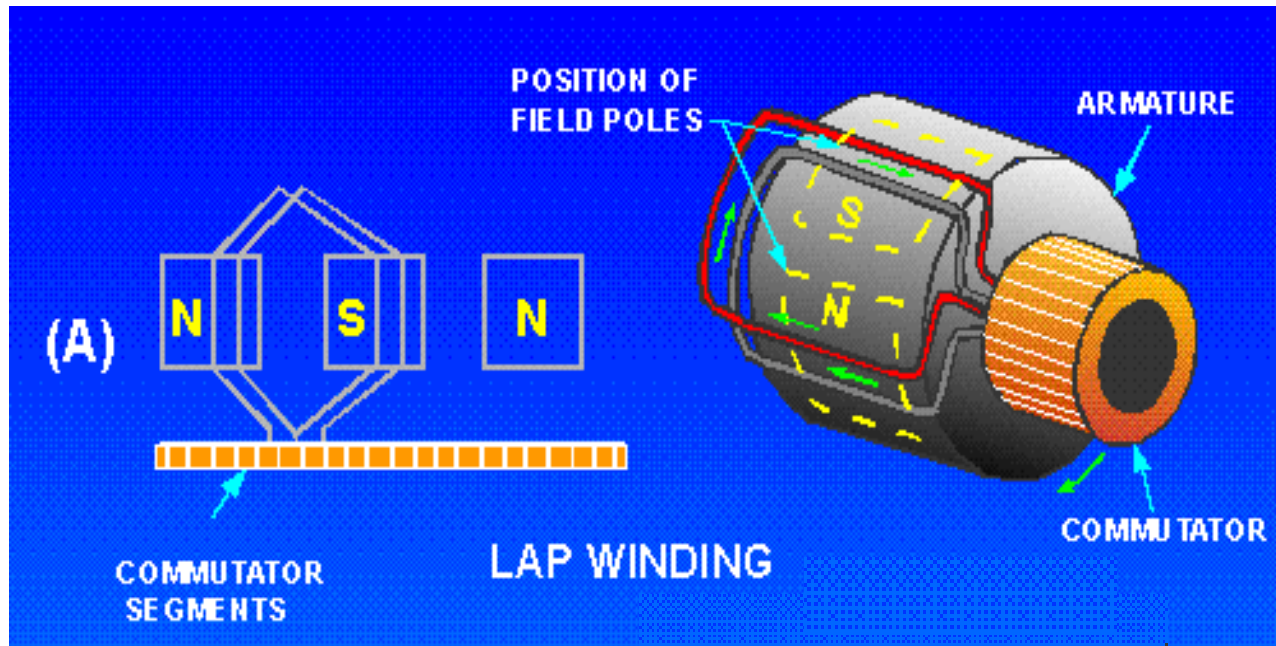


DC machine armature

# ARMATURE WINDINGS

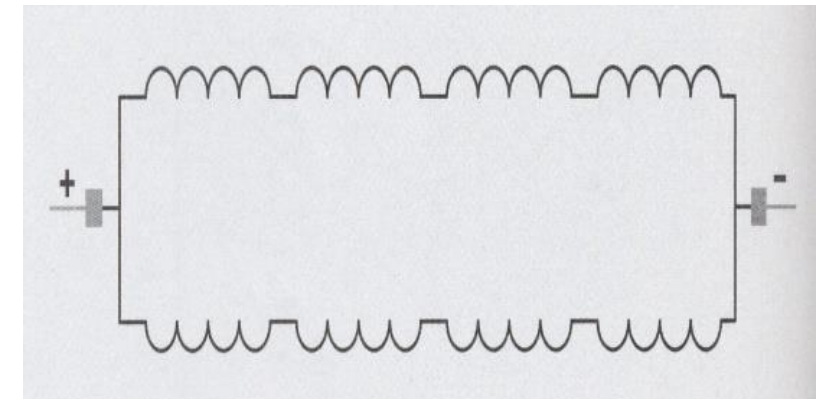
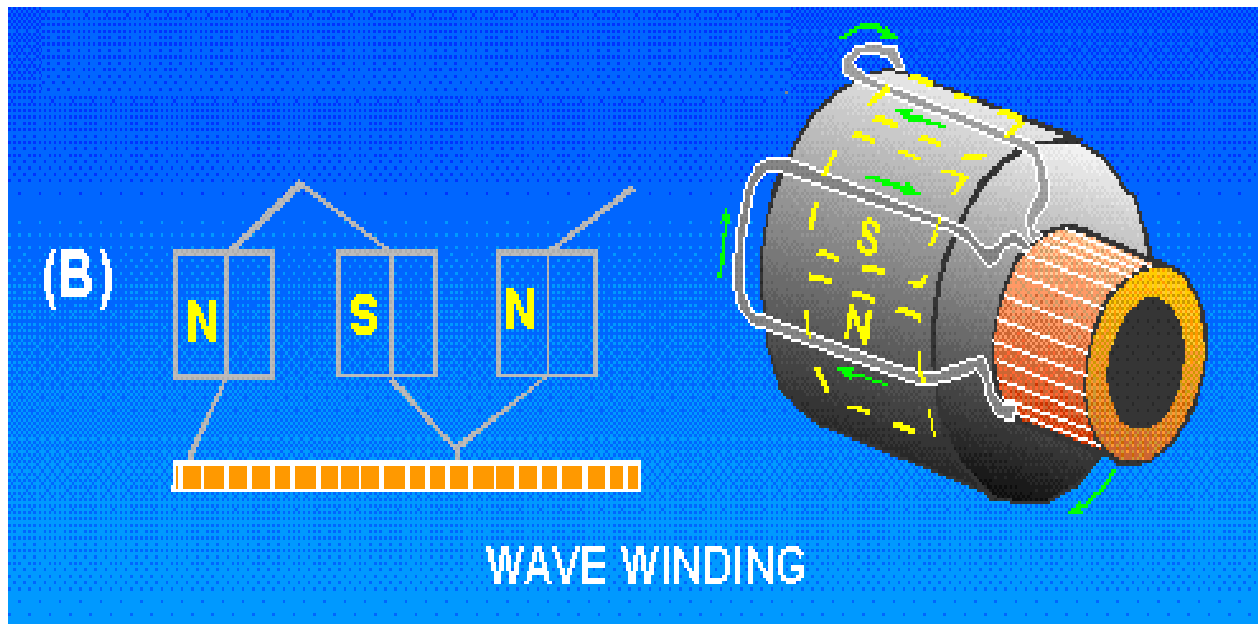
- **Lap Wound Armatures**

- are used in machines designed for low voltage and high current
- armatures are constructed with large wire because of high current
- Eg: - are used in the starter motor of almost all automobiles
- The windings of a lap wound armature are connected in parallel. This permits the current capacity of each winding to be added and provides a higher operating current
- No. of the parallel path,  $A=P$  (or equal to the number of brushes)



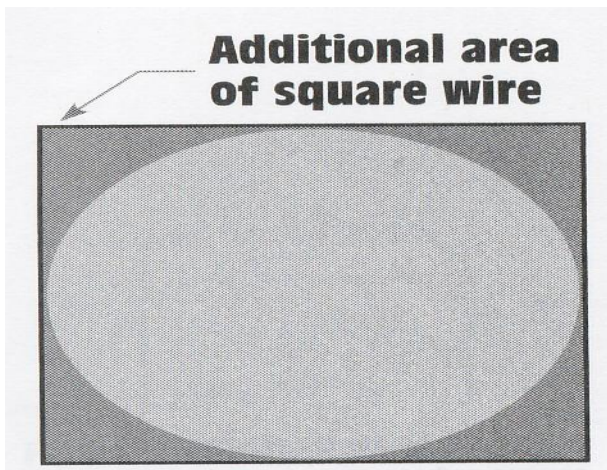
# ARMATURE WINDINGS

- Wave Wound Armatures
  - are used in machines designed for high voltage and low current
  - their windings connected in series
  - When the windings are connected in series, the voltage of each winding adds, but the current capacity remains the same
  - are used is in the small generator in hand-cranked megahmmeters
  - No. of the parallel path,  $A=2$

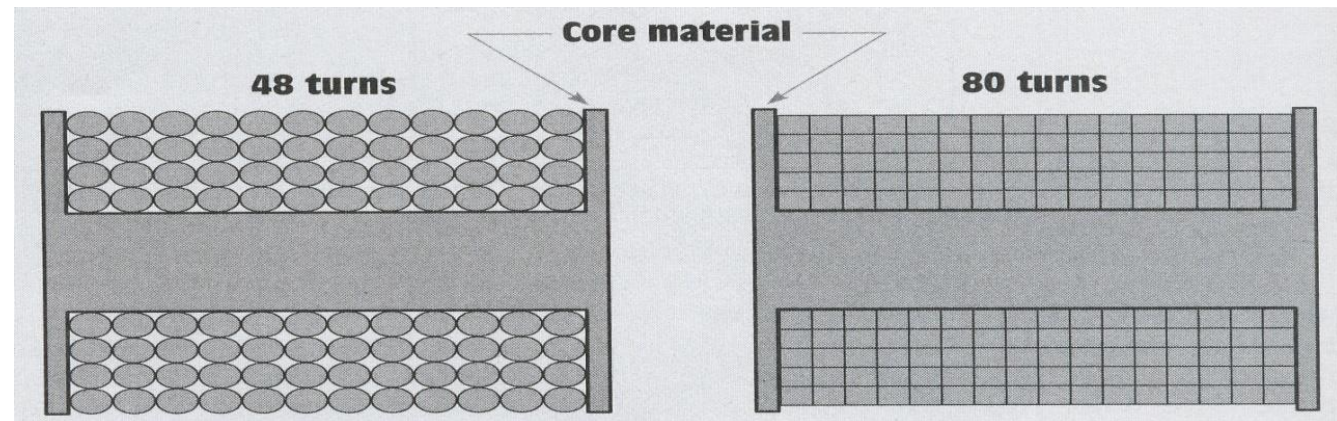


# FIELD WINDINGS

- Most DC machines use wound electromagnets to provide the magnetic field.
- **Series field windings**
  - are **so named** because they are **connected in series with the armature**
  - are **made with relatively few windings turns of very large wire and have a very low resistance**
  - usually **found in large horsepower machines wound with square or rectangular wire.**
  - The use of square wire **permits the windings to be laid closer together, which increases the number of turns that can be wound in a particular space**
  - Square and rectangular wire can also be made physically smaller than round wire and still contain the same surface area



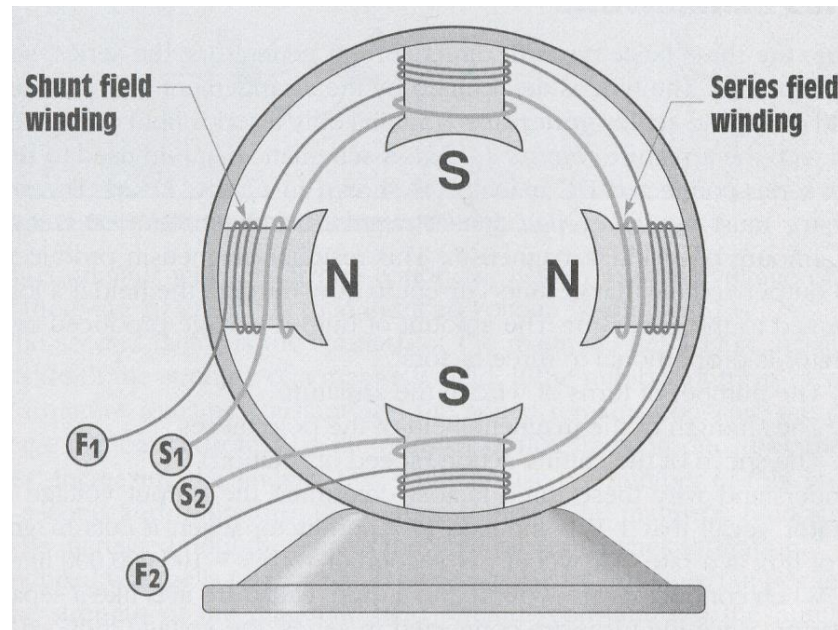
Square wire contains more surface than round wire



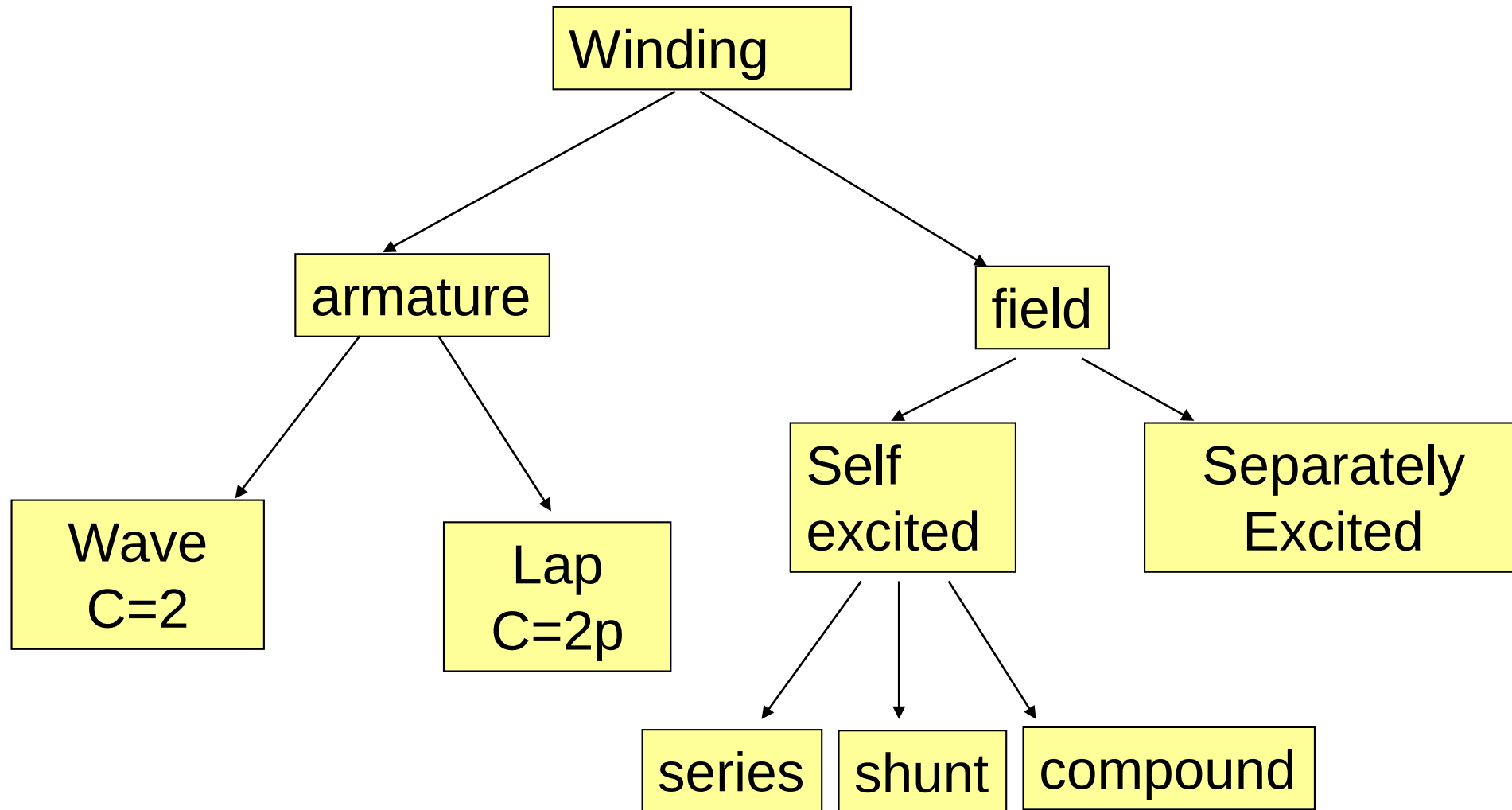
Square wire permits more turns than round wire in the same area

# FIELD WINDINGS

- **Shunt field windings**
  - is constructed with relatively many turns of small wire, thus, it has a much higher resistance than the series field.
  - is intended to be connected in parallel with, or shunt, the armature.
  - high resistance is used to limit current flow through the field.
  - When a DC machine uses both series and shunt fields, each pole piece will contain both windings.
  - The windings are wound on the pole pieces in such a manner that when current flows through the winding it will produce alternate magnetic polarities.

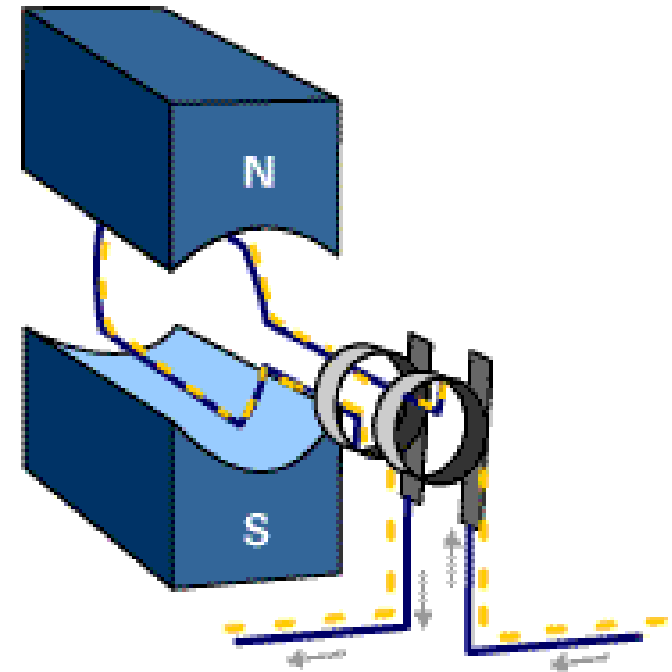
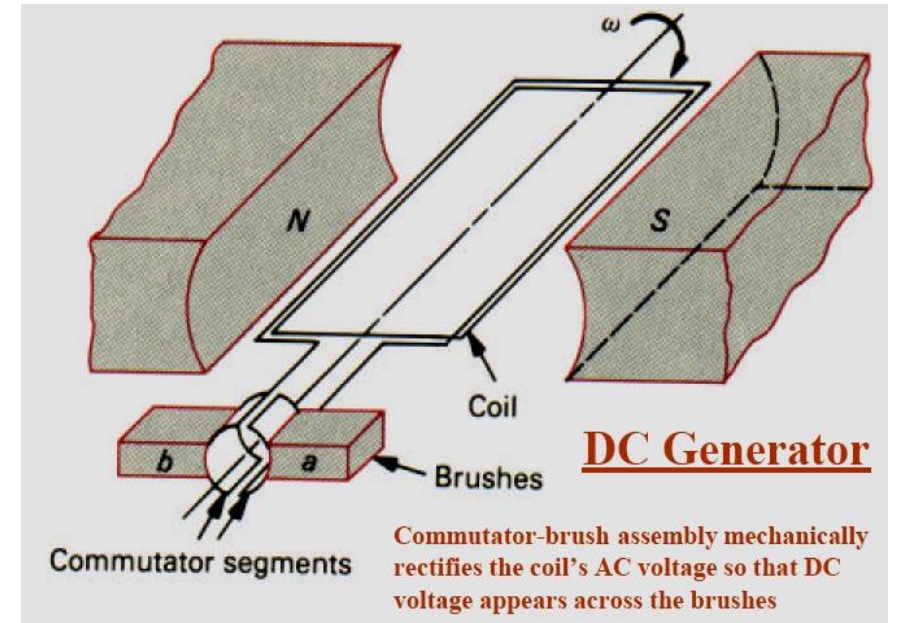


# MACHINE WINDINGS OVERVIEW



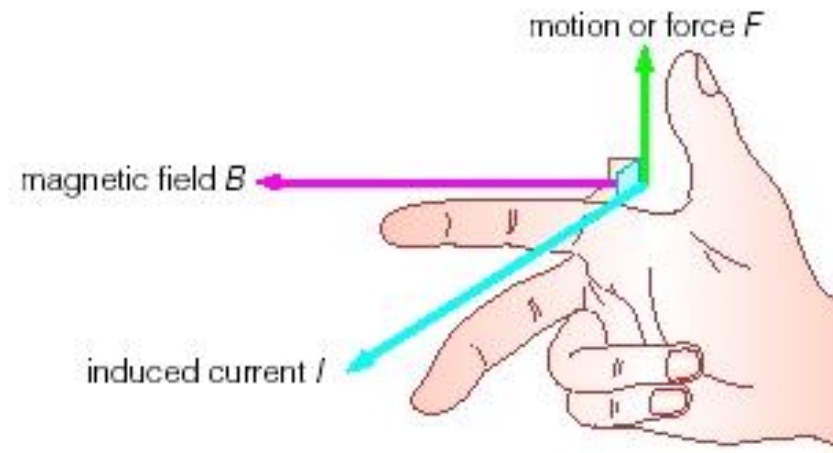
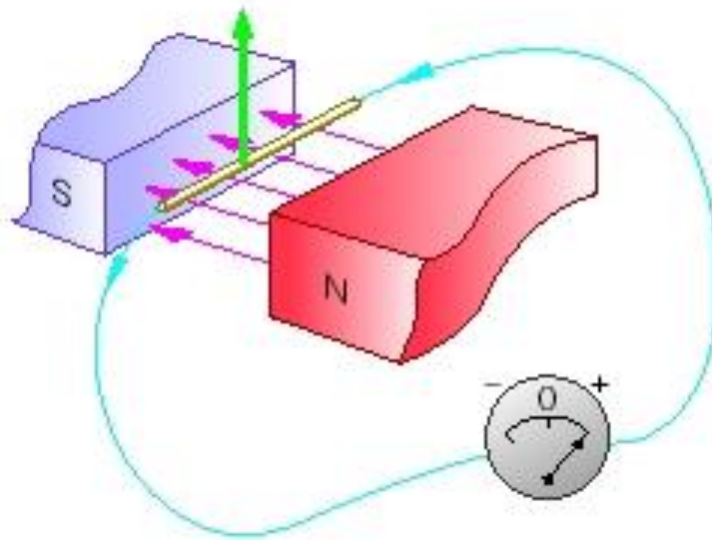
# Principle operation of Generator

- Whenever a conductor is moved within a magnetic field in such a way that the conductor cuts across magnetic lines of flux, voltage is generated in the conductor.
- The AMOUNT of voltage generated depends on:
  - i. the strength of the magnetic field,
  - ii. the angle at which the conductor cuts the magnetic field,
  - iii. the speed at which the conductor is moved, and
  - iv. the length of the conductor within the magnetic field



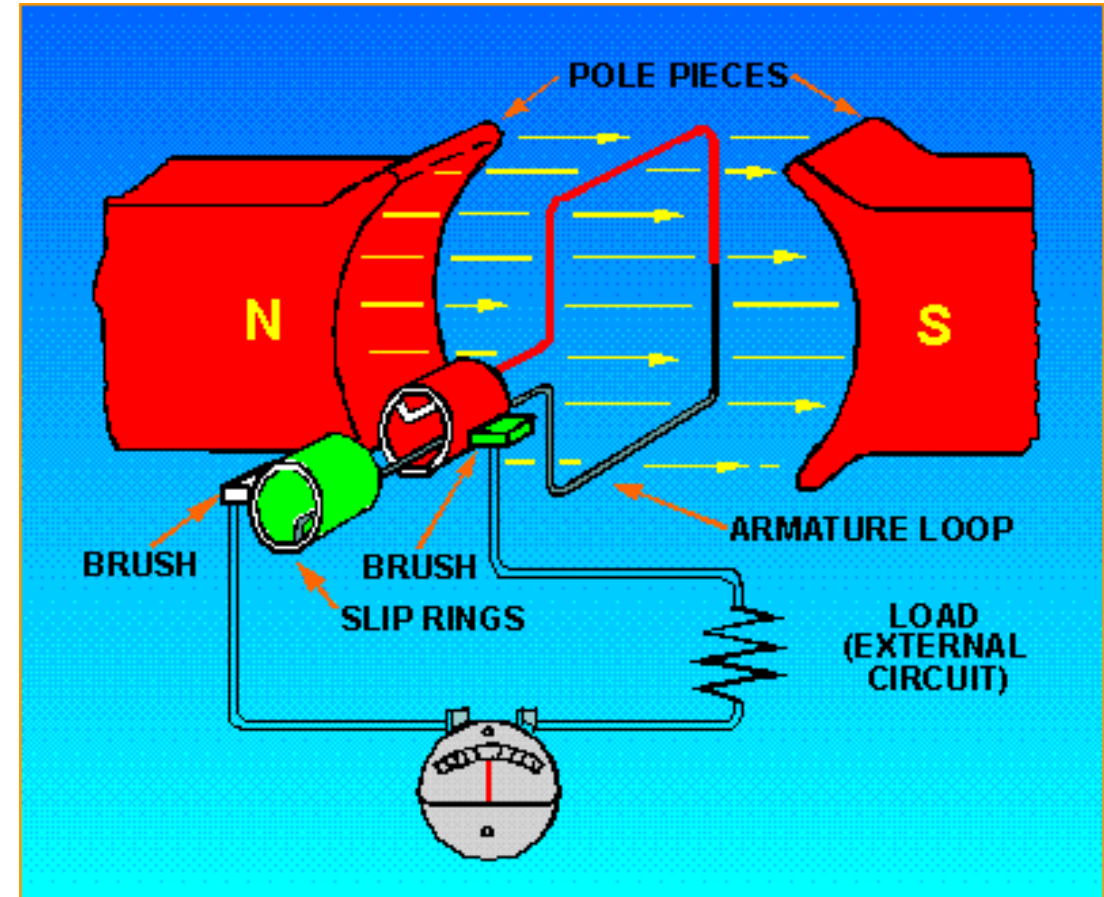
# Fleming's Right hand rule (Generator Rule)

- Use: To determine the direction of the induced emf/current of a conductor moving in a magnetic field.
- The POLARITY of the voltage depends on the direction of the magnetic lines of flux and the direction of movement of the conductor.



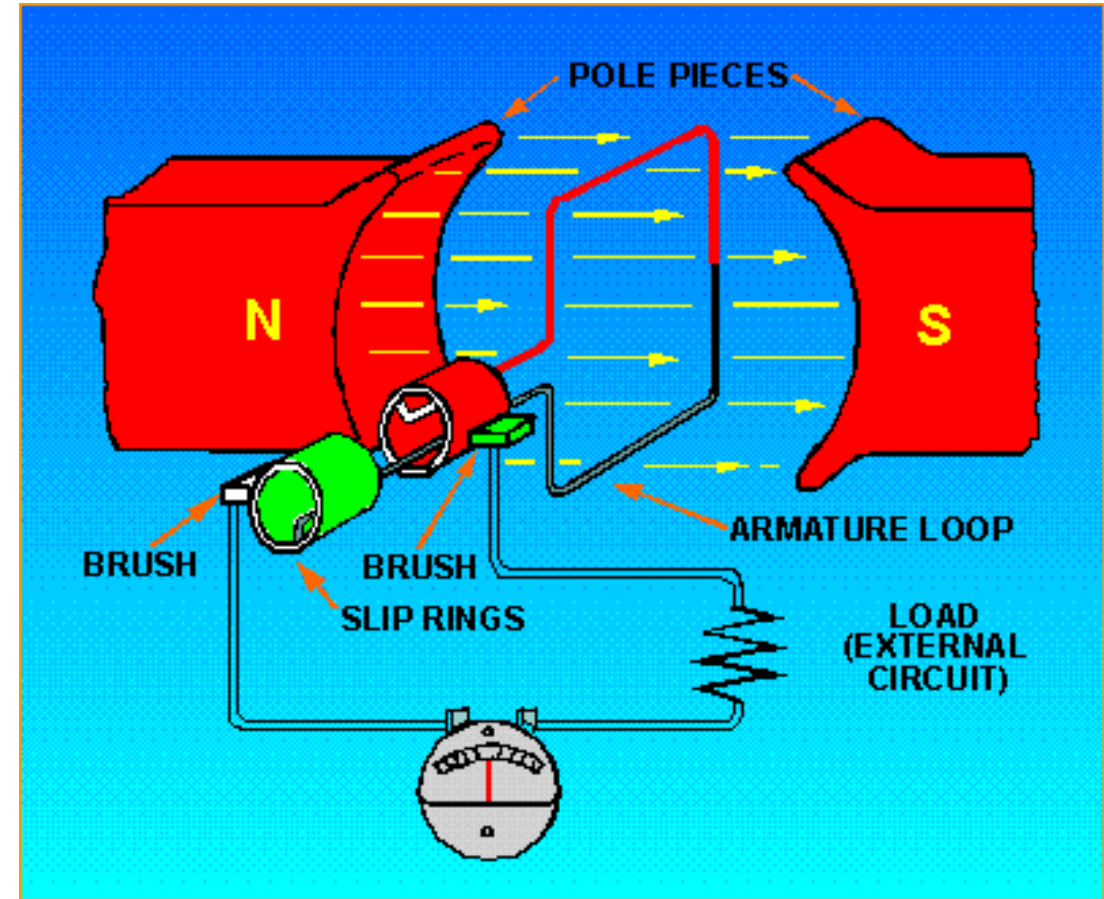
# THE ELEMENTARY GENERATOR

- The simplest elementary generator that can be built is an ac generator.
- Basic generating principles are most **easily explained** through the use of the elementary ac generator.
- For this reason, the ac generator will be discussed first. The dc generator will be discussed later.
- An elementary generator consists of a wire loop mounted on the shaft, so that it can be rotated in a stationary magnetic field.
- This will produce an induced emf in the loop.
- Sliding contacts (brushes) connect the loop to an external circuit load in order to pick up or use the induced emf.



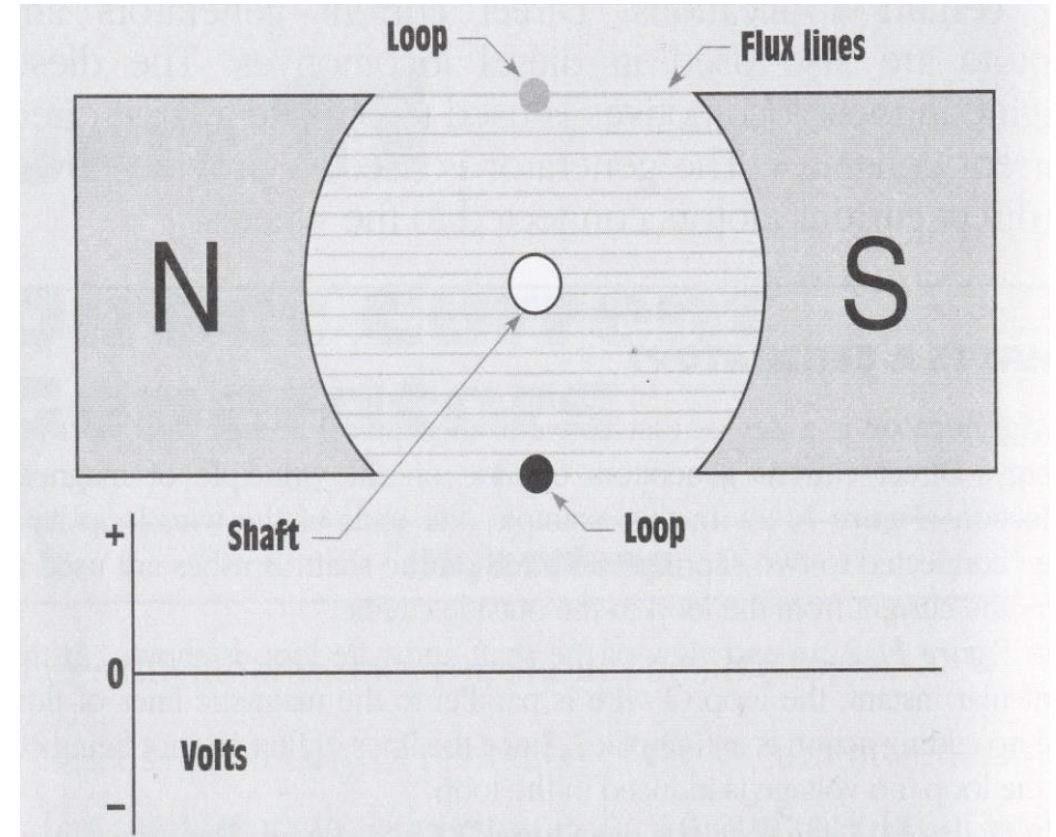
# THE ELEMENTARY GENERATOR

- The pole pieces (marked N and S) **provide the magnetic field**. The pole pieces are shaped and positioned as shown **to concentrate the magnetic field as close as possible to the wire loop**.
- The loop of wire that rotates through the field is called the **ARMATURE**. The ends of the armature loop are **connected to rings called SLIP RINGS**. They rotate with the armature.
- The brushes, usually made of carbon, with wires attached to them, ride against the rings. **The generated voltage appears across these brushes**. (These brushes transfer power from the battery to the commutator as the motor spins – discussed later in dc elementary generator).



# THE ELEMENTARY GENERATOR

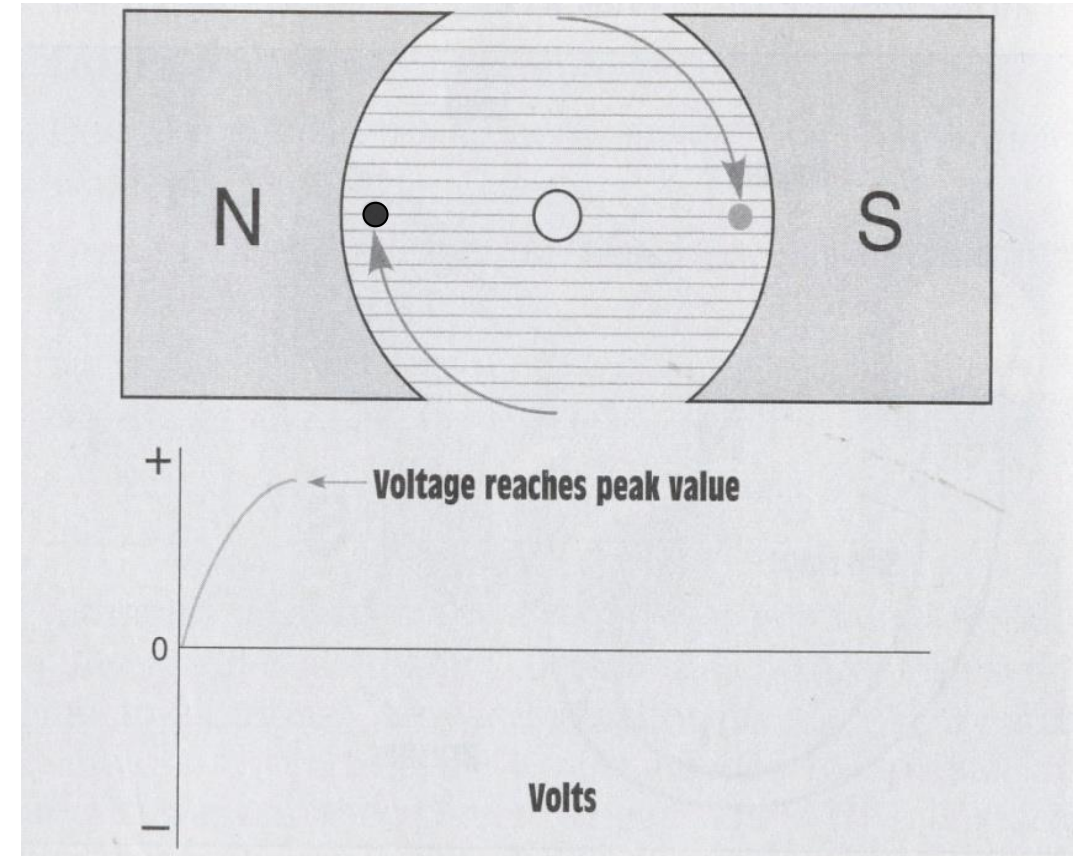
- An end view of the shaft and wire loop is shown.
- At this particular instant, the loop of wire (the black and white conductors of the loop) is parallel to the magnetic lines of flux, and no cutting action is taking place.
- Since the lines of flux are not being cut by the loop, no emf is induced in the conductors, and the meter at this position indicates zero.
- This position is called the **NEUTRAL PLANE**.



0° Position (Neutral Plane)

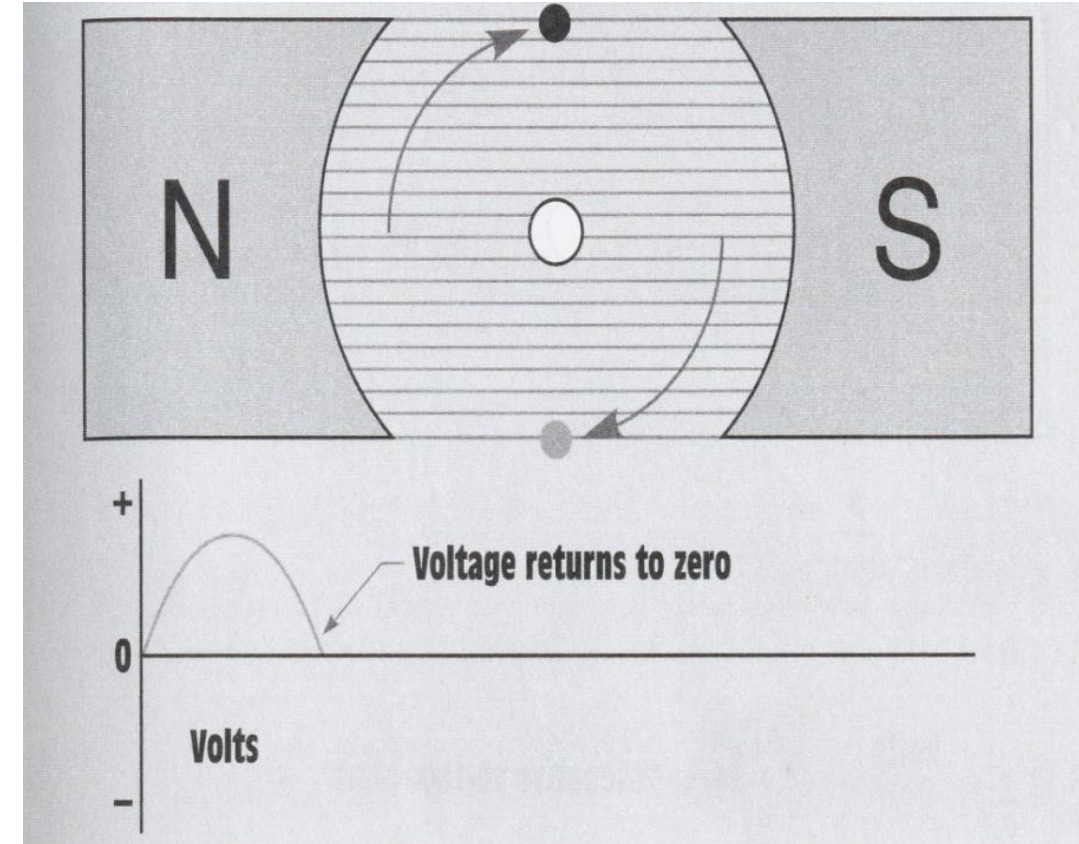
# THE ELEMENTARY GENERATOR

- The shaft has been turned  $90^\circ$  clockwise, the conductors cut through more and more lines of flux, and voltage is induced in the conductor.
- at a continually increasing angle, the induced emf in the conductors builds up from zero to a maximum value or peak value.
- Observe that from  $0^\circ$  to  $90^\circ$ , the black conductor cuts DOWN through the field.
- At the same time the white conductor cuts UP through the field.
- The induced emfs in the conductors are series-adding.
- This means the resultant voltage across the brushes (the terminal voltage) is the sum of the two induced voltages.
- The meter at position B reads maximum value.



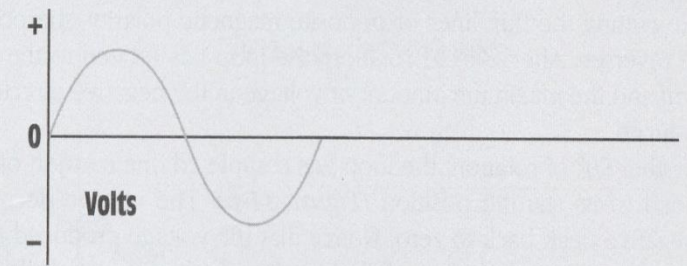
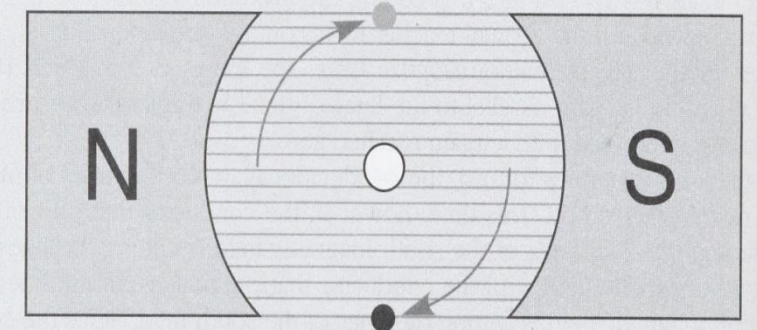
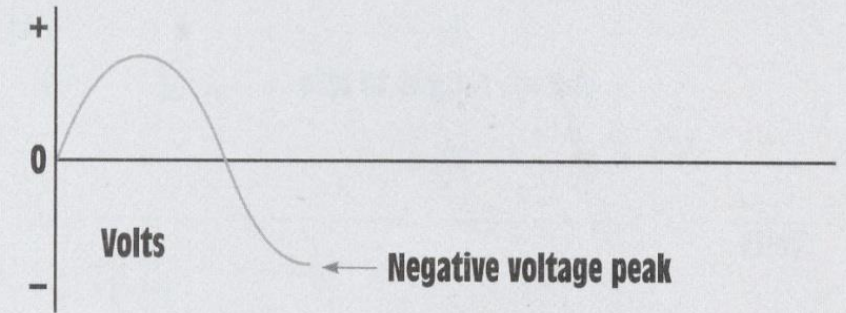
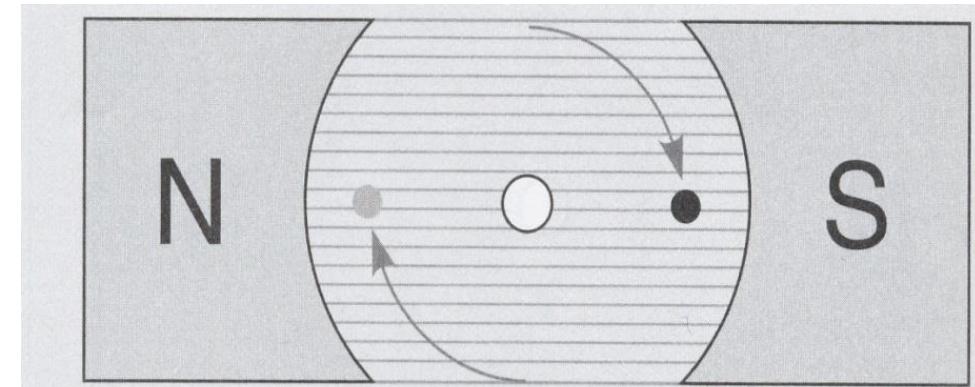
# THE ELEMENTARY GENERATOR

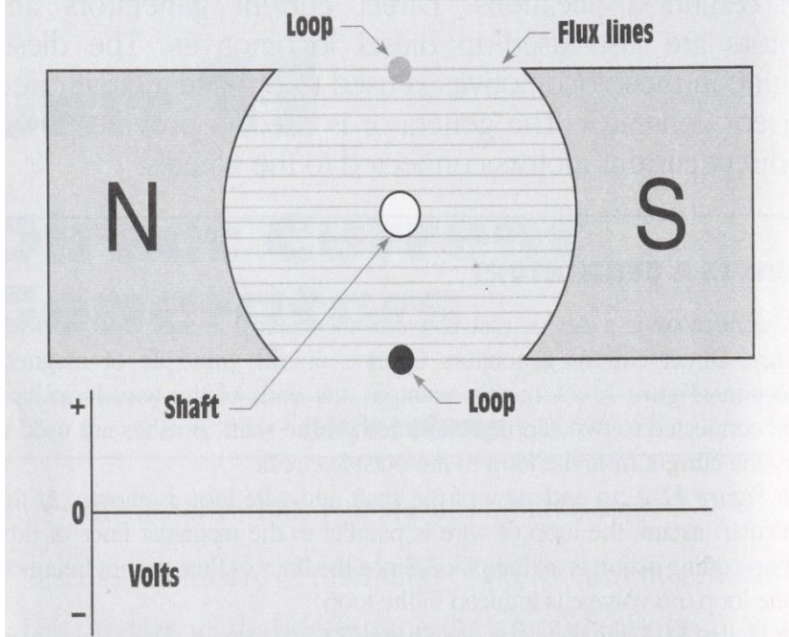
- After another  $90^\circ$  of rotation, the loop has completed  $180^\circ$  of rotation and is again parallel to the lines of flux.
- As the loop was turned, the voltage decreased until it again reached zero.
- Note that : From  $0^\circ$  to  $180^\circ$  the conductors of the armature loop have been moving in the same direction through the magnetic field.
- Therefore, the polarity of the induced voltage has remained the same



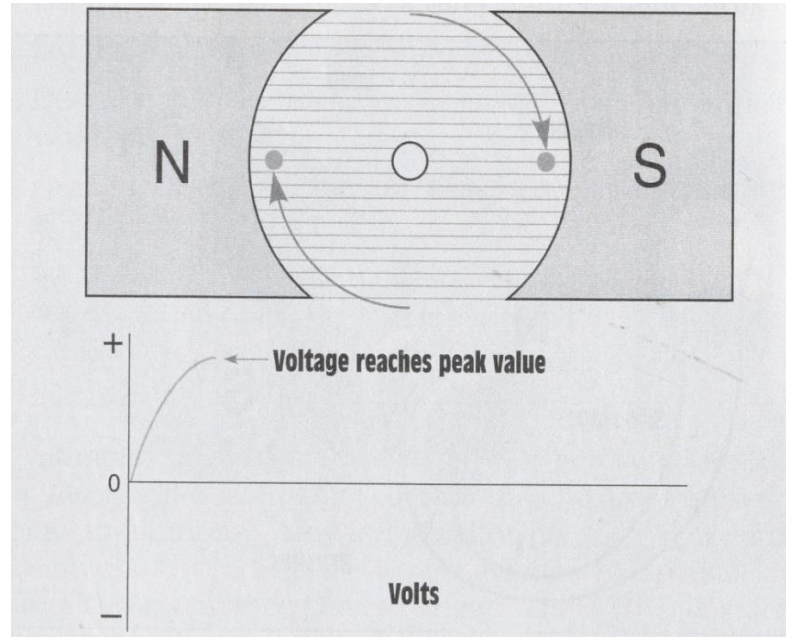
# THE ELEMENTARY GENERATOR

- As the loop continues to turn, the conductors again cut the lines of magnetic flux.
- This time, however, the conductor that previously cut through the flux lines of the south magnetic field is cutting the lines of the north magnetic field, and vice-versa.
- Since the conductors are cutting the flux lines of opposite magnetic polarity, the polarity of the induced voltage reverses.
- After 270° of rotation, the loop has rotated to the position shown, and the maximum terminal voltage will be the same as it was from A to C except that the polarity is reversed.
- After another 90° of rotation, the loop has completed one rotation of 360° and returned to its starting position.
- The voltage decreased from its negative peak back to zero.
- Notice that the voltage produced in the armature is an alternating polarity. The voltage produced in all rotating armatures is alternating voltage.

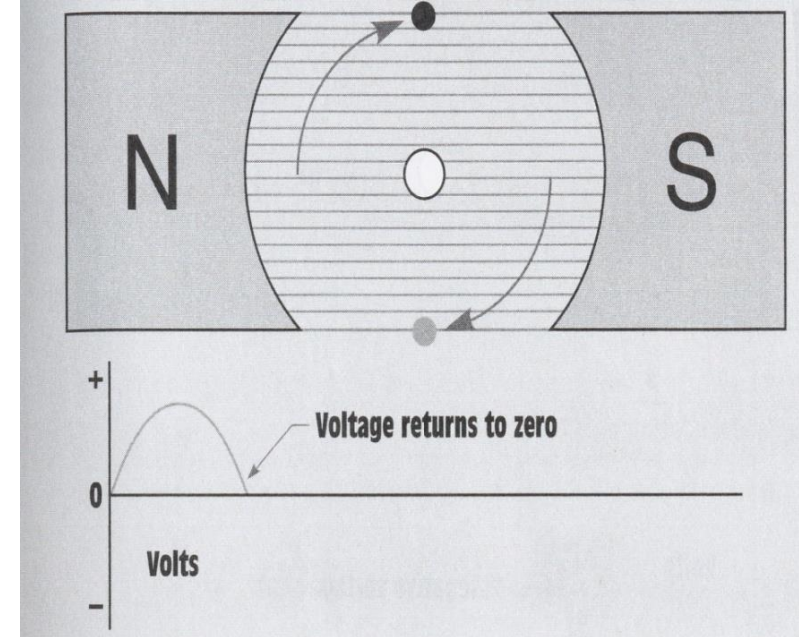




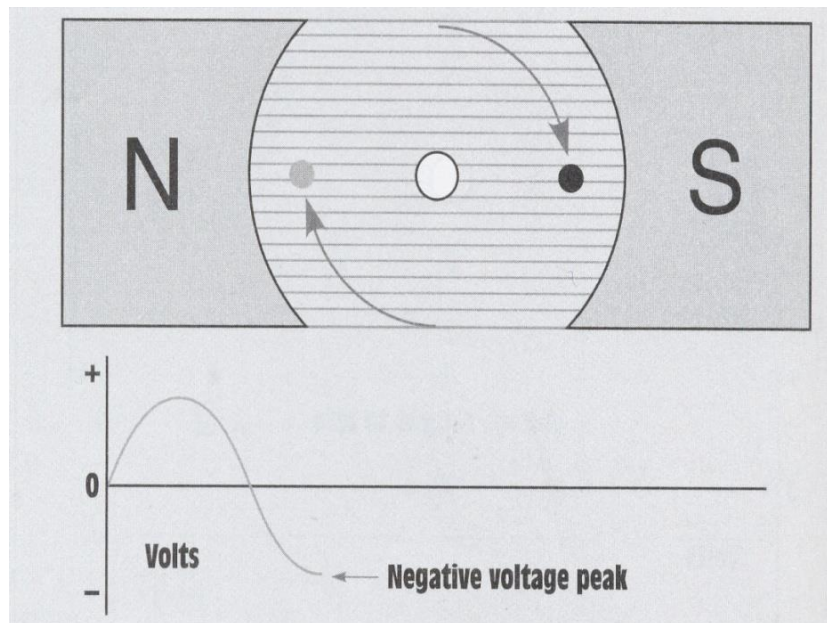
0° Position



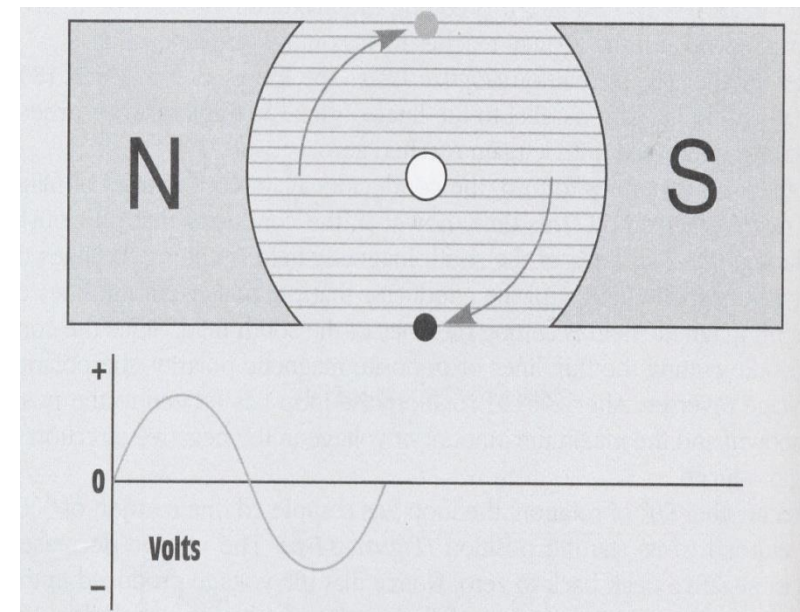
90° Position



180° Position



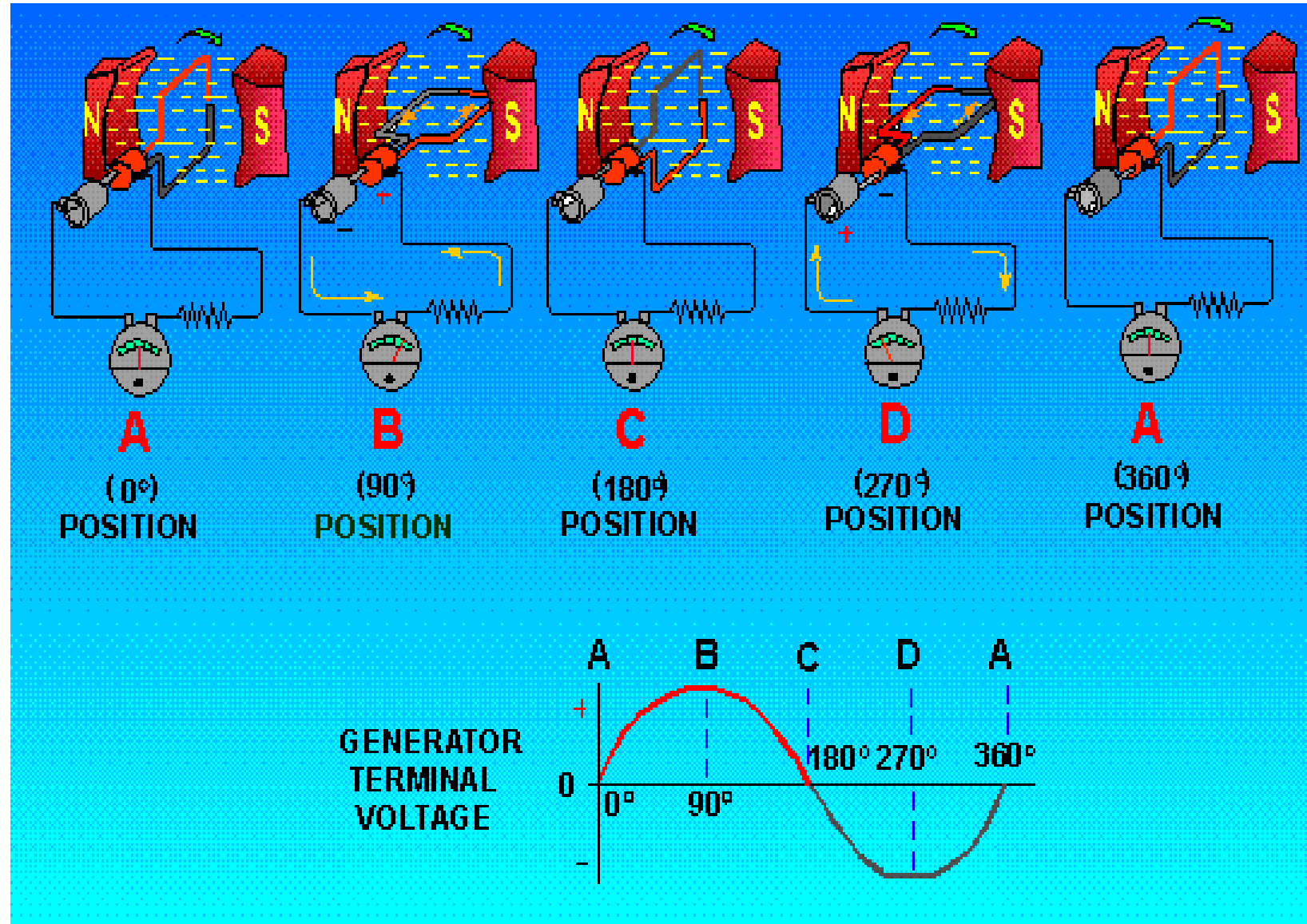
270° Position



360° Position

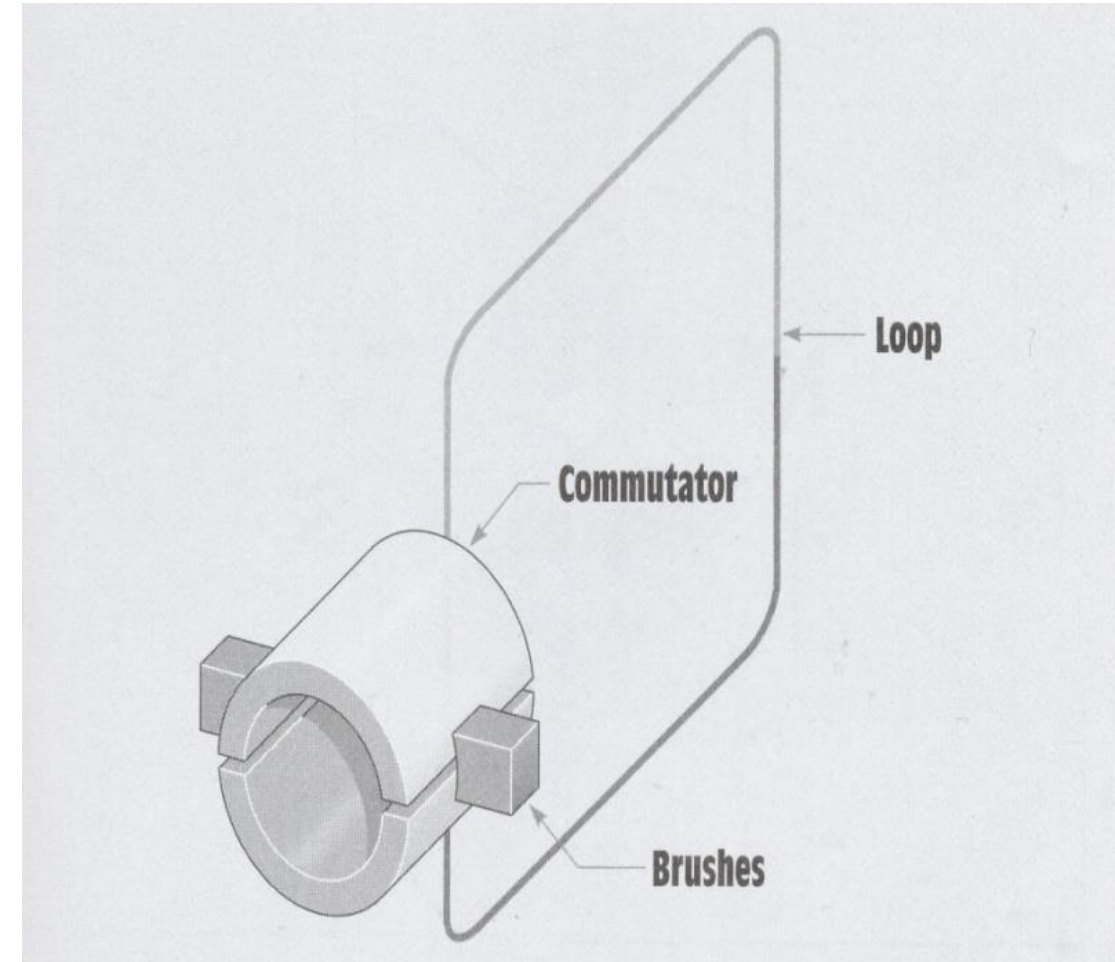
# THE ELEMENTARY GENERATOR (SUMMARY)

- Observe
  - The meter direction
  - The conductors of the armature loop
  - Direction of the current flow



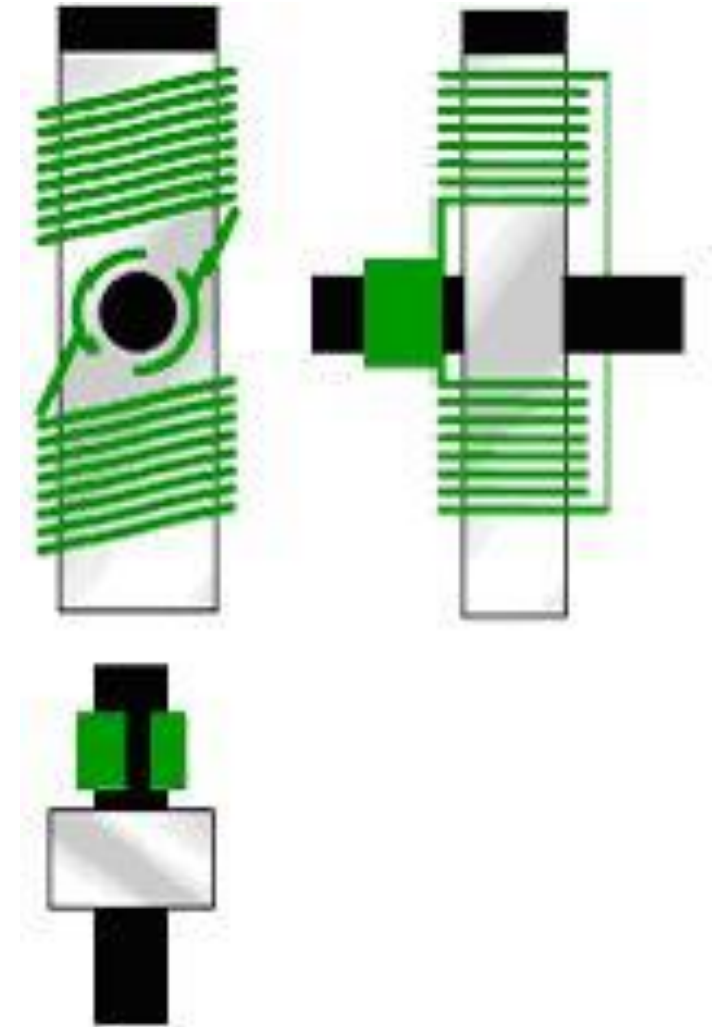
# THE DC GENERATOR

- Since DC generators must produce DC current instead of AC current, a device must be used to change the AC voltage produced in the armature windings into DC voltage.
- This job is performed by the commutator.
- The commutator is constructed from a copper ring split into segments with insulating material between the segments (See next page).
- Brushes riding against the commutator segments carry the power to the outside circuit.
- The commutator in a dc generator replaces the slip rings of the ac generator. This is the main difference in their construction.
- The commutator mechanically reverses the armature loop connections to the external circuit.



# THE DC GENERATOR (Armature)

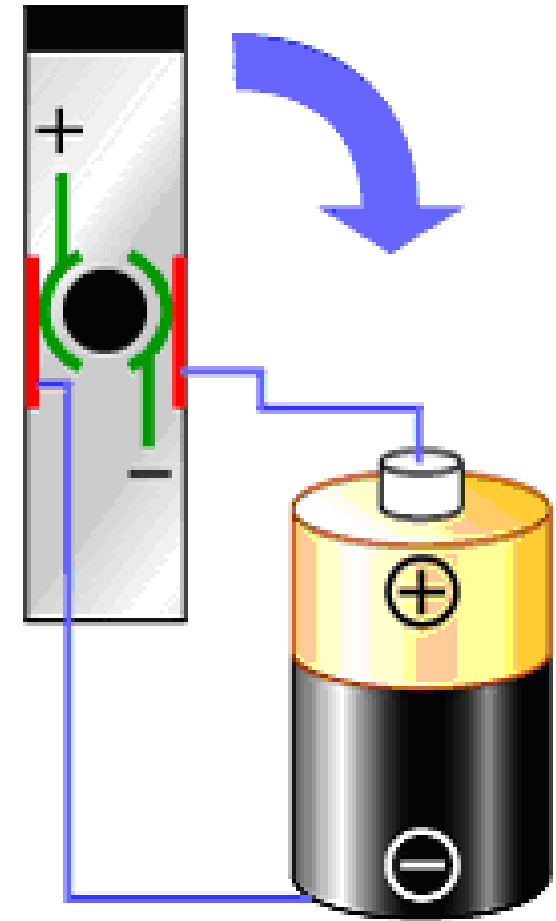
- The armature has an **axle**, and the commutator is attached to the axle.
- In the diagram to the right, you can see three different views of the same armature: **front, side and end-on**.
- In the end-on view, the winding is eliminated to make the commutator more obvious.
- We can see that the commutator is **simply a pair of plates attached to the axle**.
- These plates provide the two connections for the coil of the electromagnet.



Armature with commutator view

# How Commutator and Brushes Work

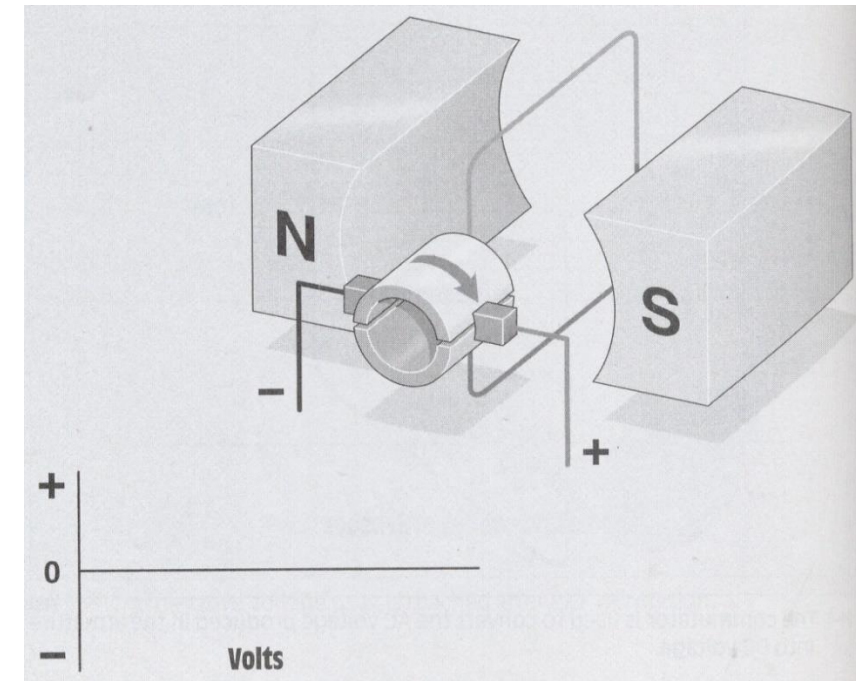
- The diagram at the right shows how the commutator and brushes work together to let current flow to the electromagnet, and also to flip the direction that the electrons are flowing at just the right moment.
- The contacts of the commutator are attached to the axle of the electromagnet, so they spin with the magnet.
- The brushes are just two pieces of springy metal or carbon that make contact with the contacts of the commutator.
- Through this process the commutator changes the generated ac voltage to a pulsating dc voltage which also known as commutation process.



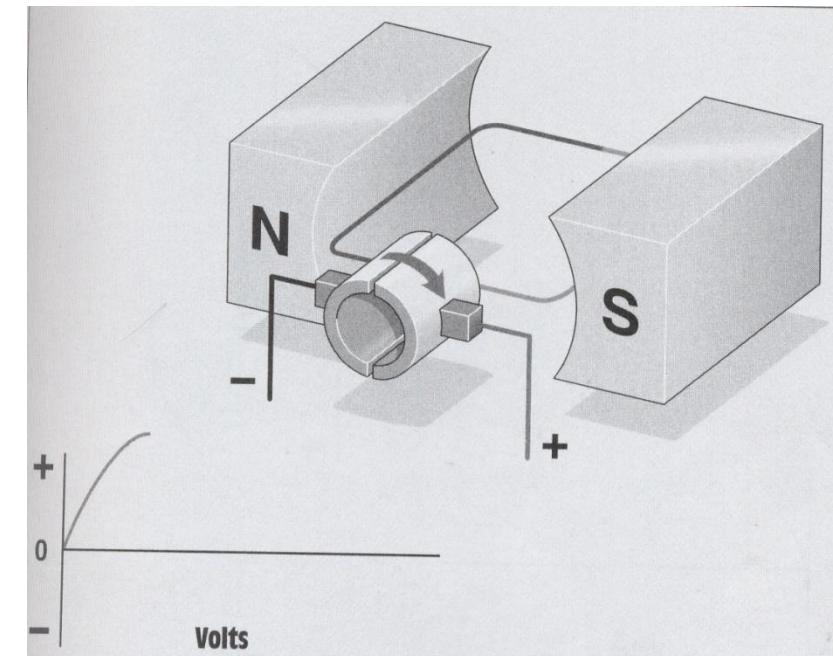
Brushes and commutator

# DC Generator

- The loop is parallel to the magnetic lines of flux, and **no voltage is induced in the loop**
- Note that the **brushes make contact with both of the commutator segments at this time**. The position is called **neutral plane**.
- As the loop rotates, **the conductors begin to cut through the magnetic lines of flux**.
- The conductor cutting through the **south magnetic field is connected to the positive brush**, and the conductor cutting through the **north magnetic field is connected to the negative brush**.
- Since the loop is cutting lines of flux, a **voltage is induced into the loop**.
- After  $90^\circ$  of rotation, the **voltage reaches its most positive point**.



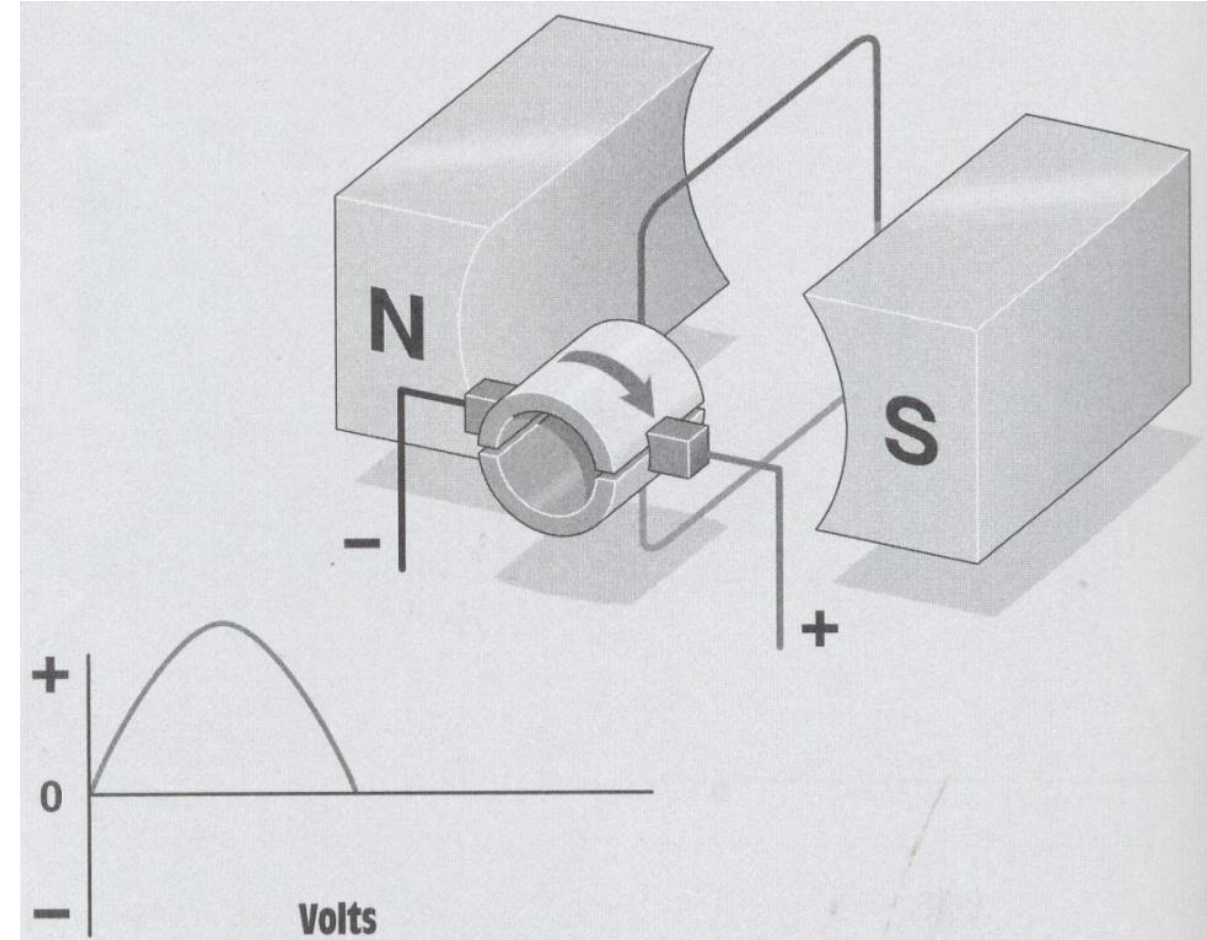
$0^\circ$  Position (DC Neutral Plane)



$90^\circ$  Position (DC)

# DC Generator

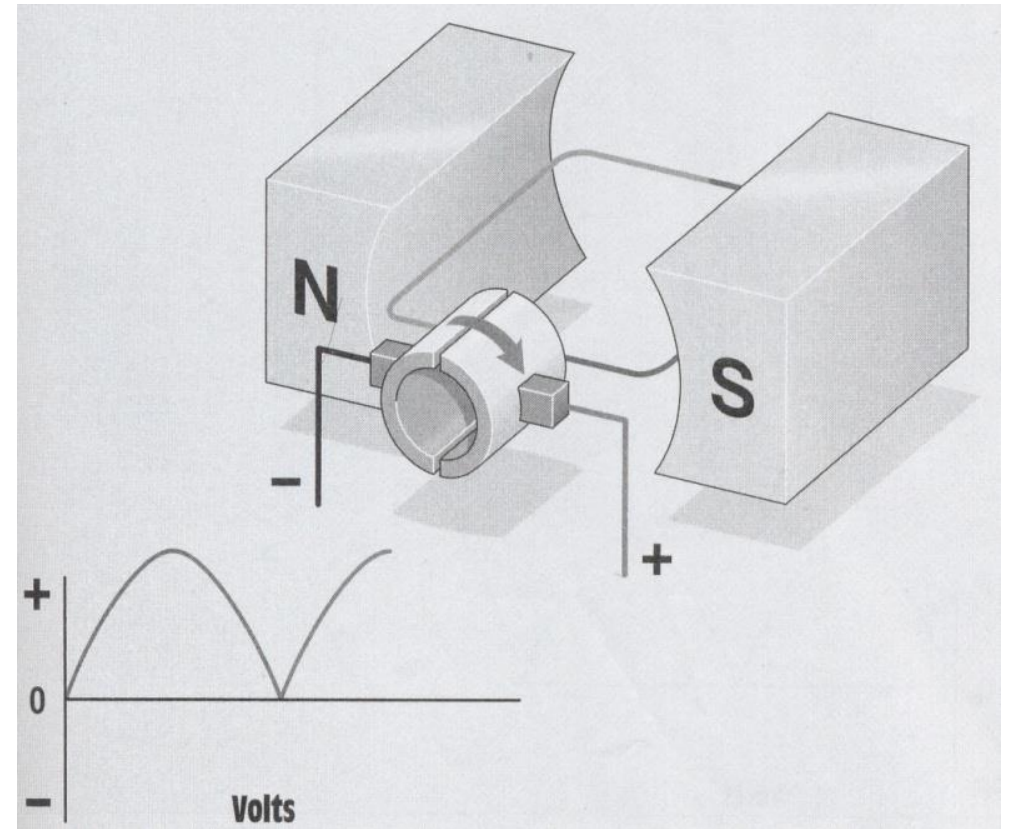
- As the loop continues to rotate, the voltage decreases to zero.
- After  $180^\circ$  of rotation, the conductors are again parallel to the lines of flux, and no voltage is induced in the loop.
- Note that the brushes again make contact with both segments of the commutator at the time when there is no induced voltage in the conductors



$180^\circ$  Position (DC)

# DC Generator

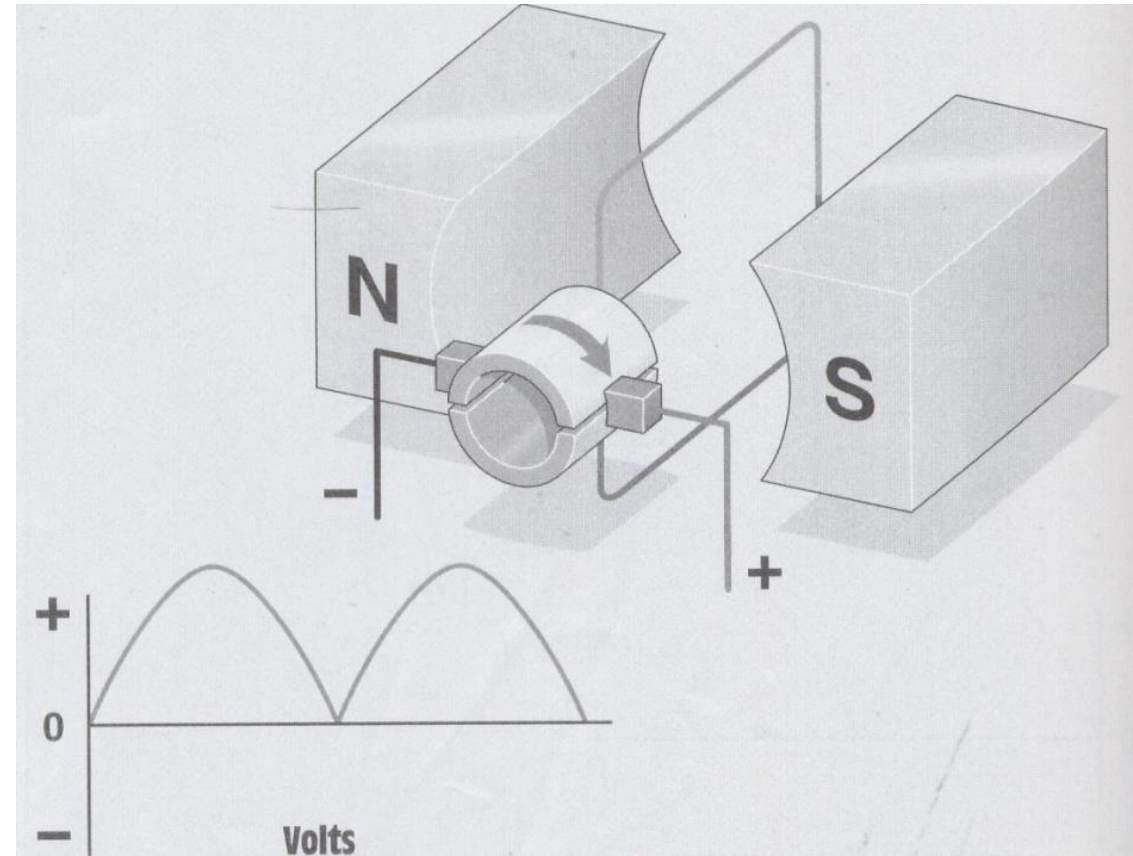
- During the next  $90^\circ$  of rotation, the conductors again cut through the magnetic lines of flux.
- This time, however, **the conductor** that previously cut through the south magnetic field is **now cutting the flux lines of the north field, and vice-versa**..
- Since these conductors are cutting the lines of flux of opposite magnetic polarities, the polarity of induced voltage is different for each of the conductors. **The commutator, however, maintains the correct polarity to each brush.**
- The conductor cutting through the north magnetic field will always be connected to the negative brush, and the conductor cutting through the south field will always be connected to the positive brush.
- **Since the polarity at the brushes has remained constant, the voltage will increase to its peak value in the same direction.**



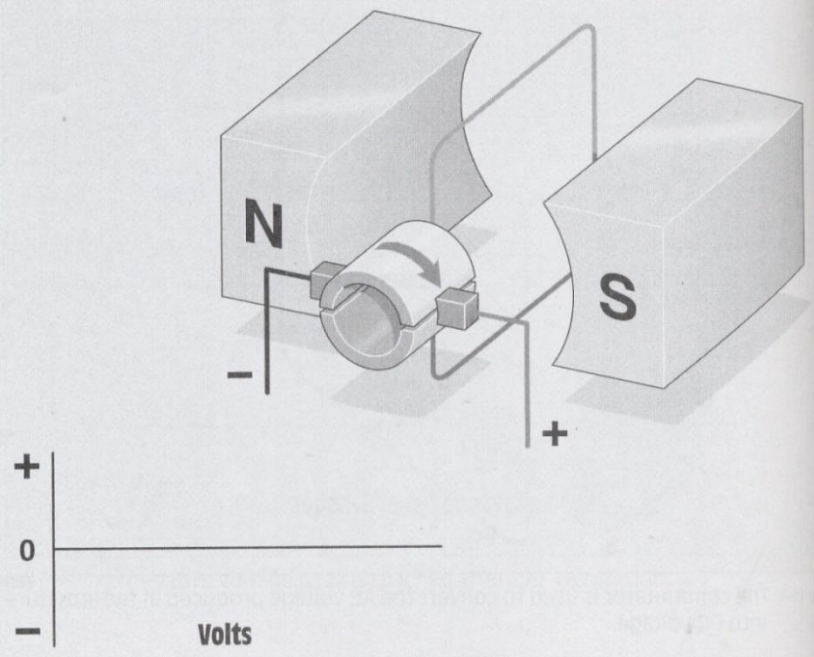
270° Position (DC)

# DC Generator

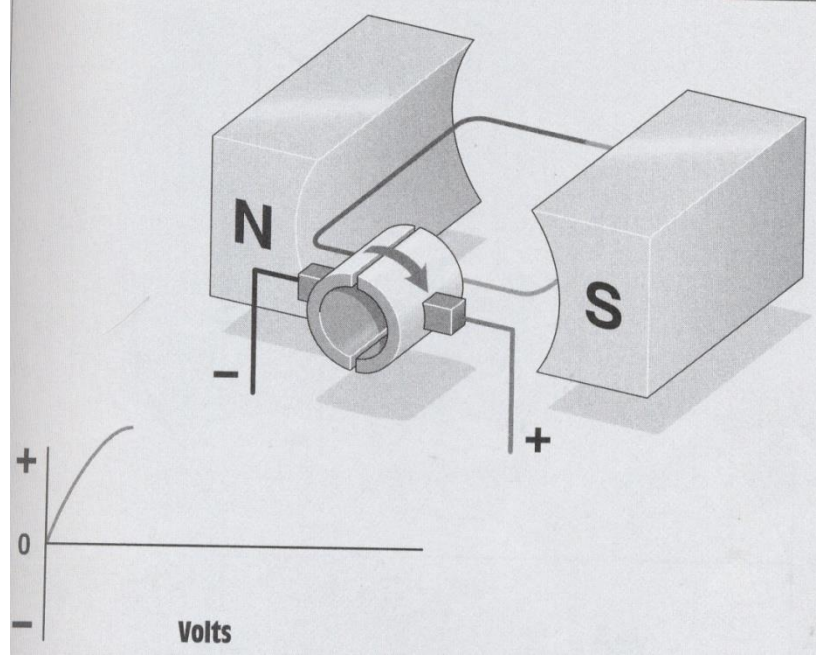
- As the loop continues to rotate, the induced voltage again decreases to zero when the conductors become parallel to the magnetic lines of flux.
- Notice that during this  $360^\circ$  rotation of the loop the polarity of voltage remained the same for both halves of the waveform. This is called rectified DC voltage.
- The voltage is pulsating. It does turn on and off, but it never reverses polarity. Since the polarity for each brush remains constant, the output voltage is DC.



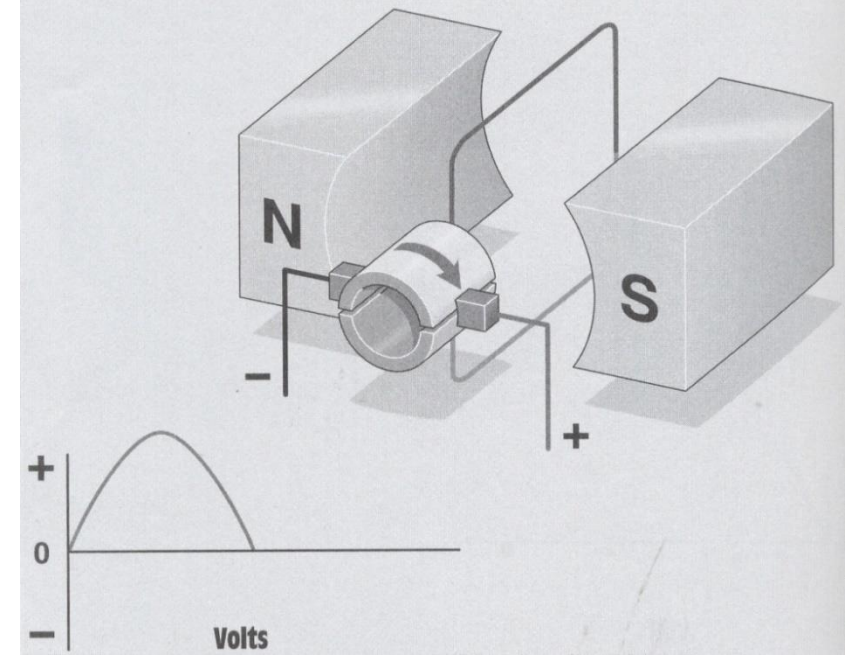
$360^\circ$  Position (DC Neutral Plane)



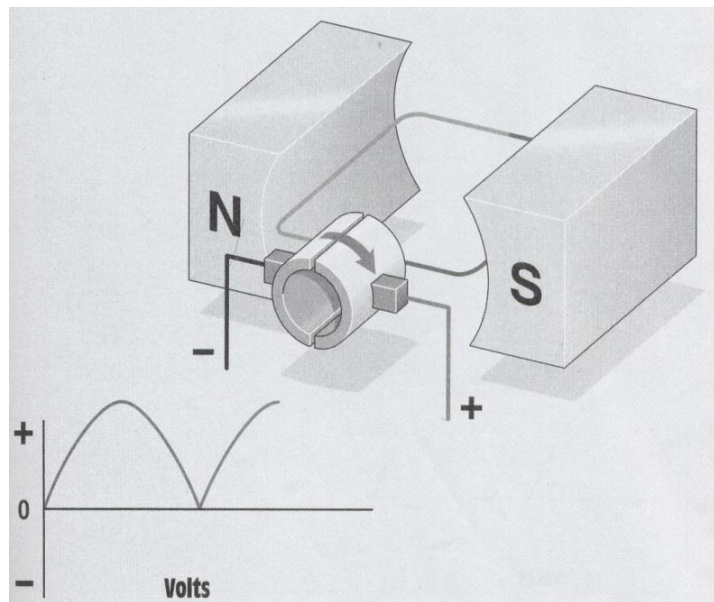
0° Position (DC Neutral Plane)



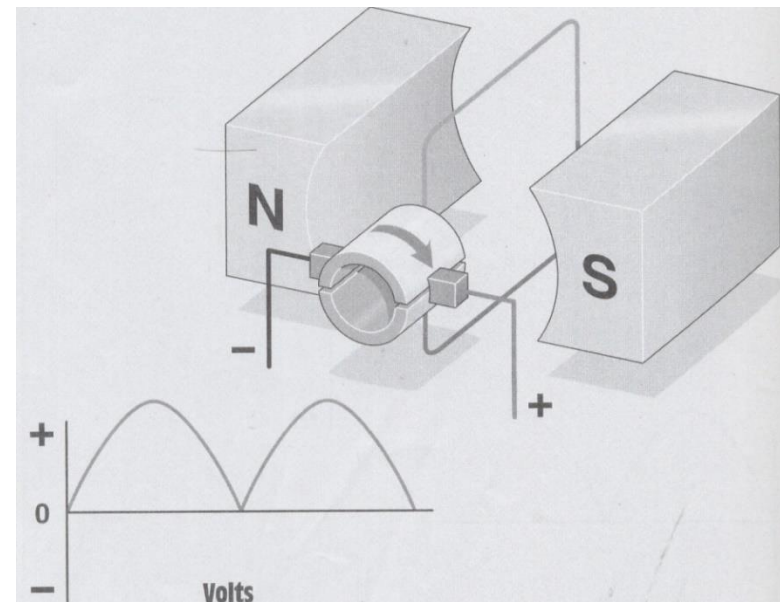
90° Position (DC)



180° Position (DC)



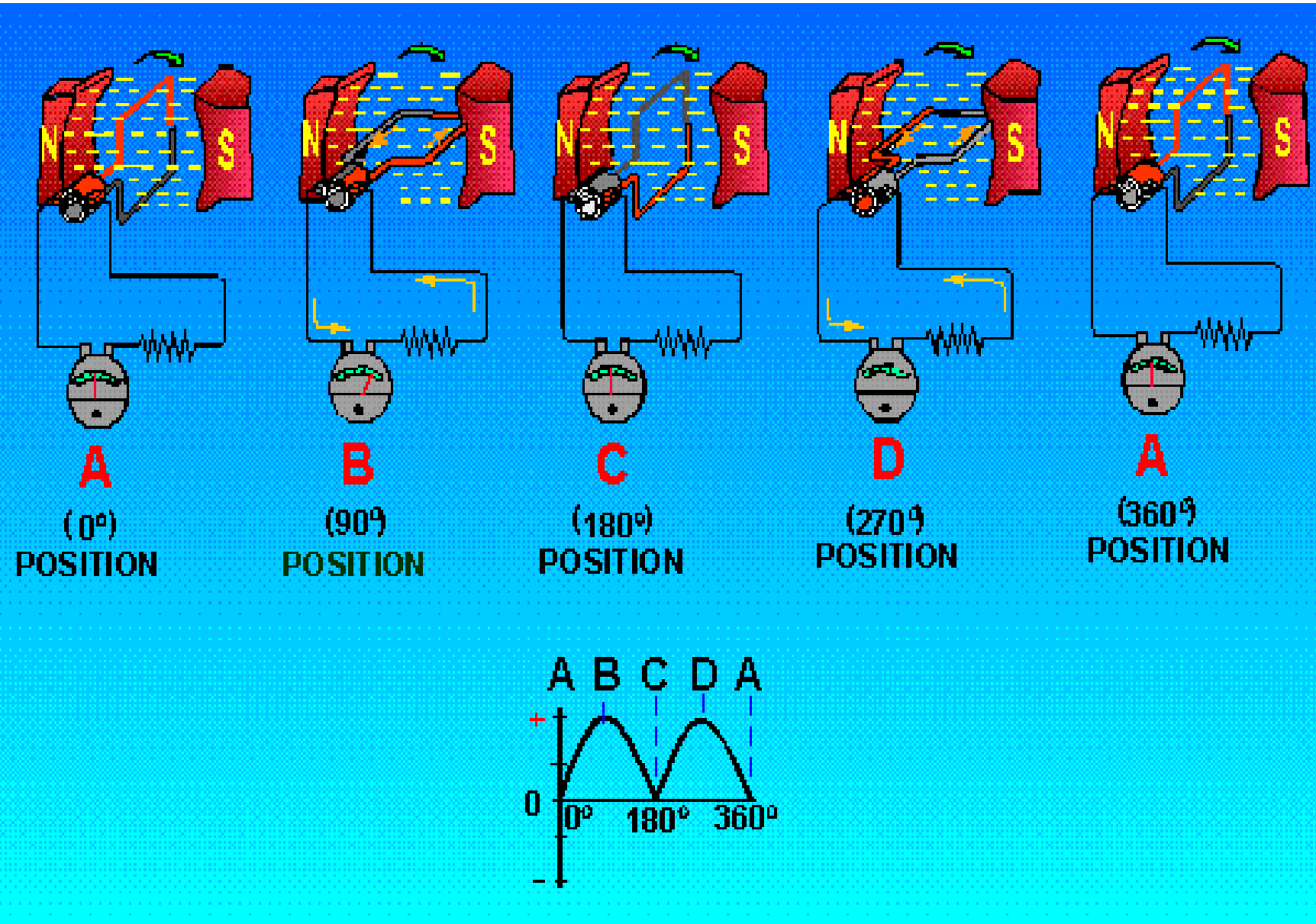
270° Position (DC)



360° Position (DC Neutral Plane)

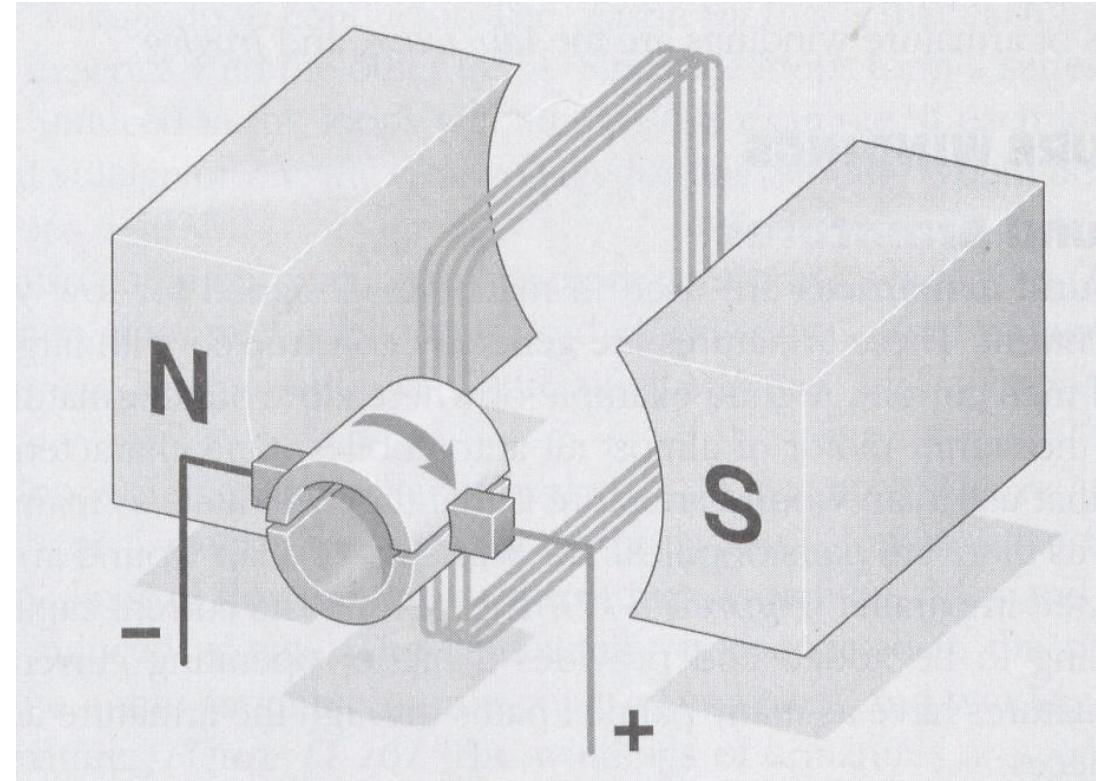
# DC Generator (Summary)

- Observe
  - The meter direction
  - The conductors of the armature loop
  - Direction of the current flow



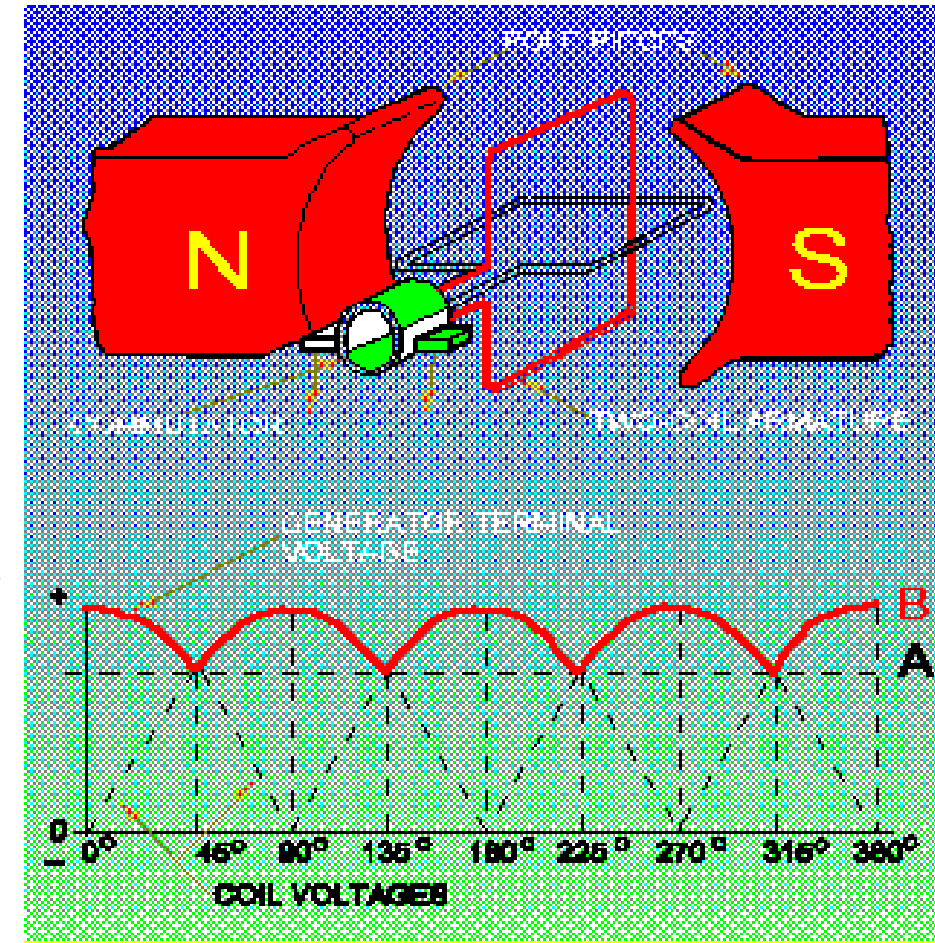
# DC Generator: Effects of additional turns

- To increase the amount of output voltage, it is common practice to increase the number of turns of wire for each loop.
- If a loop contains 20 turns of wire, the induced voltage will be 20 times greater than that for a single-loop conductor.
- The reason for this is that **each loop is connected in series** with the other loops. Since the loops form a series path, **the voltage induced in the loops will add**.
- In this example, if each loop has an induced voltage of 2V, the total voltage for this winding would be 40V (2V x 20 loops = 40 V).



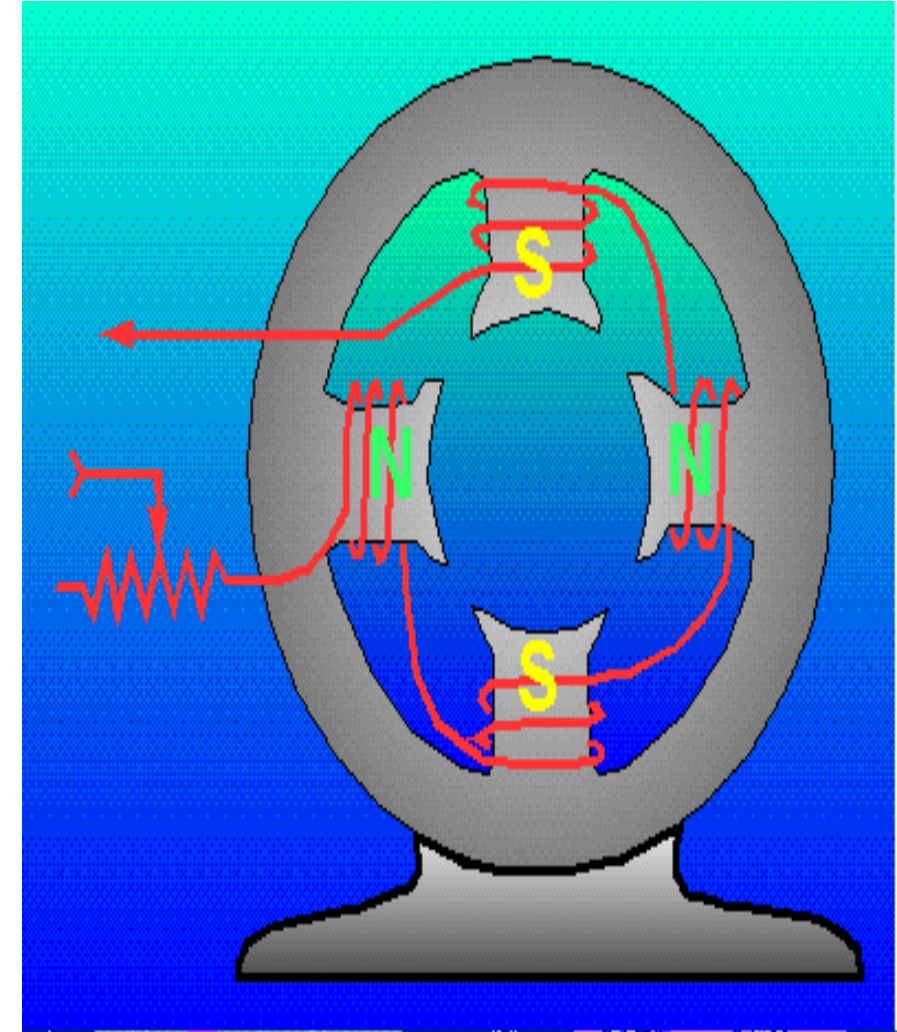
# DC Generator: Effects of additional coils

- When more than one loop is used, the average output voltage is higher and there is less pulsation of the rectified voltage.
- Since there are four segments in the commutator, a new segment passes each brush every  $90^\circ$  instead of every  $180^\circ$ .
- Since there are now four commutator segments in the commutator and only two brushes, the voltage cannot fall any lower than at point A.
- Therefore, the ripple is limited to the rise and fall between points A and B on the graph. By adding more armature coils, the ripple effect can be further reduced. **Decreasing ripple in this way increases the effective voltage of the output.**



# The Practical DC Generator

- The actual construction and operation of a practical dc generator differs somewhat from our elementary generators
- Nearly all practical generators use electromagnetic poles instead of the permanent magnets used in our elementary generator
- The main advantages of using electromagnetic poles are:
  - (1) increased field strength and
  - (2) possible to control the strength of the fields:
    - By varying the input voltage, the field strength is varied.
    - By varying the field strength, the output voltage of the generator can be controlled.



# EMF Equation a DC Generator

Let  $\Phi$  = flux/pole in Wb (weber)

$Z$  = total no. of armature conductors

$P$  = no. of generator poles

$A$  = no. of parallel paths in armature

$N$  = rotational speed of armature in revolutions per min. (rpm)

$E$  = emf induced in any parallel path in armature

Now, flux cut/conductor in one revolution  $d\Phi = \Phi P$  Wb

No. of revolutions/second =  $N/60$   $\therefore$  Time for one revolution,  $dt = 60/N$  second

Hence, according to Faraday's Laws of Electromagnetic Induction,

$$\text{EMF generated/conductor} = \frac{d\phi}{dt} = \frac{\phi PN}{60} \text{ volt}$$

No. of conductors (in series) in one path =  $Z/A$

For a simplex wave-wound generator:  $A=2$

$$\text{EMF generated/path} = \frac{\phi PN}{60} \times \frac{Z}{2} = \frac{\phi ZPN}{120} \text{ volt}$$

For a simplex lap-wound generator:  $A=P$ , No. of conductors (in series) in one path= $Z/P$

$$\text{EMF generated/path} = \frac{\phi PN}{60} \times \frac{Z}{P} = \frac{\phi NZ}{60} \text{ volt}$$

# Types of DC Generator

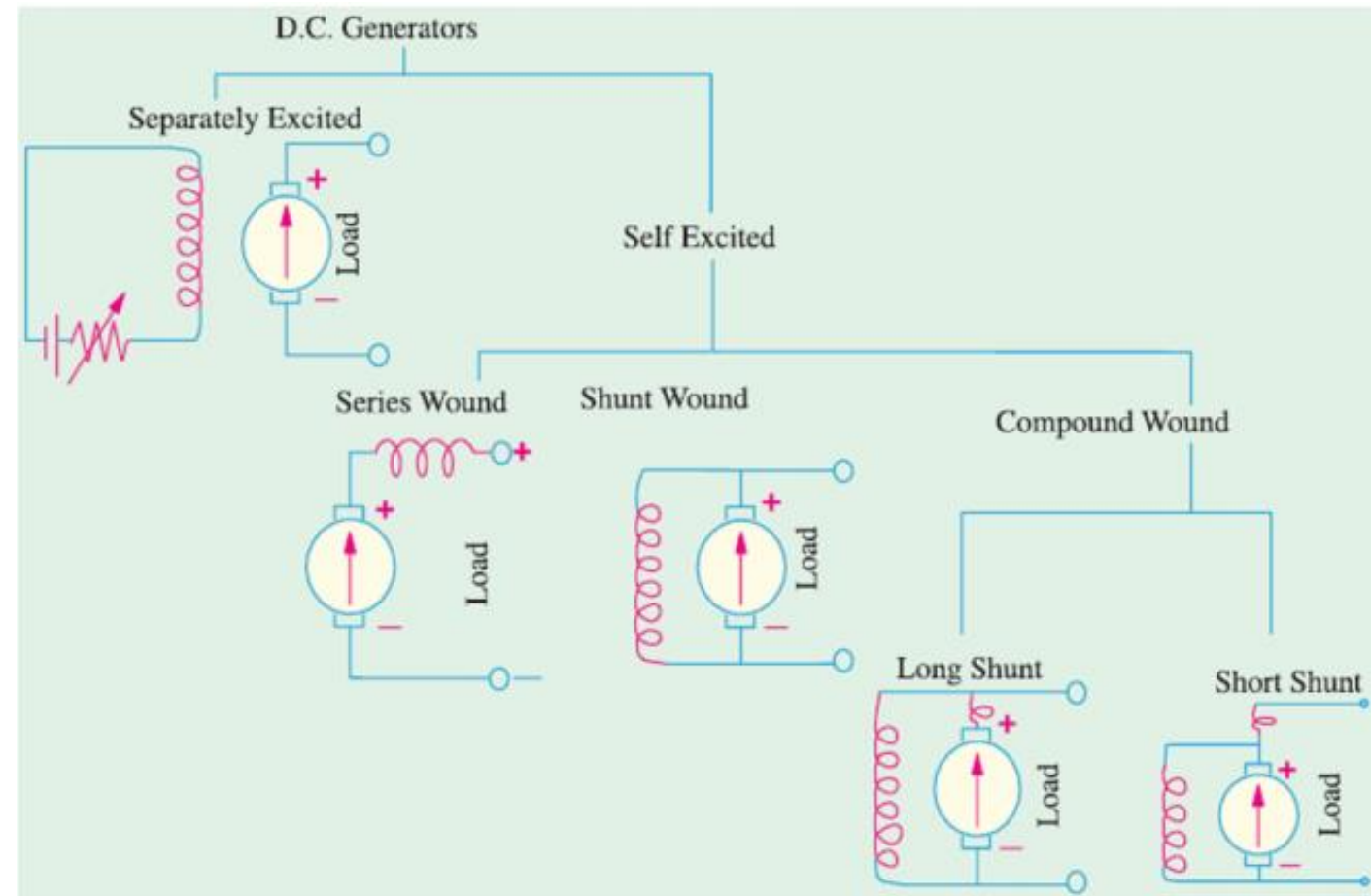
- Generators are usually classified according to the way in which their fields are excited:
  - Separately excited generators
  - Self excited generators
  - Permanent Magnet generators

## 1. Permanent magnet

- The poles are made of permanent magnets.
- No field winding required.
- Small size.
- Disadvantage is low flux density, so low torque.

## 2. Separately excited

The field flux is derived from a separate power source independent of the generator itself.



# Types of DC Generator

## 3. Self-excited

- Shunt Wound

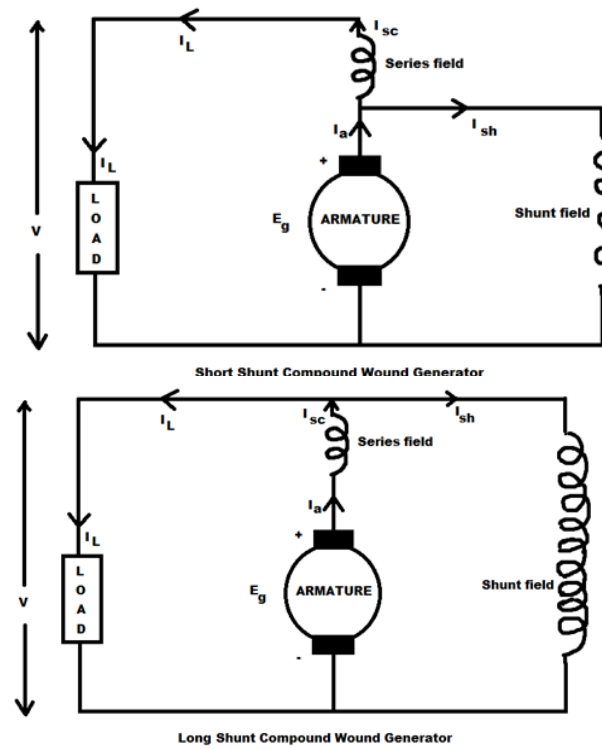
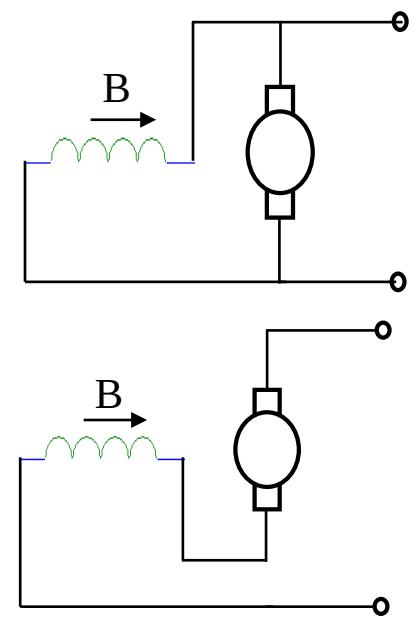
The field flux is derived by connecting the field directly across the terminals of the generator.

- Series Wound

The field windings are joined in series with the armature conductors. As they carry full load current, they consist of relatively few turns of thick wire or strips. Such generators are rarely used except for special purposes i.e. as boosters etc.

- Compound Wound

It is a combination of a few series and a few shunt windings and can be either short-shunt or long-shunt. In a compound generator, the shunt field is stronger than the series field. When series field aids the shunt field, generator is said to be cumulatively-compounded. On the other hand if series field opposes the shunt field, the generator is said to be differentially compounded.



# Losses in a DC machine

- **Copper losses**

1. Armature Cu loss =  $I_a^2 R_a$

;  $I_a$  is Armature current &  $R_a$  is Armature resistance. This loss is about **30 to 40%** of full load losses.

2. Field Cu loss =  $I_f^2 R_f$

;  $I_f$  is field current and  $R_f$  is field resistance.

Shunt field copper loss =  $I_{sh}^2 R_{sh}$

Series field copper loss =  $I_{se}^2 R_{se}$

Field copper loss is about **20 to 30%** of full load losses.

3. Loss due to brush contact resistance

; There is also brush contact loss due to brush contact resistance (i.e., resistance between the surface of brush and surface of commutator).

- **Iron Losses**

1. Hysteresis loss
2. Eddy current loss

- **Mechanical losses**

1. Friction losses
2. Windage losses

# Losses in a DC machine

3. **Iron losses** – hysteresis losses and eddy current losses. They vary as  $B^2$  (square of flux density) and as  $n^{1.5}$  (speed of rotation of the magnetic field).

**Hysteresis loss:** This loss is due to the reversal of magnetization of the armature core. The loss depends upon the volume and grade of iron, maximum value of flux density  $B_{\max}$  and frequency of magnetic reversals. For normal flux densities (i.e. up to 1.5 Wb/m<sup>2</sup>), hysteresis loss is given by Steinmetz formula. According to this formula,

$$W_h = \eta B_{\max}^{1.6} f V \text{ (watt)},$$

where  $B_{\max}$  = maximum flux density,  $V$  = volume of the core in m<sup>3</sup>,  $\eta$  = Steinmetz hysteresis coefficient. Value of  $\eta$  different for different metals.

**Eddy Current Loss:** When the armature core rotates, it also cuts the magnetic flux. Hence, an emf is induced in the body of the core according to the laws of electromagnetic induction. This emf though small, sets up large current in the body of the core due to its small resistance. This current is known as eddy current. The power loss due to the flow of this current is known as eddy current loss. This loss would be considerable if solid iron core were used. In order to reduce this loss and the consequent heating of the core to a small value, the core is built up of thin laminations, which are stacked and then riveted at right angles to the path of the eddy currents. These core laminations are insulated from each other by a thin coating of varnish. Eddy current loss ( $W_e$ ) is given by the following relation:

$$W_e = k B_{\max}^2 f^2 t^2 V^2 \text{ (watt)}, \text{ where}$$

$B_{\max}$  = maximum flux density,  $f$  = frequency of magnetic reversals,  $t$  = thickness of each lamination and  $V$  = volume of armature core.

# Losses in a DC machine

**4. Mechanical losses** – losses associated with mechanical effects: friction (friction of the bearings) and windage (friction between the moving parts of the machine and the air inside the casing). These losses vary as the cube of rotation speed  $n^3$ .

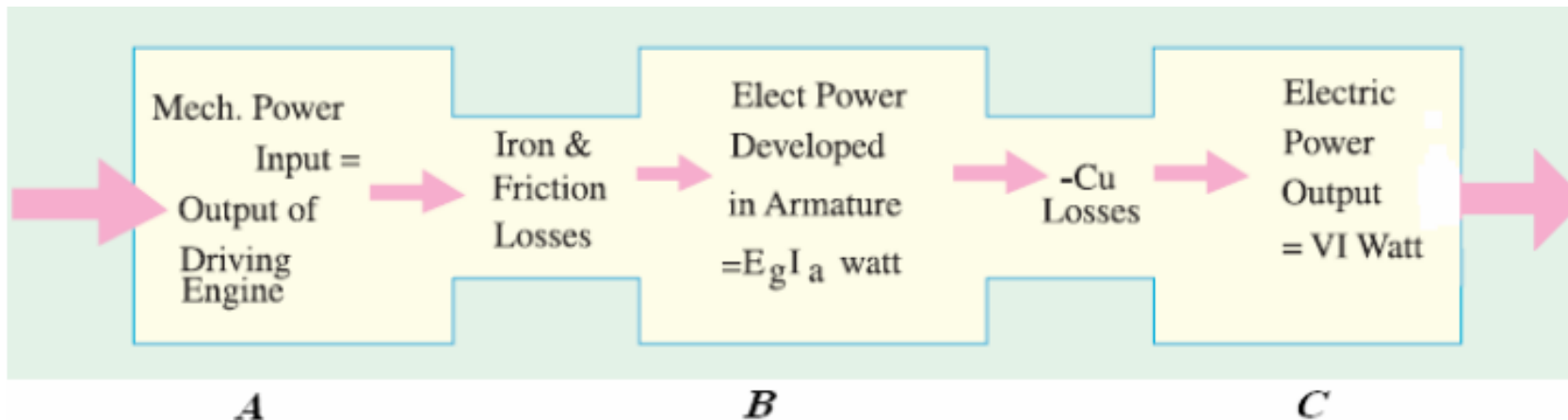
**5. Stray (Miscellaneous) losses** – losses that cannot be classified in any of the previous categories. They are usually due to inaccuracies in modeling. For many machines, stray losses are assumed as 1% of full load.

Magnetic and mechanical losses are collectively known as Stray Losses.  $W_{\text{stray}} = W_{\text{iron}} + W_{\text{mech}}$ . Field Cu loss is constant for shunt and compound generators. Hence, stray losses and shunt Cu loss are constant in their case. These losses are together known as standing or constant losses ( $W_c$ ). Hence, for shunt and compound generators:

$$\text{Total loss} = \text{armature copper loss} + W_c = I_a^2 R_a + W_c = (I + I_{\text{sh}})^2 R_a + W_c$$

Armature Cu loss  $I_a^2 R_a$  is known as variable loss because it varies with the load current.

Total loss = variable loss + constant losses ( $W_c$ )



# DC Generator Efficiencies

1. **Mechanical Efficiency,  $\eta_m$**  = total power generated in armature/mechanical input power =  $E_g I_a$  / *output of driving engine*
2. **Electrical Efficiency,  $\eta_e$**  = watts available in load / total watts generated =  $VI / E_g I_a$
3. **Overall Efficiency,  $\eta_c = \eta_m \times \eta_e$**

## Condition for Maximum Efficiency:

Generator output =  $VI$

Generator input = output + losses =  $VI + I_a^2 R_a + W_c = VI + (I + I_{sh})^2 R_a + W_c$  ( $\because I_a = I + I_{sh}$ )

However, if  $I_{sh}$  is negligible as compared to load current, then  $I_a = I$  (approx.)

$$\eta = \frac{\text{output}}{\text{input}} = \frac{VI}{VI + I_a^2 R_a + W_c} = \frac{VI}{VI + I^2 R_a + W_c} \quad (\because I_a = I)$$

Now, efficiency is maximum when denominator is minimum i.e. when

$$\frac{d}{dt} \left( \frac{IR_a}{V} + \frac{W_c}{VI} \right) = 0 \quad \text{or} \quad \frac{R_a}{V} - \frac{W_c}{VI^2} = 0 \quad \text{or} \quad I^2 R_a = W_c$$

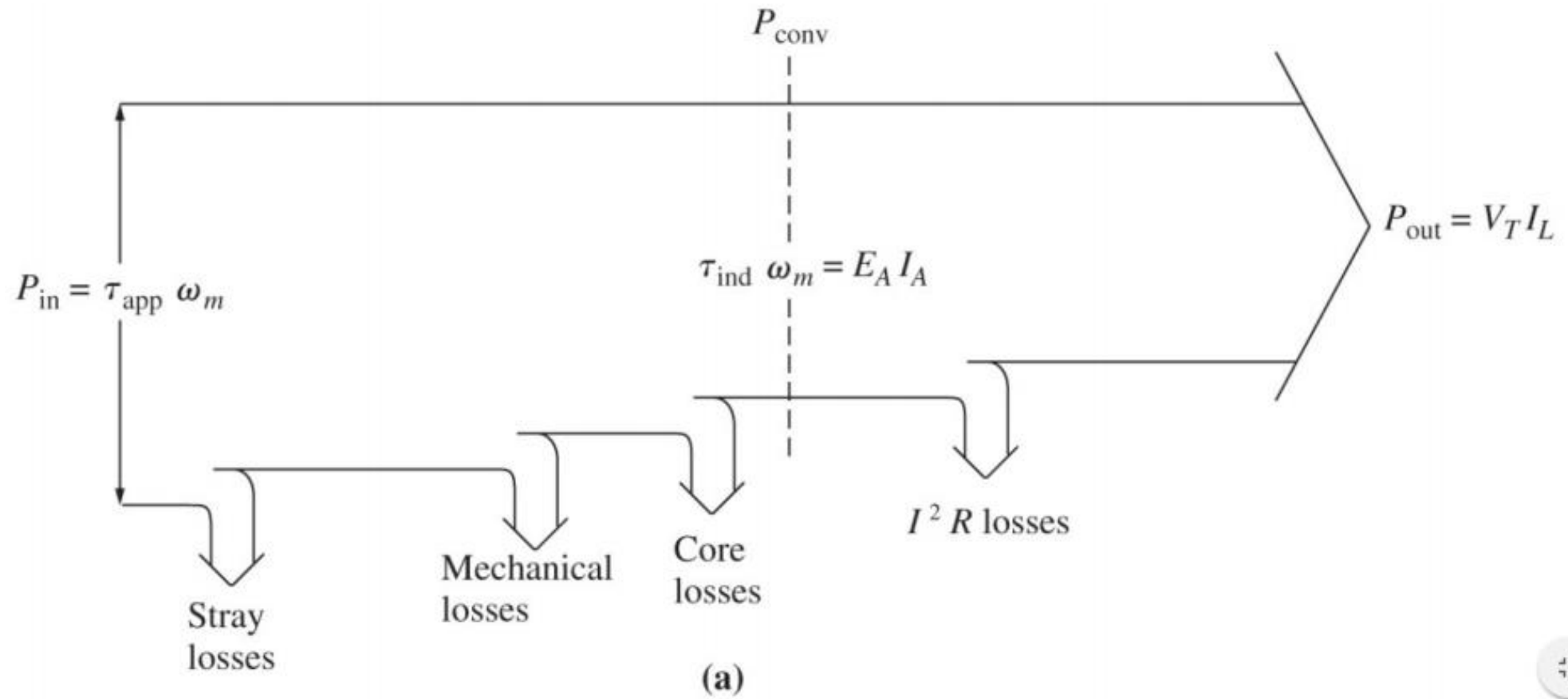
Hence, generator efficiency is maximum when

Variable loss = constant loss

The load current corresponding to maximum efficiency is given by the relation

$$I^2 R_a = W_c \quad \text{or} \quad I = \sqrt{\frac{W_c}{R_a}}$$

# Power Flow Diagram



# Equations

## VOLTAGE INDUCED IN A LOOP

To determine the total voltage  $e_{\text{tot}}$  on the loop, examine each segment of the loop separately and sum all the resulting voltages. The voltage on each segment is given by:

$$\mathbf{e}_{\text{ind}} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}$$

Thus, the total induced voltage on the loop is:

$$\mathbf{e}_{\text{ind}} = 2\mathbf{v}\mathbf{B}\mathbf{l}$$

- Examine the figure
- The tangential velocity  $v$  of the edges of the loop can be expressed as  $v = r\omega$   
Substituting this expression into the  $e_{\text{ind}}$  equation before, gives:

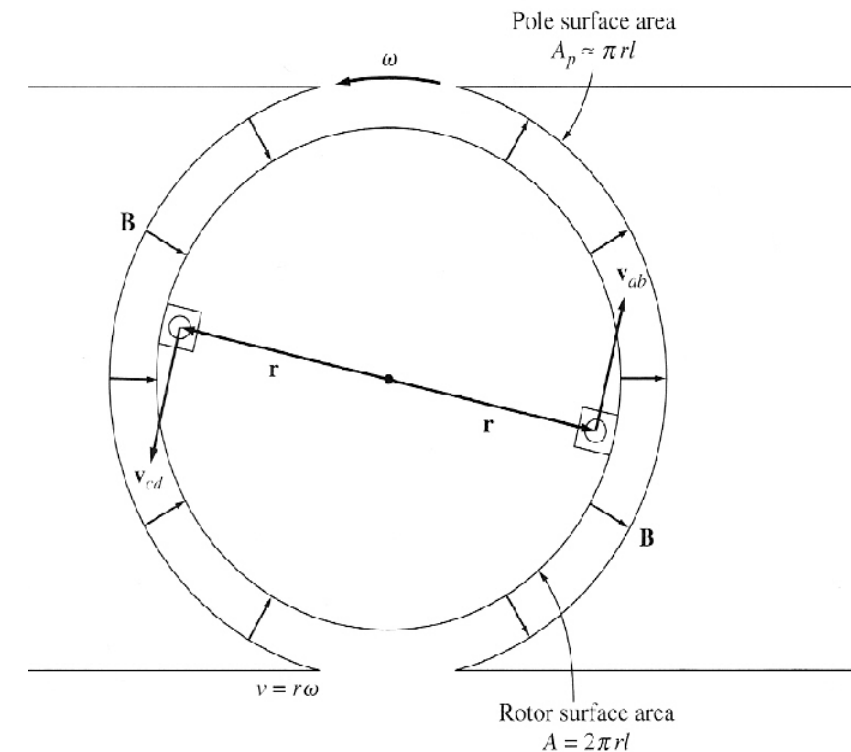
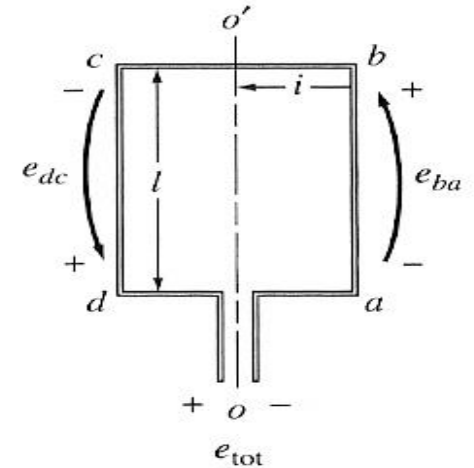
$$\mathbf{e}_{\text{ind}} = 2r\omega\mathbf{B}\mathbf{l}$$

- The rotor surface is a cylinder, so the area of the rotor surface  $A$  is equal to  $2\pi rl$
- Since there are 2 poles, the area under each pole is  $\mathbf{A}_p = \pi rl$ . Thus,

$$e_{\text{ind}} = \frac{2}{\pi} A_p B \omega$$

- the flux density  $B$  is constant everywhere in the air gap under the pole faces, the total flux under each pole is  $\phi = \mathbf{A}_p\mathbf{B}$ . Thus, the final form of the voltage equation is:

$$e_{\text{ind}} = \frac{2}{\pi} \phi \omega$$



# Equations

In general, the voltage in any real machine will depend on the same 3 factors:

- The flux in the machine
- The speed of rotation
- A constant representing the construction of the machine

## Induced Torque in the Rotating Loop

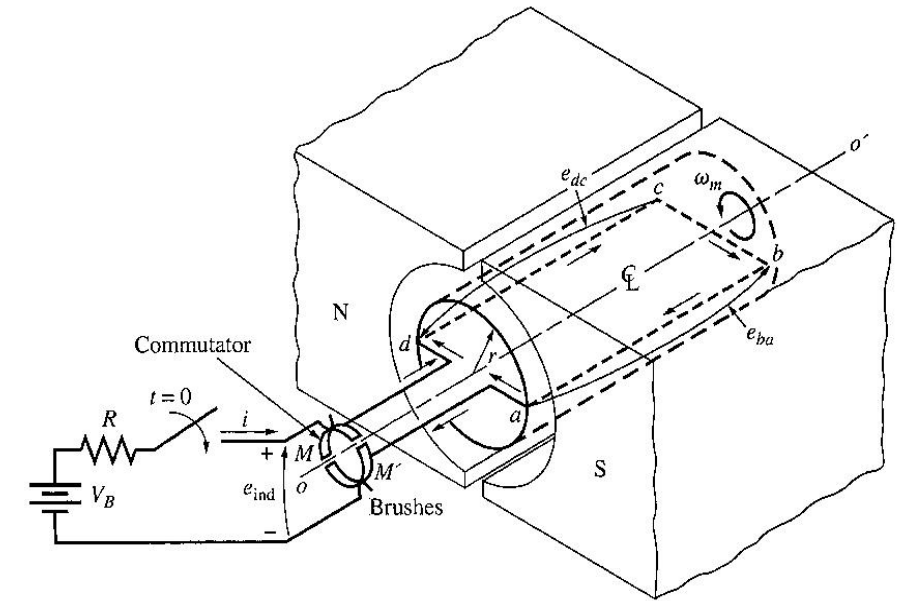
Suppose a battery is now connected to the machine as shown here, together with the resulting configuration

How much torque will be produced in the loop when the switch is closed?

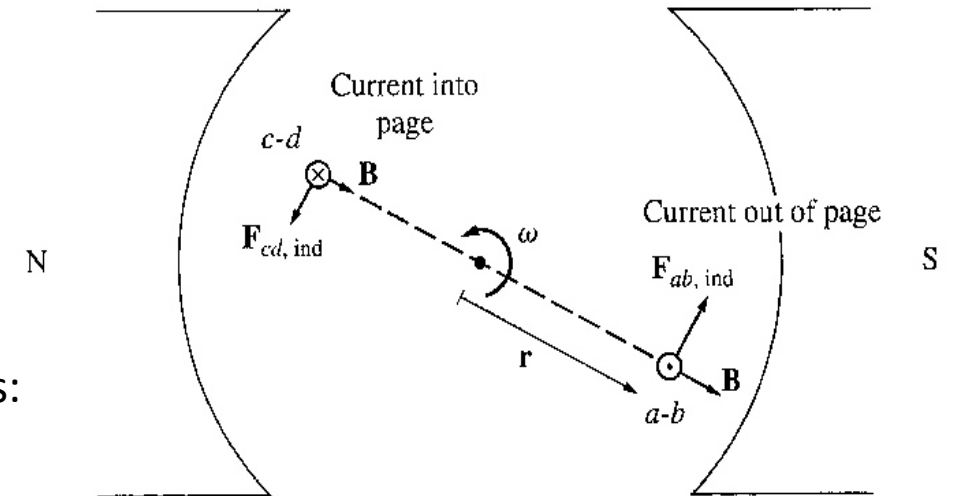
- The force on a segment of the loop is given by :  $\mathbf{F} = i (\mathbf{l} \times \mathbf{B})$  ,  
and the torque on the segment is :  $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} \sin \theta$
- The resulting total induced torque in the loop is:  $\tau_{ind} = 2 \mathbf{r} \cdot \mathbf{i} \cdot \mathbf{l} \cdot \mathbf{B}$
- By using the fact that  $\mathbf{A}_p = \pi \mathbf{r} \mathbf{l}$  and  $\boldsymbol{\phi} = \mathbf{A}_p \mathbf{B}$ , the torque expression can be reduced to:

$$\tau_{ind} = \frac{2}{\pi} \phi i$$

- In general, torque in any real machine will depend on the following 3 factors:
- The flux in the machine
- The current in the machine
- A constant representing the construction of the machine



(a)



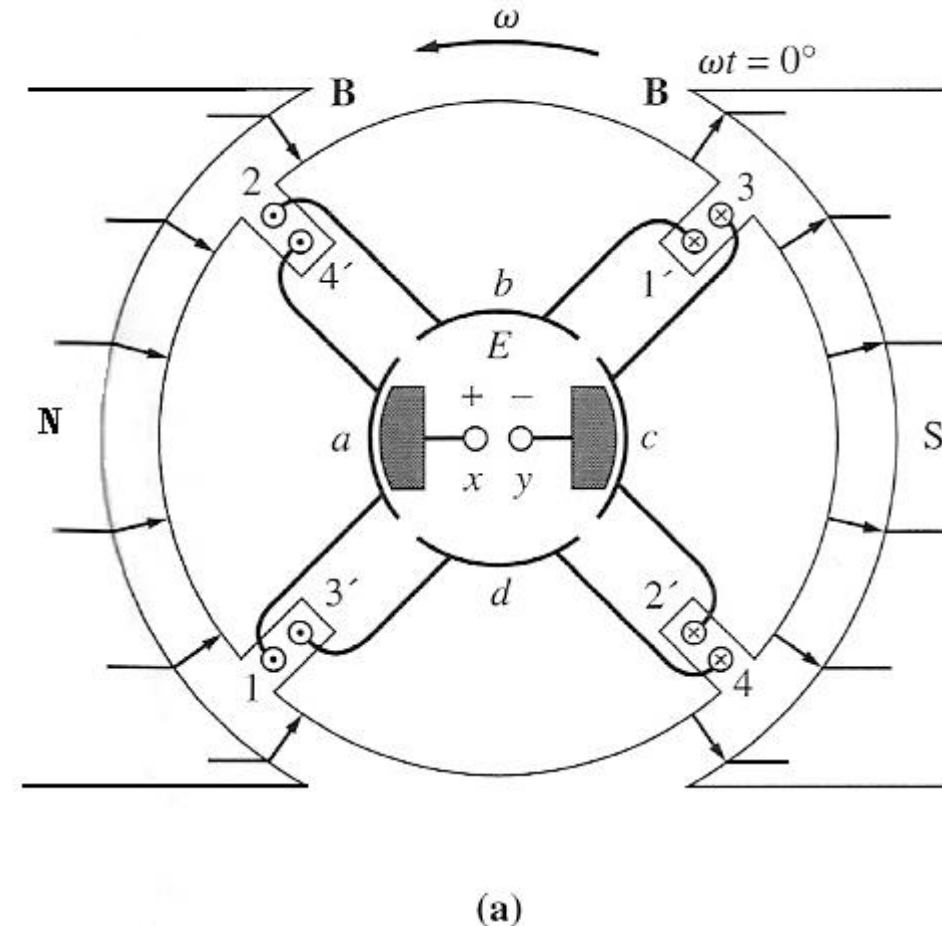
(b)

# Commutation

- In real DC machines, there are several ways in which the loops on the rotor (armature) can be connected to its commutator segments.
- Different connections affect the number of parallel current paths within the rotor, the output voltage of the rotor and the number and position of the brushes riding on the commutator segments.
- Commutation is the process for switching the loop connections on the rotor of a dc machine just as the voltage in the loop switches polarity, in order to maintain an essentially constant dc voltage

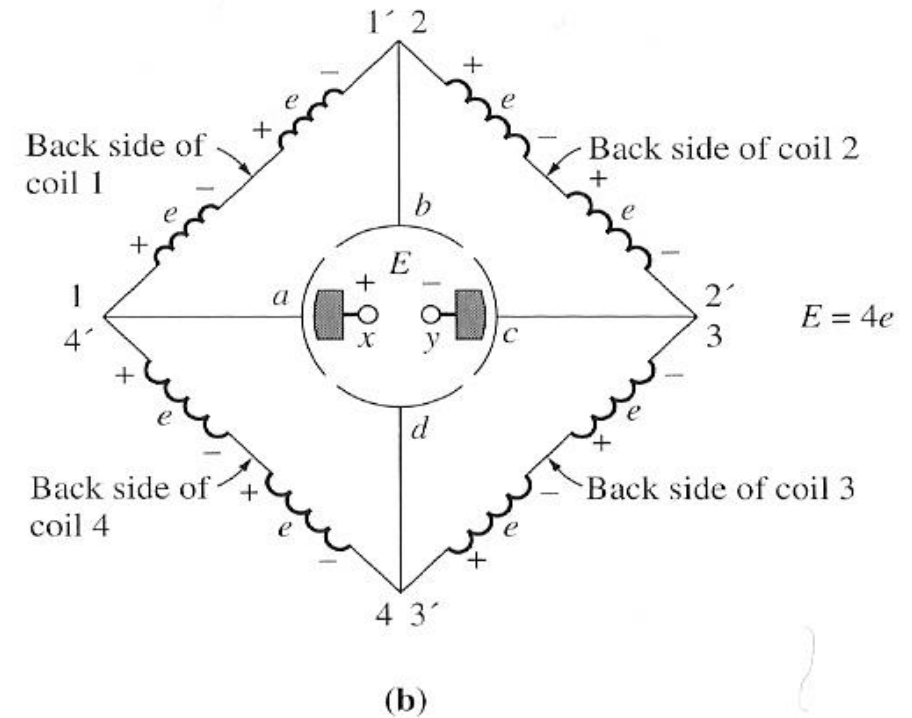
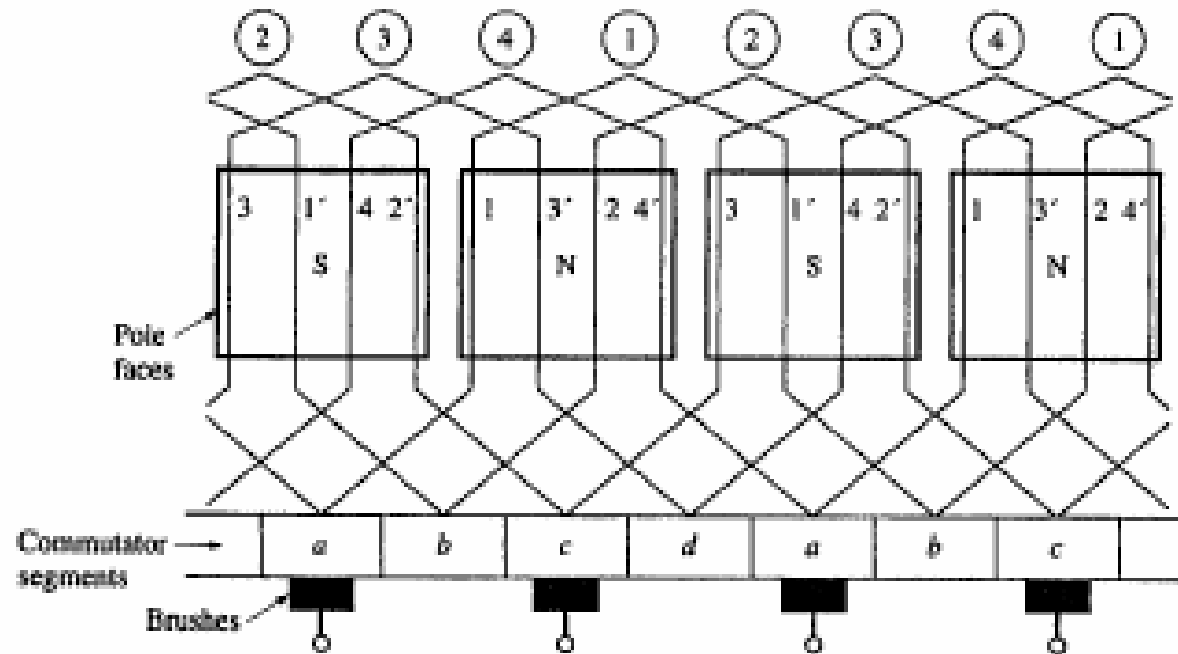
## Commutation in a Simple Four-Loop, 2 pole DC Machine

- This machine has 4 complete loops buried in slots carved in the laminated steel of its rotor
- The pole faces of the machine are curved to provide a uniform air-gap width and to give a uniform flux density everywhere under the faces
- The 4 loops of this machine are laid into the slots in a special manner
- The “unprimed” end of each loop is the outermost wire in each slot, while the “primed” end of each loop is the innermost wire in the slot directly opposite



# Commutation

- A winding Diagram showing interconnections of rotor loops



- The winding's connections to the machine's commutator are shown above:

Note : loop 1 stretches between commutator segments a and b, loop 2 stretches between segments b and c, and so forth around the rotor

# Commutation

- At the instant shown in figure (a), the 1, 2, 3' and 4' ends of the loops are under the north pole face, while the 1', 2', 3 and 4 ends of the loops are under the south pole face.

- The voltage in each of the 1, 2, 3' and 4' ends of the loops is given by

$$e_{\text{ind}} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}$$

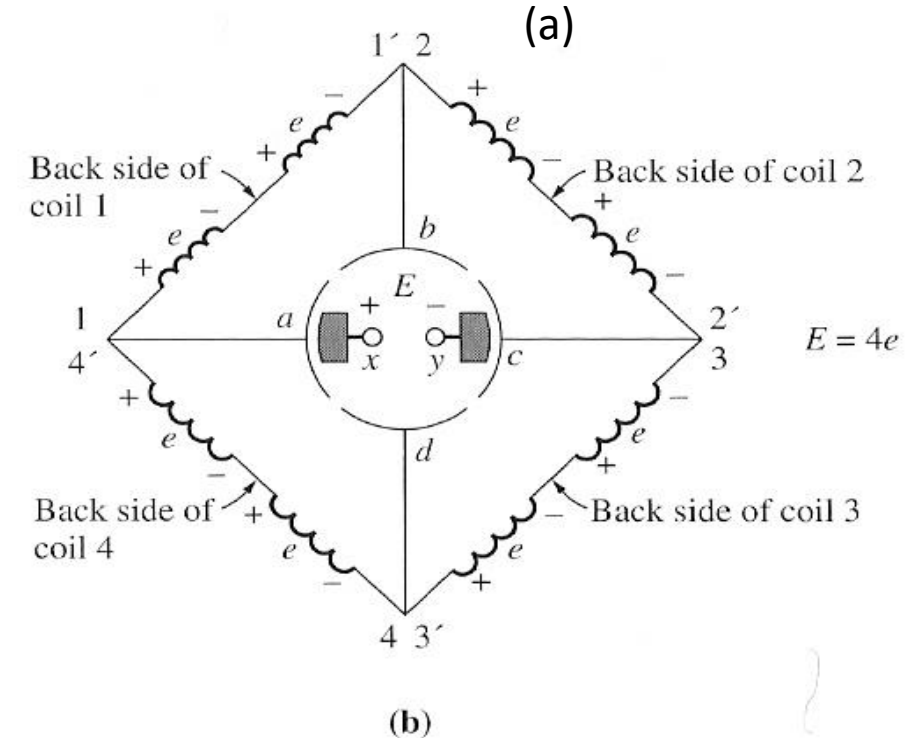
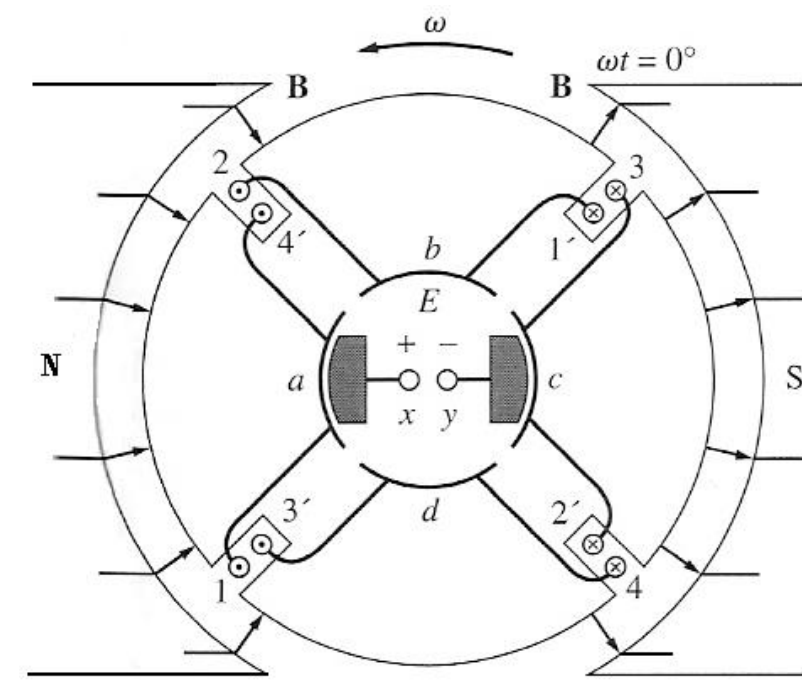
$$e_{\text{ind}} = vBl \text{ (positive out of page)}$$

- The voltage in each of the 1', 2', 3 and 4 ends of the loops is given by

$$e_{\text{ind}} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}$$

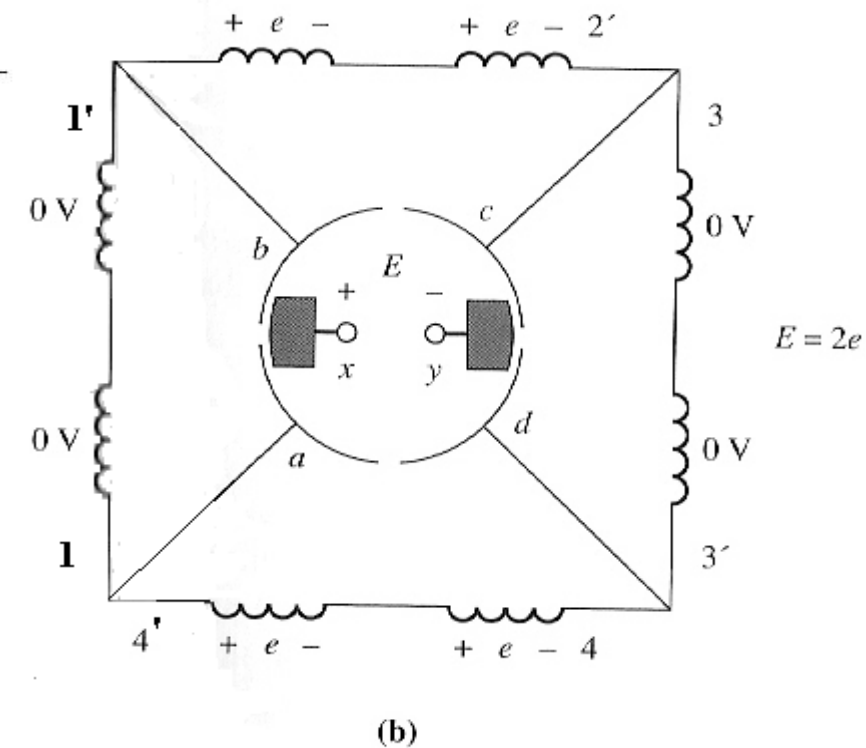
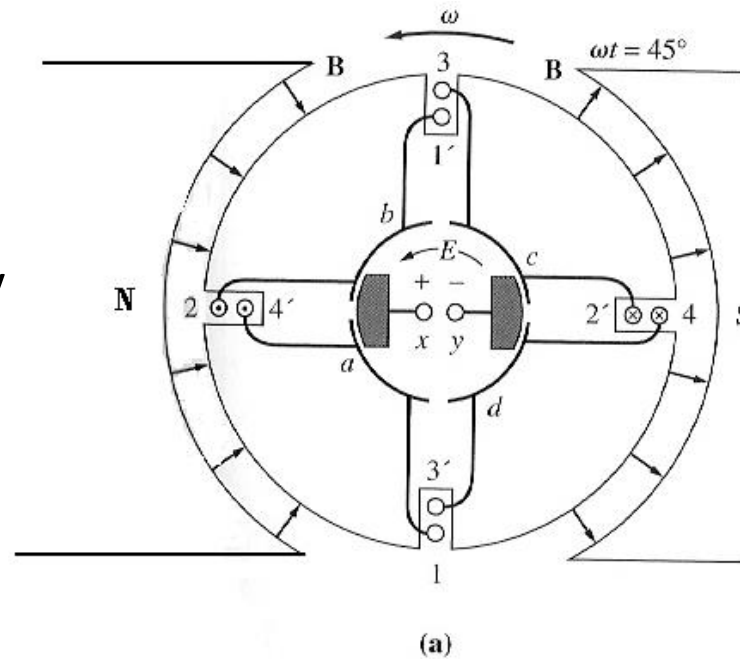
$$e_{\text{ind}} = vBl \text{ (positive into the page)}$$

- The overall result is shown in figure (b)
- Each coil represents one side (or conductor) of a loop
- If the induced voltage on any one side of a loop is called  $e = vBl$ , then the total voltage at the brushes of the machine is  $E = 4e$  ( $\omega t = 0^\circ$ )
- Note: there are two parallel paths for current through the machine
- The existence of two or more parallel paths for rotor current is a common feature of all commutation schemes



# Commutation

- What happens to the voltage  $E$  of the terminals as the rotor continues to rotate
- figure shows the machine at time  $\omega t = 45^\circ$
- At that time, loops 1 and 3 have rotated into the gap between the poles, so the voltage across each of them is zero
- Note: at this instant the brushes of the machine are shorting out commutator segments  $ab$  and  $cd$
- This happens just at the time when the loops between these segments have  $0\text{ V}$  across them, so shorting out the segments creates no problem
- At this time, only loops 2 and 4 are under the pole faces, so the terminal voltage  $E$  is given by:  
$$E = 2e \quad (\omega t = 45^\circ)$$
- Now, let the rotor continue to turn another  $45^\circ$ . The resulting situation is shown next slide.

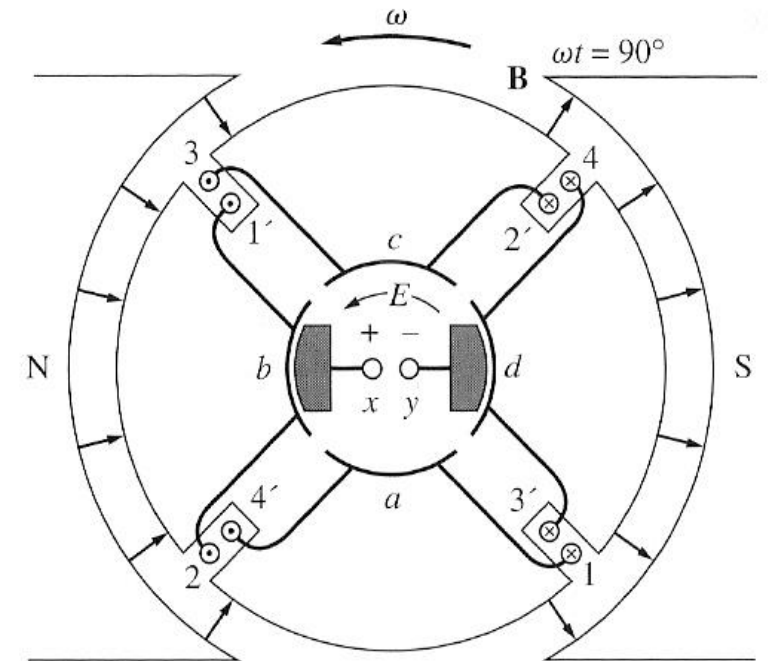
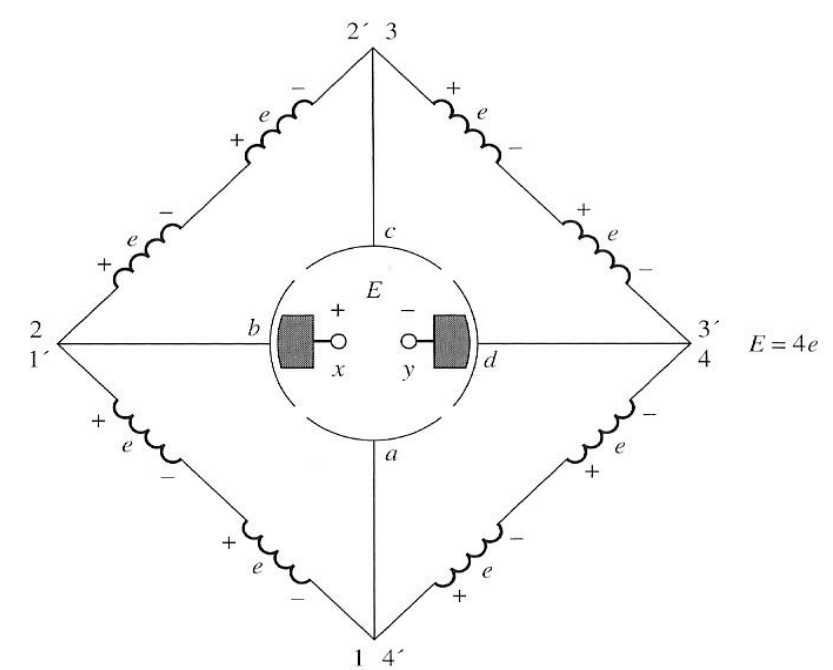


# Commutation

- Here, the 1', 2, 3, and 4' ends of the loops are under the north pole face, and the 1, 2', 3' and 4 ends of the loops are under the south pole face .
- The voltages are still built up out of the page for the ends under the north pole face and into the page for the ends under the south pole face
- There are now 4 voltage-carrying ends in each parallel path through the machine, so the terminal voltage E is given by

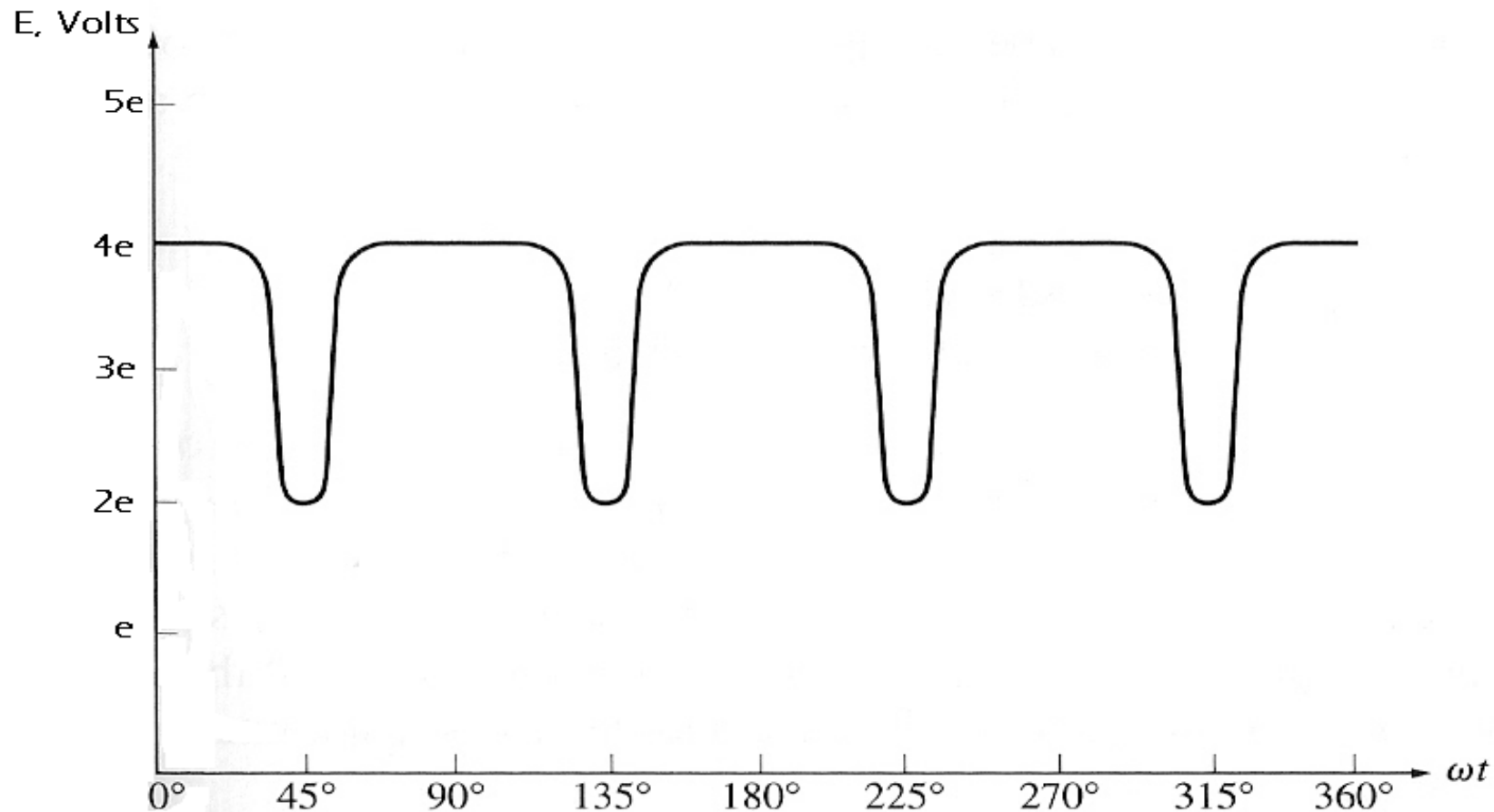
$$E = 4e \quad (\omega t = 90^\circ)$$

- Note: the voltages on loops 1 and 3 have reversed between the 2 pictures (from  $\omega t = 0^\circ$  to  $\omega t = 90^\circ$ ),
- However, since their connections have also reversed, the total voltage is still being built up in the same direction as before. This is the heart of every commutation scheme.



# Commutation

- The resulting voltage diagram is shown here:
- This is a better approximation to a constant dc level than the single rotating loop.
- As the number of loop of rotor increases, the approximation to a perfect dc voltage increases



# Armature Reaction

- In practice, there are two major effects that disturb the commutation process:

1- Armature Reaction

2-  $L \frac{di}{dt}$  voltages

## Armature Reaction

- If the magnetic field windings of a dc machine are connected to a power supply and the rotor of the machine is turned by an external source of mechanical power, then a voltage will be induced in the conductors of the rotor.
- This voltage will be rectified into dc output by the action of the machine's commutator.
- Now, connect a load to the terminals of the machine, and a current will flow in its armature windings.
- This current flow will produce a magnetic field of its own, which will distort the original magnetic field from the machine's poles.
- This distortion of the flux in a machine as the load is increased is called **armature reaction**.
- *It causes 2 serious problems in real dc machine*

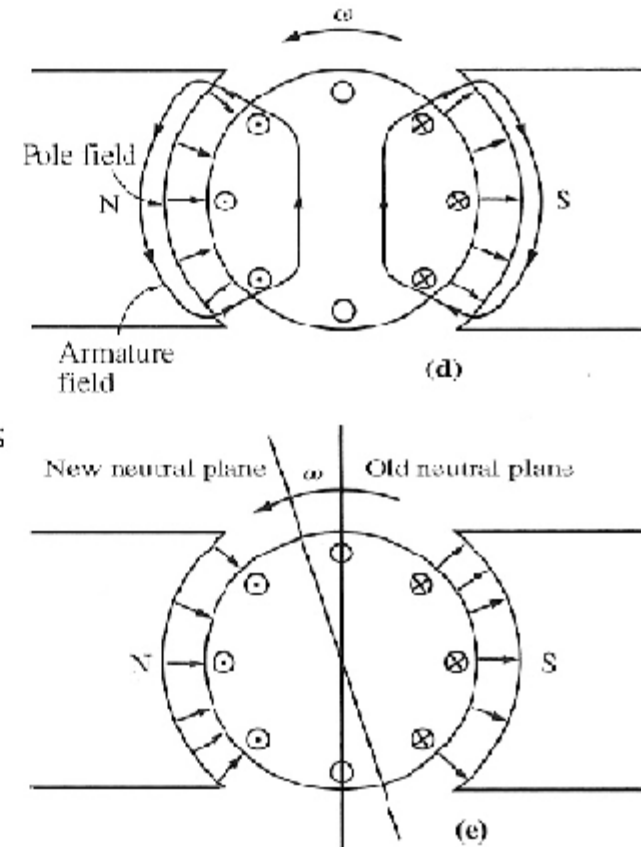
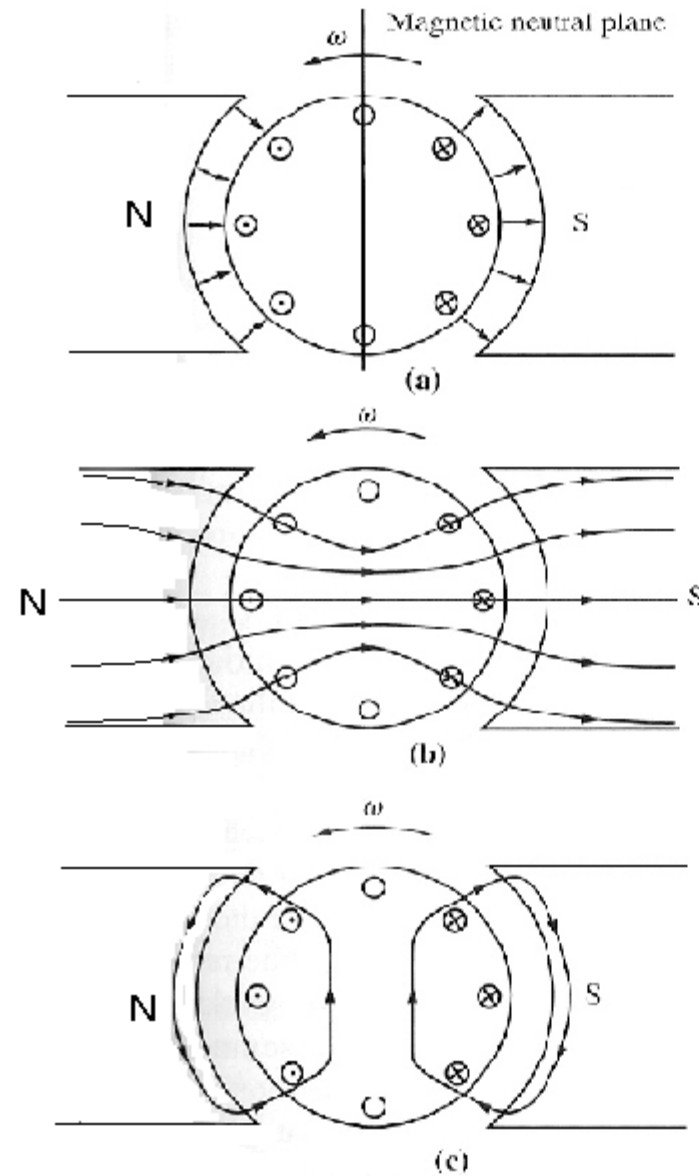
### Problem 1 : Neutral-Plane Shift

- The **magnetic neutral plane** is defined as the plane within the machine where the velocity of the rotor wires is exactly parallel to the magnetic flux lines
- *so that  $e_{ind}$  in the conductors in the plane is exactly zero*

# Armature Reaction

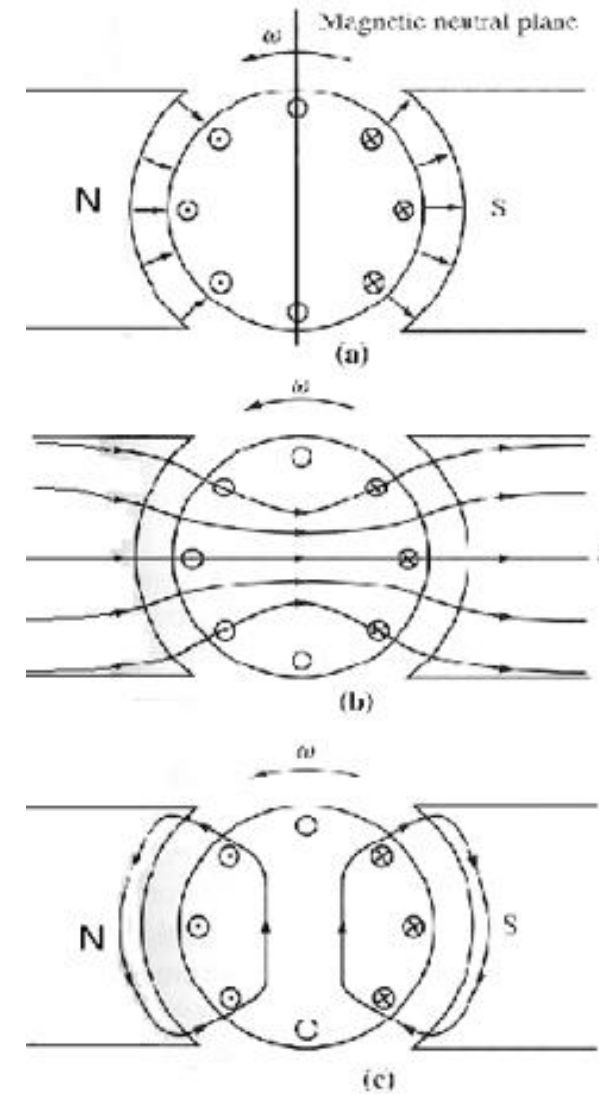
The development of armature reaction in dc generator:

- (a) Initially the pole flux is uniformly distributed & the magnetic neutral plane is vertical
- (b) The effect of the air gap on the pole flux distribution
- (c) The armature magnetic field resulting when a load is connected to the machine
- (d) Both rotor and pole fluxes are shown, indication points where they add and subtract
- (e) The resulting flux under the poles. The neutral plane has shifted in the direction of motion



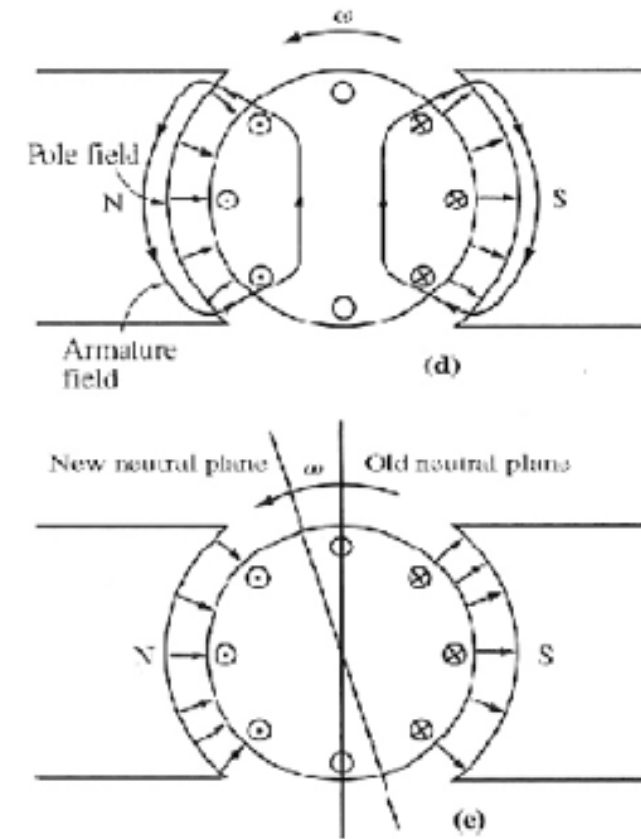
# Armature Reaction

- Figure (a) shows a two poles machine.
- Note: flux is distributed uniformly under the pole faces (in air gap)
- The rotor windings shown have voltages built up out of the page for wires under the north pole and into the page for wires under the south pole face
- The magnetic neutral plane in this machine is exactly vertical at this stage
- Now, suppose a load is connected to this machine so that it acts as a generator
- Current will flow out of the positive terminal of the generator
- Current will be flowing out of the page for wires under the north pole face and into the page for wires under the south pole face
- This current flow produces a magnetic field from the rotor windings, figure (c)
- This rotor magnetic field affects the original magnetic field from the poles that produced the generator's voltage.



# Armature Reaction

- In some places under the pole surfaces, it subtracts from the pole flux, and in other places it adds to the pole flux
- both rotor & pole fluxes shown, indicating points they add and subtract figure (d)
- The overall result is that the magnetic flux in the air gap of the machine is skewed, as shown in figure (e)
- Notice: the place on rotor where the induced voltage in a conductor would be zero (the neutral plane) has shifted.
- For the generator shown here, the magnetic neutral plane shifted in direction of rotation



In general, the neutral-plane shifts

(a) in “direction of motion” for **generator** &

(b) opposite to “direction of motion” for **a motor**

Furthermore, the amount of shift depends on the amount of rotor current and hence on the load of the machine

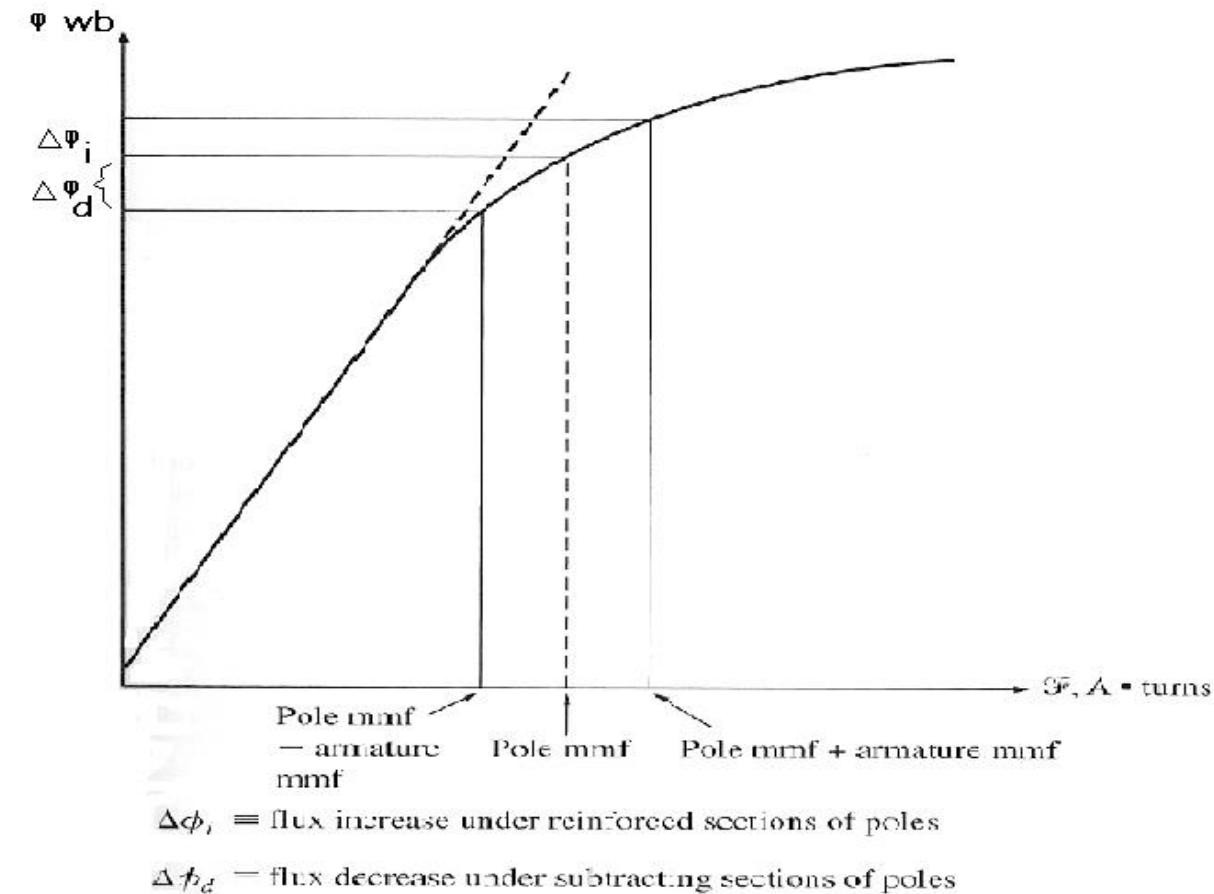
# Effect of Armature Reaction

- If brushes are set to short out conductors in the vertical plane, then voltage between segments is indeed zero until machine is loaded.
- **When machine is loaded, neutral plane shifts & brushes short out commutator segments with a finite voltage across them .**
- The result is a current flow circulating between shorted segments & large sparks at brushes when current path interrupted.
- This is a very serious problem, since it leads to drastically reduced brush life, pitting commutator segments & greatly increased maintenance cost
- **Note: this problem can not be solved even by placing brushes over full-load neutral plane, because then they would spark at no load**
- In extreme cases neutral plane shift can even lead **to flashover** in commutator segments near brushes
- Air near brushes in a machine is normally ionized as a result of sparking on brushes
- Flashover occurs when voltage of adjacent commutator segments gets large enough to sustain an arc in ionized air above them
- If flashover occurs, resulting arc can even melt commutator's surface.

# Armature Reaction

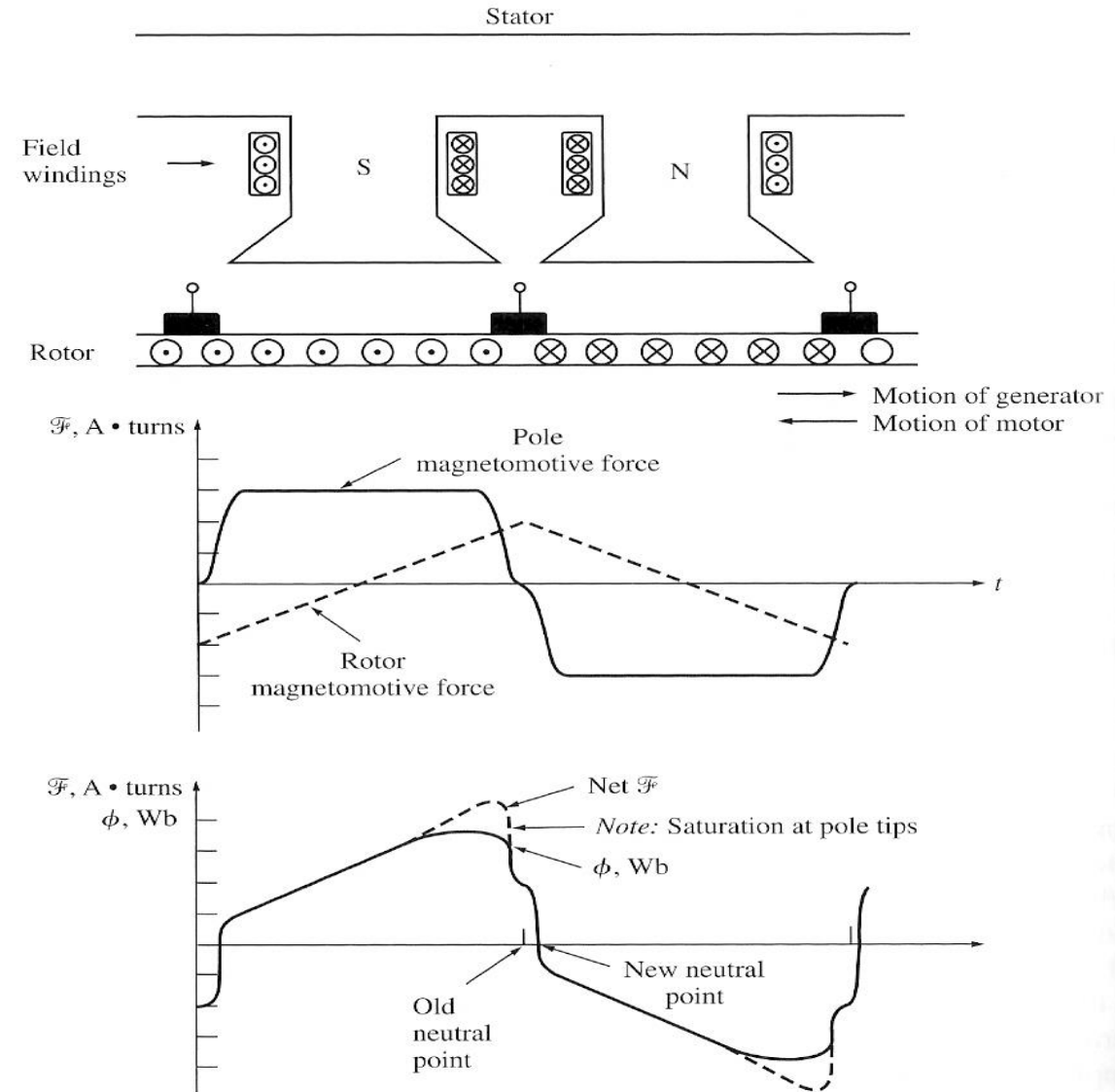
- Problem 2 : flux weakening

- Refer to magnetization curve:
- most machine operate at flux densities near saturation point
- Therefore at locations on pole surfaces, where rotor mmf adds pole mmf, only a small increase in flux occurs
- But at locations on pole surfaces where rotor mmf subtracts from pole mmf, there is a larger decrease in flux
- Net result → total average flux under entire pole face is decreased. Shown in the figure in the next slide.



# Armature Reaction

- Flux weakening causes problems in both generators & motors
- In generators effect of flux weakening is simply to reduce voltage supplied by generator for any given load
- In motors effect can be more serious
- As shown when flux in motor decreased, its speed increases
- But increasing speed of motor can increase its load, resulting in more flux weakening
- It is possible for some shunt dc motors to reach runaway condition as a result where speed of motor just keeps increasing until machine is disconnected, or been destroyed



Flux and magnetomotive force under the pole faces in a dc machine. Where mmf subtract, flux follows the net mmf force; but where mmf add, saturation limits total flux.

# Armature Reaction: Solution

3 approaches to (partially or completely) rectify problems of armature reaction

- **Brush Shifting**
- **Commutating Poles or Interpoles**
- **Compensating Windings**
- To improve process of commutation in real dc machines, we must stop the sparking at brushes caused by neutral-plane shifts and  $L \frac{di}{dt}$  effects.
- 1st approach: if neutral plane of machine shifts, why not shift the brushes with it in order to stop sparking?
- However there are several serious problems associated with it:
  - 1- neutral plane moves with every change in load , & shift direction reverses when machines goes from motor operation to generation operation, and brushes should be adjusted every time load changed
  - 2- shifting brushes may stop brush sparking, however can aggravate flux-weakening since:
    - (a) Rotor mmf now has a vector component opposes mmf of poles
    - (b) Change in armature current distribution cause flux to bunch up even more at saturated parts of pole faces

# Equivalent Circuit of DC Generators

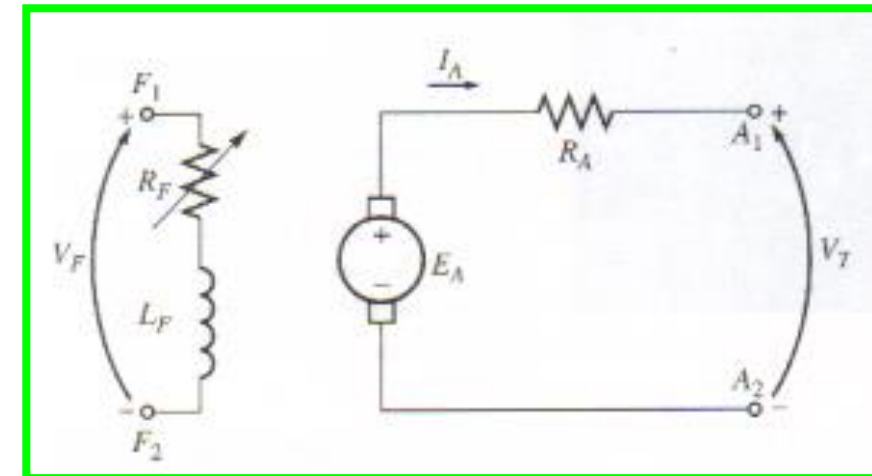
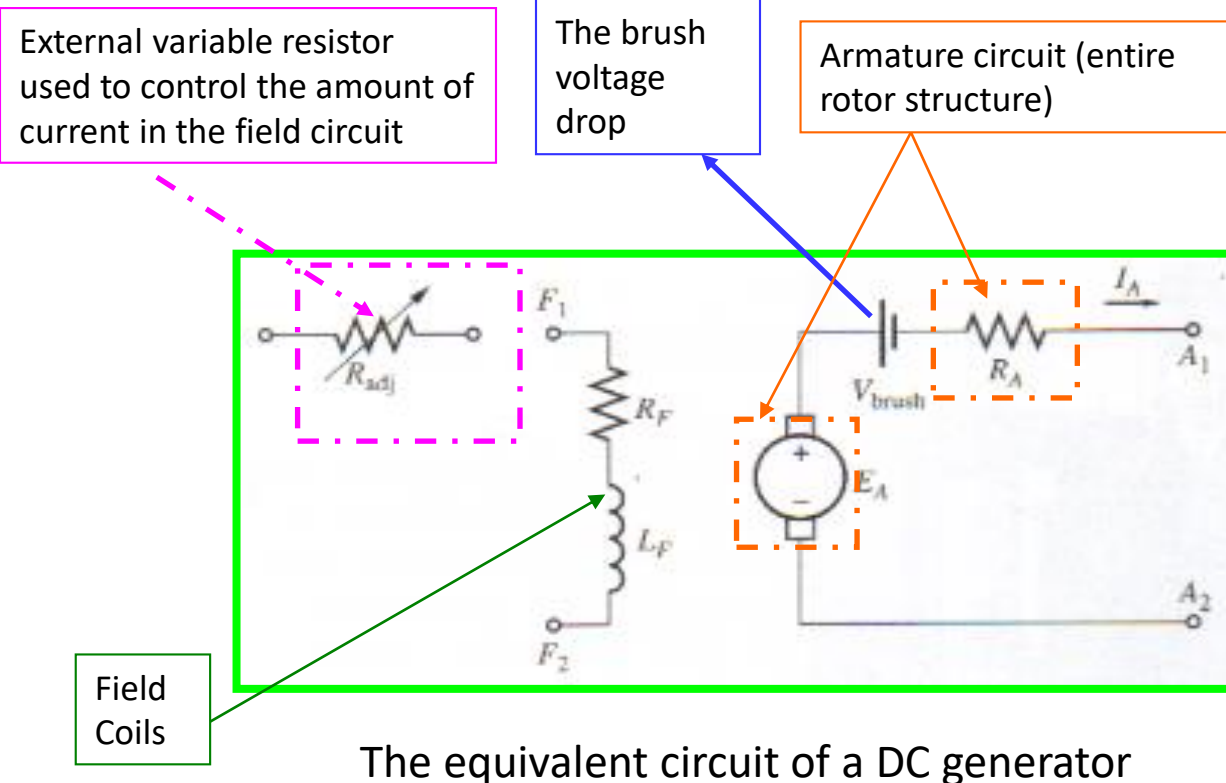
These various types of dc generator differ in their terminal (voltage-current) characteristic, and the application is depending to which is suited.

DC generators are compared by their voltages, power ratings, efficiencies and voltage regulations:

$$VR = \frac{V_{nl} - V_{fl}}{V_{fl}} \times 100\%$$

+VR = Dropping characteristics

-VR = Rising characteristic



A simplified equivalent circuit of a DC generator, with  $R_F$  combining the resistances of the field coils and the variable control resistor

# Separately Excited Generator

A separately excited DC generator is a generator whose field current is supplied by a separately external DC voltage source

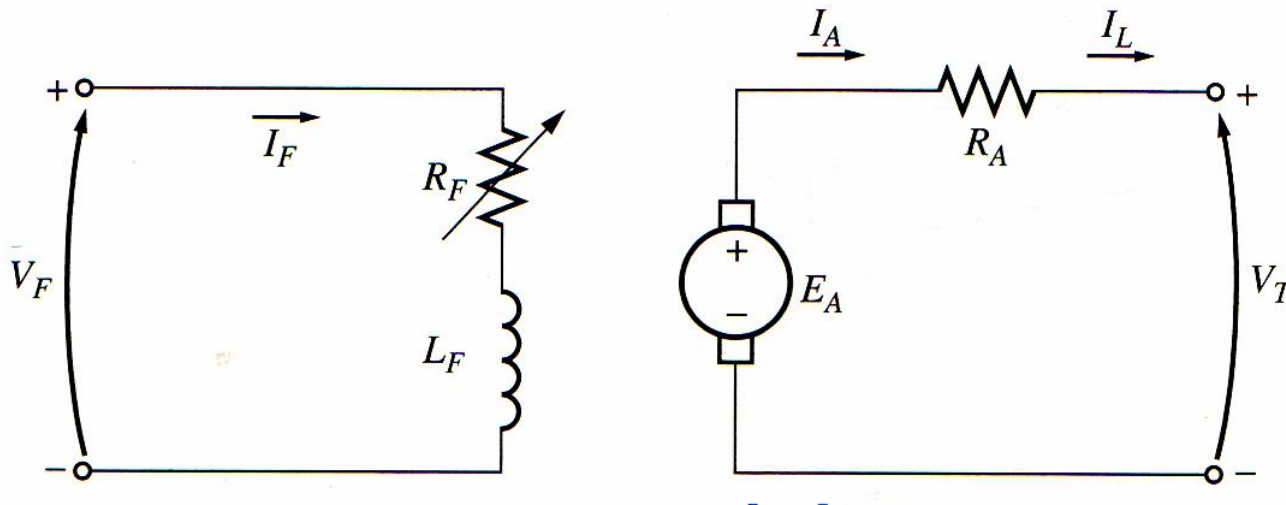
$V_T$  = Actual voltage measured at the terminals of the generator

$I_L$  = current flowing in the lines connected to the terminals.

$E_A$  = Internal generated voltage.

$I_A$  = Armature current.

$$I_L = I_A$$



$$I_F = \frac{V_F}{R_F}$$

$$V_T = E_A - I_A R_A$$

Fig : Equivalent circuit of Separately excited DC generator

# The Terminal Characteristic of A Separately Excited DC Generator

For DC generator, the output quantities are its **terminal voltage** and **line current**. The terminal voltage is

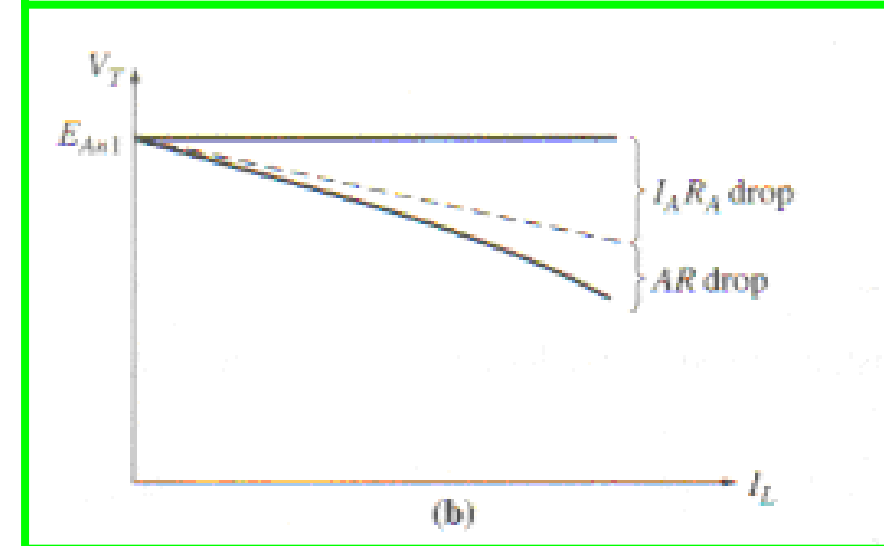
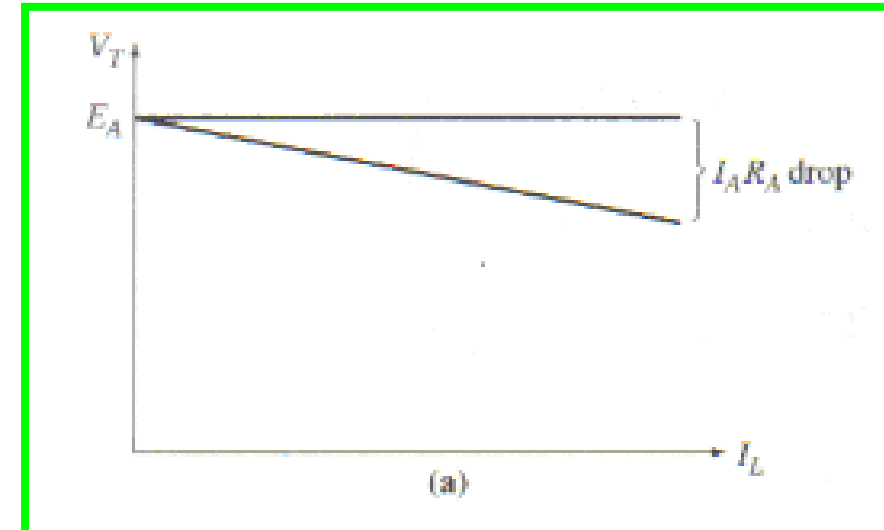
$$V_T = E_A - I_A R_A \quad (I_A = I_L)$$

Since the internal generated voltage  $E_A$  is independent of  $I_A$ , the **terminal characteristic** of the separately excited generator is a **straight line**.

- When the **load** supplied by the generator is **increased**,  $I_L$  (and therefore  $I_A$ ) **increase**.
- As the armature current increase, the  $I_A R_A$  **drop increase**, so the **terminal voltage** of the generator **falls**. **Figure (a)**

This terminal characteristic is not always entirely accurate.

- In the generators **without compensating windings**, an **increase in  $I_A$**  causes an **increase in the armature reaction**, and **armature reaction causes flux weakening**.
- This flux weakening causes a decrease in  $E_A = K\Phi\omega$  which further decreases the terminal voltage of the generator. The resulting terminal characteristic is shown in **Figure (b)**



The terminal characteristic of a separately excited dc generator (a) with and (b) without compensating windings ( $E_A = K\Phi\omega$ )

# Control of Terminal Voltage

We control torque-speed in DC Motor, while in DC Generator we control  $V_T$

- The terminal voltage of a separately excited DC generator can be controlled by changing the internal generated voltage  $E_A$  of the machine.

$$V_T = E_A - I_A R_A$$

- If  $E_A$  increases,  $V_T$  will increase, and if  $E_A$  decreases,  $V_T$  will decrease. Since the internal generated voltage =  
 $E_A = K\Phi\omega$

there are **two possible ways** to control the voltage of this generator:

1. Change the **speed of rotation**. If  $\omega$  increases, then  $E_A = K\Phi\omega \uparrow$  increases, so  $V_T = E_A \uparrow - I_A R_A$  increases too.
2. Change the **field current**. If  $R_F$  is decreased, then the field current increases ( $I_F = V_F / R_F$ ). Therefore, the flux  $\Phi$  in the machine increases. As the flux rises,  $E_A = K\Phi \uparrow \omega$  must rise too, so  $V_T = E_A - I_A R_A$  increases.

# The Shunt DC Generator

A shunt DC generator is a DC generator that supplies its **own field current** by having its field connected directly across the terminals of the machine.

- The armature current of the machine is supplies both by the field current and the load current.
- No external power supply is required for the field circuit.
- Because of generator supply it own field current, it required voltage buildup.

$$I_A = I_F + I_L$$
$$V_T = E_A - I_A R_A$$
$$I_F = \left( \frac{V_T}{R_F} \right)$$

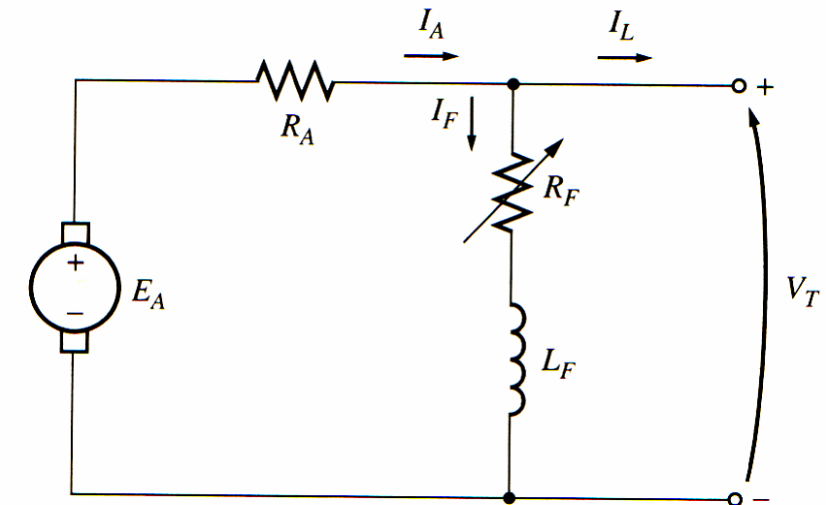


Figure : The equivalent circuit of a shunt DC generator.

# Voltage Buildup in A Shunt Generator

Assume the DC generator has no load connected to it and that the prime mover starts to turn the shaft of the generator. The voltage buildup in a DC generator depends on the presence of a *residual flux* in the poles of the generator.

This voltage is given by

$$E_A = K\Phi_{res} \omega$$

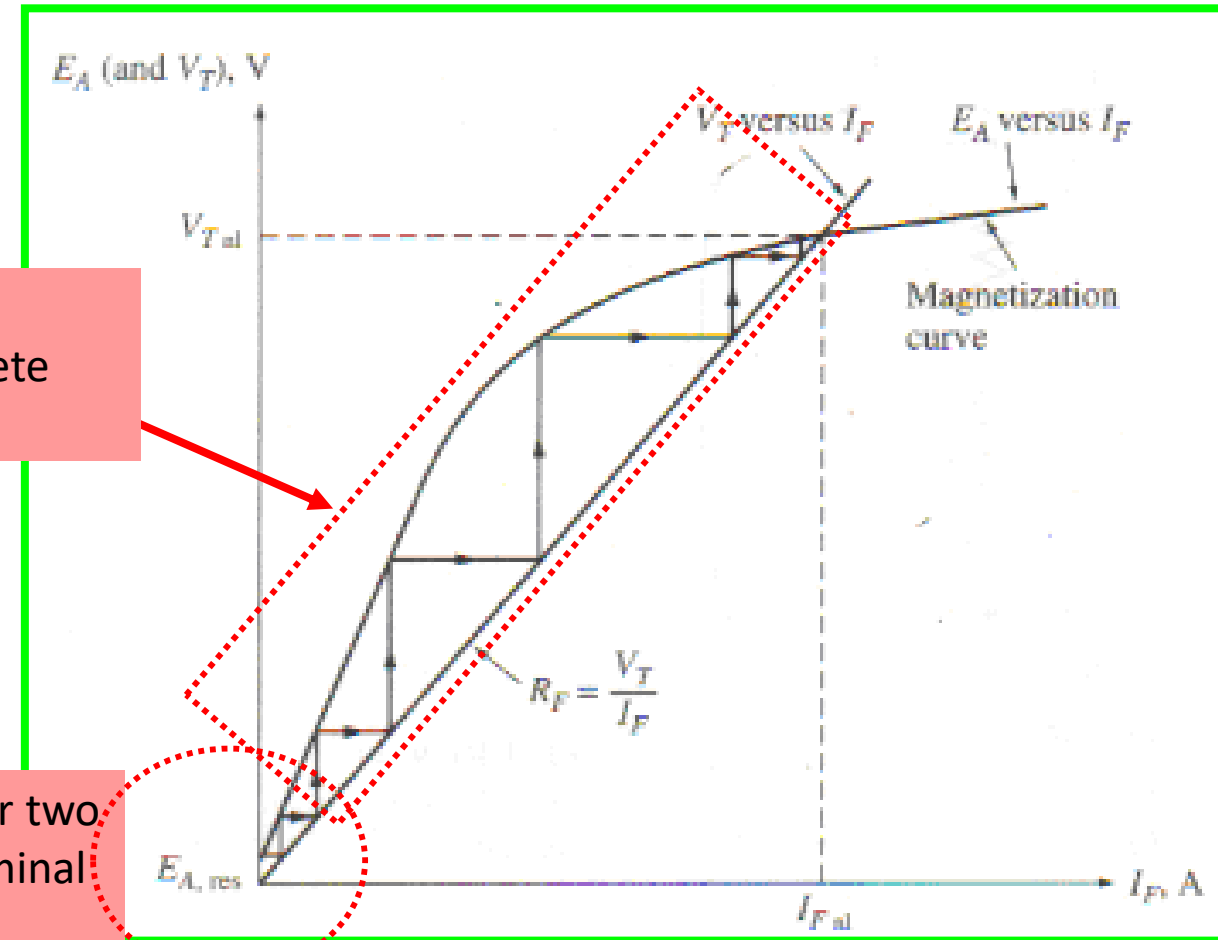
This voltage,  $E_A$  (a volt or two appears at terminal of generators), and it causes a current  $I_F$  to flow in the field coils. This field current produces a magnetomotive force in the poles, which increases the flux in them.

$\Phi \uparrow \rightarrow E_A \uparrow$  ( $E_A = K\Phi\omega$ ), then  $V_T$  increase and cause further increase  $I_F \uparrow$ , which further increasing the flux  $\Phi \uparrow$  and so on.

The final operating voltage is determined by intersection of the field resistance line and saturation curve.

Voltage buildup occurred in discrete steps

$E_A$  may be a volt or two appear at the terminal during start-up



# Voltage Buildup in A Shunt Generator

Several causes for the voltage to fail to build up during starting which are :

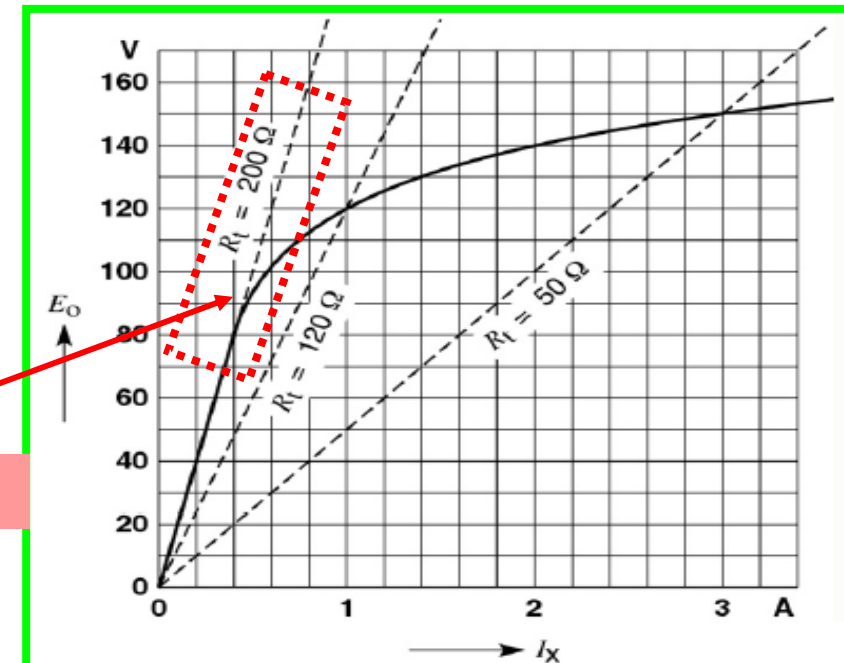
- *Zero Residual magnetism*. If there is **no residual flux** in the poles, there is no Internal generated voltage,  $E_A = 0V$  and the voltage will never build up.
- *The direction of rotation of the generator may have been reversed, or the connections of the field may have been reversed*. In either case, the residual flux produces an internal generated voltage  $E_A$ . The voltage  $E_A$  produce a field current which produces a flux **opposing the residual flux, instead of adding to it**.

Under these conditions, **the flux** actually **decreases below  $\Phi_{res}$**  and **no voltage can ever build up**.

- *Critical resistance*. Normally, the shunt generator builds up to a voltage determined by the intersection of the field resistance line and the saturation curve. If the **field resistance is greater than critical resistance**, the generator fails to build up and the voltage remains at the residual level. To solve this problem, the field resistance is reduced to a value less than critical resistance.

page 604-605 (Chapman)

Critical resistance



## What is critical speed of DC shunt generator?

- **Critical speed:**

It is the minimum speed of the armature which is required to build up EMF by the generator, so all the generators work above the critical speed.

- **Critical resistance:**

It is the maximum field resistance above which a DC generator couldn't develop EMF, so all the generator's must have their field resistance below the critical field resistance.

## How do you find the critical speed of a DC generator?

- Critical speed is defined as the speed at which the given shunt field resistance is equal to the critical resistance. It is the speed at which the shunt generator just fails to build up its voltage without any external resistance in the field circuit. It is denoted by  $N_c$ .

### Significance of Critical Speed :

- If the speed of the dc machine is reduced to a value such that the field resistance line does not pass through the O.C.C then the machine fails to induce i.e., it may not generate any appreciable voltage. The speed at which the generator fails to build up the voltage is known as the critical speed.
- Below this speed, the field resistance line does not cut the O.C.C. Hence fails to build up the voltage. Therefore the machine should not be operated below the critical speed as it fails to induce. Hence, it should always be operated above the critical speed.

# The Terminal Characteristic of a Shunt DC Generator

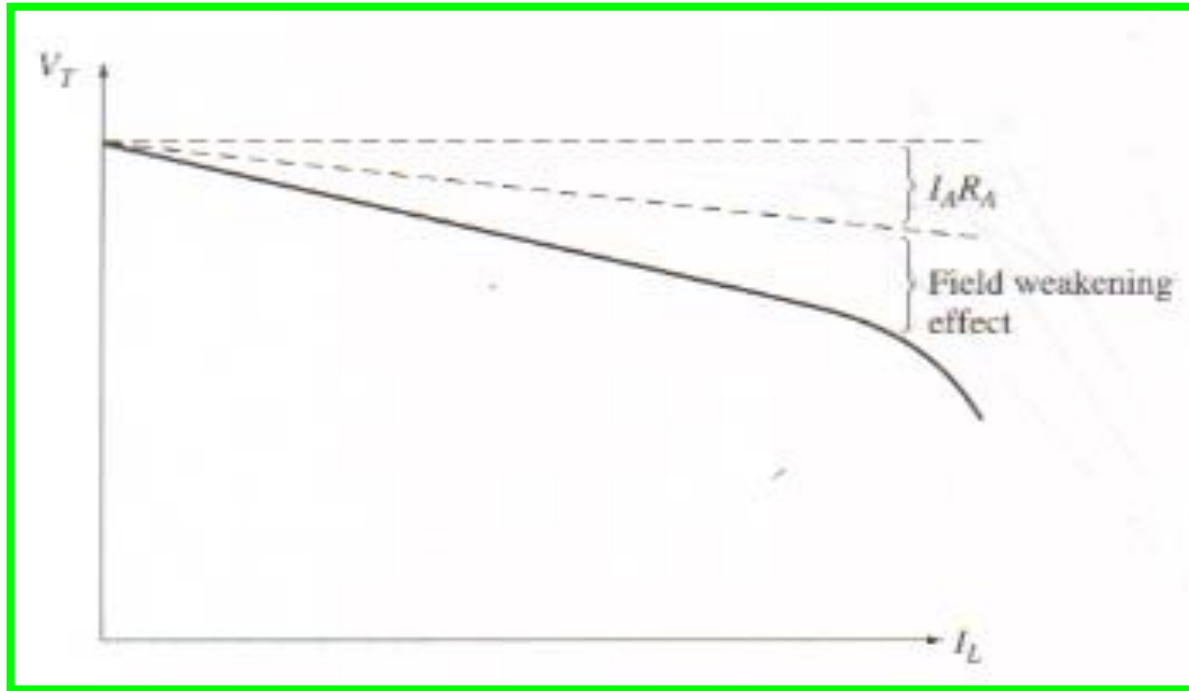


Figure : The terminal characteristic of a shunt dc generator

As the load on the generator is increased,  $I_L$  increases and so  $I_A \uparrow = I_F + I_L$  also increase. An increase in  $I_A$  increases the armature resistance voltage drop  $I_A R_A$ , causing  $V_T = E_A - I_A \uparrow R_A$  to decrease.

However, when  $V_T$  decreases, the field current  $I_F \downarrow$  in the machine decreases with it. This causes the flux in the machine to decrease; decreasing  $E_A$ . Decreasing  $E_A$  causes a further decrease in the terminal voltage,  $V_T = E_A - I_A R_A$

# Voltage Control for Shunt DC Generator

There are two ways to control the voltage of a shunt generator:

1. Change the shaft **speed**,  $\omega_m$  of the generator.
2. Change the **field resistor** of the generator, thus changing the field current.

Changing the field resistor is the **principal method** used to control terminal voltage in real **shunt generators**. If the field resistor  $R_F$  is decreased, then the field current  $I_F = V_T / R_F \downarrow$  increases.

When  $I_F \uparrow$ , the machine's flux  $\Phi \uparrow$ , causing the internal generated voltage  $E_A \uparrow$ .  $E_A \uparrow$  causes the terminal voltage of the generator to increase as well.

# The Series DC Generator

A series DC generator is a generator whose field is **connected in series with its armature**. Because the **field winding** has to **carry the rated load current**, it usually have **few turns of heavy wire**.

Clear distinction, **shunt generator** tends to maintain a **constant terminal voltage** while the **series generator** has tendency to **supply a constant load current**.

The Kirchhoff's voltage law for this equation :

$$V_T = E_A - I_A(R_A + R_S)$$

The magnetization curve of a series DC generator looks very much like the magnetization curve of any other generator.

- At **no load**, however, there is **no field current**, so  $V_T$  is **reduced to a very small level** given by the residual flux in the machine.
- As the **load increases**, the **field current rises**, so  $E_A$  rises rapidly. The  $I_A(R_A + R_S)$  drop goes up too, but at **first the increase in  $E_A$  goes up more rapidly than the  $I_A(R_A + R_S)$  drop rises**, so  $V_T$  increases.
- After a while, the machine approaches saturation, and  $E_A$  becomes almost constant. At that point, the resistive drop is the predominant effect, and  $V_T$  starts to fall.
- Armature reaction will cause very large current flow which causes arc.

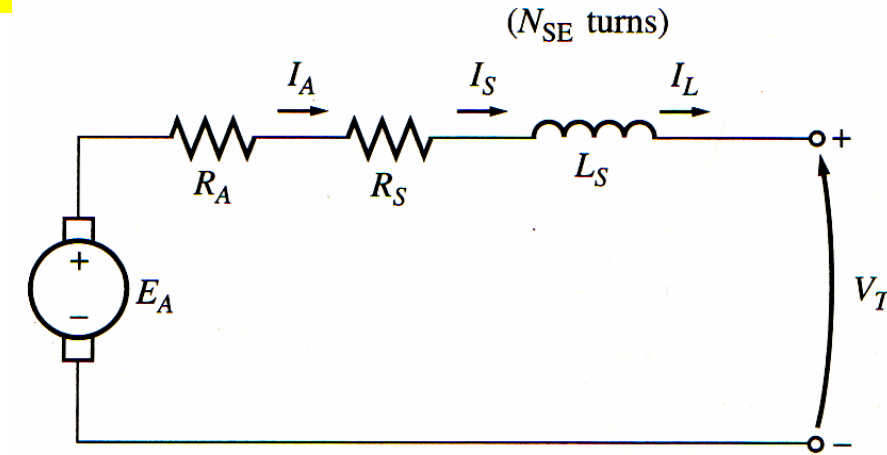


Figure : The equivalent circuit of a series dc generator

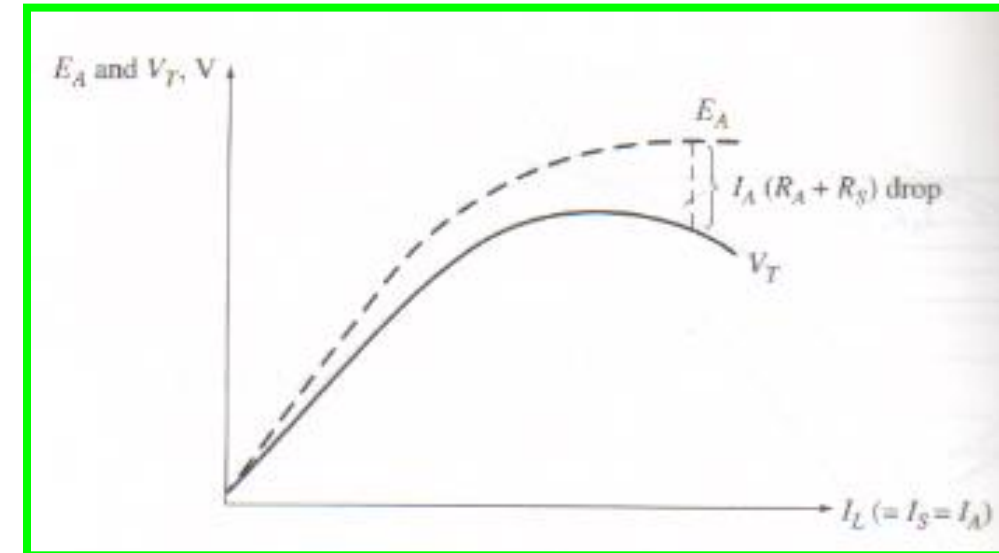


Figure : A series generator terminal characteristic with large armature reaction effects

# The Cumulatively Compounded DC Generator

A cumulatively compounded DC generator is a DC generator with both **series and shunt fields**, connected so that the **magnetomotive forces** from the two fields are **additive**.

$$I_A = I_F + I_L$$

$$V_T = E_A - I_A(R_A + R_S)$$

$$I_F = \frac{V_T}{R_F}$$

The total magnetomotive force on this machine is given by

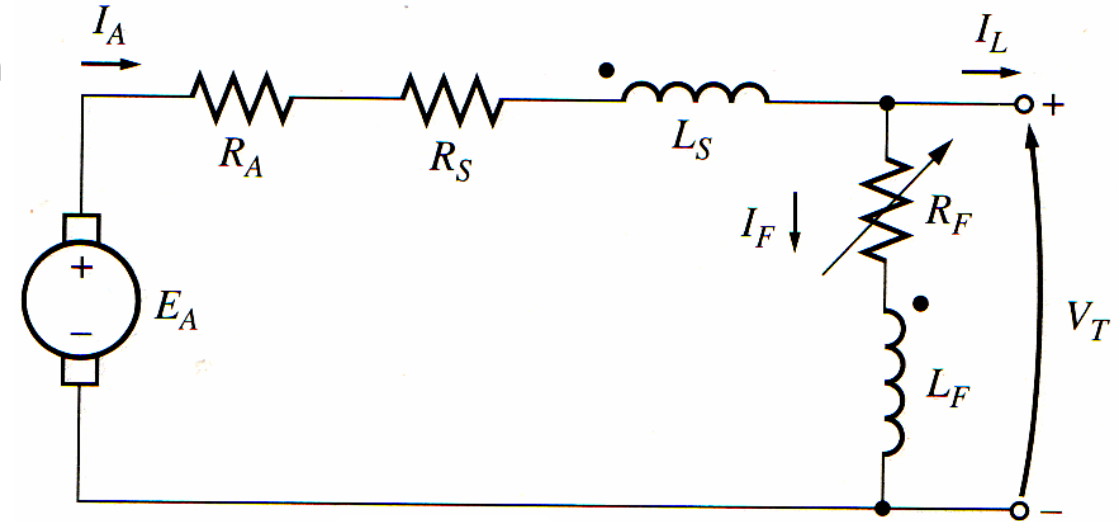
$$F_{\text{net}} = F_F + F_{SE} - F_{AR}$$

where  $F_F$  = the shunt field magnetomotive force  
 $F_{SE}$  = the series field magnetomotive force  
 $F_{AR}$  = the armature reaction magnetomotive force

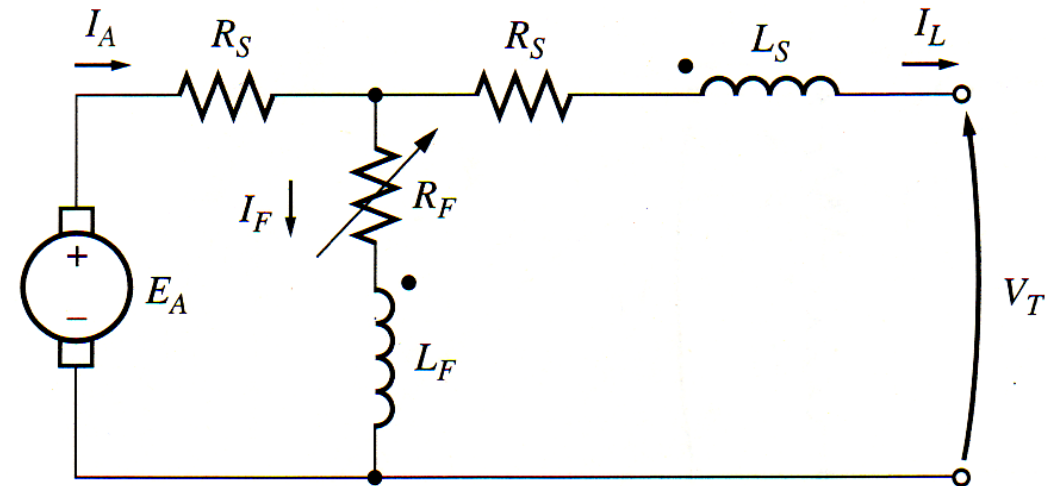
$$N_F I_F^* = N_F I_F + N_{SE} I_A - F_{AR}$$

Effective shunt field current

$$I_F^* = I_F \pm \frac{N_{SE}}{N_F} I_A - \frac{F_{AR}}{N_F}$$



Long shunt connection



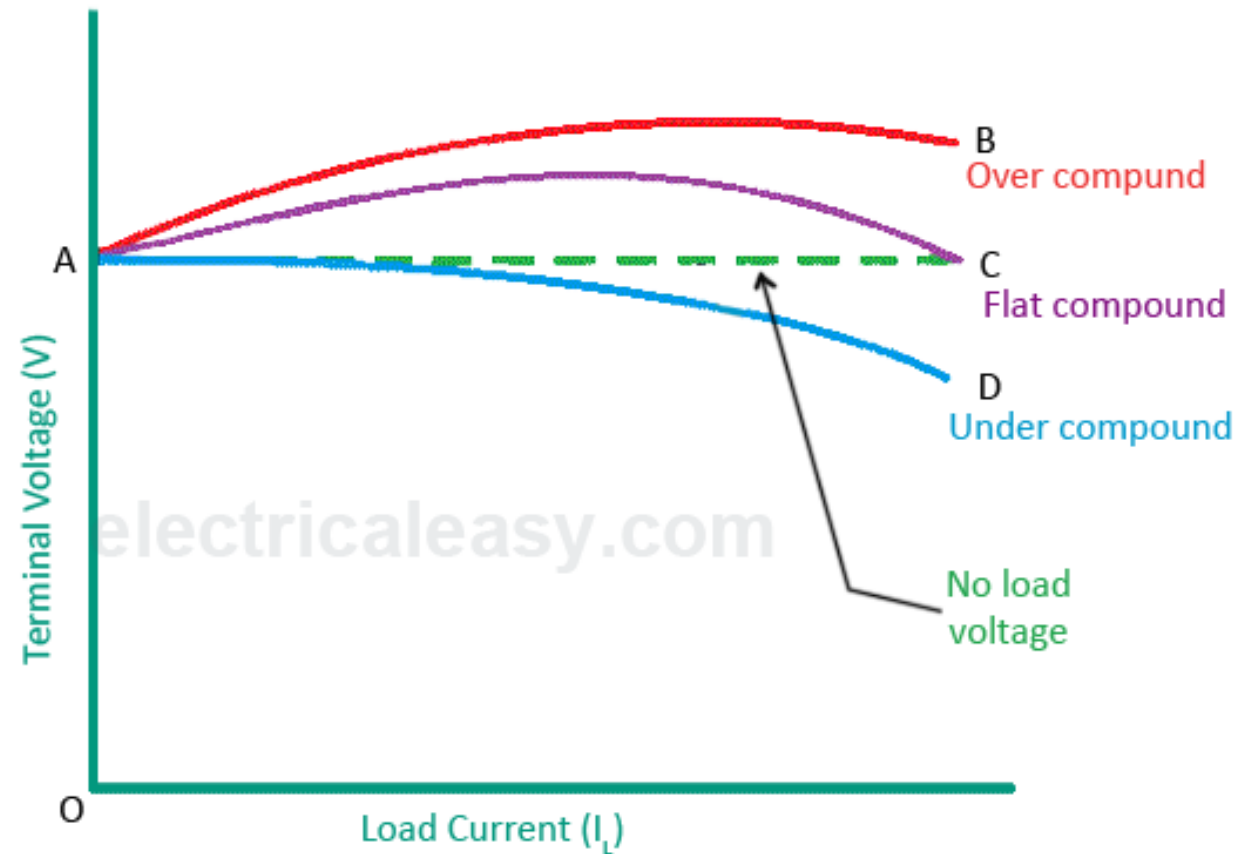
Short shunt connection

# The Terminal Characteristic of a Cumulatively Compounded DC Generator

When the **load** on the generator is **increased**, the load current  **$I_L$**  also **increases**.

Since  $I_A = I_F + I_L \uparrow$ , the armature current  **$I_A$**  **increases** too. At this point two effects occur in the generator:

1. As  $I_A$  increases, the  $I_A (R_A + R_S)$  voltage drop increases as well. This tends to cause a **decrease** in the terminal voltage,  $V_T = E_A - I_A (R_A + R_S)$ .
2. As  $I_A$  **increases**, the series field magnetomotive force,  $F_{SE} = N_{SE} I_A$  increases too. This **increases the total magnetomotive force**,  $F_{tot} = N_F I_F + N_{SE} I_A$  which **increases the flux** in the generator. The increased flux in the generator **increases  $E_A$** , which in turn tends to make  $V_T = E_A - I_A (R_A + R_S)$  **rise**.



External characteristic of DC compound generator

# The Terminal Characteristic of a Cumulatively Compounded DC Generator

The two effects above oppose each other, one increases  $V_T$  and the other decreases  $V_T$ . So which effect will predominate in a given machine?

1. Few Series turns ( $N_{SE}$  small):
  - Under-compounded.
  - Resistive voltage drop effect wins hands down.
  - Full load terminal voltage < no load terminal voltage.
2. More series turns ( $N_{SE}$  larger):
  - Flat-compounded.
  - The flux-strengthening effect wins and the terminal voltage rises with load.
  - As load increases, magnetic saturation kicks in; the resistive drop becomes stronger.
  - Full load terminal voltage = no load terminal voltage.
3. Even more series turns ( $N_{SE}$  largest):
  - Over-compounded.
  - Flux-strengthening effect predominates for larger period of time before resistive drop takes over.
  - Full load terminal voltage > no load terminal voltage.

# Voltage Control of Cumulatively Compounded DC Generator

The techniques available for controlling the terminal voltage of a cumulatively compounded DC generator are exactly the same as the technique for controlling the voltage of a shunt DC generator:

1. Change the **speed of rotation**. An increase in  $\omega$  causes  $E_A = K\Phi\omega$  to increase, increasing the terminal voltage  $V_T = E_A \uparrow - I_A (R_A + R_S)$ .
2. Change the **field current**. A decrease in  $R_F$  causes  $\uparrow I_F = V_T / R_F \downarrow$  to increase, which increase the total magnetomotive force in the generator. As  $F_{\text{tot}}$  increases, the flux  $\Phi$  in the machine increases, and  $E_A = K\Phi\omega$  increases. Finally, an increase in  $E_A$  raises  $V_T$ .

# The Differentially Compounded DC Generator

A differentially compounded DC generator is a generator with both shunt and series fields, but this time their magnetomotive forces subtract from each other.

The net magnetomotive force is

$$F_{\text{net}} = F_F - F_{SE} - F_{AR}$$

$$F_{\text{net}} = N_{FIF} - N_{SE}I_A - F_{AR}$$

And the equivalent shunt field current due to the series field and armature reaction is given by :

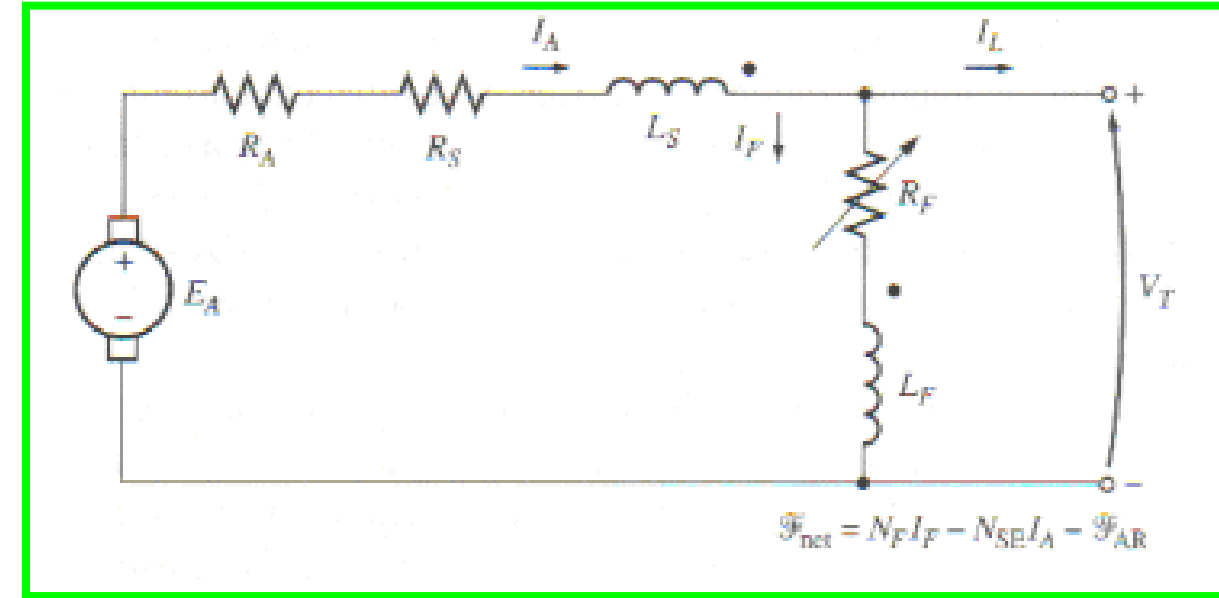
$$I_{eq} = -\frac{N_{SE}}{N_F} I_A - \frac{F_{AR}}{N_F}$$

The total effective shunt field current in this machine is

$$I_F^* = I_F + I_{eq}$$

or

$$I_F^* = I_F - \frac{N_{SE}}{N_F} I_A - \frac{F_{AR}}{N_F}$$



The equivalent circuit of a differentially compounded DC generator

$$I_A = I_L + I_F$$

$$I_F = \frac{V_T}{R_F}$$

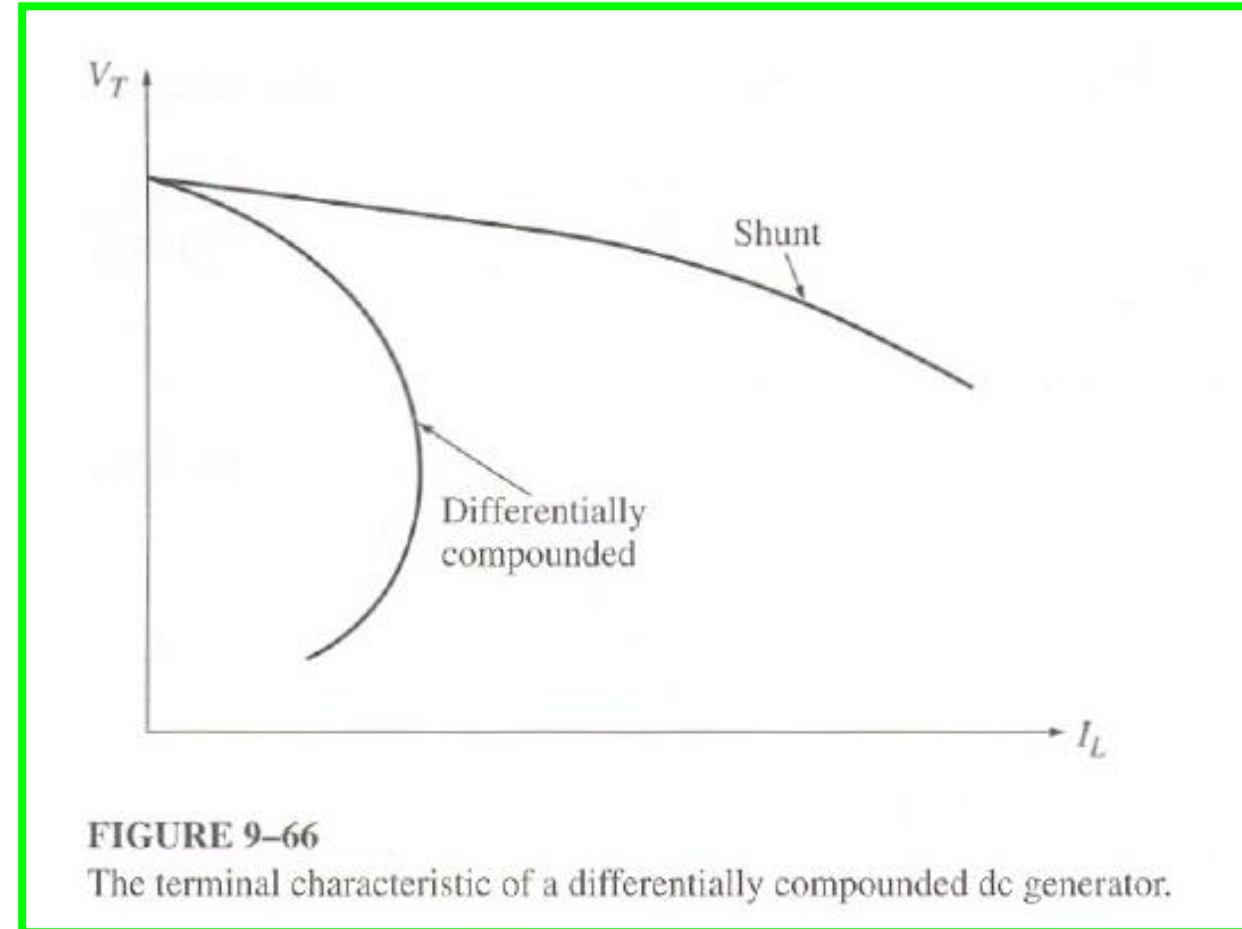
$$V_T = E_A - I_A(R_A + R_F)$$

# Terminal characteristics of Differentially Compounded DC Generator

Two effects occur in the terminal characteristic of a differentially compounded DC generator are

1. As  $I_A$  increases, the  $I_A (R_A + R_S)$  voltage drop increases as well. This increase tends to cause the terminal voltage to decrease  $V_T = E_A - I_A \uparrow (R_A + R_S)$ .
2. As  $I_A$  increases, the series field magnetomotive  $F_{SE} = N_{SE}I_A$  increases too. This increases in series field magnetomotive force reduces the net magnetomotive force on the generator, ( $F_{tot} = N_F I_F - N_{SE} I_A \uparrow$ ), which in turn reduces the net flux in the generator. A decrease in flux decreases  $E_A$ , which in turn decreases  $V_T$ .

Since both effects tend to decrease  $V_T$ , the voltage drop drastically as the load is increased on the generator.



# Voltage Control of Differentially Compounded DC Generator

The techniques available for adjusting terminal voltage are exactly the same as those for shunt and cumulatively compounded DC generator:

1. Change the speed of rotation,  $\omega_m$ .
2. Change the field current,  $I_F$ .

# DC MOTOR

# DC Motor

A same DC machine can be used as a motor or generator. Construction of a DC motor is same as that of a DC generator.

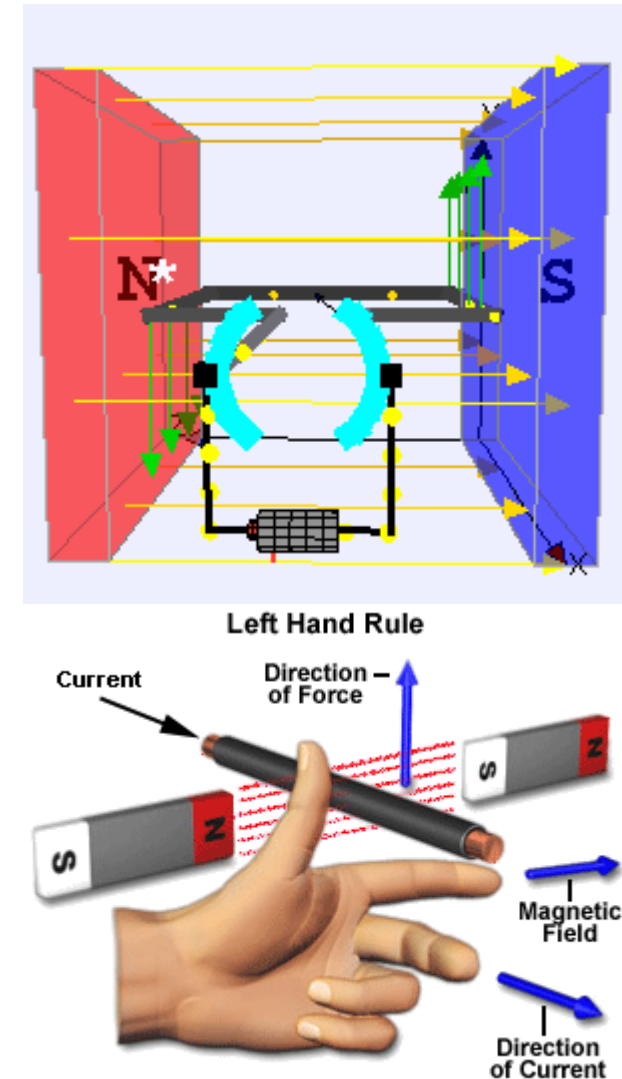
- A motor is an electrical machine which converts electrical energy into mechanical energy.
- The **principle of working of a DC motor** is that "*whenever a current carrying conductor is placed in a magnetic field, it experiences a mechanical force*".
- The direction of this force is given by Fleming's left hand rule and it's magnitude is given by

$$F = BIL.$$

Where, B = magnetic flux density, I = current and L = length of the conductor within the magnetic field.

- When armature windings are connected to a DC supply, current sets up in the winding.
- The current carrying armature conductors experience force due to the field magnetic field.
- Commutator is made segmented to achieve unidirectional torque. Otherwise, the direction of force would have reversed every time when the direction of movement of conductor is reversed the magnetic field.

**Fleming's left hand rule:** If we stretch the first finger, second finger and thumb of our left hand to be perpendicular to each other AND direction of magnetic field is represented by the first finger, direction of the current is represented by second finger then the thumb represents the direction of the force experienced by the current carrying conductor.



# Working of DC Motor

Consider a part of a multipolar dc motor as shown in Fig. (1).

When the terminals of the motor are connected to an external source of dc supply:

(i) The field magnets are excited developing N and S poles;

(ii) The armature conductors supplied with current.

- All conductors under N pole carry currents downwards while all the conductors under S-pole carry currents upwards direction as shown in Fig.(1).
- Since each armature conductor is carrying current and is placed in the magnetic field, mechanical force acts on it. Referring to Fig.(1) and applying Fleming's left hand rule, it is clear that force on each conductor is tending to rotate the armature in anticlockwise direction.
- All these forces add together to produce a driving torque which sets the armature rotating.
- When the conductor moves from one side of a brush to the other, the current in that conductor is reversed using **commutator** and at the same time it comes under the influence of next pole which is of opposite polarity. Consequently, the direction of force on the conductor remains the same.

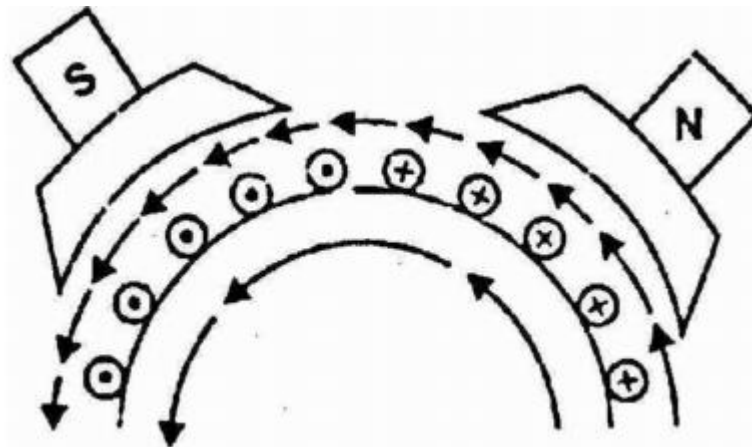
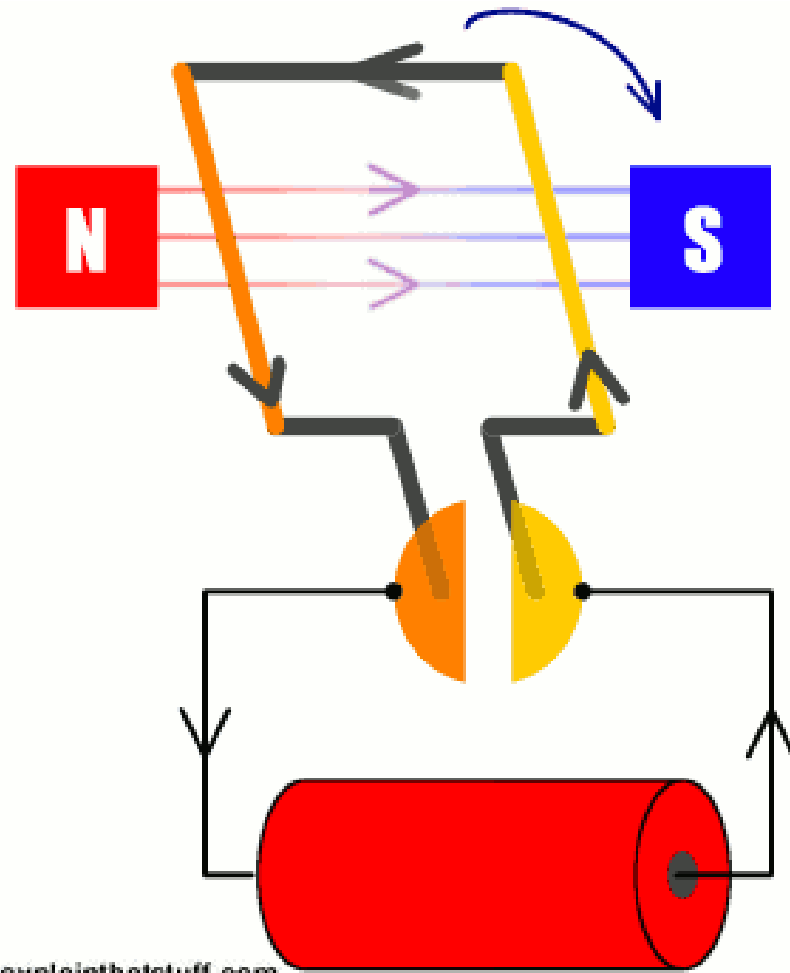
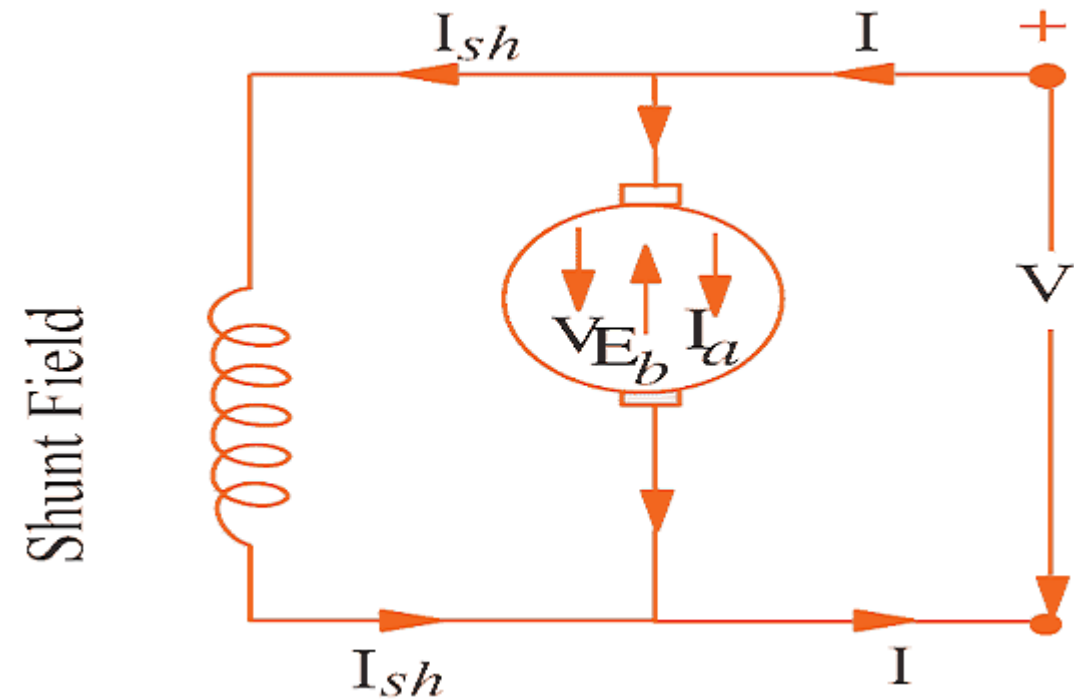


Fig.(1)



# Back or Counter EMF

- When the armature of a dc motor rotates under the influence of the driving torque, the armature conductors move through the magnetic field and hence emf is induced in them as in a generator.
- The induced emf acts in opposite direction to the applied voltage  $V$  (Lenz's law) and is known as back or counter emf  $E_b$ .
- The back emf  **$E_b (= P\phi ZN/60 \text{ A})$**  is always less than the applied voltage  $V$ , although this difference is small when the motor is running under normal conditions.
- A shunt wound dc motor shown in figure.
- When dc voltage  $V$  is applied across the motor terminals, the field magnets are excited and armature conductors are supplied with current.
- Therefore, driving torque acts on the armature which begins to rotate.
- As the armature rotates, back emf  $E_b$  is induced which opposes the applied voltage  $V$ .
- The applied voltage  $V$  has to force current through the armature against the back emf  $E_b$ .
- The electric work done in overcoming and causing the current to flow against  $E_b$  is converted into mechanical energy developed in the armature. It follows, therefore, that energy conversion in a d.c. motor is only possible due to the production of back emf  $E_b$ .



Net voltage across armature circuit =  $V - E_b$

If  $R_a$  is the armature circuit resistance, then,  $I_a = (V - E_b)/R_a$

# Significance of Back EMF

The presence of back emf makes the d.c. motor a self-regulating machine i.e., it makes the motor to draw as much armature current as is just sufficient to develop the torque required by the load.

$$\text{Armature Current, } I_a = (V - E_b)/R_a$$

- When the motor is running on no load, small torque is required to overcome the friction and windage losses. Therefore, the armature current  $I_a$  is small and the back emf is nearly equal to the applied voltage.
- If the motor is suddenly loaded, the first effect is to cause the armature to slow down. Therefore, the speed at which the armature conductors move through the field is reduced and hence the back emf  $E_b$  falls. The decreased back emf allows a larger current to flow through the armature and larger current means increased driving torque. Thus, the driving torque increases as the motor slows down. The motor will stop slowing down when the armature current is just sufficient to produce the increased torque required by the load.
- If the load on the motor is decreased, the driving torque is momentarily in excess of the requirement so that armature is accelerated. As the armature speed increases, the back emf  $E_b$  also increases and causes the armature current  $I_a$  to decrease. The motor will stop accelerating when the armature current is just sufficient to produce the reduced torque required by the load.

It follows, therefore, that back emf in a d.c. motor regulates the flow of armature current i.e., it automatically changes the armature current to meet the load requirement.

# Losses in a DC Motor

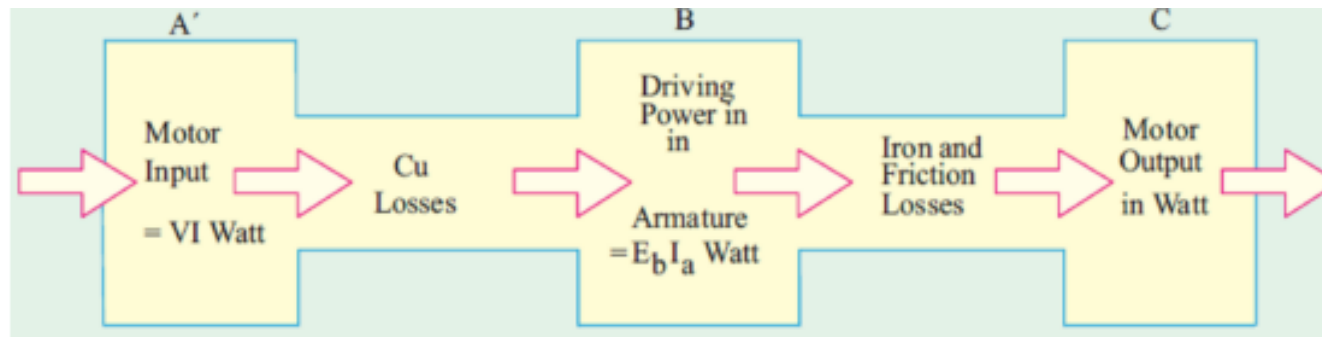
The losses occurring in a dc motor are the same as in a dc generator. These are:

- (i) copper losses and Iron losses or magnetic losses
- (ii) Mechanical losses

As in a generator, these losses cause (a) an increase of machine temperature and (b) Reduction in the efficiency of the dc motor.

The following points may be noted:

1. Apart from armature Cu loss, field Cu loss and brush contact loss, Cu losses also occur in interpoles (commutating poles) and compensating windings. Since these windings carry armature current ( $I_a$ ),  
Loss in interpole winding =  $I_a^2 \times \text{Resistance of interpole winding}$   
Loss in compensating winding =  $I_a^2 \times \text{Resistance of compensating winding}$
2. Since dc machines (generators or motors) are generally operated at constant flux density and constant speed, the iron losses are nearly constant.
3. The mechanical losses (i.e. friction and windage) vary as the cube of the speed of rotation of the dc machine (generator or motor). Since dc machines are generally operated at constant speed, mechanical losses are considered to be constant.



# Equations of DC Motor

## Voltage and power Equations of DC Motor:

$$V = E_b + I_a R_a$$
$$I_a V = I_a E_b + I_a^2 R_a$$

## Armature Torque of a Motor

Let  $T_a$  be the torque developed by the armature of a motor running at  $N$  r.p.m. If  $T_a$  is in N.m, then

- Power developed =  **$T_a \cdot 2\pi N/60$  watt**
- Electrical power converted into mechanical power in the armature =  **$E I_a$  watt**
- So we get :  **$T_a \cdot 2\pi N/60 = E I_a$**
- We know  **$E = \phi ZNP/(60A)$  Volt**
- So,

$$T_a = \frac{1}{2\pi} \phi Z I_a \left( \frac{P}{A} \right) \text{ N.m}$$

$\therefore T_a \propto I_a$ .

- In the case of a series motor,  $\phi$  is directly proportional to  $I_a$  (before saturation) because field windings carry full armature current.

$$\therefore T_a \propto I_a^2$$

- For shunt motors,  $\phi$  is practically constant.

$$\therefore T_a \propto I_a$$

# Equations of DC Motor

## Shaft Torque (T<sub>sh</sub>)

The torque which is available for doing useful work is known as shaft torque T<sub>sh</sub>. It is so called because it is available at the shaft:

$$T_{sh} = \frac{P_{out}}{2\pi N / 60} \text{ N.m}$$

The difference (T<sub>a</sub>-T<sub>sh</sub>) is known as lost torque and is due to iron and friction losses of the motor.

## speed regulation

DC motors are compared by their speed regulation:

$$SR = [\omega_{nl} - \omega_{fl}] / \omega_{fl} \times 100\%$$

- It is a rough measure of shape of motor's torque-speed characteristic.
- A positive regulation means speed drops with increasing load & a negative speed regulation means speed increases with increasing load.
- Magnitude of S.R. approximately show how steep is the slope of torque-speed
- Dc motors driven from a dc power supply (unless specified) and input voltage assumed constant)

# Types of DC Motor

Five major types of dc motor:

1- separately excited dc motor

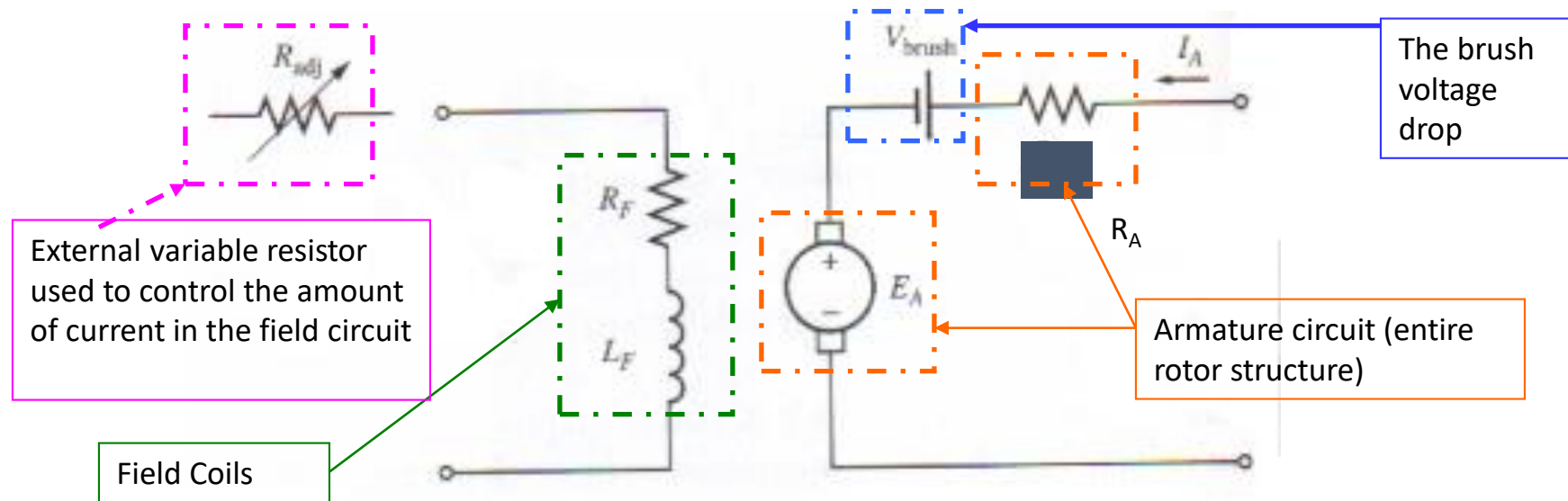
3- permanent-magnet dc motor

5-compounded dc motor

2-shunt dc motor

4- series dc motor

## DC Motor Equivalent Circuit



*Note: Because a dc motor is the same physical machine as a dc generator, its equivalent circuit is exactly the same as generator except for the direction of current flow.*

# The Magnetization Curve of a DC machine

The internal generated voltage in the motor  $E_A = K\Phi\omega$

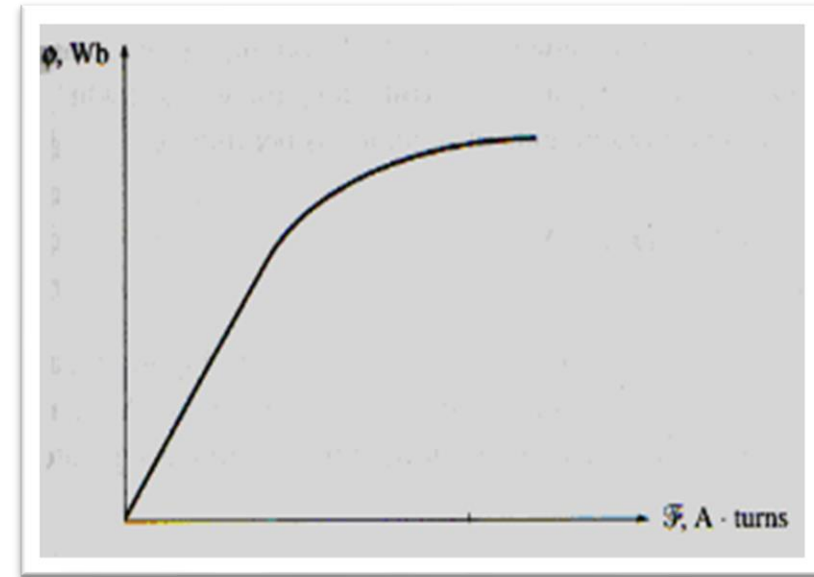
From the equation,

$E_A$  is directly proportional to the flux ( $\Phi$ ) in the motor and speed of the motor ( $\omega$ ).

The field current ( $I_F$ ) in dc machines produces a field magnetomotive force (mmf)

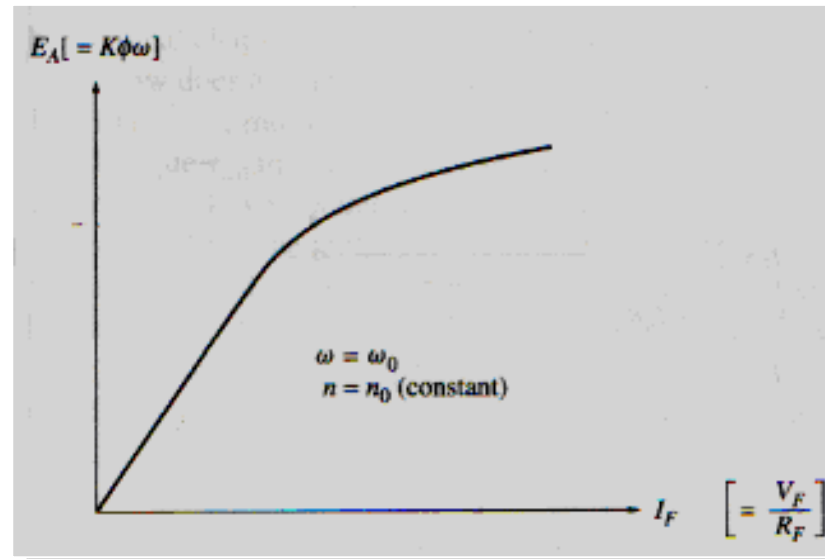
This magnetomotive force (mmf) produces a flux ( $\Phi$ ) in the motor in accordance with its magnetization curve.

$$I_F \rightarrow \text{mmf} \rightarrow \text{flux}$$



The magnetization curve of a ferromagnetic material ( $\Phi$  vs  $F$ )

Since the field current ( $I_F$ ) is directly proportional to magnetomotive force (mmf) and  $E_A$  is directly proportional to the flux, the magnetization curve is presented as a plot  $E_A$  versus field current for a given speed.



*Note: To get the maximum possible power, the motors and generators are designed to operate near the saturation point on the magnetization curve (at the knee of the curve).*

The induced torque developed by the motor is given as

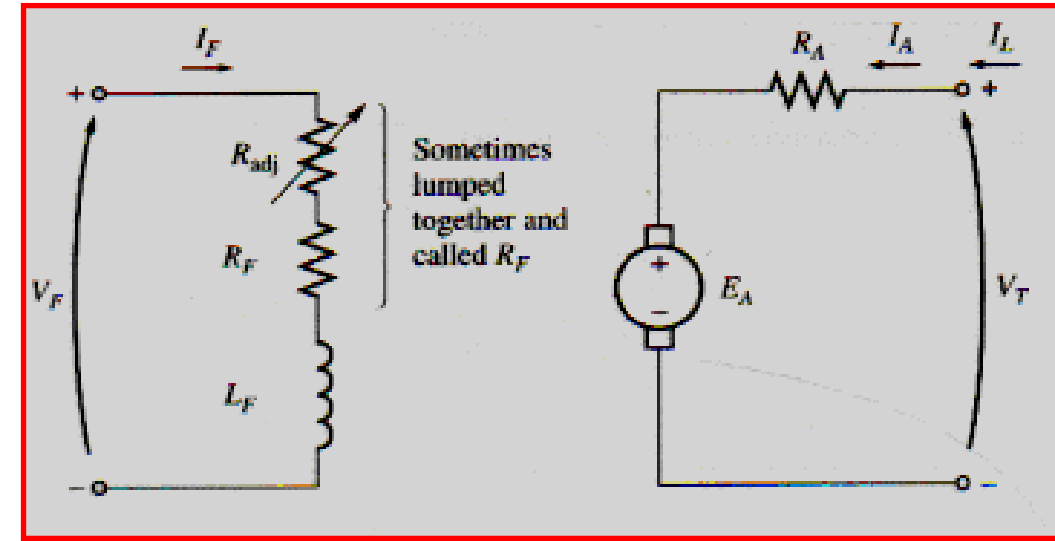
$$\tau_{ind} = K\Phi I_A$$

The magnetization curve of a dc machine expresses as a plot of  $E_A$  versus  $I_F$ , for a fixed speed  $\omega_0$

# The equivalent circuit of Separately Excited DC Motor

Separately excited motor is a motor whose field current is supplied from a separate constant-voltage power supply.

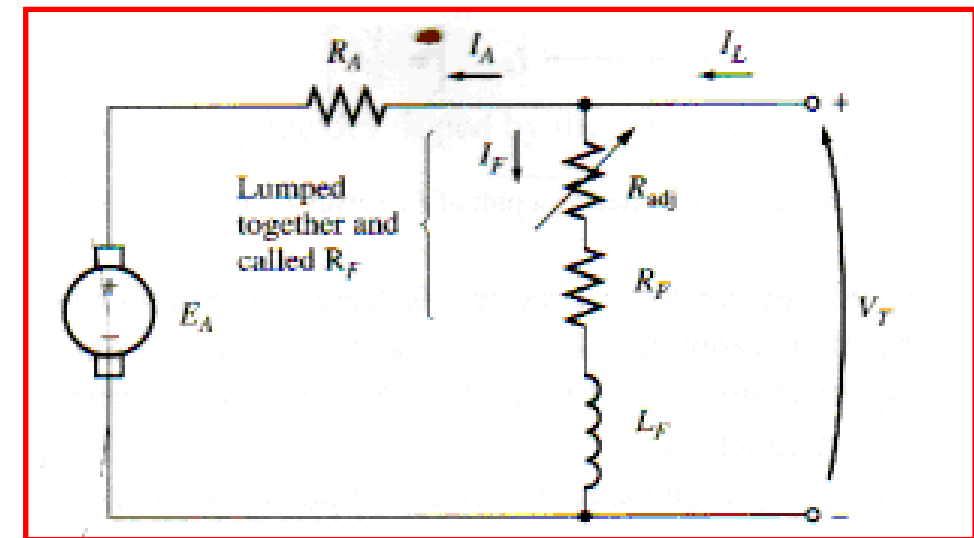
$$I_F = \frac{V_F}{R_F} \quad V_T = E_A + I_A R_A \quad I_L = I_A$$



# The equivalent circuit of a Shunt DC Motor

A shunt dc motor is a motor whose field circuit get its power directly across the armature terminals of the motor.

$$I_F = \frac{V_T}{R_F}$$
$$V_T = E_A + I_A R_A$$
$$I_L = I_A + I_F$$



# Shunt DC Motor Characteristics

Following are the three important characteristics of a dc motor:

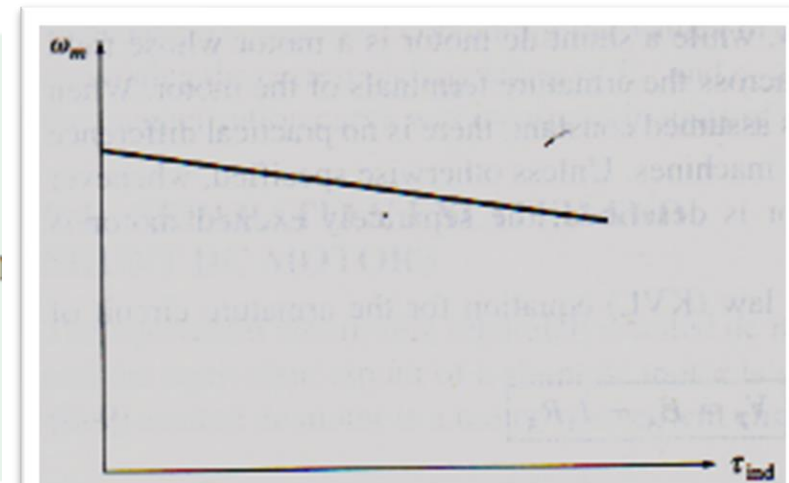
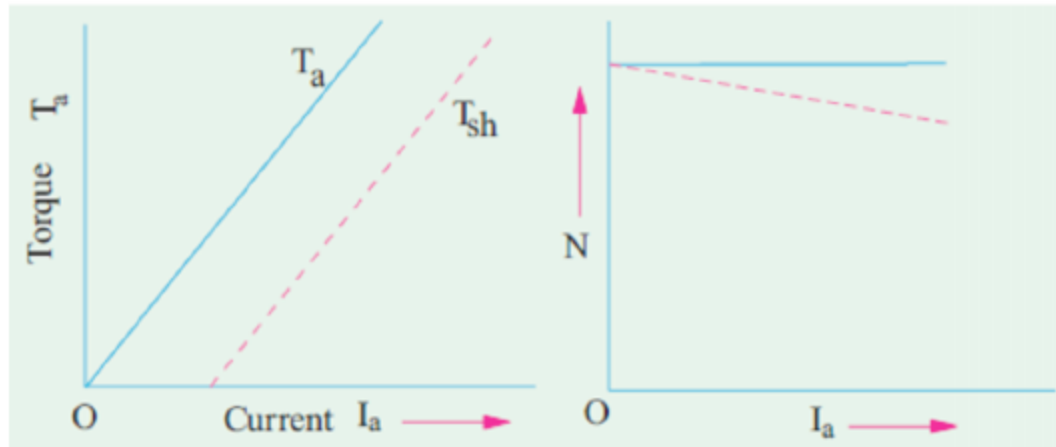
- **Torque and Armature current characteristic ( $T_a/I_a$ ):** It is known as electrical characteristic of the motor.
  - Assuming  $\Phi$  to be practically constant (though at heavy loads,  $\phi$  decreases somewhat due to increased armature reaction) we find that  $T_a \propto I_a$ .
  - Hence, the electrical characteristic, is practically a straight line through the origin. Shaft torque is shown dotted.
  - Since a heavy starting load will need a heavy starting current, shunt motor should never be started on (heavy) load.
- **Speed and armature current characteristic ( $N/I_a$ ):** It is very important characteristic as it is often the deciding factor in the selection of the motor for a particular application.
  - If  $\Phi$  is assumed constant, then  $N \propto E$ . As  $E$  is also practically constant, speed is, for most purposes, constant.
- **Speed and torque characteristic ( $N/T_a$  or  $\omega/T_{ind}$ ):** It is also a mechanical characteristic.

$$V_T = E_A + I_A R_A \quad I_A = \frac{\tau_{ind}}{K\Phi}$$

$$E_A = K\Phi \omega$$

$$V_T = K\Phi \omega + I_A R_A \quad V_T = K\Phi \omega + \frac{\tau_{ind}}{K\Phi} R_A$$

$$\omega = \frac{V_T}{K\Phi} - \frac{R_A}{(K\Phi)^2} \tau_{ind}$$



# Speed Control of Shunt DC Motor

Two common ways in which the speed ( $\omega$ ) of a shunt dc machine can be controlled.

1. Adjusting the field resistance  $R_F$  (and thus the field flux)
2. Adjusting the terminal voltage applied to the armature.

The less common method of speed control is by

3. Inserting a resistor in series with armature circuit.

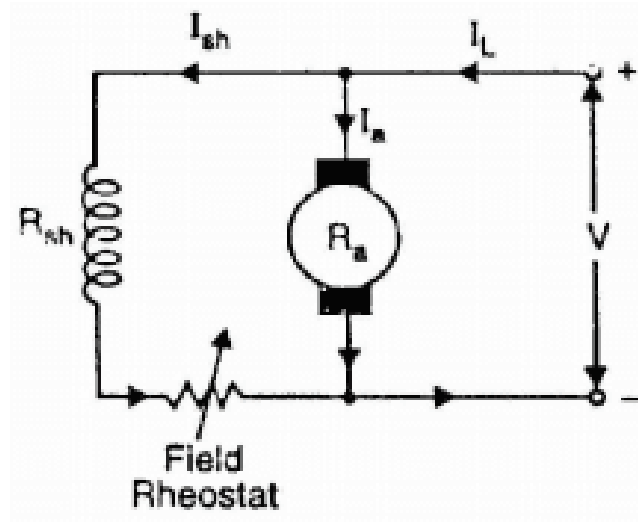
# Speed Control of Shunt DC Motor

The speed of a shunt motor can be changed by

- (i) **Flux control method**
- (ii) **Armature control method**
- (iii) **Voltage control method**

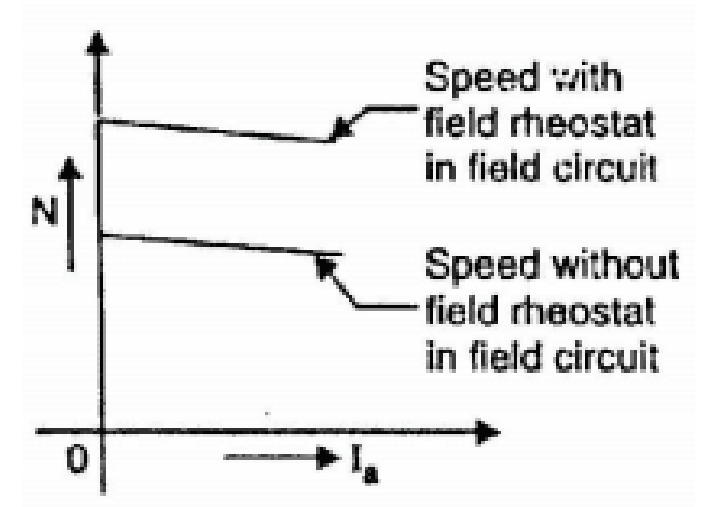
# Speed Control of Shunt DC Motor

## 1: Changing The Field Resistance



**Fig. (5.1)**

1. Increasing  $R_F$  causes  $I_F = \left( \frac{V_T}{R_F \uparrow} \right)$  to decrease.
2. Decreasing  $I_F$  decreases  $\Phi$ .
3. Decreasing  $\Phi$  lowers  $E_A (= K\Phi \downarrow \omega)$
4. Decreasing  $E_A$  by increasing  $I_A = \left( \frac{V_T - E_A \downarrow}{R_A} \right)$

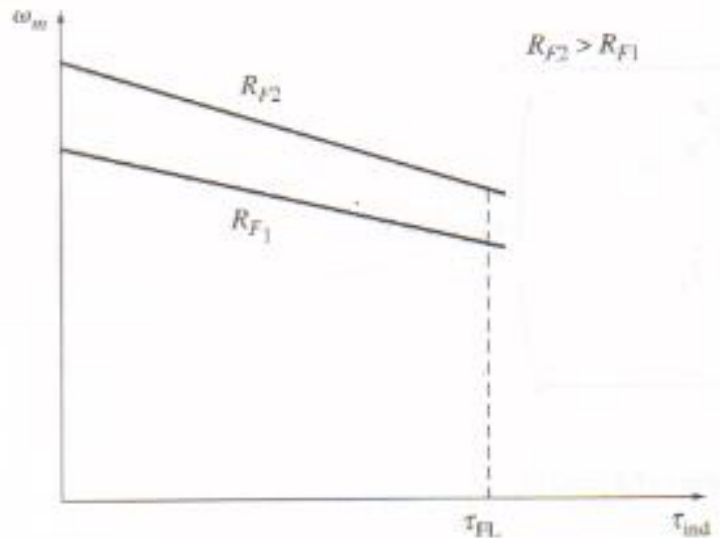


**Fig. (5.2)**

# Speed Control of Shunt DC Motor

## 1: Changing The Field Resistance

5. Increase  $I_A$  by increasing  $\tau_{ind} = (K\Phi \downarrow I_A \uparrow \uparrow)$   
with the change in  $I_A$  dominant over the change in flux ( $\Phi$ ).
6. Increasing  $\tau_{ind}$  makes  $\tau_{ind} > \tau_{load}$  and the speed  $\omega$  increases.
7. Increasing speed to increases  $E_A = K\Phi\omega \uparrow$  again.
8. Increasing  $E_A$  decreases  $I_A$ .
9. Decreasing  $I_A$  decreases  $\tau_{ind}$  until  $\tau_{ind} = \tau_{load}$  at a higher speed  $\omega$

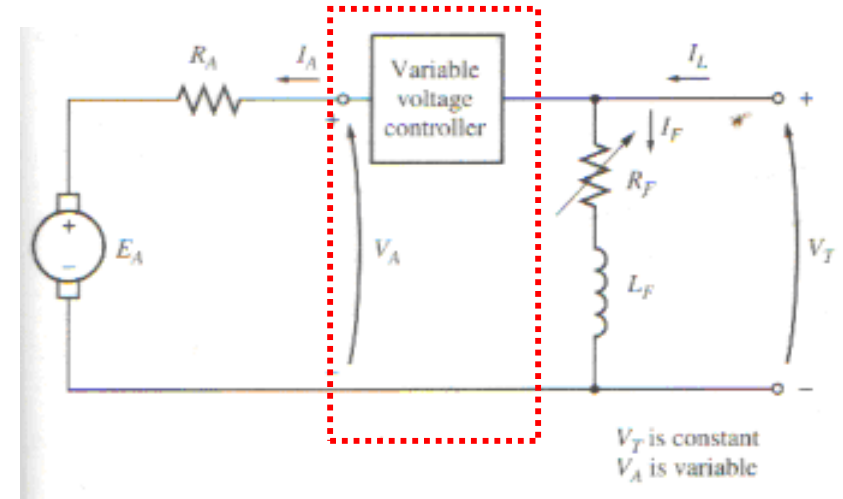


The effect of field resistance speed control on a shunt motor's torque speed characteristic: over the motor's normal operating range

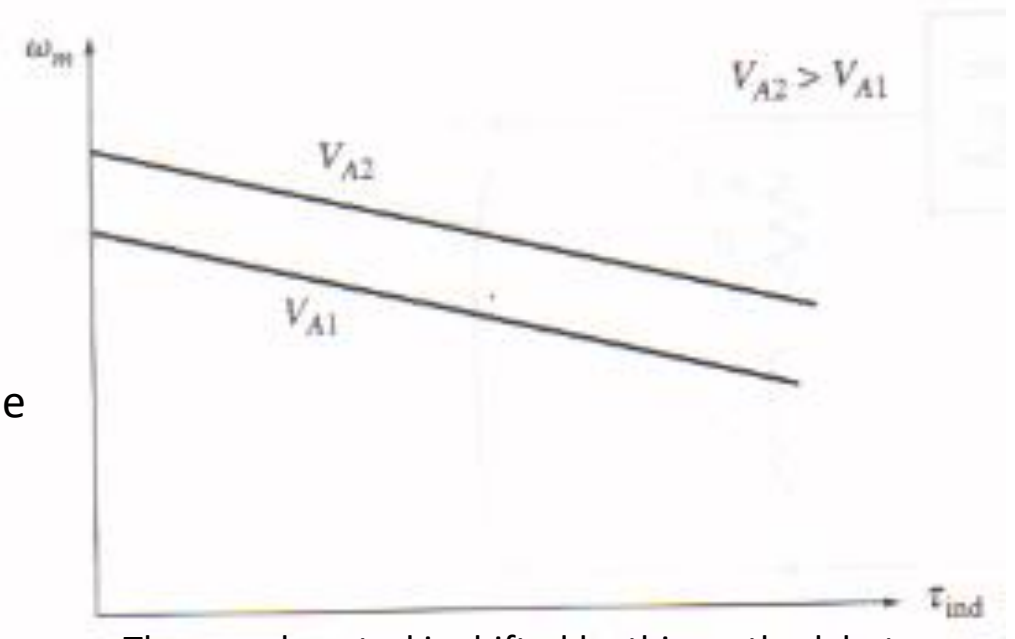
**Note:** The field of a shunt motor in operation should never be opened because its speed will increase to an extremely high value.

## 2: Changing The Armature Voltage

1. An increase in  $V_A$  by increasing  $I_A = [(V_A \uparrow - E_A)/R_A]$
2. Increasing  $I_A$  increases  $\tau_{ind} = (K\Phi I_A \uparrow)$
3. Increasing  $\tau_{ind}$  makes  $\tau_{ind} > \tau_{load}$  increasing  $\omega$ .
4. Increasing  $\omega$  increases  $E_A = (K\Phi\omega \uparrow)$
5. Increasing  $E_A$  by decreasing  $I_A = [(V_A - E_A \uparrow)/R_A]$
6. Decreasing  $I_A$  decreases  $\tau_{ind}$  until  $\tau_{ind} = \tau_{load}$  at a higher  $\omega$ .



Armature voltage control of a shunt (or separately excited) dc motor.



The speed control is shifted by this method, but the slope of the curve remains constant

The effect of armature voltage speed control on a shunt motor's torque speed characteristic

### 3 : Inserting Resistor in Series with Armature Circuit

Additional resistor in series will drastically increase the slope of the motor's characteristic, making it operate more slowly if loaded

$$\omega = \frac{V_T}{K\Phi} - \frac{R_A}{(K\Phi)^2} \tau_{ind}$$

The above equation shows if  $R_A$  increase, speed will decrease

This method is very wasteful method of speed control, since the **losses in the inserted resistor is very large**. For this it is rarely used.

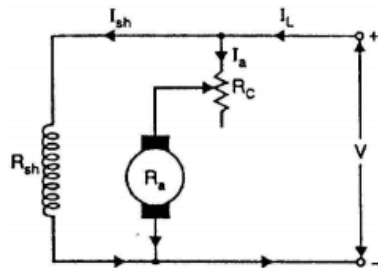


Fig. (5.3)

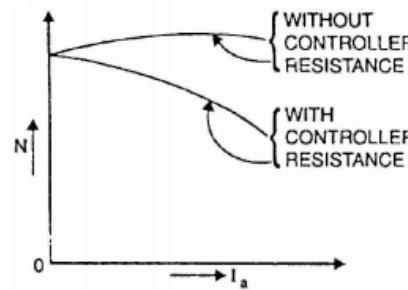


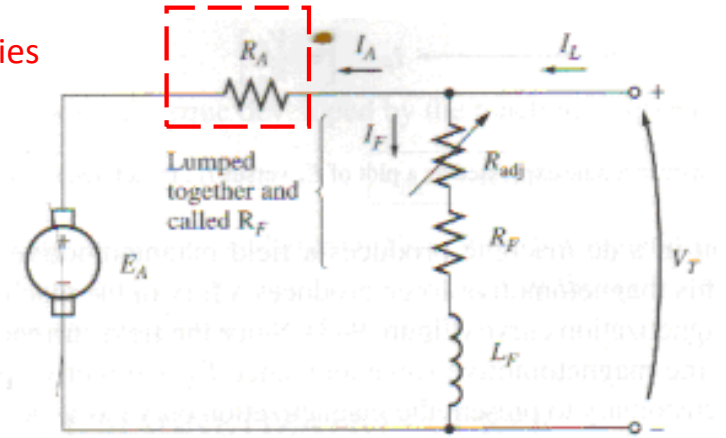
Fig. (5.4)

$$N \propto V - I_a(R_a + R_c)$$

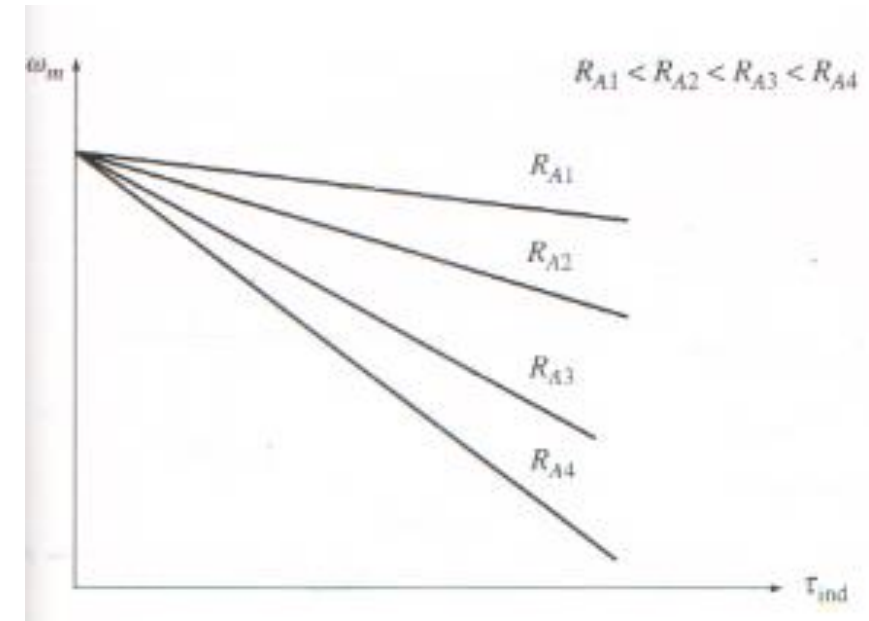
ere

$R_c$  = controller resistance

Add resistor in series with  $R_A$



Equivalent circuit of DC shunt motor



The effect of armature resistance speed control on a shunt motor's torque - speed characteristic

# Series DC Motor Characteristics

The Kirchhoff's voltage law equation for this motor

$$V_T = E_A + I_A(R_A + R_S)$$

From the equation;  $\tau_{ind} = K\Phi I_A = KcI_A^2$  the armature current can be expressed as:

$$I_A = \sqrt{\frac{\tau_{ind}}{Kc}}$$

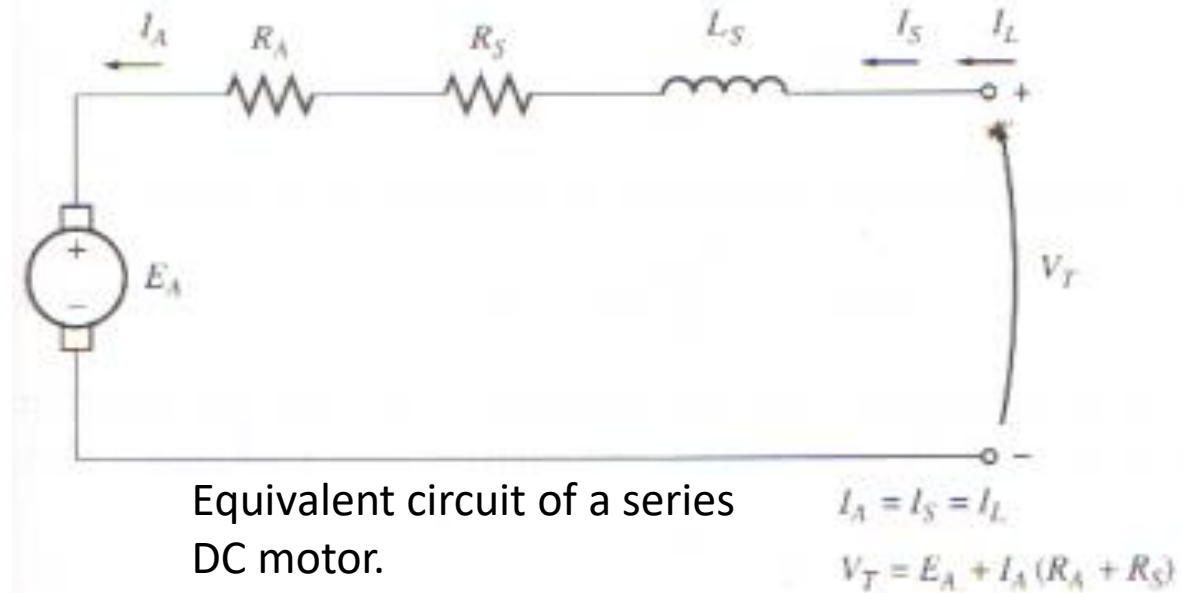
Also,  $E_A = K\Phi\omega$ , substituting these expression yields:

$$V_T = K\Phi\omega + \sqrt{\frac{\tau_{ind}}{Kc}}(R_A + R_S)$$

$$\Phi = cI_A \ ; \ I_A = \frac{\Phi}{c} \ \text{so,} \ \Phi = \sqrt{\frac{c}{K}} \sqrt{\tau_{ind}}$$

$$V_T = K \sqrt{\frac{c}{K}} \sqrt{\tau_{ind}} \omega + \sqrt{\frac{\tau_{ind}}{Kc}}(R_A + R_S)$$

$$\omega = \frac{V_T}{\sqrt{Kc}} \frac{1}{\sqrt{\tau_{ind}}} - \frac{R_A + R_S}{Kc}$$



One disadvantage of series motor can be seen immediately from this equation. When the **torque** on this motor goes to **zero**, its **speed goes to infinity**.

In practice, the torque can never go entirely to zero, because of the mechanical, core and stray losses that must be overcome.

However, if **no other load** is connected to the motor, it can **turn fast enough** to seriously **damage itself**.

NEVER completely unload a series motor, and never connect one to a load by a belt or other mechanism that could break.

# Series DC Motor Characteristics

- **Ta/Ia Characteristic:** We have seen that  $T_a \propto \Phi I_a$ . In this case, as field windings also carry the armature current,  $\Phi \propto I_a$  up to the point of magnetic saturation. Hence, before saturation,

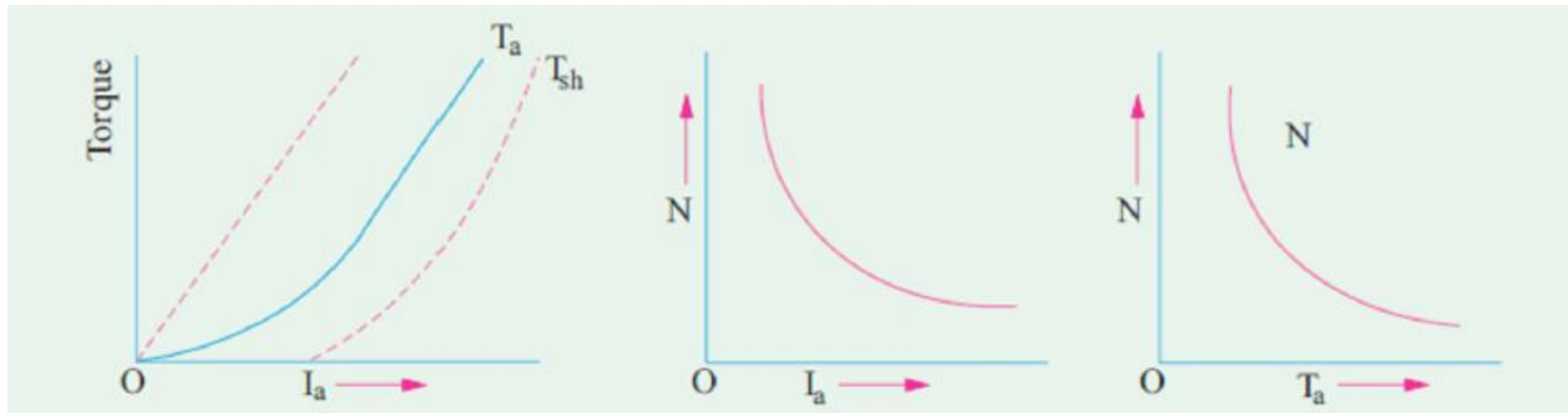
$$T_a \propto \Phi I_a \text{ and } T_a \propto I_a^2$$

At light loads,  $I_a$  and hence  $\Phi$  is small. But as  $I_a$  increases,  $T_a$  increases as the square of the current. Hence,  $T_a/I_a$  curve is a parabola.

- **N/Ia Characteristics:** Variations of speed can be deduced from the formula:  $N \propto E_b$   
Change in  $E_b$ , for various load currents is small and hence may be neglected for the time being. With increased  $I_a$ ,  $\Phi$  also increases. Hence, speed varies inversely as armature current. When load is heavy,  $I_a$  is large. Hence, speed is low (this decreases  $E_b$  and allows more armature current to flow). But when load current and hence  $I_a$  falls to a small value, speed becomes dangerously high.

- **N/Ta or mechanical characteristic:** Determined in the previous slide.

$$\omega = \frac{V_T}{\sqrt{Kc}} \frac{1}{\sqrt{\tau_{ind}}} - \frac{R_A + R_S}{Kc}$$



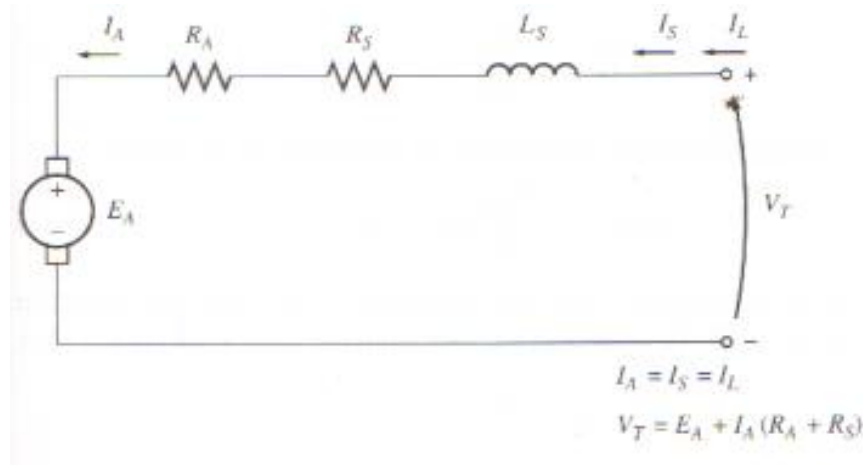
# Speed Control of Series DC Motor

Method of controlling the speed in series motor.

1. Change the terminal voltage of the motor. If the terminal voltage is increased, the speed also increased, resulting in a higher speed for any given torque. This is only one efficient way to change the speed of a series motor.

$$\omega = \frac{V_T}{\sqrt{Kc}} \frac{1}{\sqrt{\tau_{ind}}} - \frac{R_A + R_S}{Kc}$$

2. By the insertion of a series resistor into the motor circuit, but this technique is very wasteful of power and is used only for intermittent period during the start-up of some motor.



# The Compounded DC Motor

The Kirchhoff's voltage law equation for a compound dc motor is:

$$V_T = E_A + I_A(R_A + R_S)$$

$$I_A = I_L - I_F \quad I_F = \frac{V_T}{R_F}$$

The net magnetomotive force given by

$$F_{\text{net}} = F_F \pm F_{SE} - F_{AR}$$

$F_F$  = magnetomotive force (shunt field)

$F_{SE}$  = magnetomotive force (series field)

$F_{AR}$  = magnetomotive force (armature reaction)

The effective shunt field current in the compounded DC motor given by:

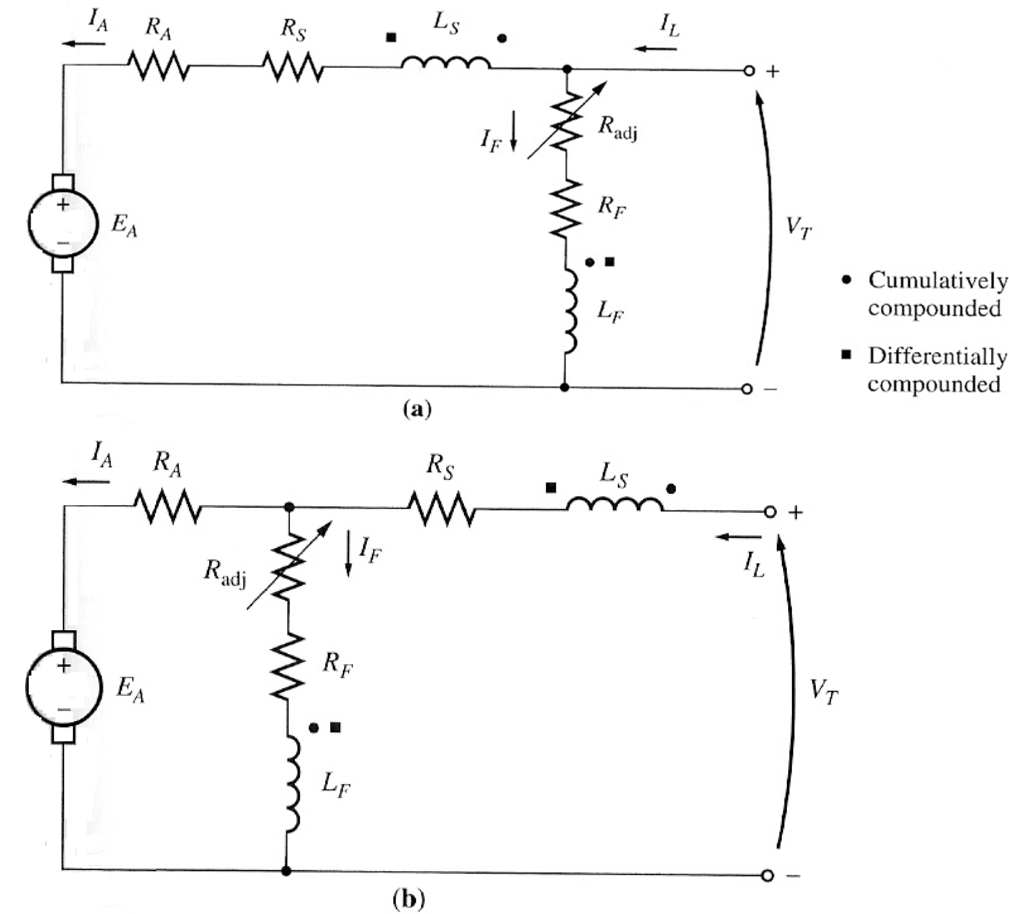
$$I_F^* = I_F \pm \frac{N_{SE}}{N_F} I_A - \frac{F_{AR}}{N_F}$$

$N_{SE}$  = winding turn per pole on series winding

$N_F$  = winding turn per pole on shunt winding

The positive (+) sign is for **cumulatively** compound motor

The negative (-) sign is for **differentially** compound motor



The equivalent compound DC motor (a) Long-shunt connection (b) Short-shunt connection

# The Torque Speed Characteristic of a Cumulatively Compounded DC Motor

- The **cumulatively** compounded motor has a **higher starting torque than a shunt motor** (whose flux is constant) but a **lower starting torque than a series motor** (whose entire flux is proportional to armature current).
- It combines the best features of both **the shunt** and **the series motors**. Like a series motor, it has **extra torque for starting**; like a **shunt motor, it does not over speed at no load**.
- At **light loads**, the **series field** has a very **small effect**, so the motor behaves approximately **as a shunt dc motor**.
- As **the load gets very large**, the **series flux** becomes quite **important** and the torque speed curve begins to look like a series motor's characteristic.
- A comparison of these torque speed characteristics of each types is shown in next slide.

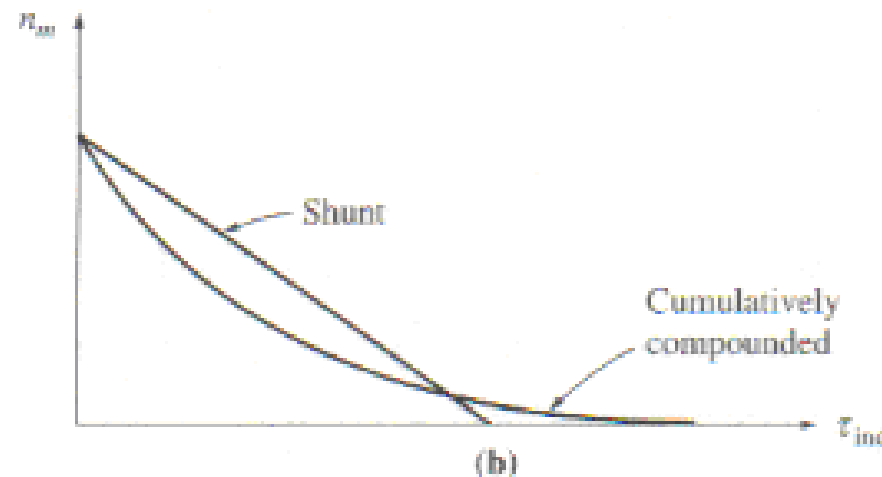
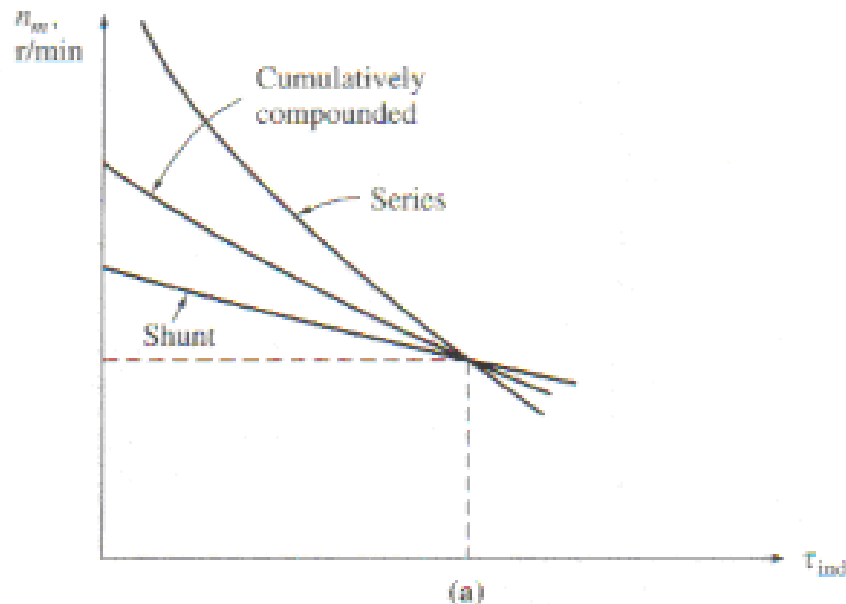


Fig (a) The torque-speed characteristic of a cumulatively compounded dc motor compared to series and shunt motors with the **same full-load rating**.

Fig. (b) The torque-speed characteristic of a cumulatively compounded dc motor compared to a shunt motor with the **same no-load speed**.

# The Torque Speed Characteristic of a Differently Compounded DC Motor

In a differentially compounded DC motor, *the shunt magnetomotive force and series magnetomotive force subtract from each other.*

- This means that as the load on the motor increase,  $I_A$  increase and the flux in the motor decreased, ( $I_A \uparrow \Phi \downarrow$ )
- As the flux decrease, the speed of the motor increase, ( $\Phi \downarrow \omega \uparrow$ )
- This speed increase causes an-other increase in load, which further increase  $I_A$ , Further decreasing the flux, and increasing the speed again.

All the phenomena resulting the differentially compounded motor is unstable and tends to run away.

This instability is much worse than that of a shunt motor with armature reaction, and make it unsuitable for any application.

The techniques available for control of speed in a cumulatively compounded dc motor are the same as those available for a shunt motor:

1. Change the field resistance,  $R_F$
2. Change the armature voltage,  $V_A$
3. Change the armature resistance,  $R_A$

The arguments describing the effects of changing  $R_F$  or  $V_A$  are very similar to the arguments given earlier for the shunt motor.

# DC Motor Starter

In order for a dc motor to **function properly** on the job, it must have some special control and protection equipment associated with it. The purposes of this equipment are:

1. To protect the motor against damage due to short circuits in the equipment
2. To protect the motor against damage from long term overloads
3. To protect the motor against damage from excessive starting currents
4. To provide a convenient manner in which to control the operating speed of the motor

## DC Motor Problem on Starting

DC motor must be protected from physical damage during the starting period.

At starting conditions, the motor is not turning, and so  $E_A = 0$  V.

Since the **internal resistance** of a normal dc motor is **very low**, a very **high current flows**, hence the **starting current will be dangerously high**, could severely damage the motor, even if they last for only a moment.

Consider the dc shunt motor: 
$$I_A = \frac{V_T - E_A}{R_A} = \frac{V_T}{R_A}$$

When  $E_A = 0$  and  $R_A$  is very small, then the current  $I_A$  will be very high.

Two methods of limiting the starting current :

- Insert a starting resistor in series with armature to limit the current flow (until  $E_A$  can build up to do the limiting).  
The **resistor must not be permanent to avoid excessive losses and cause torque speed to drop excessively with increase of load.**
- Manual DC motor starter, totally human dependant

# Inserting a Starting Resistor in Series & Manual DC Motor

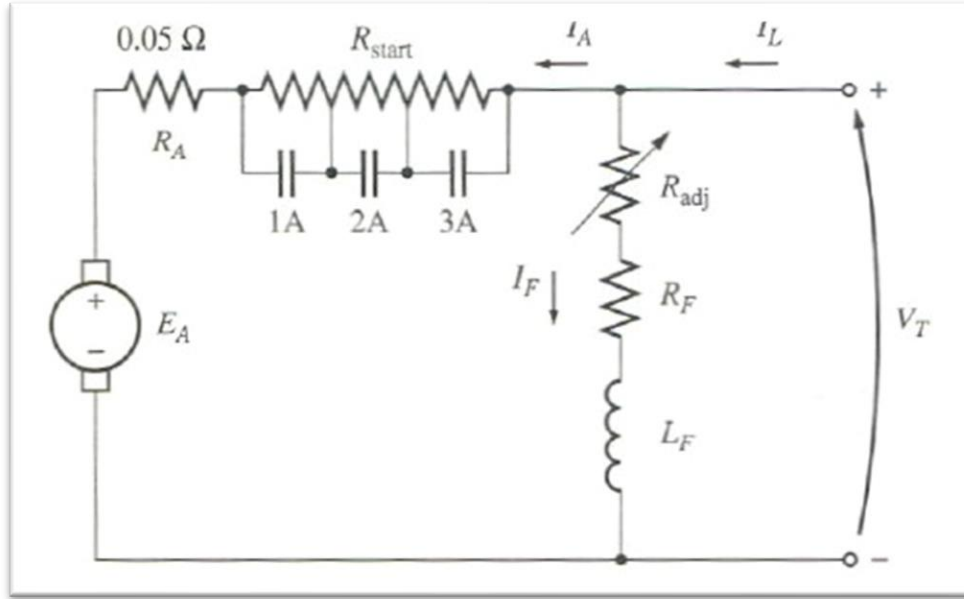


Fig : A shunt motor with a starting resistor in series with an armature. Contacts 1A, 2A and 3A short circuit portions of the starting resistor when they close

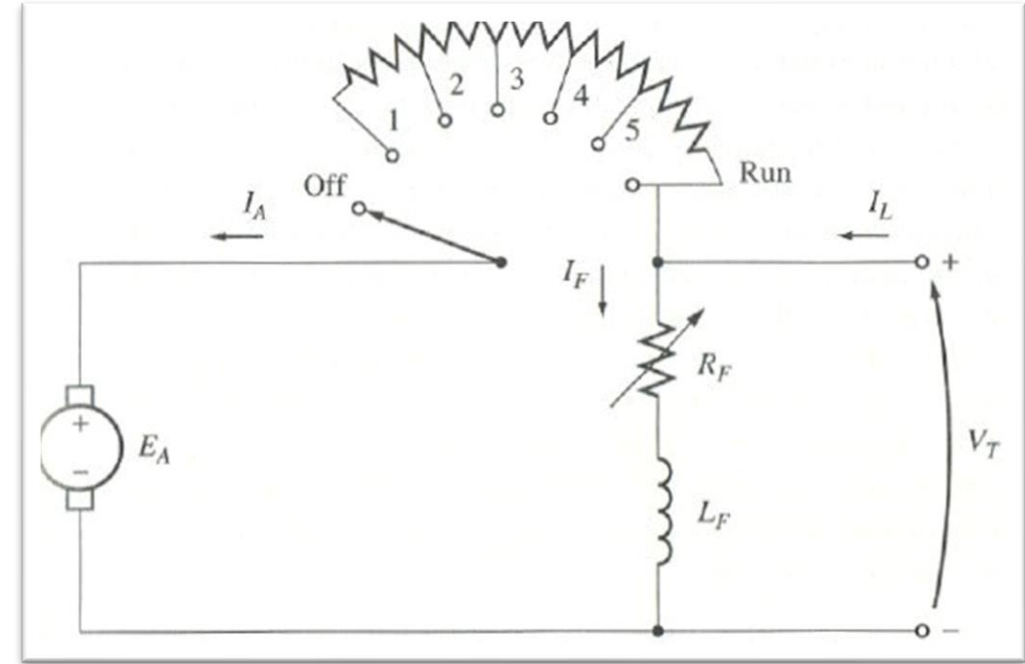


Fig : A Manual DC Motor

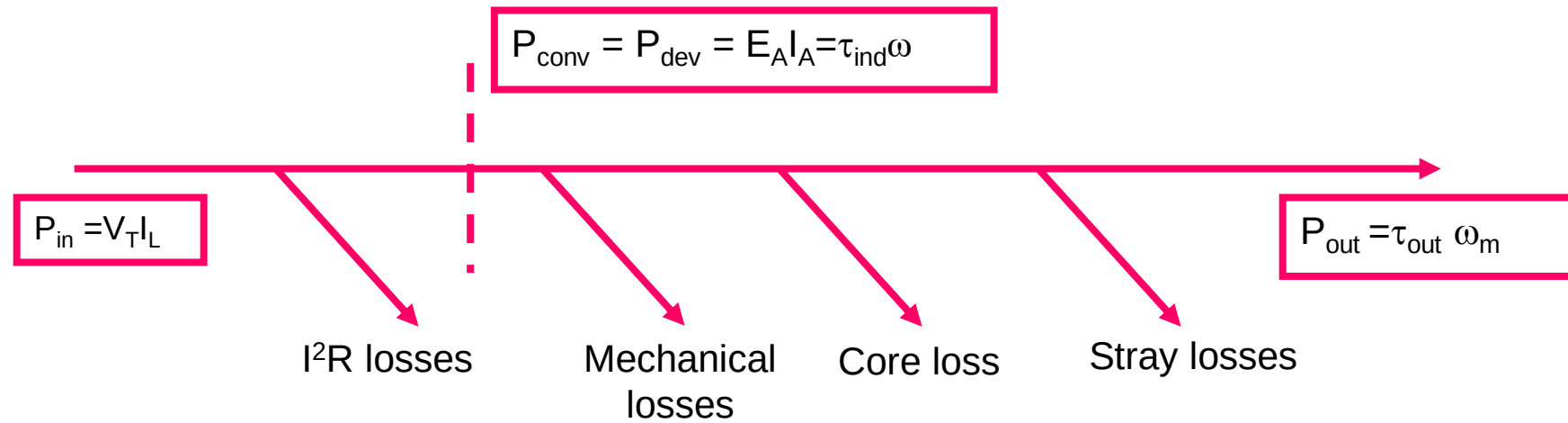
Human dependant:

- Too quickly, the resulting current flow would be too low.
- Too slowly, the starting resistor could burn-up

# DC Motor Efficiency Calculations

To calculate the efficiency of a dc motor, the following losses must be determined :

- Copper losses ( $I^2R$  losses)
- Brush drop losses
- Mechanical losses
- Core losses
- Stray losses



$$\eta = \frac{P_{output}}{P_{input}} \times 100\%$$

$$\eta = \frac{P_{input} - P_{losses}}{P_{input}} \times 100\%$$

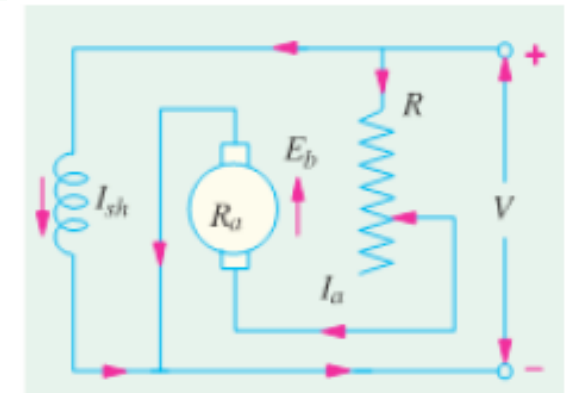
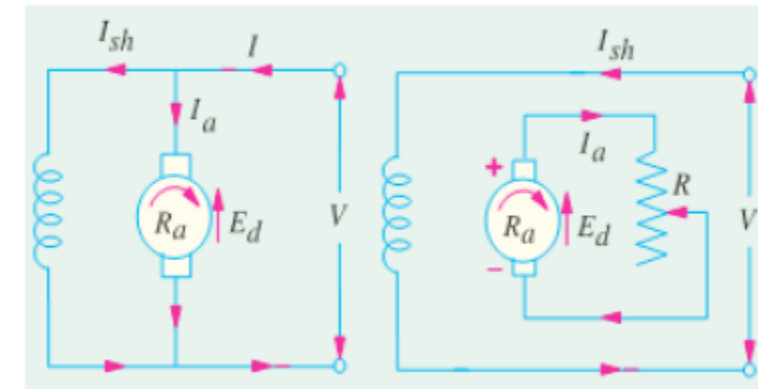
# Electric Braking

- A motor and its load may be brought to rest quickly by using either (i) Friction Braking or (ii) Electric Braking.
- Mechanical brake has one drawback: it is difficult to achieve a smooth stop because it depends on the condition of the braking surface as well as on the skill of the operator.
- The excellent electric braking methods are available which eliminate the need of brake lining levers and other mechanical gadgets. Electric braking, both for shunt and series motors, is of the following three types:
  1. Rheostatic or dynamic braking
  2. Plugging i.e., reversal of torque so that armature tends to rotate in the opposite direction.
  3. Regenerative braking.

Obviously, friction brake is necessary for holding the motor even after it has been brought to rest.

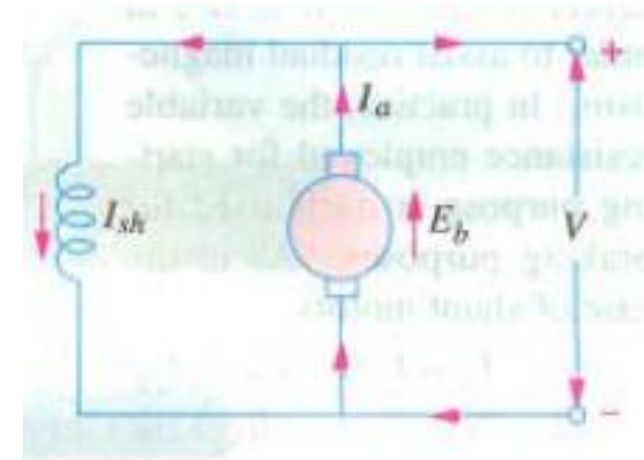
**1. Rheostatic or Dynamic Braking:** In this method, the armature of the shunt motor is disconnected from the supply and is connected across a variable resistance  $R$ . The field winding is left connected across the supply. The braking effect is controlled by varying the series resistance  $R$ . Obviously, this method makes use of generator action in a motor to bring it to rest.

**2. Plugging or Reverse Current Braking:** This method is commonly used in controlling elevators, rolling mills, printing presses and machine tools etc. In this method, connections to the armature terminals are reversed so that motor tends to run in the opposite direction. Due to the reversal of armature connections, applied voltage  $V$  and  $E$  start acting in the same direction around the circuit. In order to limit the armature current to a reasonable value, it is necessary to insert a resistor in the circuit while reversing armature connections.



# Electric Braking

- **3.Regenerative Braking:**
- **Regenerative braking** is used where, load on the motor has very high inertia (e.g in electric trains). When applied voltage to the motor is reduced to less than back emf  $E_b$ , obviously armature current  $I_a$  will get reversed, and hence armature torque is reversed. Thus speed falls. As generated emf is greater than applied voltage (machine is acting as a DC generator DC), power will be returned to the line, this action is called as regeneration. Speed keeps falling, back emf  $E_b$  also falls until it becomes lower than applied voltage and direction of armature current again becomes opposite to  $E_b$ .
- This method is used when the load on the motor has over-hauling characteristic as in the lowering of the cage of a hoist or the downgrade motion of an electric train. Regeneration takes place when  $E_b$  becomes greater than  $V$ . This happens when the overhauling load acts as a prime mover and so drives the machines as a generator. Consequently, direction of  $I_a$  and hence of armature torque is reversed and speed falls until  $E$  becomes lower than  $V$ . It is obvious that during the slowing down of the motor, power is returned to the line which may be used for supplying another train on an upgrade, thereby relieving the powerhouse of part of its load.



These are braking for shunt motors. Text book for braking for series motors

# Synchronous Machines

# SYNCHRONOUS GENERATOR CONSTRUCTION

- SYN. GEN.s are USED to CONVERT MECHANICAL ENERGY TO AC ELECTRIC ENERGY: GENERATORS IN POWER PLANTS

## GENERATOR CONSTRUCTION

- in synchronous generator, rotor winding energized by dc source to develop rotor magnetic field
- rotor is turned by a prime mover, producing a rotating magnetic field which induce 3 phase voltages in stator windings

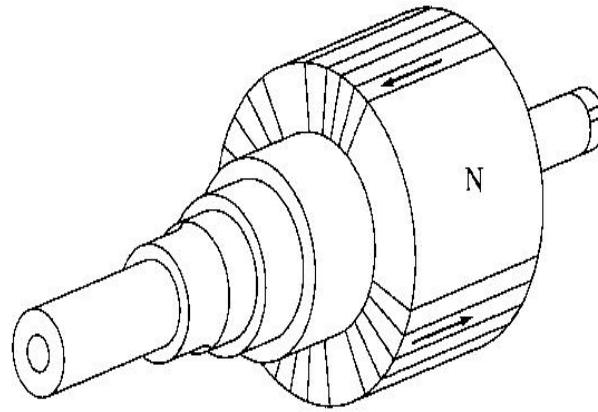
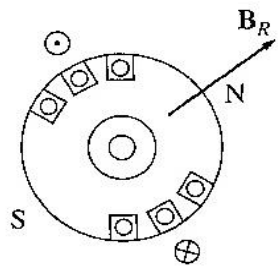
**In general rotor carry the “field windings” , while “armature windings” (or “stator windings”) carry the main voltages of machine**

- **therefore:**
  - rotor windings  $\equiv$  field windings
  - stator windings  $\equiv$  armature windings

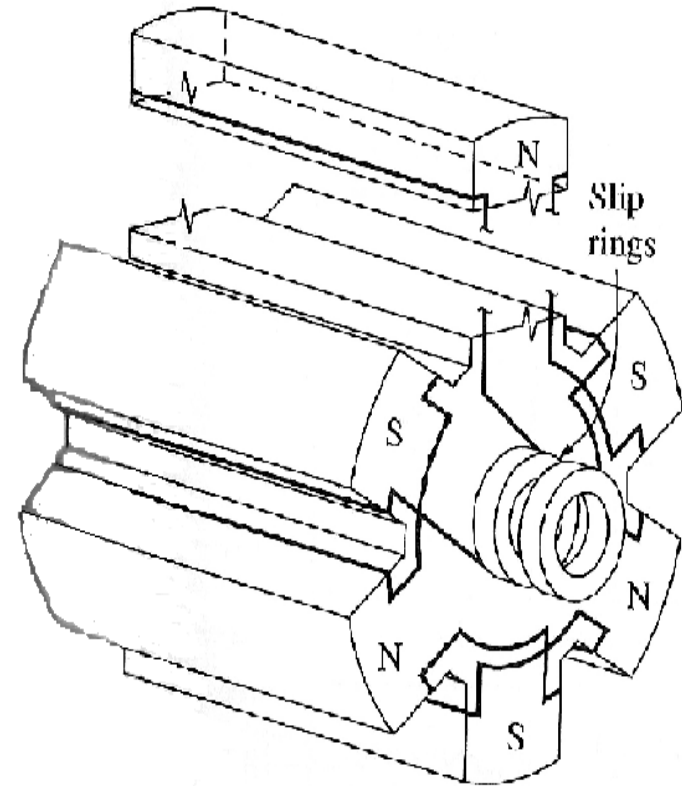
# SYNCHRONOUS GENERATOR CONSTRUCTION

- Rotor of synchronous machine can be

**Nonsalient:** 2 pole rotor    **Salient:** six-pole rotor



Side View



# SYNCHRONOUS GENERATOR CONSTRUCTION

- **Rotor** experience varying magnetic fields, therefore is constructed of thin laminations to reduce eddy current losses
- To supply the rotor winding while it is rotating, special arrangement employed to connect its terminal to dc supply
  1. supply dc power from an external dc source to rotor by means of slip rings
  2. supply dc power from a special dc power source mounted on shaft of rotor
- **SLIP RINGS:** are metal rings encircling shaft and are insulated from it
  - one end of rotor winding is connected to each of the 2 slip rings
  - and a stationary brush mounted on the machine casing ride on each slip ring
- **Brush:** a block of graphite like carbon compound that conducts and has low friction
- same dc voltage is applied to field winding during rotation
- Problems associated with slip rings and brushes:
  - 1- increase the required maintenance (brushes should be examined for wear regularly)
  - 2- brush voltage drop results in significant power losses if field current is high
- Despite of above problems, SLIP RINGS & BRUSHES used for smaller synchronous machines since is cost-effective.

What is a brushless exciter?

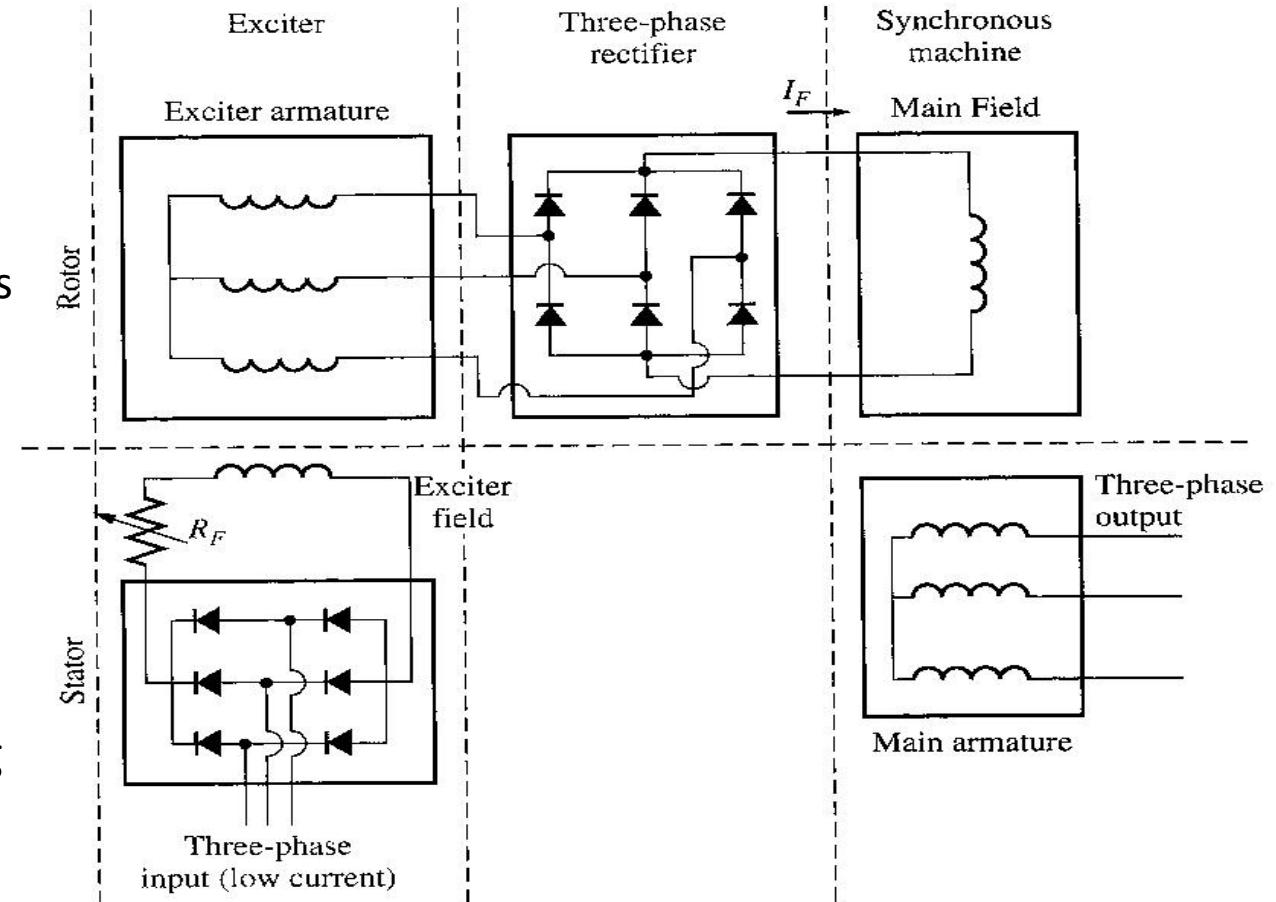
A brushless excitation system is, essentially, an inside-out ac generator that delivers its ac voltage to the rotor of the main generator, and receives its excitation from the stator of that same generator.

What are the advantages of a brushless exciter?

The main advantage of the brushless excitation system is there is no contact between Mechanical and Electrical parts of the system and there is no use of circuit breakers and field breakers. Reduced in maintenance time, outage time, and costs for replacements parts.

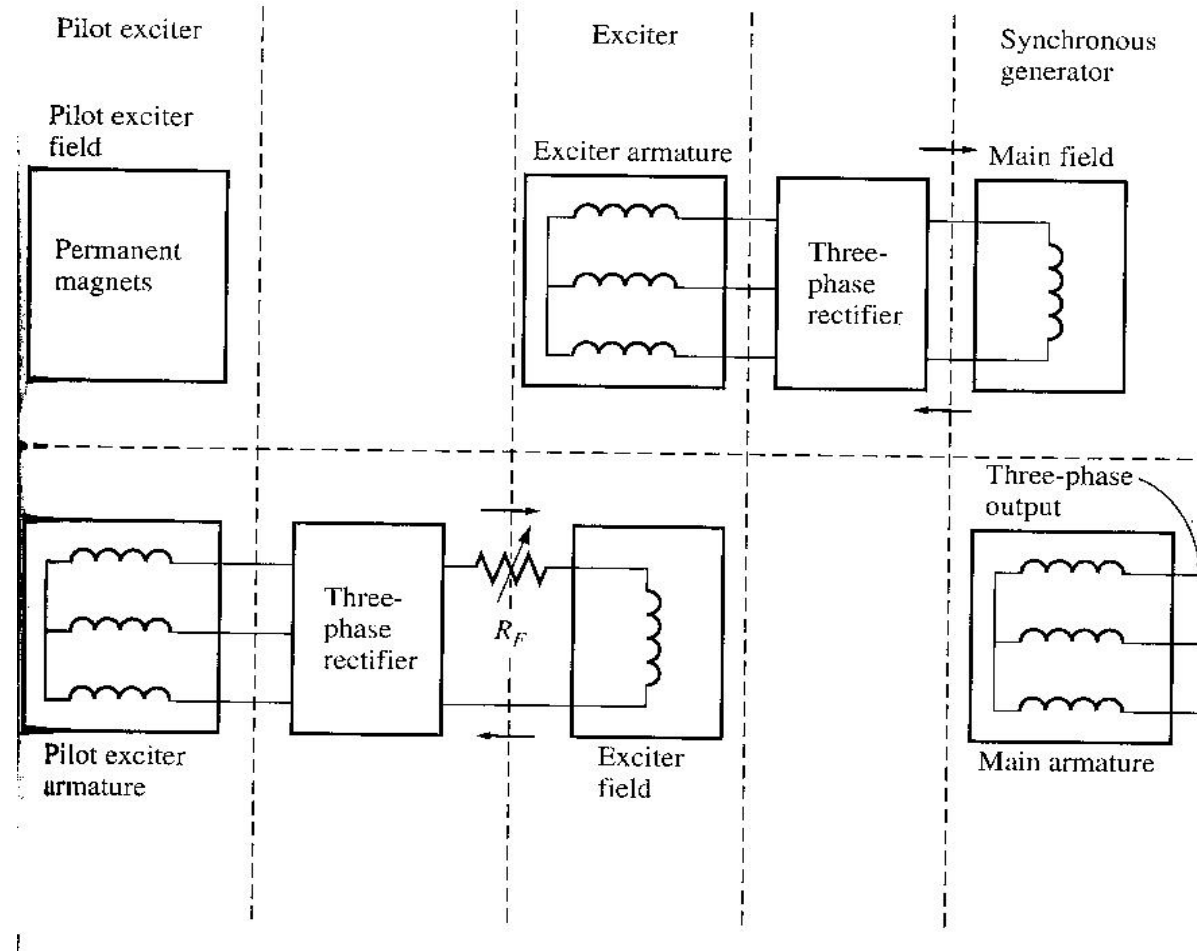
# SYNCHRONOUS GENERATOR CONSTRUCTION

- on larger generator & motors, brushless exciters are used
- **Brushless Exciter:** is a smaller ac generator with its field circuit mounted on stator & its armature circuit mounted on rotor shaft
- 3 phase output of exciter generator rectified by a 3 phase rectifier mounted also on shaft
- By controlling small dc field current of exciter generator, it is possible to fed (and also adjust) field current of main machine without slip rings and brushes
- Schematic arrangement of a brushless exciter
- A small pilot exciter often included in system to have the excitation of generator independent of any external power sources
- A pilot exciter is a small ac generator with permanent magnets mounted on rotor shaft & a 3 phase winding on stator
- It produces power for field circuit of exciter, which in turn controls the field circuit of main machine
- With pilot exciter on shaft of generator, no external electric power is required to run generator
- Many Syn. Gen.s with brushless exciters also have slip rings and brushes, as an auxiliary source of dc field in emergencies



# SYNCHRONOUS GENERATOR CONSTRUCTION

- Brushless exciter including a pilot exciter



## Speed of rotation of synchronous generator

- synchronous generators are *synchronous, during their operation* means: electrical frequency is synchronized with mechanical speed of rotor
- Relation between electrical frequency of stator and mechanical speed of rotor as shown before:

$$f_e = n_m p / 120$$

$f_e$  : electrical frequency in Hz

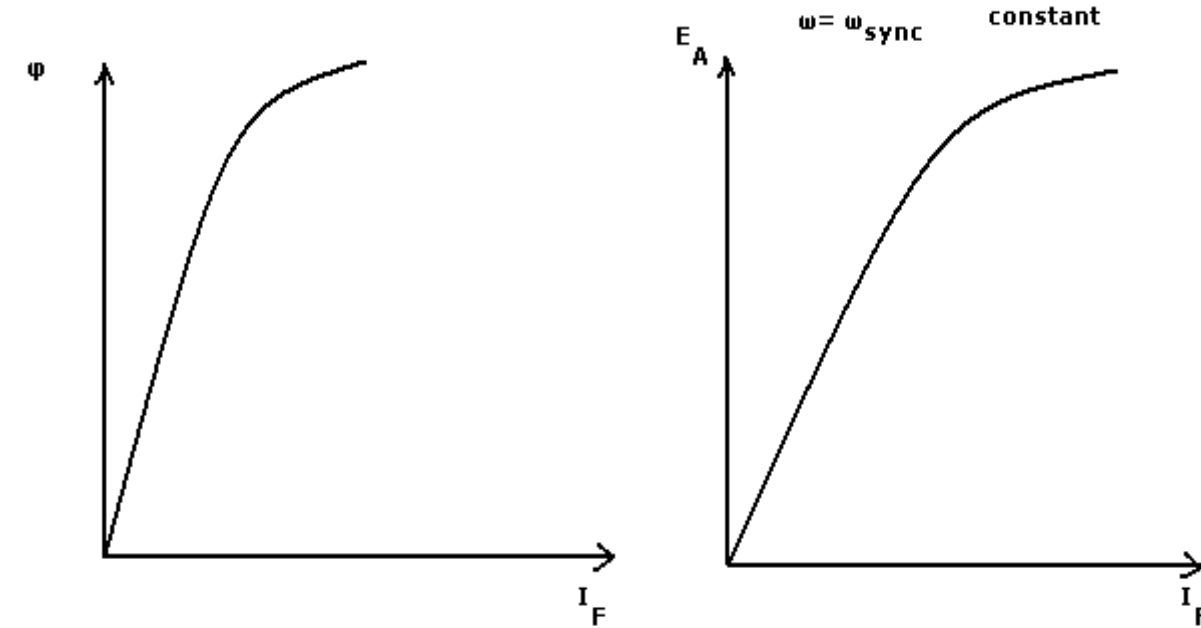
$n_m$ : speed of rotor in r/min

$p$ : number of poles

- Electric power generated at 50 or 60 Hz, so rotor must turn at fixed speed depending on number of poles on machine
- To generate 60 Hz in 2 pole machine, rotor must turn at 3600 r/min, and to generate 50 Hz in 4 pole machine, rotor must turn at 1500 r/min
- **INTERNAL GENERATED VOLTAGE OF A SYNCHRONOUS GENERATOR**
- magnitude of induced voltage in one phase determined in last section:  $E_A = \sqrt{2} \pi N C \phi f$

# INTERNAL GENERATED VOLTAGE

- Induced voltage depends on flux  $\phi$ , frequency or speed of rotation  $f$ , & machine's construction
- Last equation can be rewritten as:  
$$E_A = K \phi \omega$$
- Note:  $E_A$  proportional to flux & speed, while flux depend on current in rotor winding  $I_F$ , therefore  $E_A$  is related to  $I_F$  & its plot named: magnetization curve, or O/C characteristic



- Plots of flux vs  $I_F$  and magnetization curve

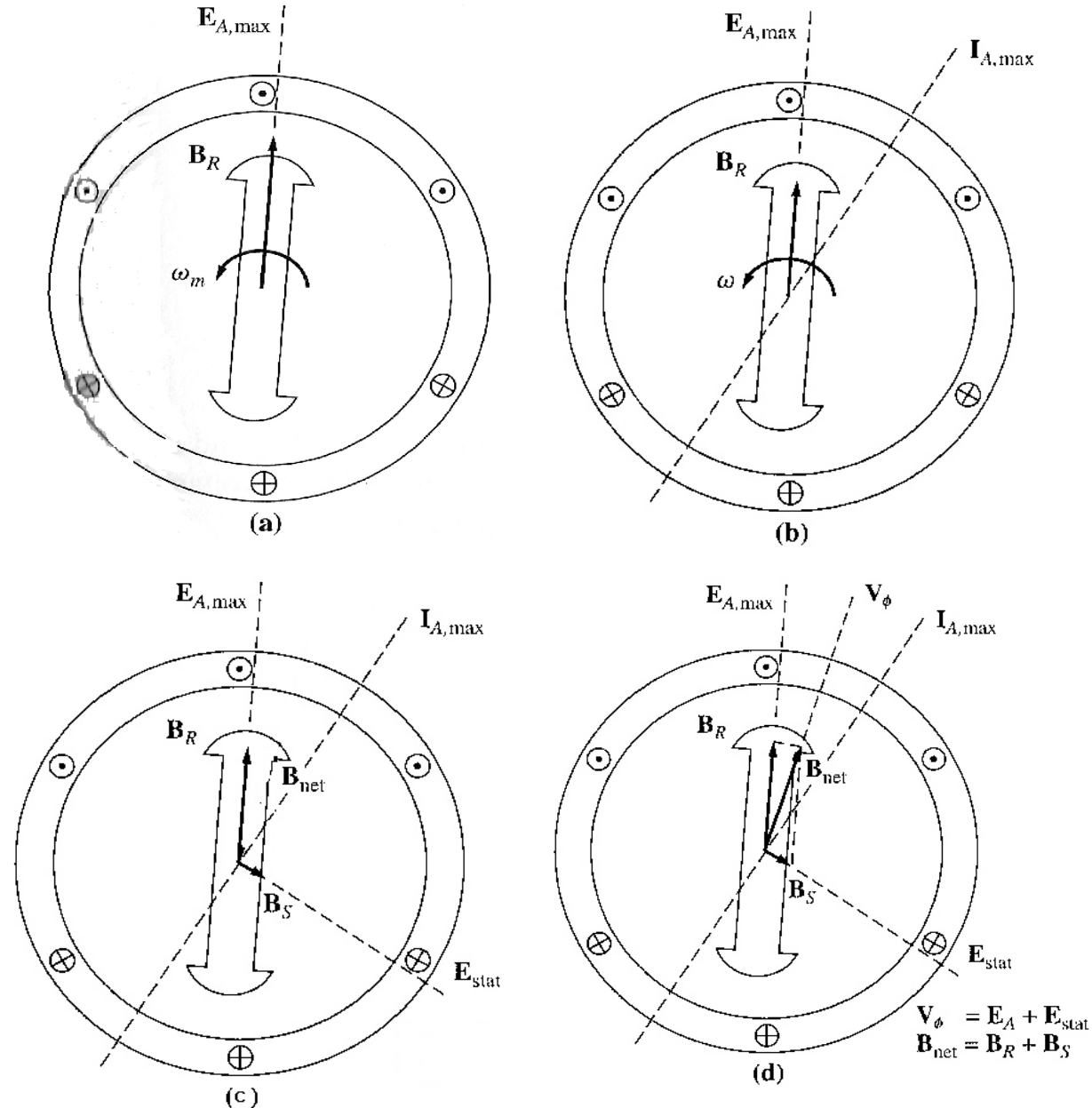
# SYNCHRONOUS GENERATOR EQUIVALENT CIRCUIT

- To develop a relation for  $V_\phi$  as terminal voltage of generator which is different from internal voltage  $E_A$  equivalent circuit is needed
- Reasons for  $V_\phi$  to be different from  $E_A$ 
  - 1- distortion of air-gap magnetic field due to current flowing in stator, called *armature reaction*
  - 2- self-inductance of armature coils
  - 3- resistance of armature coils
  - 4- effect of salient-pole rotor shapes (ignored as machines have cylindrical rotors)

At first we will explore the effects of the first three factors and derive a machine model from them. The effects of a salient-pole shape will be ignored: in other words, all the machines here are non salient or cylindrical.

# SYNCHRONOUS GENERATOR ARMATURE REACTION

- Last figure shows a 2 pole rotor spinning inside a 3 phase stator, without load
- Rotor magnetic field  $B_R$  develop a voltage  $E_A$  as discussed in last chapter voltage is positive out of conductors, at top, and negative into the conductors at bottom of figure
- When there is no load on generator, armature current zero,  $E_A = V_\phi$
- If generator be connected to a lagging load, peak current occur at an angle behind peak voltage as in fig (b)
- Current flowing in stator windings produces its magnetic field
- Stator magnetic field named  $B_S$  & its direction found by R.H.R. as shown in fig(c) this  $B_S$  produces another voltage in stator, named  $E_{stat}$  and shown in figure



# SYNCHRONOUS GENERATOR ARMATURE REACTION

- Having these 2 voltage components in stator windings, total voltage in one phase is sum of  $E_A$  and  $E_{stat}$  :

$$V_\phi = E_A + E_{stat}$$

and

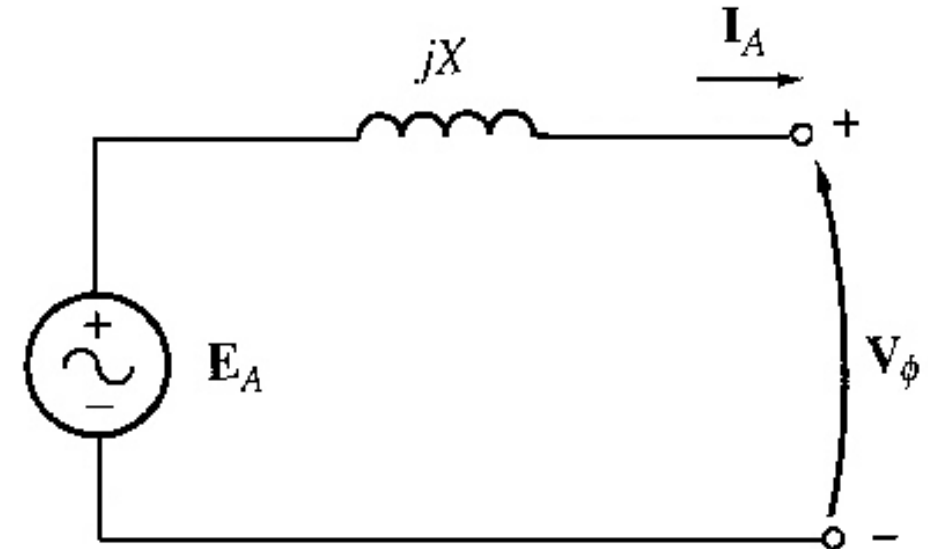
$$B_{net} = B_R + B_S$$

angle of  $B_{net}$  coincide with angle of  $V_\phi$  shown in fig (d)

- To model the effect of armature reaction, note:
  - $E_{stat}$  lies at an angle of  $90^\circ$  behind plane of maximum current  $I_A$
  - $E_{stat}$  directly proportional to  $I_A$  and  $X$  is constant of proportionality

→  $E_{stat} = -j X I_A$

→ voltage in one phase  
 $V_\phi = E_A - j X I_A$



# SYNCHRONOUS GENERATOR ARMATURE REACTION

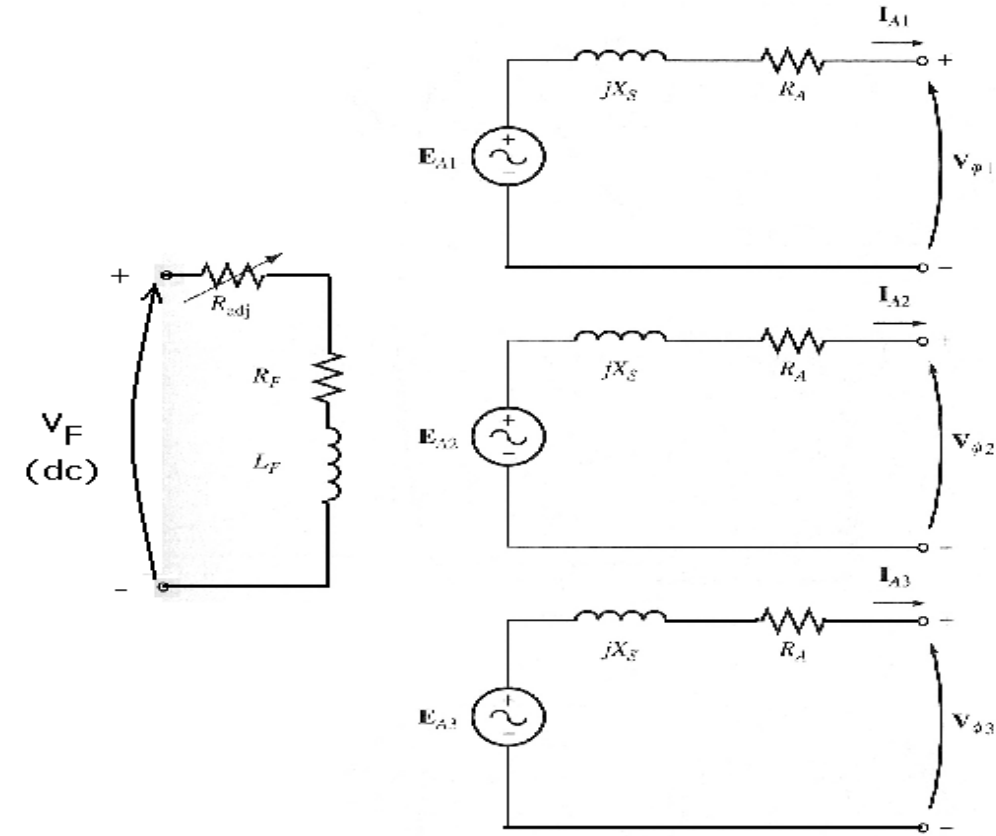
- Armature reaction voltage can be modeled as an inductor in series with internal induced voltage
- In addition to armature reaction, stator coils have a self-inductance and a resistance
- stator self-inductance named  $L_A$  (its reactance  $X_A$ ) and stator resistance is  $R_A$  :

$$V_\phi = E_A - jX_L I_A - jX_A I_A - R_A I_A$$

- Armature reaction & self-inductance in machine both represented by reactance, normally they are combined to a single reactance as :

$$X_S = X_L + X_A$$

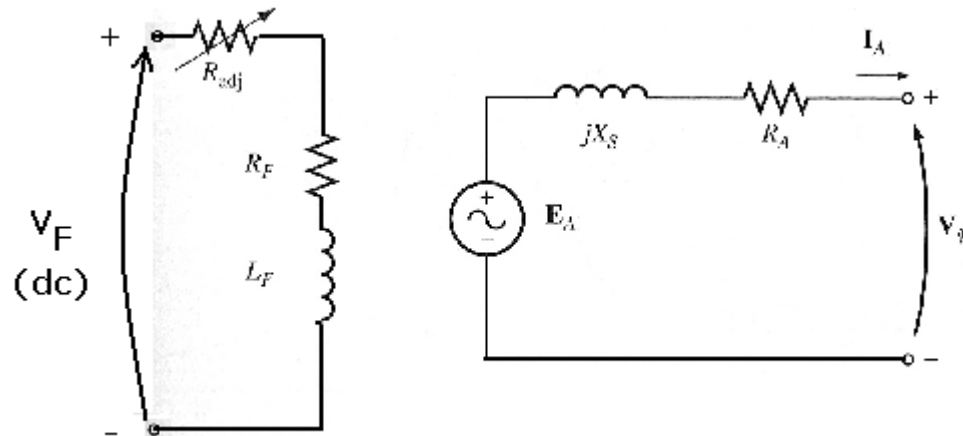
$$V_\phi = E_A - jX_S I_A - R_A I_A$$



equivalent circuit of a 3 phase synchronous generator

# SYNCHRONOUS GENERATOR EQUIVALENT CIRCUIT

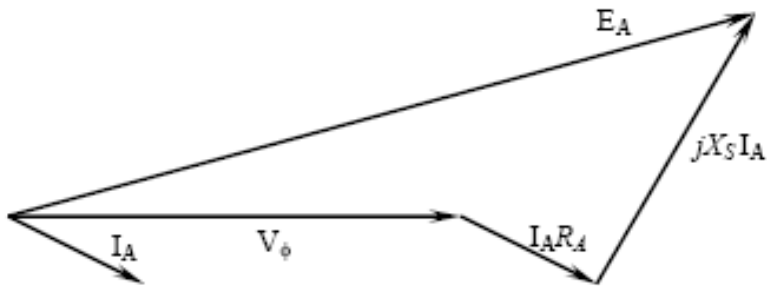
- Figure shows a dc source, supplying rotor winding, modeled by coil inductance & resistance in series with an adjustable resistor  $R_{adj}$  that controls current
- Rest of equivalent circuit consists of model for each phase
- the voltages and currents of each phase are  $120^\circ$  apart with same magnitude
- Three phases can be connected in Y or  $\Delta$
- If connected in Y :  $V_T = \sqrt{3} V_\phi$
- If connected in  $\Delta$ :  $V_T = V_\phi$



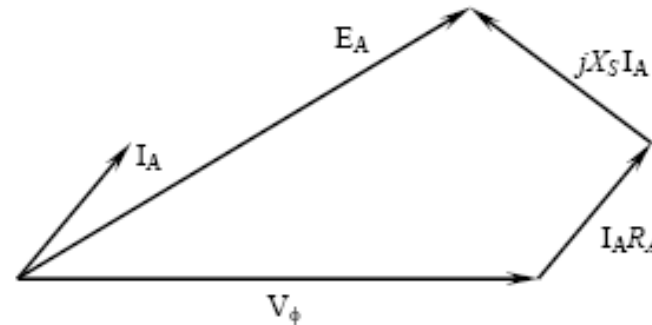
The per phase equivalent circuit

# Phasor Diagram

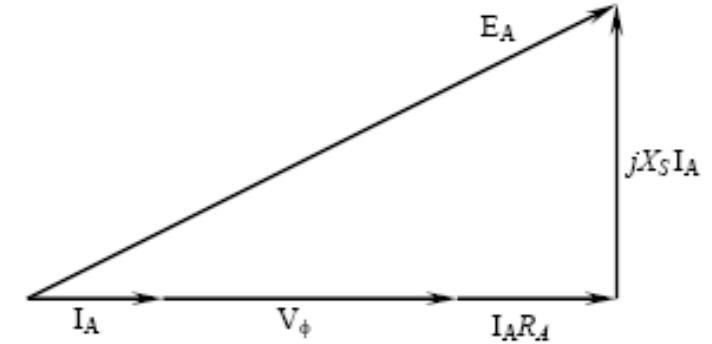
- Voltages in a synchronous generator are expressed as **phasors** because they are AC voltages. Since we have **magnitude** and **angle**, the relationship between voltage and current must be expressed by a two-dimensional plot.
- It is noticed that, for a given **phase voltage** and armature current, a larger induced voltage  $E_A$  is required for lagging loads than leading loads.



*Phasor diagram of a synchronous generator at lagging factor (Inductive Load).*



*Phasor diagram of a synchronous generator at leading factor (Capacitive Load).*

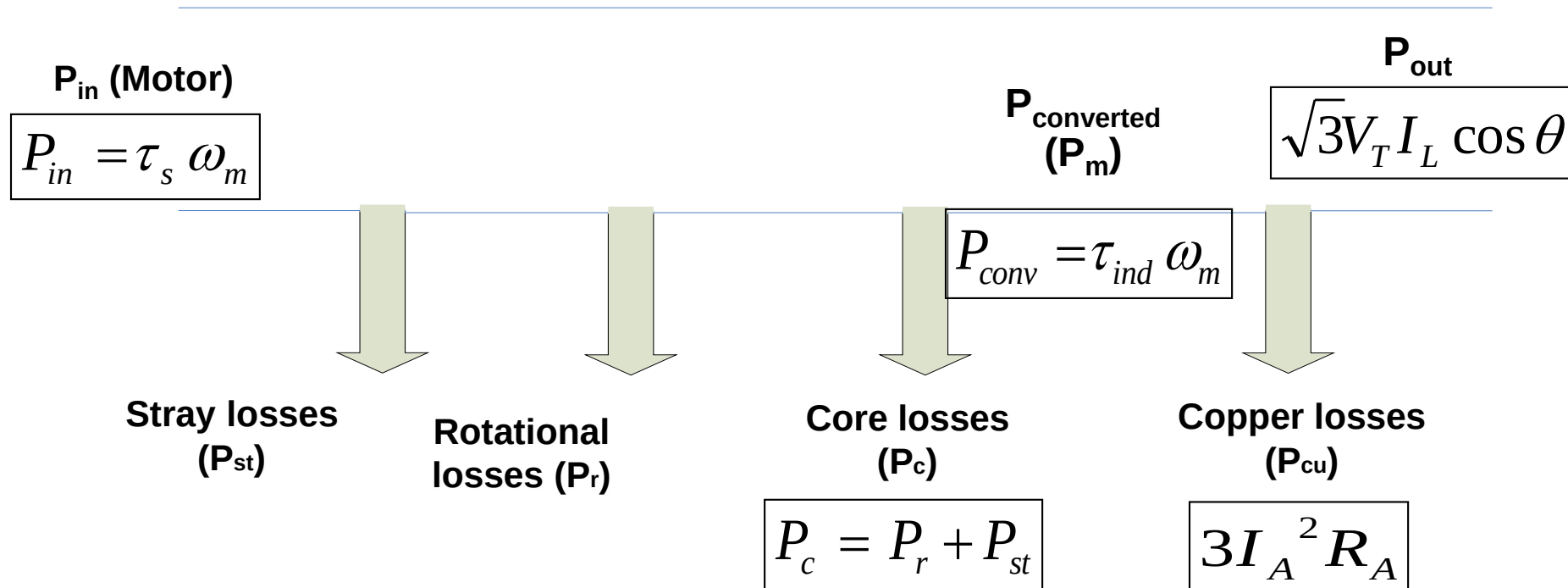


*Phasor diagram of a synchronous generator at unity power factor (purely resistive Load).*

*Notice that larger internal voltage is needed for lagging loads, therefore, larger field currents is needed with lagging loads to get same terminal voltage*

# Power Relationships

- Not all the mechanical power going into a synchronous generator becomes electrical power out of the machine. The difference between input power and output power represents the losses of the machine. The input mechanical power is the shaft power in the generator.



# Power Relationships

The power converted from mechanical to electrical is given by;

$$P_{conv} = \tau_{ind} \omega_m = 3E_A I_A \cos \gamma$$

Where  $\gamma$  is the angle between  $E_A$  and  $I_A$ :

If the armature resistance  $R_A$  is ignored ( $X_S \gg R_A$ ), Therefore:

$$I_A \cos \theta = \frac{E_A \sin \delta}{X_S}$$

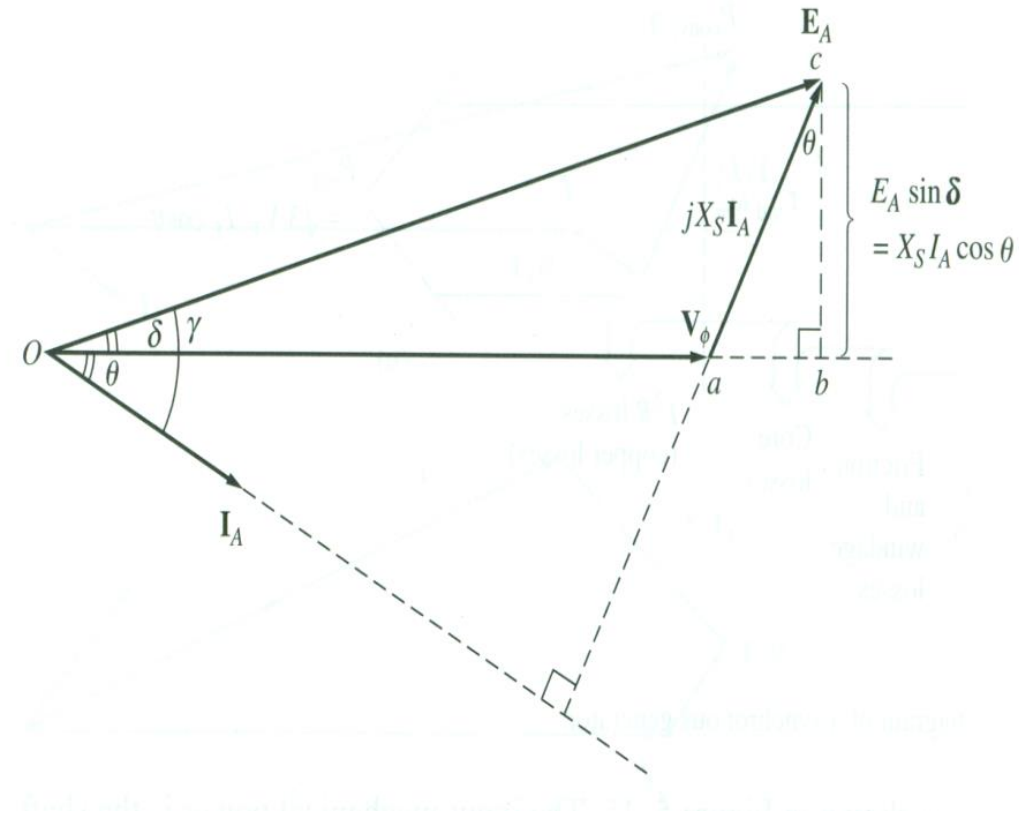
Substituting this equation into **Pout**, gives;

$$\therefore P = \frac{3V_\phi E_A \sin \delta}{X_S}$$

Where  $\delta$  is the angle between  $E_A$  and  $V_T$ .

The induced torque can be express as;

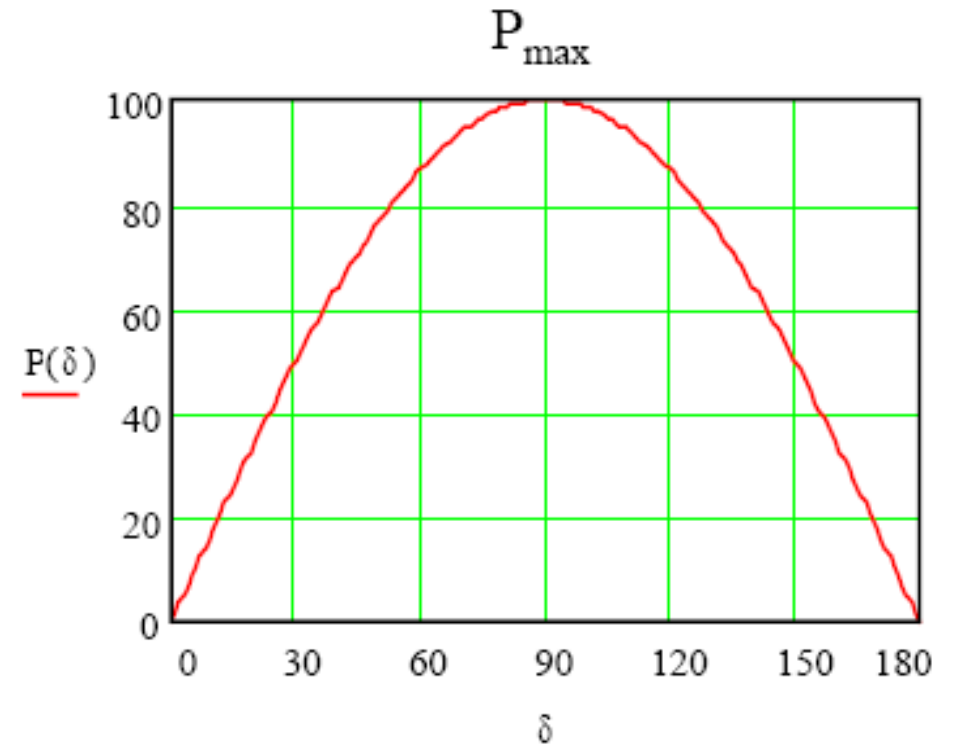
$$\therefore \tau_{ind} = \frac{3V_\phi E_A \sin \delta}{\omega_m X_S}$$



# Power Angle Characteristics

- The  $P(\delta)$  curve shows that the increase of power increases the angle between the induced voltage and the terminal voltage.
- The power is maximum when  $\delta=90^\circ$
- The further increase of input power forces the generator out of synchronism. This generates large current and mechanical forces.
- The maximum power is the **static stability limit** of the system.
- Safe operation requires a 15-20% power reserve.

## Round Rotor Machine



$$\therefore P_{\max} = \frac{3V_{\phi}E_A}{X_S}$$

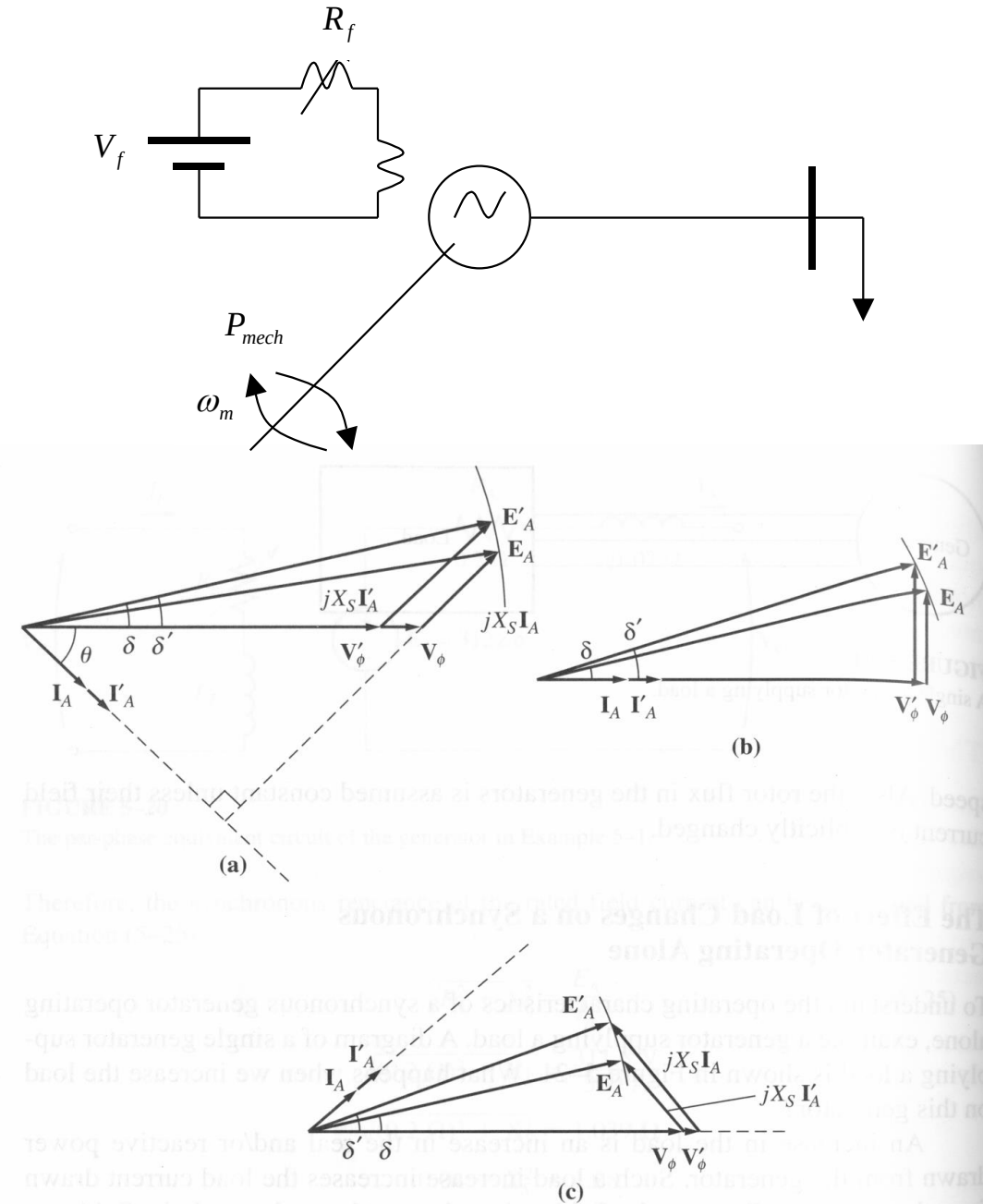
# Efficiency

$$\eta = \frac{P_{out}}{P_{in}} \times 100 \%$$

$$P_{in} = P_{out} + P_{losses}$$

# The Synchronous Generator Operating Alone- Variable Loads

- The behavior of Synch. Generator depend on the power factor of the load and whether the generator operating alone or parallel . By assuming SG operating alone , **what happens when we increase the load on this generator?**
  - At lagging power factor the increase of load current will decrease the terminal voltage significantly.
  - At unity power factor, the increase of load current will decrease the terminal voltage only slightly.
  - At leading power factor the increase of load current will increase the terminal voltage.



# Voltage Regulation

- As the load on the generator increases, the terminal voltage drops (lagging and unity PF loads cases). But, the terminal voltage, must be maintained constant, and hence the excitation on the machine is varied, or input power to the generator is varied. That means, EA has to be adjusted to keep the terminal voltage  $V_T$  constant.

∴ Voltage Regulation, VR;

$$\frac{V_{NL} - V_{FL}}{V_{FL}} \times 100 \%$$

- If SG operate at lagging power factor the VR is very high.(Positive voltage regulation).
- If SG operate at unity power factor just small positive VR
- At leading power factor VR is negative.

## How the terminal voltage is corrected?

Normally, it is desirable to keep the voltage supplied to a load constant, even though the load itself varies. How can terminal voltage variations be corrected for?

- Recall:  $V_\phi = E_A - jX_s I_A$   
and  $E_A = K\phi\omega$
- Since the frequency ( $\omega$ ) should not be changed, then  $\Phi$  must be changed.

### *The procedure:*

- Decreasing the field resistance will increase its field current.
- The increase of field current will increase the flux and increase the  $E_A$ , and the  $V_\phi$  will increase.

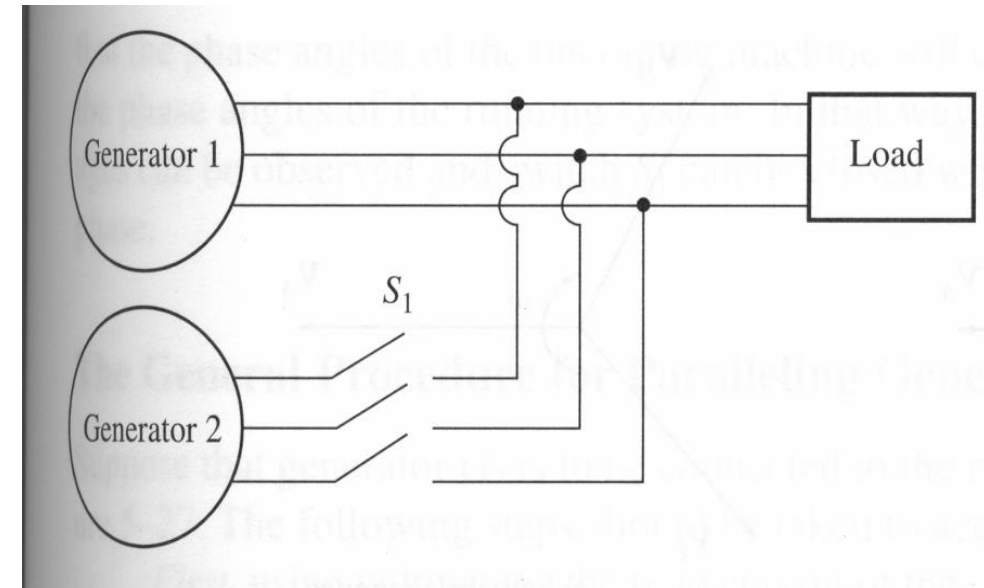
# Parallel Operation of Synch Generators

## Benefits:

- Increases the real and reactive power supply in the system.
- Increase the reliability of the power system.
- Allows shut down and preventive maintenance for some generators.
- Allows the operation near full load then maximum efficiency can be obtained.

**The following requirements have to be satisfied prior to connecting an alternator to other generator.**

1. The rms line voltage of the two generators must be equal.
2. The two generators must have the same phase sequence (aa' bb' cc').
3. The frequency of the oncoming alternator must be slightly higher than the frequency of the running system.



# The General Procedure for Paralleling Generators

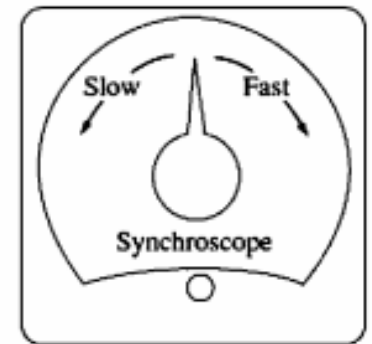
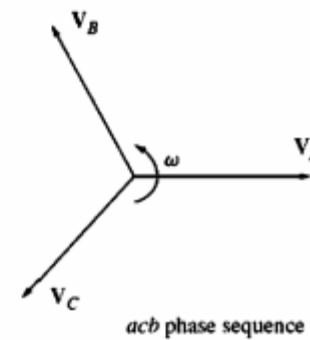
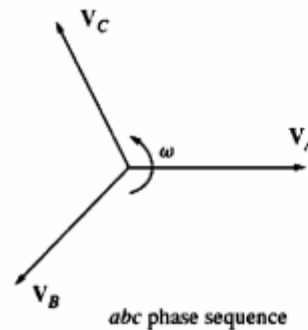
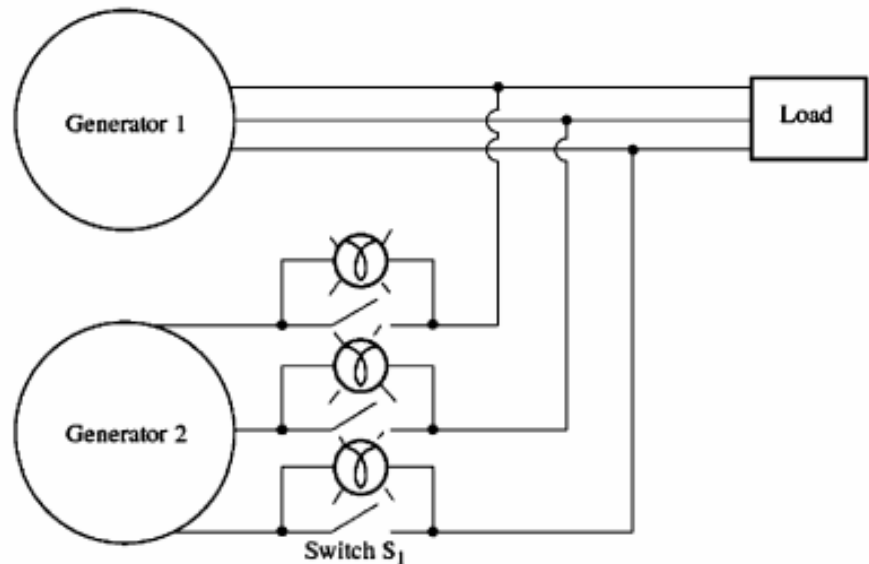
Suppose that generator  $G_2$  is to be connected to the running system shown in Figure below. The following steps should be taken to accomplish the paralleling.

- **First**, using voltmeters, the field current of the oncoming generator should be adjusted until its terminal voltage is equal to the line voltage of the running system.
- **Second**, the phase sequence of the oncoming generator must be compared to the phase sequence of the running system.
  - Alternately connect a small induction motor to the terminals of each of the two generators. If the motor rotates in the same direction each time, then the phase sequence is the same for both generators. If the motor rotates in opposite directions, then the phase sequences differ, and two of the conductors on the incoming generator must be reversed.
  - **The three-light-bulb method:** In this approach, three light bulbs are stretched across the open terminals of the switch connecting the generator to the system as shown in Figure 5- 27b. As the phase changes between the two systems, the light bulbs first get bright (large phase difference) and then get dim (small phase difference). If all three bulbs get bright and dark together, then the systems have the same phase sequence. If the bulbs brighten in succession, then the systems have the opposite phase sequence, and one of the sequences must be reversed.
- Next, the frequency of the oncoming generator is adjusted to be slightly higher than the frequency of the running system.

# The General Procedure for Paralleling Generators

Once the frequencies are very nearly equal, the voltages in the two systems will change phase with respect to each other very slowly. The phase changes are observed, and when the phase angles are equal, the switch connecting the two systems together is shut.

- How can one tell when the two systems are finally in phase? A simple way is to watch the three light bulbs described above in connection with the discussion of phase sequence. When the three light bulbs all go out, the voltage difference across them is zero and the systems are in phase. This simple scheme works, but it is not very accurate. A better approach is to employ a synchroscope.
- A synchroscope is a meter that measures the difference in phase angle between the phases of the two systems.



# Frequency Power and Voltage

**The speed droop of prime mover:**

$$SD = \frac{n_{nl} - n_{fl}}{n_{fl}} \times 100 \%$$

where:  $n_{nl}$  : No load speed  
 $n_{fl}$  : No load speed

**The relation between power and frequency:**

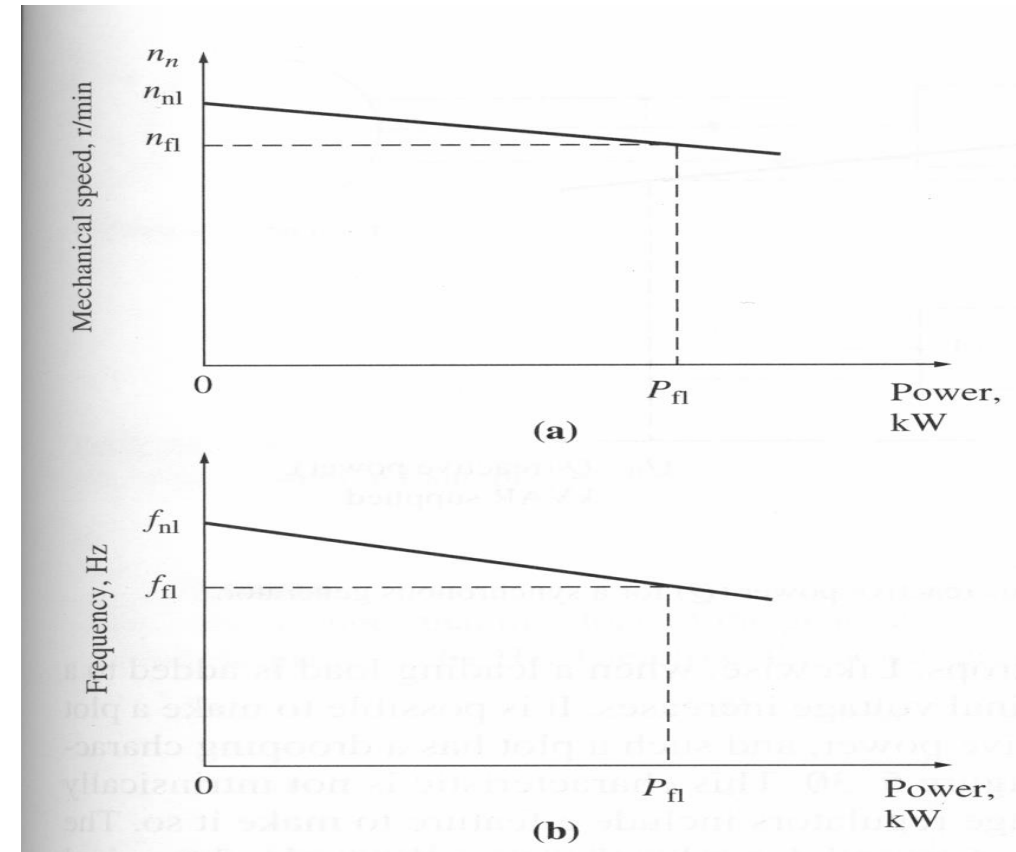
$$P = s_p (f_{nl} - f_{sys})$$

where: P: output power  
 $s_p$ : slope of the curve in kwh/Hz  
 $f_{nl}$ : No load frequency  
 $f_{fl}$ : Full load frequency

**The relation between reactive power and voltage:**

$$Q = s_p (V_{nl} - V_{sys})$$

Q: output reactive power  
 $s_p$ : slope of the curve in kvar/Hz  
 $V_{nl}$ : No load voltage  
 $V_{fl}$ : Full load voltage



# Connection with infinite bus

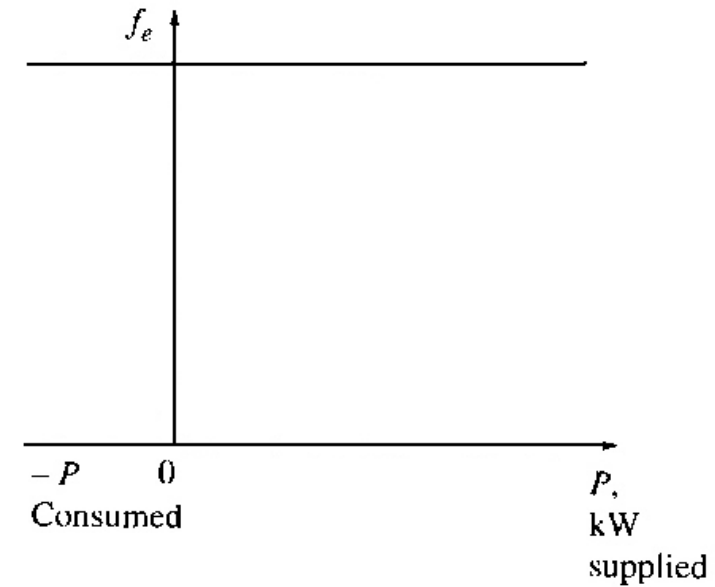
The following requirements have to be satisfied prior to connecting an alternator to the **infinite bus (connection line)**.

1. The line voltage of the (incoming) alternator must be equal to the constant voltage of the of the infinite bus.
2. The frequency of the incoming alternator must be exactly equal to that of the infinite bus.
3. The phase sequence of the incoming alternator must be identical to the phase sequence of the infinite bus.

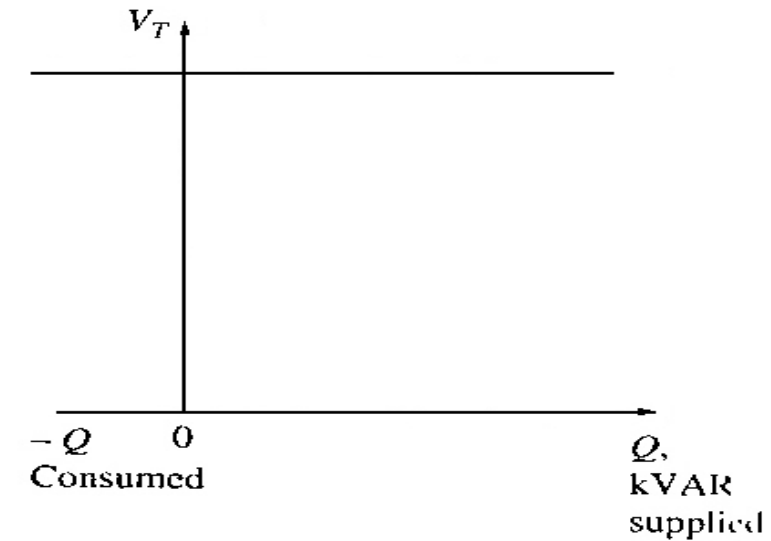
- When a syn. Gen. connected to a power system:

1-The real power versus frequency characteristic of such a system is shown in figure a

2-And the reactive power-voltage characteristic is shown in figure b

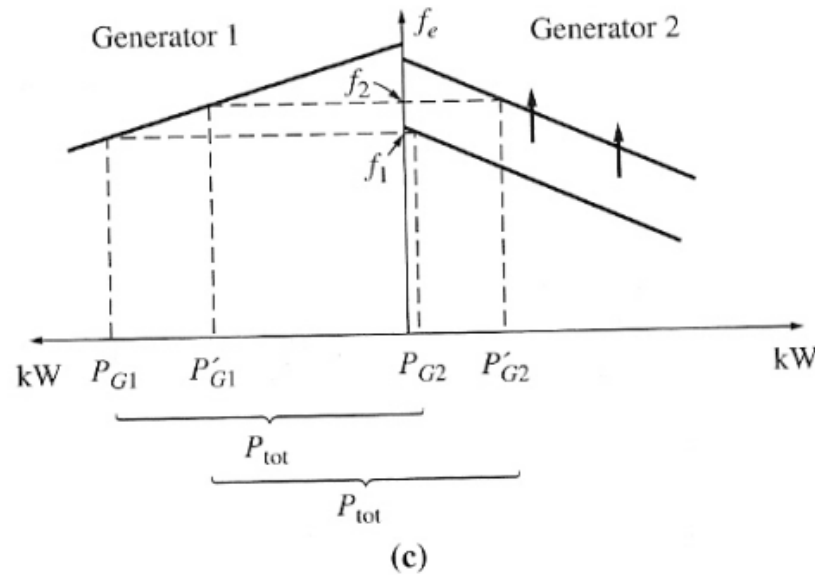


(a)



(b)

## Parallel operation of two Syn. generators of the same size. *(the governor set points of G2 are increased)*



The effect of increasing  $G_2$ 's governor set points on the operation of the system.

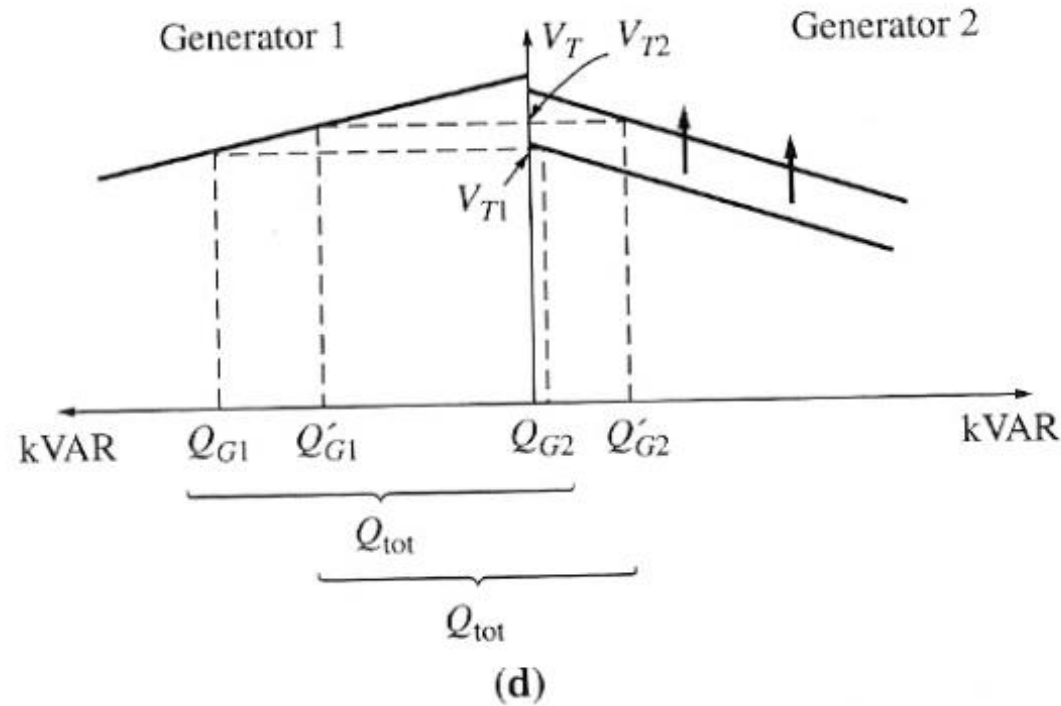
The total power supplied to the load must not change. At the original frequency  $f_1$ , the power supplied by  $G_1$  and  $G_2$  will now be larger than the load demand, so the system cannot continue to operate at the same frequency as before. In fact, there is only one frequency at which the sum of the powers out of the two generators is equal to  $P_{load}$ . That frequency  $f_2$  is higher than the original system operating frequency. At that frequency,  $G_2$  supplies more power than before, and  $G_1$  supplies less power than before.

Thus, when 2 generators are operating together, an increase in governor set points on one of them

1. increases the system frequency.
2. increases the power supplied by that generator, while reducing the power supplied by the other one.

## Parallel operation of two Syn. generators of the same size. *(the field current of G2 is increased)*

The resulting behaviour is analogous to the real-power situation as shown below:



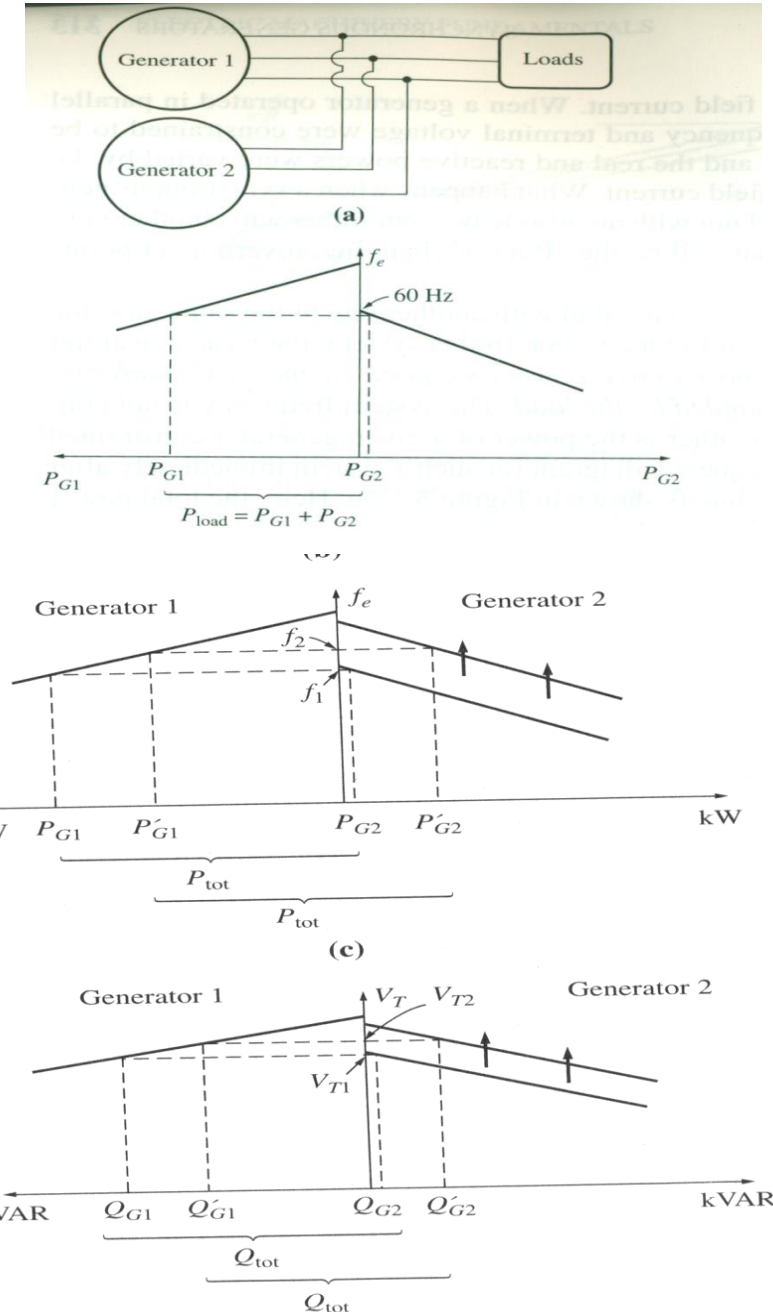
The effect of increasing  $G_2$ 's field current on the operating of the system.

When 2 generators are operating together and the field current of  $G_2$  is increased,

1. The system terminal voltage is increased.
2. The reactive power  $Q$  supplied by that generator is increased, while the reactive power supplied by the other generator is decreased.

# Parallel operation of two Syn. generators of the same size

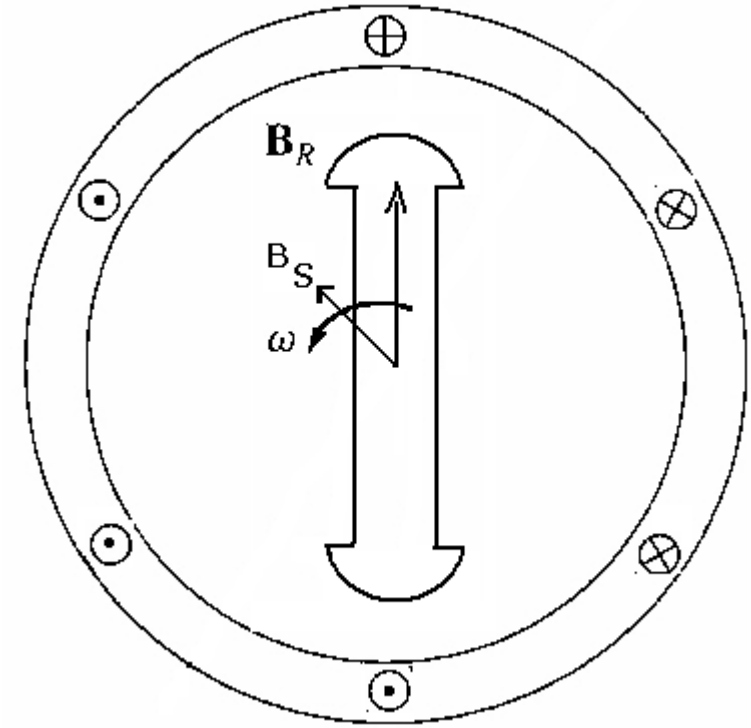
- The sum of the real and reactive powers supplied by the two generators must equal to the P and Q demanded by the load. **This will not change unless demand change**
- The system frequency is not constrained to constant, and neither is the power of a given generator is constrained to constant.
- The increase of the governor set point will increase the system frequency, **increase the real power supplied by G1 and reduce the power of second G2.**
- The increase of field current will increase the system terminal voltage, **increase reactive power of G1 and reduce reactive power of G2.**



# SYNCHRONOUS MOTOR

# SYNCHRONOUS MOTOR

- They are Synchronous machines employed to convert electric energy to mechanical energy
- To present the principles of Synchronous motor, a 2-pole synchronous motor considered
- The field current  $I_f$  of the motor produces a steady-state magnetic field  $B_R$ .
- A three phase voltage on the stator produces a three phase current flow in the windings.
- As a three phase set of currents in an armature produce uniform rotating magnetic field  $B_S$ . Therefore there are two magnetic fields present in the machine and the rotor field will tend to line up with the other.
- Since the stator magnetic field is rotating, the rotor magnetic field will constantly try to catch up.
- The larger the angle between two fields, the greater the torque.
- So, the rotor chases the rotating stator magnetic field but never quite catches up to it
- It has the same basic speed, power, & torque equations as Syn. Gen.



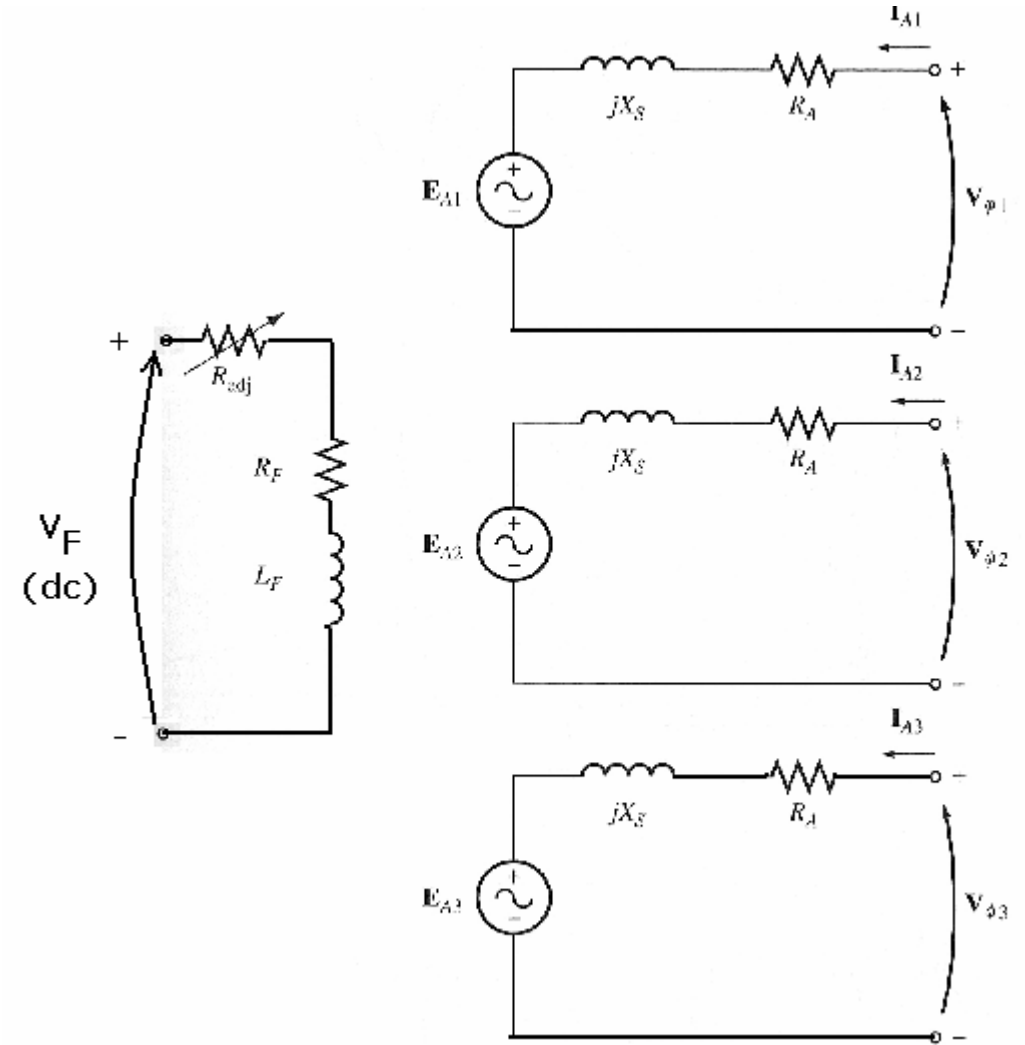
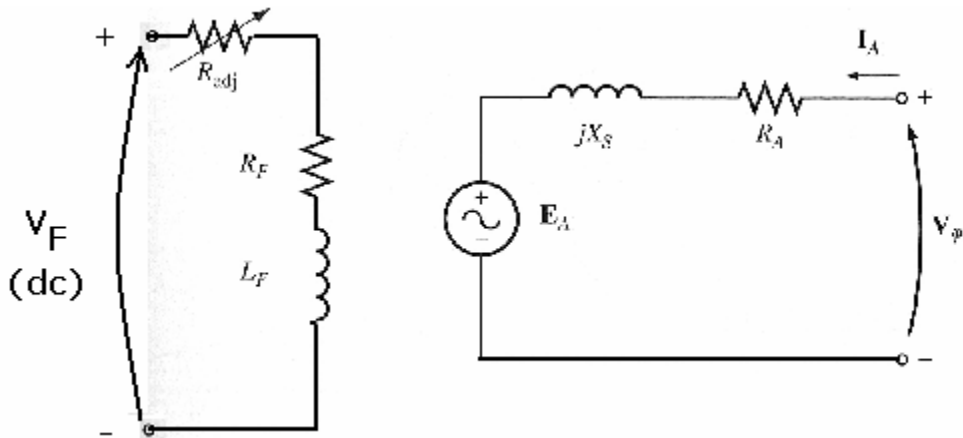
$$\tau_{ind} = k B_R \times B_S = \text{counterclockwise}$$

# Synchronous Motors Equivalent Circuit

- Syn. Motor is the same in all respects as Syn. Gen., except than direction of power flow
- Since the direction of power flow reversed, direction of current flow in stator of motor may also be reversed
- Therefore its equivalent circuit is exactly as Syn. Gen. equivalent circuit, except that the reference direction of  $I_A$  is reversed

• 3 phase Eq. cct. →

• Per phase Eq. cct.



# The Phasor Diagram and The Magnetic Field Diagram

- The related KVL equations:

$$V_{\phi} = E_A + jX_S I_A + R_A I_A$$

$$E_A = V_{\phi} - jX_S I_A - R_A I_A$$

- Operation From Magnetic Field Perspective**

- FIGURE (1) for Generator**

**FIGURE (2) for Motor**

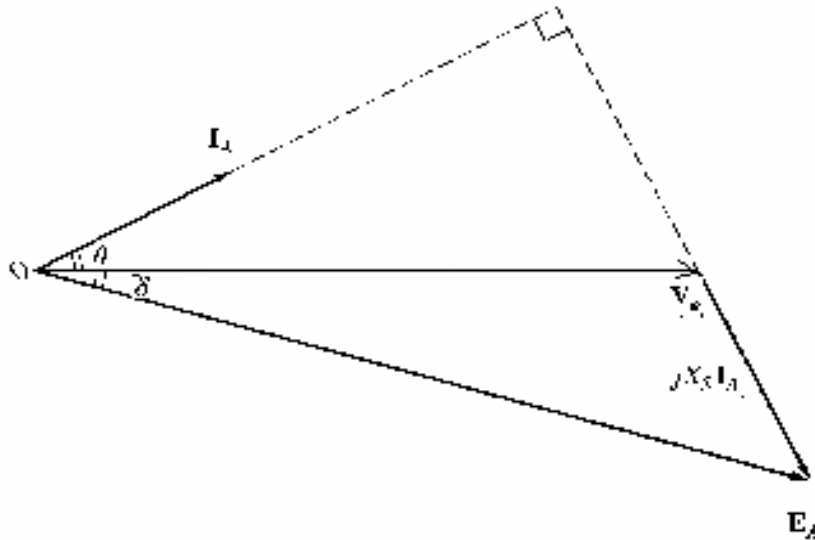


FIGURE (2)

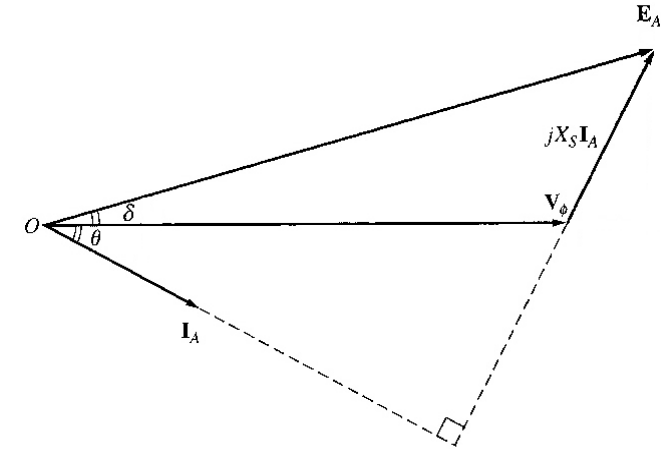
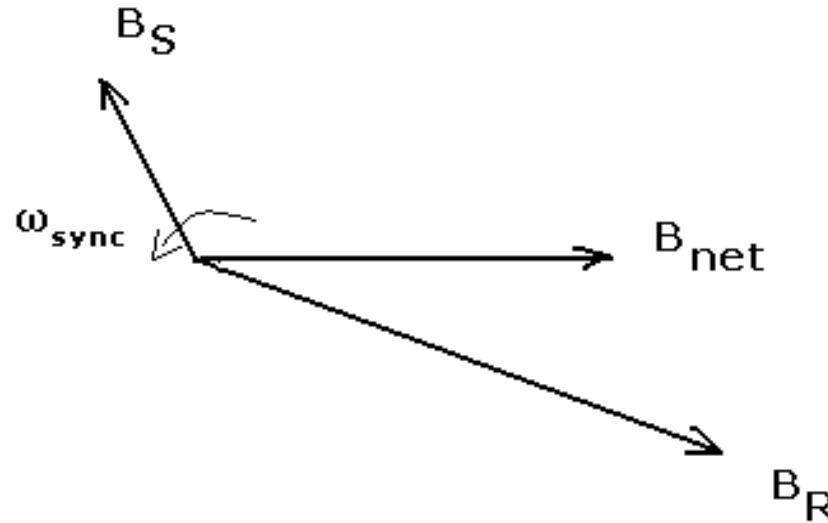


FIGURE (1)

# Synchronous Motors Operation

- Induced torque is given by:

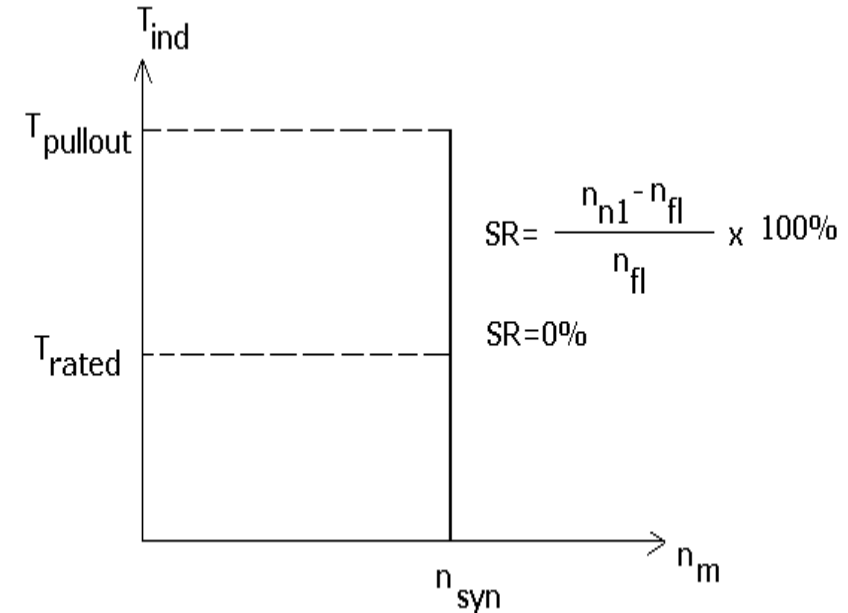
$$T_{ind} = k B_R \times B_{net} \quad (1)$$

$$T_{ind} = k B_R B_{net} \sin \delta \quad (2)$$

- Note: from magnetic field diagram, induced torque is clock wise, opposing direction of rotation in Generator related diagram
- In other words; induced torque in generator is a countertorque, opposing rotation caused by external applied torque  $T_{app}$ .
- Suppose, instead of turning shaft in direction of motion, prime mover lose power & starts to drag on machine's shaft
- What happens to machine? Rotor slows down because of drag on its shaft and falls behind net magnetic field in machine  $\rightarrow B_R$  slows down & falls behind  $B_{net}$ , operation of machine suddenly changes
- Using Equation (1), when  $B_R$  behind  $B_{net}$ , torque's direction reverses & become counterclockwise
- Now, machine's torque is in direction of motion
- Machine is acting as a motor
- With gradual increase of torque angle  $\delta$ , larger & larger torque develop in direction of rotation until finally motor's induced torque equals load torque on its shaft
- Then machine will operate at steady state & synchronous speed again, however as a motor

# Synchronous Motors Steady-state Operation

- Will study behavior of synchronous motors under varying conditions of load & field current , also its application to power-factor correction
- In discussions, armature resistance ignored for simplicity
- **Torque-Speed Characteristic**
- Syn. Motors supply power to loads that are constant speed devices
- Usually connected to power system, and power systems appear as infinite buses to motors
- Means that terminal voltage & system frequency will be constant regardless of amount of power drawn by motor
- Speed of rotation is locked to applied electrical frequency
- so speed of motor will be constant regardless of the load
- Resulting torque-speed characteristic curve is shown here →
- S.S. speed of motor is constant from no-load up to max. torque that motor can supply (named : pullout torque) so speed regulation of motor is 0%.
- Torque equation:
- $T_{ind} = k B_R B_{net} \sin \delta$
- $T_{ind} = 3 V_\phi E_A \sin \delta / (\omega_m X_S)$
- Pullout torque occurs when  $\delta = 90^\circ$
- Full load torque is much less than that, may typically be 1/3 of pullout torque
- When torque on shaft of syn. Motor exceeds pullout torque, rotor can not remain locked to stator & net magnetic fields. Instead rotor starts to slip behind them.



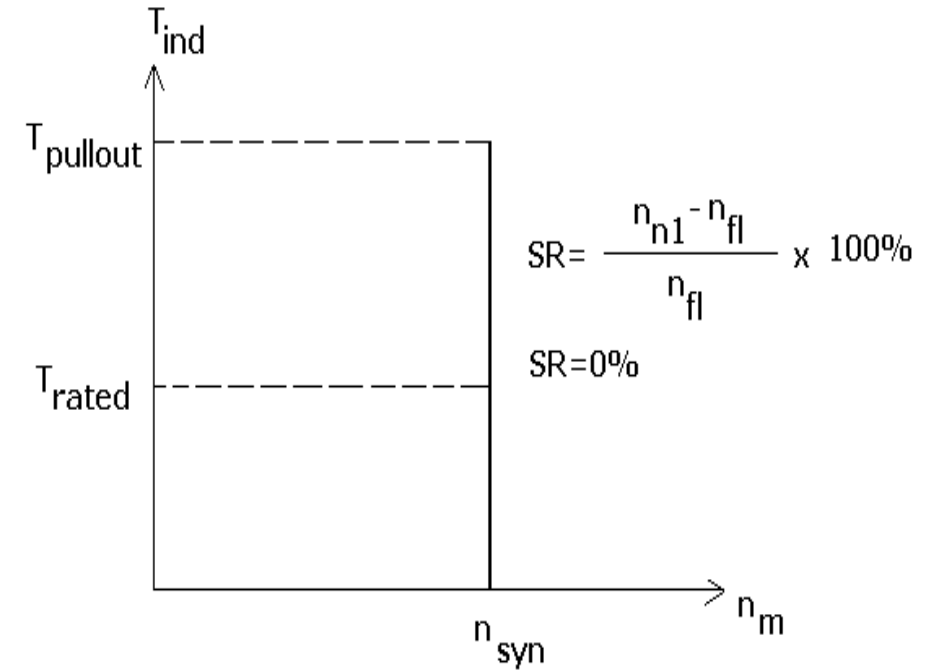
# SYNCHRONOUS GENERATOR CONSTRUCTION

- As rotor slows down, stator magnetic field “laps” it repeatedly, and direction of induced torque in rotor reverses with each pass
- Resulting huge torque surges, (which change direction sequentially) cause whole motor to vibrate severely
- Loss of synchronization after pullout torque is exceeded known as “slipping poles”
- Maximum or pullout torque of motor is:

$$T_{\max} = k B_R B_{\text{net}}$$

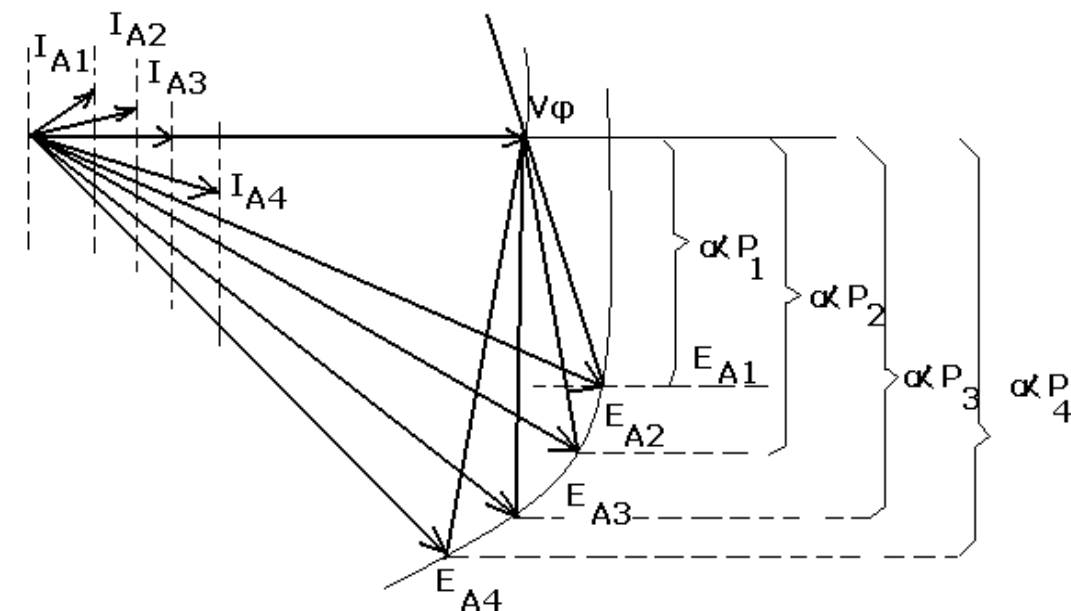
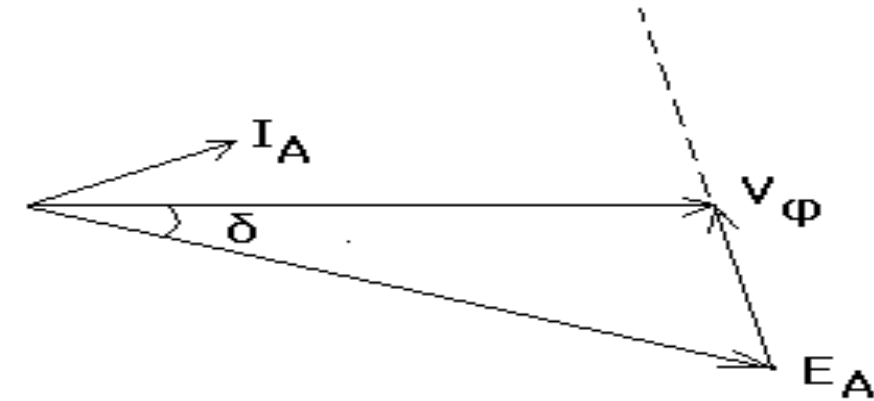
$$T_{\max} = 3 V_{\phi} E_A / (\omega_m X_S)$$

- From last equation, the larger the field current, larger  $E_A$ , the greater the torque of motor
- Therefore there is a stability advantage in operating motor with large field current or  $E_A$



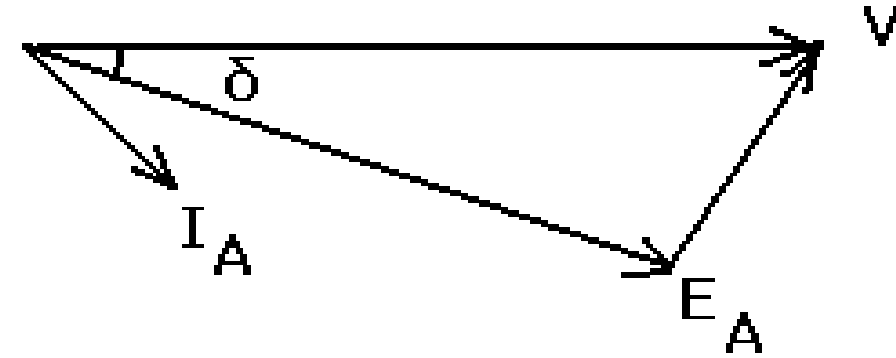
# Effect of load changes on motor operation

- Effect of load changes on motor operation
- when load attached to shaft, syn. Motor develop enough torque to keep motor & its load turning at syn. Speed
- Now if load changed on syn. motor, let examine a syn. motor operating initially with a leading power factor
- If load on shaft increased, rotor will initially slow down
- As it does, torque angle  $\delta$  becomes larger & induced torque increases
- Increase in induced torque speeds the rotor back up, & rotor again turns at syn. Speed but with a larger torque angle  $\delta$
- last figure show the phasor diagram before load increased
- Internal induced voltage  $E_A = K\phi\omega$  depends on field current & speed of machine
- Speed constrained to be constant by input power supply, and since no one changed field current it is also constant
- $|E_A|$  remain constant as load changes
- Distances proportional to power increase ( $E_A \sin\delta$  or  $I_A \cos\theta$ ) while  $E_A$  must remain constant
- As load increases  $E_A$  swings down as shown &  $jX_S I_A$  has to increase & Consequently  $I_A$  Increase, Note: p.f. angle  $\theta$  change too, causing less leading & gradually lagging



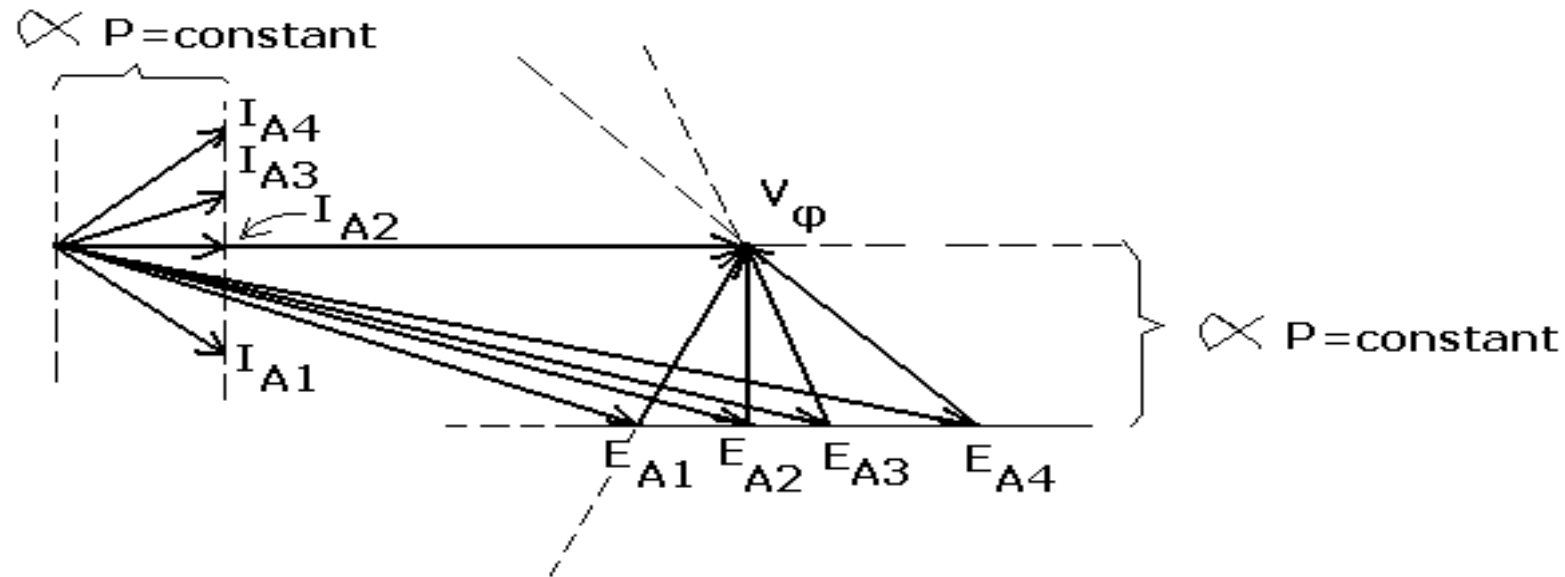
# Effect of Field current changes on a synchronous motor

- It was shown how change in shaft load affects motor torque angle and the supply current
- Effect of field current change:
- Above phasor diagram shows a motor operating at a lagging p.f.
- Now increase its  $I_F$  & see what happens to motor
- This will increase  $E_A$ , however don't affect real mechanical power supplied by motor. Since this power only changes when shaft load torque change
- Since change in  $I_F$  does not affect shaft speed  
and, since load attached to shaft is unchanged, real mechanical power supplied is unchanged
- $V_T$  is constant (by power source supply)
- power is proportional to following parameters in phasor diagram :  $E_A \sin \delta$  &  $I_A \cos \theta$  and must be constant
- When  $I_F$  increased,  $E_A$  must increase, however it can be done along line of constant power as shown in next slide



# Effect of Field current changes on a synchronous motor

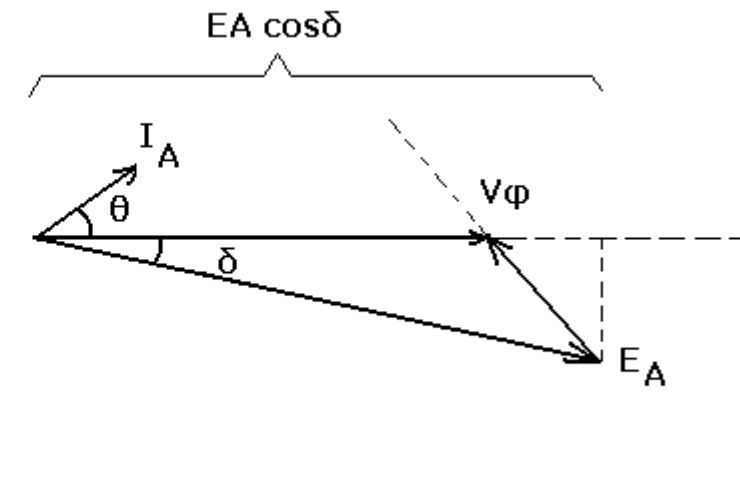
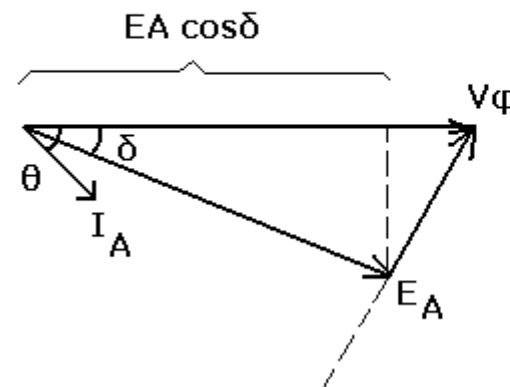
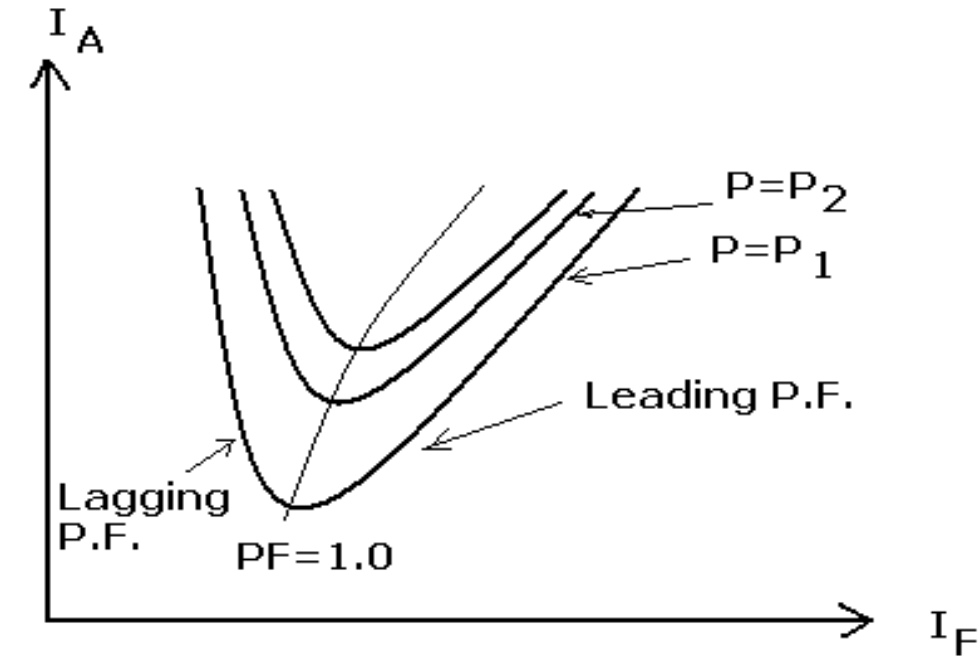
## Effect of an increase in field current



- Note: as  $E_A$  increases first  $I_A$  decreases and then increases again
- At low  $E_A$  armature current is lagging, and motor is an inductive load, consuming reactive power  $Q$
- As field current increased  $I_A$  lines up with  $V_\phi$  motor like a resistor, & as  $I_F$  increased further  $I_A$  become leading and motor become a capacitive load (capacitor-resistor) supplying reactive power

# Effect of Field current changes on a synchronous motor

- A plot of  $I_A$  versus  $I_F$  for a synchronous motor shown below known as V curves
- In each curve, minimum  $I_A$  occur at unity P.F. when only real power is being supplied to the motor
- for field current less than value related to  $I_{A,\min}$  armature current is lagging, consuming reactive power, while when field current is greater than value related to  $I_{A,\min}$  armature current is leading, and supplying Q to power system
- Therefore, by adjusting the field current, reactive power supplied to (or consumed by) power system can be controlled
- If  $E_A \cos\delta = V_\phi$  is the projection of  $E_A$  onto  $V_\phi$   $< V_\phi$  syn. motor has a lagging current & consumes Q & since  $I_f$  is small, is said to be: *underexcited*
- If  $E_A \cos\delta = V_\phi$  is the projection of  $E_A$  onto  $V_\phi$   $> V_\phi$  it has a leading current & supply Q (since  $I_f$  is large, motor named *overexcited*)

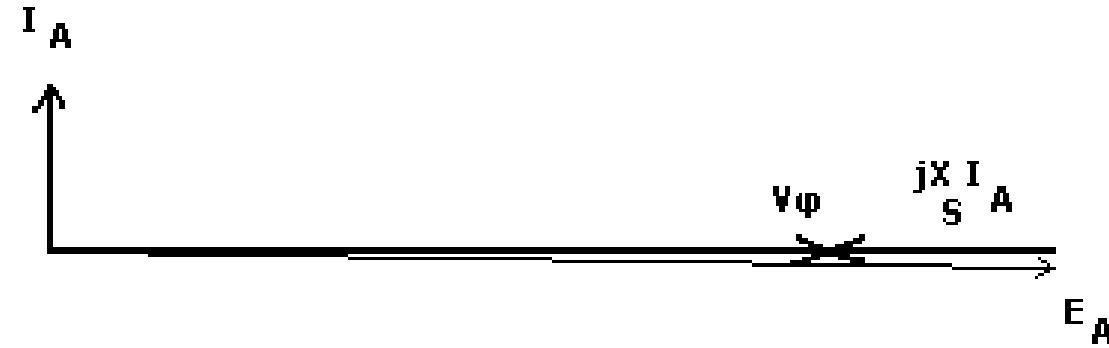


# SYNCHRONOUS MOTOR & PF CORRECTION

- Imagine an infinite bus supply an industrial plant containing several motors (as load), through a transmission line
- Two of loads are induction motors with lagging PF, and third load is a synchronous motor with variable PF
- What does the ability to set PF of one load, do for power system?
- The ability to adjust PF of loads in a power system affects operating efficiency significantly
- Most loads in typical power system are induction motors. So, lagging PF
- Having leading loads (overexcited syn. motor) is useful due to following reasons:
  - 1-leading load, supply reactive power  $Q$  for nearby lagging loads, instead of coming from Generators & thus transmission losses reduce.
  - 2- since less current pass transmission lines; lower rating and lower investment is required for a rated power flow
  - 3- requiring a synchronous motor to operate with leading PF means it must be overexcited this mode increase motor's maximum torque & reduces chance of accidentally pullout torque
- Application of syn. Motor or other equipment to improve PF is called *power-factor correction*
- Many loads that accept constant speed motor are driven by syn. Motors & due to PF correction capability save money in industrial plants
- Any syn. Motor exists in a plant run overexcited to improve PF & to increase its pullout torque, which need high  $I_f$  &  $\phi$  & heat

# Synchronous Capacitor or Synchronous Condenser

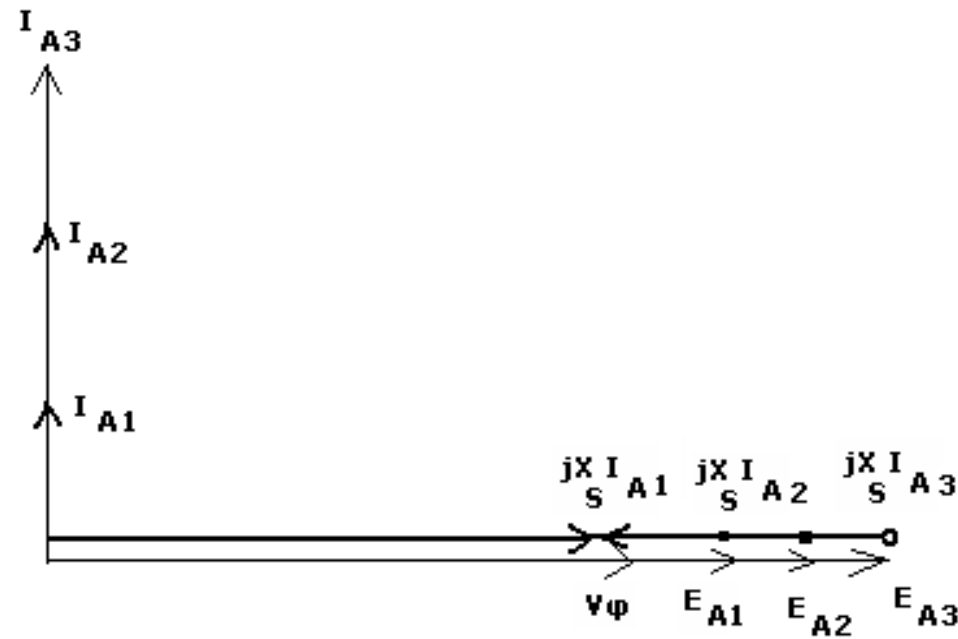
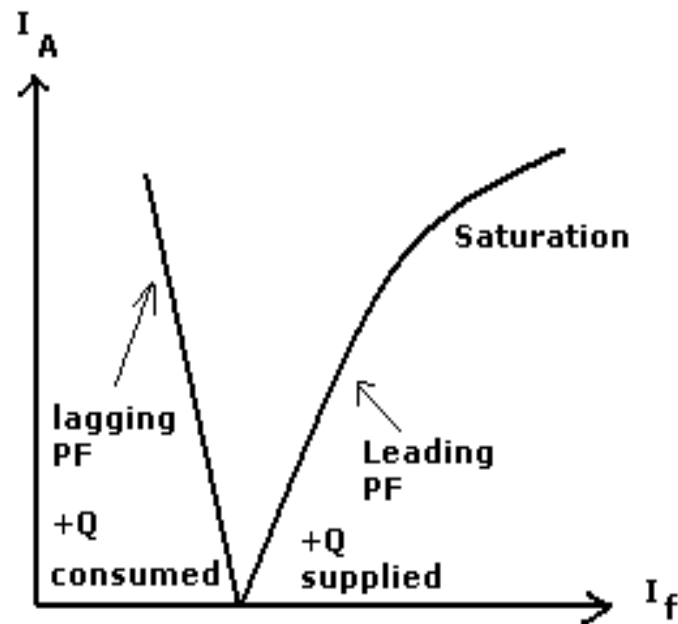
- A syn. Motor can operate overexcited to supply Q, some times in past syn. motor employed to run without a load & simply for power-correction
- This mode of operation shown in below phasor diagram



- $V_\phi = E_A + jX_S I_A$  since no power being drawn from motor  $\rightarrow EA \sin\delta$  &  $IA \cos\theta$  are zero
- $jX_S I_A$  points to left  $\rightarrow IA$  points straight up
- If  $V_\phi$  and  $I_A$  examined voltage-current relationship is similar to a capacitor
- Some syn. Motors are used specifically for PF correction & can not be connected to any load, these are called syn. Condensers or syn. Capacitors
- Today, conventional capacitors are more economical to be used than syn. Capacitors, however some syn. Capacitors may still be used in older industrial plants

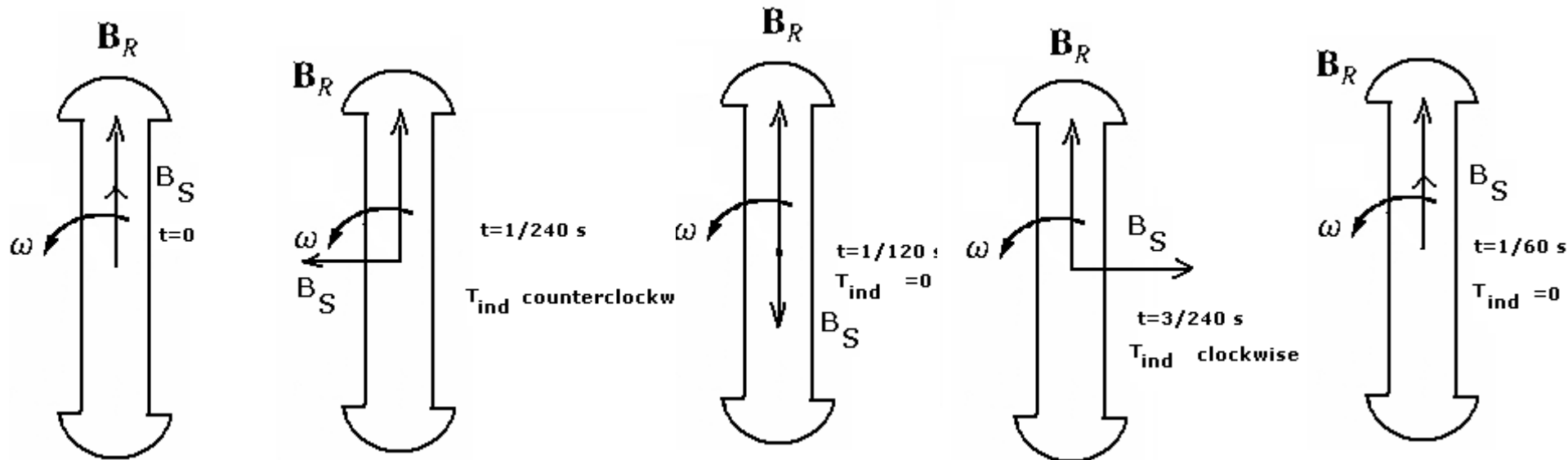
# Synchronous Capacitor or Synchronous Condenser

- V curve of Synchronous Capacitor & corresponding phasor Diagram
- Since real power supplied to machine zero (except for losses) at unity PF,  $I_A=0$



# STRATING SYNCHRONOUS MOTORS

- So far, in study of syn. motor, it is assumed that motor is initially turning at syn. Speed
- However, how did motor get to synchronous speed?
- realizing that for a 60 Hz motor at moment power applied to stator, rotor is stationary, and  $B_R$  is stationary
- Stator magnetic field  $B_S$  is starting to sweep around at syn. speed
- At  $t=0$   $B_R$  and  $B_S$  are exactly lined up.
- $T_{ind} = k B_R \times B_S$  would be zero
- Torque alternates rapidly in magnitude & direction,  $\rightarrow$  net starting torque is zero



# STARTING SYNCHRONOUS MOTORS

- The average torque over complete cycle cycle was zero
- Motor vibrates heavily with each electrical cycle & overheats
- This approach to syn. Motor starting is not satisfactory, burning up the expensive equipment
- The 3 basic approach to safely start the syn. Motor
- 1- *Reduce speed of stator magnetic field* to low enough value that rotor can accelerate & lock in with it during half-cycle of magnetic field rotation, by reducing frequency of applied electric power
- 2- *use an external prime mover* to accelerate motor up to syn. Speed, follow the paralleling procedure and bring machine on the line as a generator, turning off the prime mover will make syn. Machine a motor
- 3- *use damper windings or amortisseur windings.*

# STARTING SYNCHRONOUS MOTORS

## 1 – Starting by Reducing Electrical Frequency

- If stator B rotate at low enough speed, there will be no problem for rotor to accelerate & will lock in with stator
- Speed of BS then can be increased gradually to normal 50 or 60 Hz
- Shortcoming: how to provide a variable electrical frequency source, this needs a dedicated generator
- This requirement is obviously impractical.
- Today, rectifier-inverter & cycloconverter can be used to convert a constant frequency to any desired output frequency
- With modern solid-state variable frequency drive packages, it is perfectly possible to continuously control electrical frequency applied to motor from a fraction of Hz up to and above rated frequency
- If such a variable-frequency drive unit included in motor-control circuit to achieve speed control, then starting syn. motor is very easy
- When syn. Motor operated at a speed lower than rated speed, its internal generated voltage  $E_A = K\phi\omega$  will be smaller than normal
- & If  $E_A$  reduced, voltage applied to motor must reduced to keep stator current at safe levels.
- Voltage in any variable-frequency drive (or variable-frequency starter cct) must vary roughly linearly with applied frequency

# Effect of Field current changes on a synchronous motor

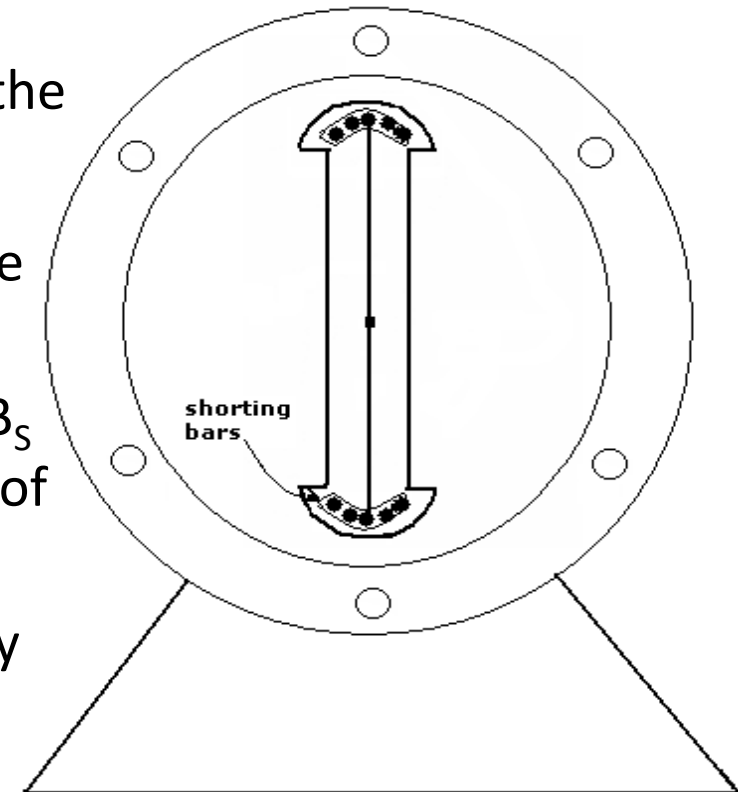
## 2- Starting With an External Prime Mover

- Attaching an external motor to it to bring syn. Machine up to full speed
- Then syn. Machine be paralleled with its power system as a generator
- Now starting motor can be detached from machine shaft, then its slow down
- BR fall behind Bnet & machine change its mode to be motor
- Once paralleling completed syn. Motor can be loaded down in an ordinary fashion
- Since starting motor should overcome inertia of syn. machine without a load & starting motor can have much smaller rating
- since most syn. motors have brushless excitation systems mounted on their shaft, often these exciters can be used as starting motors
- For many medium-size to large syn. motors, an external starting motor or starting by using exciter may be the only possible solution , because the connected power system source may not be able to feed the required starting current for amortisseur winding (next)

# STRATING SYNCHRONOUS MOTORS

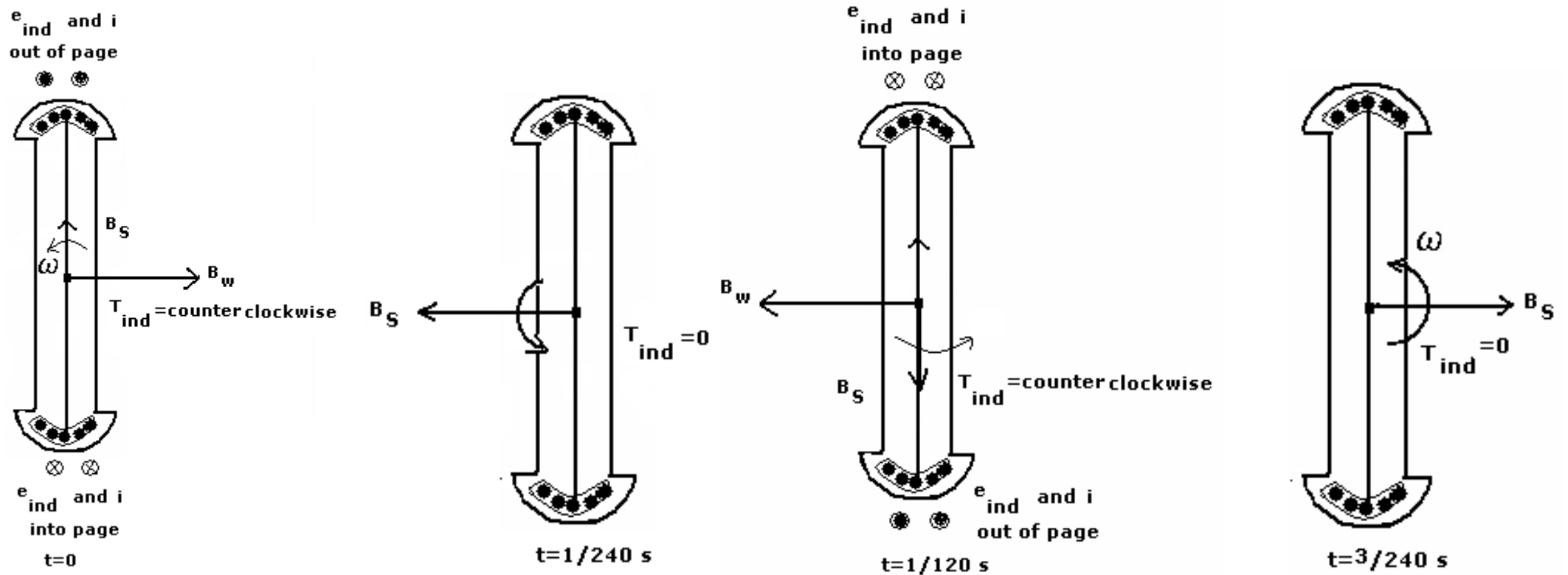
## 3- Starting by Using Amortisseur Windings

- most popular method is to employ amortisseur or damper winding
- amortisseur windings are special bars laid into notches carved in face of a syn. motor's rotor & then shorted out on each end by a large shorting ring
- pole face shown in next slide
- To understand what a set of amortisseur windings does in a syn. motor, examine salient 2 pole rotor shown next
- Simplified diagram of salient 2 pole machine (amortisseur winding with the shorting bars on the end of the two rotor poles connected by wires).
- Assume initially main rotor field winding is disconnected & that a 3 phase set of voltages applied to stator
- assume when power is first applied at  $t=0$ ,  $B_s$  is vertical as shown, & as  $B_s$  sweeps along in counter-clockwise direction, it induces a voltage in bars of amortisseur winding :
- $e_{ind} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{L}$  ;  $v$ =velocity of bar relative to  $B$  ;  $B$ =magnetic flux density vector and  $L$ =length of conductor magnetic field



# STRATING SYNCHRONOUS MOTORS

- Development of a unidirectional torque with syn. Motor amortisseur windings



- 1- at  $t=0$  Bars at top of rotor moving to right relative to magnetic field of stator, so induced voltage is out of page
- And similarly induced voltage in bottom bars into page
- Voltages produced a current flow out of top bars & into bottom bars, therefore this winding (bars) magnetic field  $B_w$  pointing to right
- Employing induced torque equation:  $T_{ind} = k B_w \times B_s$
- Direction of resulting torque on bars (& rotor) counterclockwise

# STRATING SYNCHRONOUS MOTORS

2- at  $t=1/240$  s,

$B_s$  now rotated  $90^\circ$ , while rotor has barely moved (simply can not speed up in short a time), since  $v$  is in parallel with  $B$  no induced voltage & current is zero

3- at  $t=1/120$  s

- stator magnetic field rotated  $90^\circ$  and is downward, and rotor still not moved
- Induced voltage in damping winding out of page in bottom bars & into page in top bars
- Resulting current also out of page in bottom bars & into page at top bars which cause BW pointing to left
- Induced torque :  $T_{ind} = k \mathbf{B}_w \times \mathbf{B}_s$   
is counterclockwise

4- at  $t=3/240$  s

- Here as  $t=1/240$  induced torque is zero
- Note: during these four steps, sometimes torque is counterclockwise & sometimes zero, however always unidirectional and the net is nonzero, motor speed up
- Although rotor speed up, never reach syn. Speed
- Since if rotor turn at syn. Speed, there would be no relative motion between rotor and  $B_s$  consequently induced voltage and passing current in bars zero and no torque will be developed to maintain rotor rotating, however it get close to syn. Speed, then regular field current turned on, and rotor will pull into step with stator magnetic fields

# STARTING SYNCHRONOUS MOTORS

- In Real Machines, field windings not open-circuited during starting procedure
- If field windings were O.C. then very high voltages would be produced during starting
- If field winding be sh. cct. During starting no dangerous voltage developed, and induced field current contribute extra torque to motor

## **Starting procedure for machines with amortisseur winding:**

- 1- disconnect field windings from dc power source & sh. Them
- 2- apply a 3 phase voltage to stator winding, let rotor accelerate up to near-syn. Speed, motor should have no load to get close to  $n_{syn}$
- 3- connect dc field circuit to its power source, after this motor get to syn. Speed and loads then may be added to shaft

## **Effects of Amortisseur Windings on Motor Stability**

- There is another advantage when there is an amortisseur winding, i.e. increase machine stability
- Stator magnetic field rotates at a constant speed  $n_{syn}$  which varies only when system frequency varies
- If rotor turns at  $n_{syn}$  amortisseur winding have no induced voltage
- If rotor turn slower than  $n_{syn}$  there will be relative motion between rotor &  $B_s$  & a voltage will be induced, consequently current pass and magnetic field produced that develop a torque which tend to speed machine up again
- On the other hand if rotor turn faster than  $B_s$  a torque develop to slow rotor down. These windings dampen out load or other transients on machine and this the reason that this winding named Damping Winding

# Motors VS Generators

1- syn. Gen.  $E_A$  lies ahead of  $V_\phi$  while for motor  $E_A$  lies behind  $V_\phi$

2- machine supplying  $Q$  have  $E_A \cos \delta > V_\phi$  (regardless of being motor or generator) and machine consuming reactive power  $Q$  has  $E_A \cos \delta < V_\phi$

**Synchronous motors commonly used for low speed , high power loads**

When connected to power system, frequency and terminal

voltage of syn. motor is fixed

$$n_m = n_{\text{sync}} = 120 f_e / p$$

$$P_{\text{max}} = 3 V_\phi E_A / X_s$$

this is maximum power of machine and if exceeded, motor slip poles

|                                                       | Supply reactive power $Q$<br>$E_A \cos \delta > V_\phi$ | Consume reactive power $Q$<br>$E_A \cos \delta < V_\phi$ |
|-------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|
| Supply power $P$<br>Generator<br>$E_A$ leads $V_\phi$ |                                                         |                                                          |
| Consume power $P$<br>Motor<br>$E_A$ lags $V_\phi$     |                                                         |                                                          |